

Chapter 14 : Operational Amplifiers

14.1 Introduction

So far, we have studied many circuits consisting of discrete components. In this chapter, we will study the most important analog *integrated circuit* (IC), namely, the operational amplifier or shortly op-amp. An **operational amplifier** is a high-gain amplifier circuit with two high-impedance input terminals and one low-impedance output terminal. Nowadays operational amplifiers are constructed using integrated circuit technology and are referred to as basic linear or analog IC. As operational amplifiers are built in IC, they possess many advantages like: small size, low cost, high reliability, low offset voltage and current, temperature tracking properties etc. Although some high-power op-amps are available, most are low-power devices with a maximum power rating of less than a watt. Some op-amps are optimized for bandwidth, others for low input offsets, others for low noise, and so on. This is why the variety of commercially available op-amps is so large. The objective of this chapter is to expose the reader to some terminal characteristics and some circuit applications of op-amps.

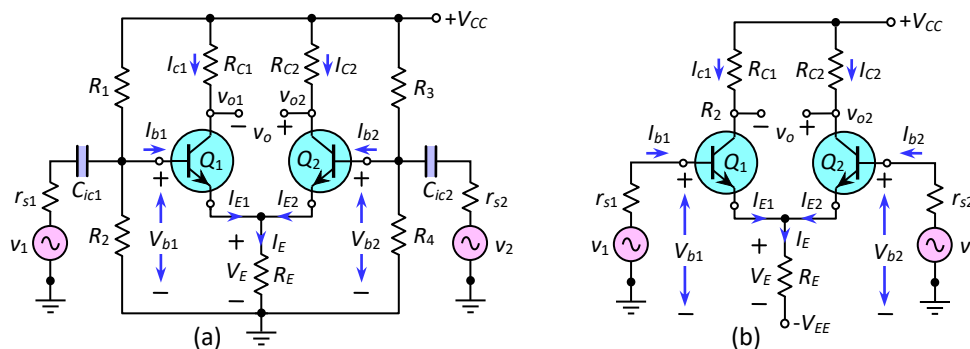


Fig.14-1: Differential amplifier using BJT with: (a) Single power supply, and (b) Split power supply

The main use of an op-amp is to amplify AC and DC input signals and was initially used for basic mathematical operations such as addition, subtraction, multiplication, differentiation and integration. As many mathematical operations can be done using this amplifier, it is called the *operational amplifier*. Nowadays, the application of op-amps varies from AC and DC signal amplification to use in active filters, oscillators, comparators, voltage regulators, instrumentation and control systems, pulse generators, square wave generators and many more electronic circuits.

The operational amplifier circuit consists of differential amplifier, active load, current mirror, and emitter follower circuits. First of all we will study these circuits.

14.2 Differential Amplifier Using BJT

The *differential amplifier* is a high gain amplifier that amplifies the difference of two input signals applied to its two input terminals. Differential amplifier can be designed either using BJT or FET. However, due to the lower gain of FET circuits, BJT is most common in differential amplifier. Differential amplifiers are widely used in IC, because it has both bias stability and good voltage gain without the use of large bypass capacitors. Fig.14-1 shows the circuit diagram of differential amplifiers using BJTs. The first circuit [Fig.14-1(a)] is designed for single power supply so the transistors are biased with voltage divider networks. The second circuit [Fig.14-1(b)] is designed for split power supply so the signal sources are connected directly to the inputs (bases) of the transistors. The emitters of both Q_1 and Q_2 are connected to a common resistor (R_E), so that, the outputs v_{o1} and v_{o2} get affected by the input signals v_1 and v_2 . For this reason, this circuit is also called an *emitter-coupled amplifier*. Input signal can be applied to both the input terminals (called dual input) or to either of the two input terminals (called single-input). Similarly output can be taken from the collector of one transistor with respect to ground (called unbalanced output) or as the voltage developed across the two collectors of the transistors (called balanced output). Thus, the differential amplifier can be used in four modes:

- Dual-input balanced output
- Dual-input unbalanced output
- Single-input balanced output
- Single-input unbalanced output

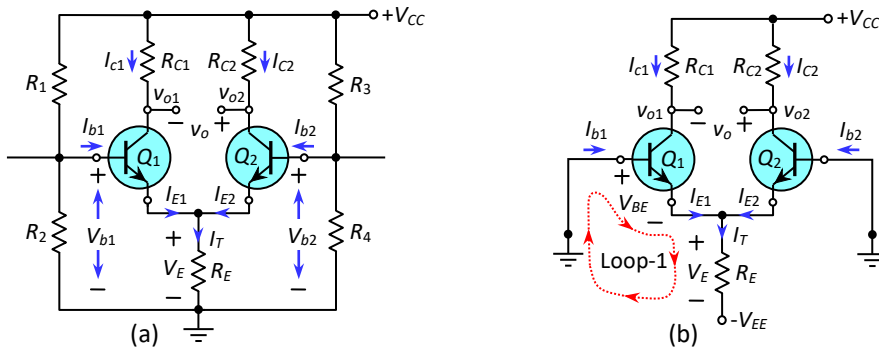


Fig.14-2: Operating point of differential amplifier: (a) Single power supply, and (b) Split power supply

14.2.1 DC Operating Point and Tail Current

In a differential amplifier, transistors Q_1 and Q_2 are made identical in all respects. To determine the DC operating point, we have to short the signal sources and open the coupling capacitors as shown in Fig.14-2. In the circuit of Fig.14-2(a), the voltage divider networks will set the base voltages while in Fig.14-2(b), the base voltages will be set by the split power supply. The biasing components are identical, that is, $R_1 = R_3$, $R_2 = R_4$, and $R_{C1} = R_{C2} = R_C$. Therefore, V_{B1} and V_{B2} will be equal. If V_{B1} and V_{B2} are equal, I_{E1} and I_{E2} will also be equal as the transistors are perfectly matched. The current flowing through the emitter resistor is called the *tail current* I_T . From the circuits we find,

$$I_T = I_{E1} + I_{E2}$$

or,

$$I_{E1} = I_{E2} = I_E = \frac{I_T}{2} \quad (14-1)$$

For the circuit of Fig.14-2(a),

The base voltage will be,

$$V_{B1} = V_{B2} = V_B = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{V_{CC}R_4}{R_3 + R_4} \quad (14-2)$$

The emitter voltage can be calculated as,

$$V_E = V_B - V_{BE}$$

or,

$$I_T R_E = V_B - V_{BE}$$

∴

$$I_T = \frac{V_B - V_{BE}}{R_E} \quad (14-3)$$

Using Equ.(14-1) and Equ.(14-3) we get,

$$I_{E1} = I_{E2} = I_E = \frac{V_B - V_{BE}}{2R_E} \approx I_{C1} = I_{C2} = I_C \quad (14-4)$$

The collector voltage will be,

$$V_{C1} = V_{C2} = V_C = V_{CC} - I_C R_C \quad (14-5)$$

Therefore, the collector-emitter voltage will be

$$V_{CE1} = V_{CE2} = V_{CE} = V_C - V_E$$

∴

$$V_{CE1} = V_{CE2} = V_{CE} = V_C - I_T R_E \quad (14-6)$$

Using Equ.(14-4) and Equ.(14-6) the DC operating point of Fig.14-2(a) can be determined.

For the circuit of Fig.14-2(b):

Using KVL in the input loop (Loop-1), we get,

$$-V_{BE} - I_T R_E + |V_{EE}| = 0$$

or,

$$I_T = \frac{|V_{EE}| - V_{BE}}{R_E} \quad (14-7)$$

From Equ.(14-1) and Equ.(14-7) we can write,

$$I_{E1} = I_{E2} = I_E = \frac{|V_{EE}| - V_{BE}}{2R_E} \approx I_{C1} = I_{C2} = I_C \quad (14-8)$$

The collector voltage will be,

$$V_{C1} = V_{C2} = V_C = V_{CC} - I_C R_C \quad (14-9)$$

Therefore, the collector-emitter voltage will be

$$V_{CE1} = V_{CE2} = V_{CE} = V_C - V_E$$

But for this circuit the emitter voltage will be, $V_E = -V_{BE}$

$$\therefore \boxed{V_{CE1} = V_{CE2} = V_{CE} = V_C + V_{BE}} \quad (14-10)$$

From Equ.(14-8), and Equ.(14-10) the DC operating point for the circuit of Fig.14-2(b) can be determined.

In Chapter 4, it is found that, voltage divider bias gives the most stability of the DC operating point. Therefore, the operating point of the differential amplifier will remain constant irrespective to the change in temperature and h_{fe} .

The emitter currents of the transistors, I_{E1} and I_{E2} will be identical only when the devices are perfectly matched. But it is difficult to fabricate fully matched devices. Therefore, I_{E1} and I_{E2} may not be equal. To adjust the variation in I_{E1} and I_{E2} , a small-value potentiometer is sometimes included between the emitters as shown in Fig.14-3. Adjustment of the potentiometer R_P will increase the resistance to the emitter of one transistor and that of the other transistor will decrease. Therefore, the emitter current of one transistor will decrease and that of the other transistor will increase. But the total tail current I_T will remain constant.

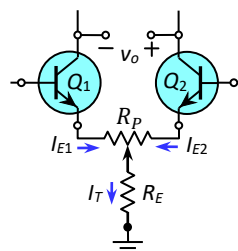


Fig.14-3: Potentiometer used to adjust I_{E1} and I_{E2}

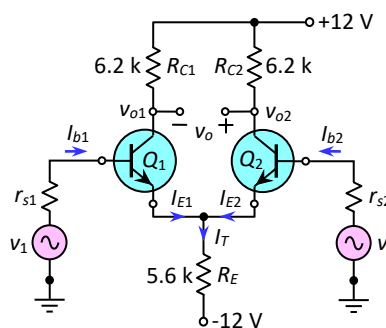


Fig.14-4: Differential amplifier for Example 13-1

Example 13-1

For the differential amplifier of Fig.14-4 determine the DC operating point and the value of tail current.

Solution:

To determine the DC operating point, we have to short the signal sources, that is, the bases are connected to ground. This circuit is same as that of Fig.14-2(b).

$$\text{Using Equ.(14-7),} \quad I_T = \frac{|V_{EE}| - V_{BE}}{R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{5.6 \text{ k}} \approx 2 \text{ mA [Ans.]}$$

$$\text{Using Equ.(14-1),} \quad I_{E1} = I_{E2} = \frac{I_T}{2} = \frac{2 \text{ mA}}{2} = 1 \text{ mA}$$

$$\therefore \quad I_{C1} = I_{C2} = I_C \approx 1 \text{ mA [Ans.]}$$

Using Equ.(14-9), $V_{C1} = V_{C2} = V_{CC} - I_C R_C = 12 \text{ V} - 1 \text{ mA} \times 6.2 \text{ k} = 5.8 \text{ V}$

Using Equ.(14-10), $V_{CE1} = V_{CE2} = V_C + V_{BE} = 5.8 \text{ V} + 0.7 \text{ V} = 6.5 \text{ V}$ [Ans.]

Comments: Remember that, the voltage dropped across R_E is 11.3 V but the emitter voltage is -0.7 V with respect to ground.

14.3 AC Operation of Differential Amplifier

From Fig.14-1, it is clear that the amplifiers are emitter-coupled CE amplifiers. First we will consider the effect of v_1 assuming $v_2 = 0$ as shown in Fig.14-5. When the input voltage v_1 increases in positive direction, collector current i_{C1} increases and the voltage drop across R_{C1} increases. So the output voltage (v_{o1}), taken from the collector of Q_1 , will decrease due to

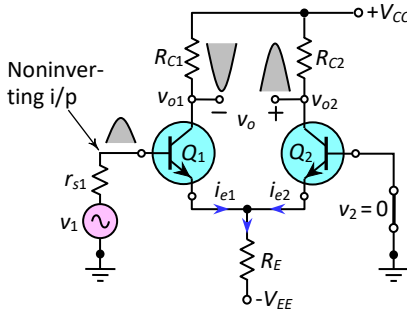


Fig.14-5: (a) A positive half-cycle at the input of Q_1 , produces negative half-cycle at Q_{C1} and positive half-cycle at Q_{C2}

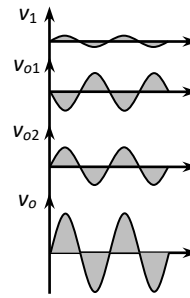


Fig.14-5: (b) Differential output, v_o , is in phase with the input v_1

180° phase shift of CE amplifier. But with the increase in v_1 , i_{e1} will also increase. So the voltage drop across R_E will increase. As the base voltage of Q_2 is fixed to zero or ground, with the increase emitter voltage, v_{be2} will decrease and hence i_{c2} will decrease. With the decrease in i_{c2} , voltage drop across R_{C2} will decrease and hence $v_{c2} (= v_{o2})$ will increase. Thus the positive half-cycle of v_1 will be amplified by both the transistors and v_{o1} will be 180° out of phase but v_{o2} will be in phase with v_1 as shown in Fig.14-5(a). The individual output signals and the differential output signal for the input signal v_1 are shown in Fig.14-5(b). From this figure, it is clear that the overall output, $v_o (= v_{o2} - v_{o1})$ will be just double of the individual output (v_{o1} or v_{o2}) and in phase with the input signal v_1 .

Now, we will consider the effect of v_2 assuming $v_1 = 0$ as shown in Fig.14-6. When the input voltage v_2 increases in positive direction, collector current i_{c2} increases and the voltage drop across R_{C2} increases. So the output voltage (v_{o2}), will go in negative direction. Here also, with the increase in v_2 , i_{e2} will increase. So the voltage drop across R_E will increase. As the base voltage of Q_1 is fixed by the biasing network, with the increase in emitter voltage, v_{be1} will decrease and hence i_{c1} will decrease. With the decrease in i_{c1} , voltage drop across R_{C1} will decrease and hence $v_{o1} (= v_{c2})$ will increase. Thus, the positive half-cycle of v_2 will be amplified by both the transistors and v_{o2} will be 180° out of phase but v_{o1} will be in phase with v_2 as shown in Fig.14-6(a), and

Fig.14-6(b). From Fig.14-6(b), again we find that the output voltage, $v_o (= v_{o2} - v_{o1})$ will be just double of the individual output (v_{o1} or v_{o2}) and is out of phase with the input signal v_2 .

From this AC operation of differential amplifier, we find that, if the signal (v_1) is applied to the input of Q_1 , the output (v_o) is not inverted (in phase with v_1). However, if the signal (v_2) is applied to the input of Q_2 , the output (v_o) is inverted (out of phase with v_2). For this reason, the input (base) of Q_1 is called *noninverting input terminal* and the input (base) of Q_2 is called *inverting input terminal*.

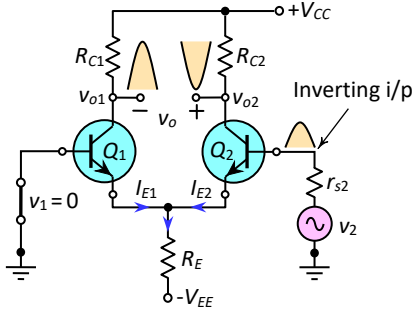


Fig.14-6:(a) A positive half-cycle at the input of Q_2 produces positive half-cycle at Q_{C1} and negative half-cycle at Q_{C2}

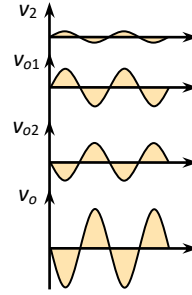


Fig.14-6:(b) Differential output, v_o , is out of phase with the input v_2

14.3.1 Voltage Gain

In Chapter 5, it is shown that the voltage gain of a CE amplifier with unbypassed emitter resistance and without load ($R_L = \infty$) is,

$$\text{From Eq.(5-72),} \quad A_v = -\frac{h_{fe}R_C}{h_{ie} + (1 + h_{fe})R_E} \quad (14-11)$$

From Fig.14-7, it is found that the resistance looking into the emitter of Q_2 is h_{ib} . So, the equivalent unbypassed emitter resistance for the first CE amplifier (with Q_1) will be $R_E || h_{ib}$. However, R_E is very large compared to h_{ib} . So we can neglect R_E . Therefore, the effective unbypassed resistance to the emitter of Q_1 will be just h_{ib} . Putting this value into Eq.(14-11) we find,

$$A_v = -\frac{h_{fe}R_C}{h_{ie} + (1 + h_{fe})h_{ib}}$$

But we know that, $(1 + h_{fe})h_{ib} = h_{ie}$. So,

$$\boxed{A_v = -\frac{h_{fe}R_C}{2h_{ie}}} \quad (14-12)$$

Equ.(14-12) gives the voltage gain from the base of Q_1 to its collector. This equation will also be applicable if the signal is applied to the base of Q_2 and the output is taken from the collector of Q_2 . In this circuit as the emitter resistor is unbypassed, the voltage gain should have been as low as given by Equ.(14-11), but the calculated voltage gain given by Equ.(14-12) is just half the voltage gain of a single-stage CE amplifier with R_E bypassed by a capacitor [see Equ.(5-54)]. Thus, we can say that the differential amplifier provides high voltage gain even though there used no bypass capacitor. This is an important advantage of differential amplifier, because bypass capacitors are large and expensive.

Therefore, if a voltage v_1 is applied to the base of Q_1 the output voltage at its collector (v_{c1}) will be, $v_{o1} = -v_1 h_{fe} R_C / 2h_{ie}$.

As the two transistors are coupled through the emitter resistor, same amount of voltage with opposite polarity will be developed at the collector of Q_2 , that is, $v_{c2} = v_{o2} = v_1 h_{fe} R_C / 2h_{ie}$. To understand this, let us consider the Fig.14-8. Applying KVL and going through two $B-E$ junctions (the dashed line), we find that v_1 will be equally divided between the junctions. For the positive half-cycle of v_1 , the base of Q_1 will be positive while the emitter of Q_2 will be positive. As the base of Q_2 is connected to ground, we can consider this amplifier as a common base (CB) amplifier. The gain of the CB amplifier (Q_2) is given by Equ.(5-86) (in Chapter 5) which is,

$$\text{Equ.(5-86), } A_v(Q_2) = \frac{h_{fb}}{h_{ib}} \left(\frac{R_C R_L}{R_C + R_L} \right)$$

But here, there is no load resistance, i.e., $R_L = \infty$. Therefore,

$$A_v(Q_2) = \frac{h_{fb} R_C}{h_{ib}} \quad (14-13)$$

From Table 5.1 (in Chapter 5), we find that $h_{ib} = h_{ie} / (1 + h_{fe})$ or, $h_{ib} \approx h_{ie} / h_{fe}$. Moreover, $h_{fb} = \alpha \approx 1$. Therefore, Equ.(14-13) can be written as,

$$A_v(Q_2) = \frac{h_{fb} R_C}{h_{ib}} = \frac{(1) R_C}{h_{ie} / h_{fe}} = \frac{h_{fe} R_C}{h_{ie}} \quad (14-14)$$

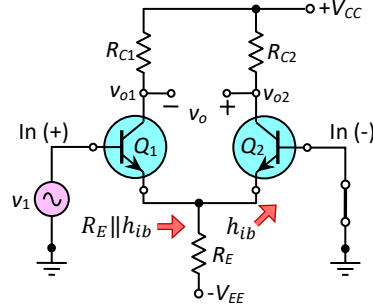


Fig.14-7: Equivalent resistance seen by Q_1 with unbypassed emitter resistance

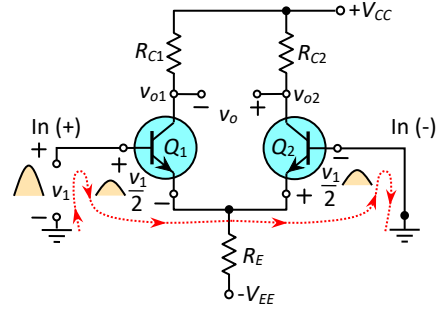


Fig.14-8: v_1 divided equally between two $B-E$ junctions

However, for the input voltage v_1 applied to the base of Q_1 , the voltage across $B-E$ junction of Q_1 is $v_1/2$ and the voltage across the $B-E$ junction of Q_2 is $v_1/2$. As the gain of a CE amplifier is $-h_{fe}R_C/h_{ie}$, the output voltage will be,

$$\text{Using Eq.(5-56) and } R_L = \infty, \quad v_{c1} = v_{o1} = \frac{v_1}{2} \times \left(-\frac{h_{fe}R_C}{h_{ie}} \right) \quad (14-15)$$

Similarly, as the gain of a CB amplifier is $h_{fe}R_C/h_{ie}$ [Equ.(14-14)], the output voltage will be,

$$\text{Using Eq.(5-86) and } R_L = \infty, \quad v_{c2} = v_{o2} = \frac{v_1}{2} \times \left(\frac{h_{fe}R_C}{h_{ie}} \right) \quad (14-16)$$

14.3.2 Input Impedance

As described in Chapter 5, the input impedance of a CE amplifier with unbypassed emitter resistance (seen from base) is,

$$\text{Eq.(5-74)} \quad Z_b = h_{ie} + (h_{fe} + 1)R_E$$

From Fig.14-7, it is found that the effective unbypassed resistance at the emitter is $R_E \parallel h_{ib} \approx h_{ib}$. Putting this value into the above equation,

$$Z_b = h_{ie} + (h_{fe} + 1)h_{ib}$$

$$\because (h_{fe} + 1)h_{ib} = h_{ie} \quad Z_b = 2h_{ie} \quad (14-17)$$

Equ.(14-17) shows that the impedance seen from the base of any terminal is just double the input impedance of a CE amplifier.

Generally, the bias resistor (if any) comes in parallel with Z_b . Therefore, the input impedance of the differential amplifier will be,

$$Z_i = R_B \parallel Z_b = R_B \parallel 2h_{ie} \quad (14-18)$$

14.3.3 Output Impedance

As described in Chapter 5, the output impedance of a CE amplifier with and without emitter bypass capacitor and that of the CB amplifier are almost same. For all the three amplifiers the output impedance is $r_o \parallel R_C$. Therefore, the output impedance of the differential amplifier will be,

$$Z_o = r_o \parallel R_C = \frac{1}{h_{oe}} \parallel R_C \quad (14-19)$$

14.4 Voltage Gain of Differential Amplifier in Different Mode

14.4.1 Single-input unbalanced output

Fig.14-9 shows the **single-input unbalanced output** mode of operation of differential amplifier. It is also called **single-ended input** and **single-ended output**. Here, the input

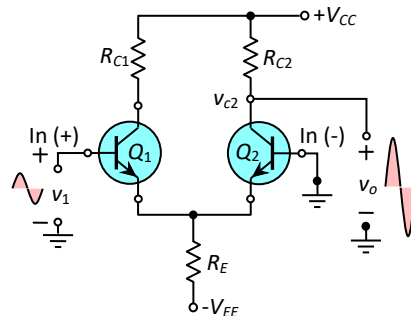


Fig.14-9: Single-input unbalanced output differential amplifier

signal is applied to only one of the two inputs (either inverting or noninverting) keeping the other input shorted to ground, and the output is taken from the collector of Q_2 with respect to the ground. As shown in the figure if the input signal is applied to the base of Q_1 , a non-inverted amplified output will be found. The value of the output voltage can be found using Equ.(14-16),

$$v_o = v_{c2} = \frac{v_1}{2} \times \left(\frac{h_{fe} R_C}{h_{ie}} \right)$$

Therefore, the gain of the differential amplifier will be,

$$A_v = \frac{v_o}{v_1} = \frac{h_{fe} R_C}{2h_{ie}} \quad (14-20)$$

But if the input signal is applied to the base of Q_2 , an inverted amplified output will be found. Then the gain of the differential amplifier will be,

$$A_v = \frac{v_o}{v_1} = -\frac{h_{fe} R_C}{2h_{ie}} \quad (14-21)$$

14.4.2 Single-input balanced output

Fig.14-10 shows the *single-input balanced output* mode of operation of differential amplifier. This configuration is also called *single-ended input and double-ended output*. Here also, the input signal is applied to only one of the two inputs, and the output is taken as difference of two collector voltages. As shown in the figure if the input signal is applied to the base of Q_1 , it is divided equally between the B-E junctions of the transistors. The voltage developed at the collector of Q_1 will be [Equ.(14-15)],

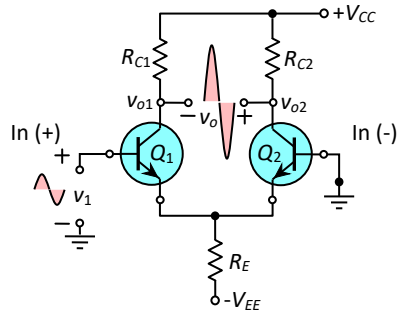


Fig.14-10: Single-input balanced output differential amplifier

$$v_{o1} = v_{c1} = \frac{v_1}{2} \times \left(-\frac{h_{fe} R_{C1}}{h_{ie}} \right) \quad (14-22)$$

And the voltage developed at the collector of Q_2 will be [Equ.(14-16)],

$$v_{o2} = v_{c2} = \frac{v_1}{2} \times \left(\frac{h_{fe} R_{C2}}{h_{ie}} \right) \quad (14-23)$$

Therefore, the overall output voltage will be,

$$v_o = v_{o2} - v_{o1}$$

Using Equ.(14-22), and Equ.(14-23),

$$v_o = \frac{v_1}{2} \times \left(\frac{h_{fe} R_{C2}}{h_{ie}} \right) + \frac{v_1}{2} \times \left(\frac{h_{fe} R_{C1}}{h_{ie}} \right)$$

Generally, $R_{C1} = R_{C2} = R_C$. Thus, we can write,

$$v_o = v_1 \left(\frac{h_{fe} R_{C2}}{h_{ie}} \right)$$

Therefore, the voltage gain of the differential amplifier will be,

$$A_v = \frac{v_o}{v_1} = \frac{h_{fe} R_C}{h_{ie}} \quad (14-24)$$

But if the input signal is applied to the base of Q_2 , an inverted amplified output will be found. Then the gain of the differential amplifier will be,

$$A_v = \frac{v_o}{v_1} = -\frac{h_{fe} R_C}{h_{ie}} \quad (14-25)$$

From Equ.(14-24) and Equ.(14-25) it is found that the voltage gain of a differential amplifier in *single-input balanced output* mode is same as that of a single-stage CE amplifier with bypassed emitter resistance, although there is no bypass capacitor used in the circuit.

14.4.3 Dual-input unbalanced output

Fig.14-11 shows the **dual-input unbalanced output** mode of operation of differential amplifier. This configuration is also called **double-ended input and single-ended output**.

Here, two input signals are applied to the two input terminals, that is, to the bases of Q_1 and Q_2 and the output is taken from the collector of Q_2 with respect to ground. Both the input signals will develop output voltage at the collector of Q_2 . The total output voltage can be calculated using superposition theorem. For the input voltage v_1 , the output voltage, taken from the collector of Q_2 , will be [Equ.(14-16)],

$$v'_o = v_{c2} = \frac{v_1}{2} \times \left(\frac{h_{fe} R_C}{h_{ie}} \right) \quad (14-26)$$

For the input voltage v_2 , the output voltage, taken from the collector of Q_2 , will be inverted. So, its value will be [Equ.(14-15)],

$$v''_o = v_{c2} = \frac{v_2}{2} \times \left(-\frac{h_{fe} R_C}{h_{ie}} \right) \quad (14-27)$$

According to the superposition theorem, the overall output voltage will be,

$$v_o = v'_o + v''_o = \frac{v_1}{2} \times \left(\frac{h_{fe} R_C}{h_{ie}} \right) - \frac{v_2}{2} \times \left(\frac{h_{fe} R_C}{h_{ie}} \right)$$

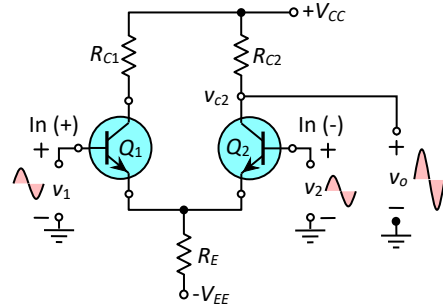


Fig.14-11: Dual-input unbalanced output differential amplifier

$$v_o = (v_1 - v_2) \frac{h_{fe} R_C}{2h_{ie}} \quad (14-28)$$

This equation shows that the difference of v_1 and v_2 will be amplified at the output. The differential gain will be,

$$A_v = \frac{v_o}{v_1 - v_2} = \frac{h_{fe} R_C}{2h_{ie}} \quad (14-29)$$

14.4.4 Dual-input balanced output

Fig.14-12 shows the *dual-input balanced output* mode of operation of differential amplifier. This configuration is also called *double-ended input and double-ended output*. Here, two input signals are applied to two input terminals, that is, to the bases of Q_1 and Q_2 , and the output is taken as difference of two collector voltages. Both the input signals will produce output voltages at the collectors of Q_2 and Q_1 . Using superposition theorem, the output voltage at the collector of Q_1 will be found as,

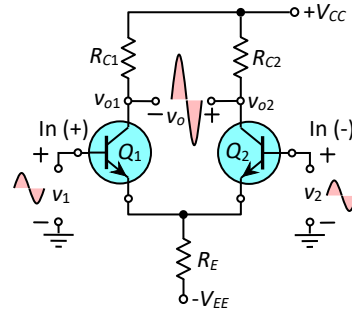


Fig.14-12: Dual-input balanced output differential amplifier

$$v_{o1} = v_{c1} = -(v_1 - v_2) \frac{h_{fe} R_C}{2h_{ie}} \quad (14-30)$$

Similarly using superposition theorem, the output voltage at the collector of Q_2 will be found as,

$$v_{o2} = v_{c2} = (v_1 - v_2) \frac{h_{fe} R_C}{2h_{ie}} \quad (14-31)$$

Therefore, the overall output voltage will be,

$$v_o = v_{o2} - v_{o1}$$

Using Equ.(14-30), and Equ.(14-31) into the above equation,

$$v_o = (v_1 - v_2) \frac{h_{fe} R_C}{2h_{ie}} + (v_1 - v_2) \frac{h_{fe} R_C}{2h_{ie}} = (v_1 - v_2) \frac{h_{fe} R_C}{h_{ie}}$$

In this case the gain will be,

$$A_v = \frac{v_o}{v_1 - v_2} = \frac{h_{fe} R_C}{h_{ie}} \quad (14-32)$$

Example 13-2

For the differential amplifier of Example 13-1, calculate the input impedance, output impedance and gain in four modes of operation. Assume $h_{fe} = 100$, $r_o = 200 \text{ k}$.

Solution:

In Example 13-1, we have found that $I_{E1} = I_{E2} = 1 \text{ mA}$. Therefore, the value of $h_{ie1} = h_{ie2} = h_{ie}$ can be calculated as,

$$h_{ie1} = h_{ie2} = h_{ie} \approx h_{fe} \times \frac{26 \text{ mV}}{I_E} = 100 \times \frac{26 \text{ mV}}{1 \text{ mA}} = 2.6 \text{ k}$$

There is no biasing resistor. So, $R_B = \infty$.

Using Equ.(14-18), $Z_i = R_B \parallel 2h_{ie} = 2h_{ie} = 2 \times 2.6 \text{ k} = 5.2 \text{ k}$ [Ans.]

Using Equ.(14-19), $Z_o = r_o \parallel R_C = 200 \text{ k} \parallel 6.2 \text{ k} \approx 6.2 \text{ k}$ [Ans.]

Single-input unbalanced output voltage gain:

Using Equ.(14-21), $A_v = -\frac{h_{fe}R_C}{2h_{ie}} = \frac{100 \times 6.2 \text{ k}}{2 \times 2.6 \text{ k}} \approx 119$ [Ans.]

Single-input balanced output voltage gain:

Using Equ.(14-24), $A_v = \frac{h_{fe}R_C}{h_{ie}} = \frac{100 \times 6.2 \text{ k}}{2.6 \text{ k}} \approx 238.5$ [Ans.]

Dual-input unbalanced output voltage gain:

Using Equ.(14-29), $A_v = \frac{h_{fe}R_C}{2h_{ie}} = \frac{100 \times 6.2 \text{ k}}{2 \times 2.6 \text{ k}} \approx 119$ [Ans.]

Dual-input balanced output voltage gain:

Using Equ.(14-32), $A_v = \frac{h_{fe}R_C}{h_{ie}} = \frac{100 \times 6.2 \text{ k}}{2.6 \text{ k}} \approx 238.5$ [Ans.]

Comments: Gains for dual-input and single-input are same. Gain in balanced output is just double the gain in unbalanced output.

14.5 Gain Calculation from Equivalent Circuit

Already we have seen that there is no difference in voltage gain for single-input and dual-input operation of differential amplifier. We have also seen that the gain in unbalanced output is just half the gain in balanced output. Here, we will discuss only the single-input balanced output configuration as shown in Fig.14-10. The hybrid equivalent circuit of this configuration is shown in Fig.14-13. Here, the transistors are assumed identical in all respects.

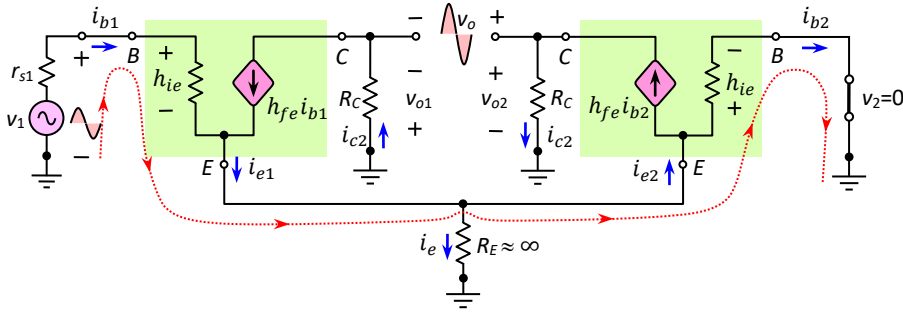


Fig.14-13: Equivalent circuit of differential amplifier

This equivalent circuit is further simplified in Fig.14-14. From this equivalent circuit, it is clear that R_E and h_{ie} (of Q_2) work in parallel. But as R_E is very large compared to h_{ie} , R_E can be neglected. Applying KVL (passing through two B-E junctions) in the equivalent circuit, we get,

$$v_1 - i_{b1}h_{ie} - i_{b2}h_{ie} = 0$$

As, v_1 almost equally divided across the two B-E junctions, the base currents will be equal, that is, $i_{b1} = i_{b2} = i_b$. Therefore,

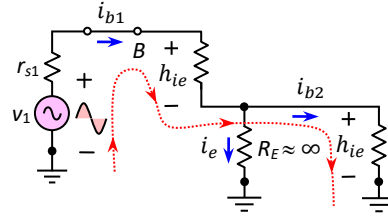


Fig.14-14: Equivalent circuit of differential amplifier

$$v_1 - i_b h_{ie} - i_b h_{ie} = 0$$

$$\therefore i_b = \frac{v_1}{2h_{ie}}$$

The collector current of the transistors will be,

$$i_{c1} = i_{c2} = i_c = h_{fe} i_b = \frac{v_1 h_{fe}}{2h_{ie}}$$

Therefore, the output voltages (taken from the collector with respect to ground of Fig.14-13) will be,

$$v_{o1} = -i_c R_C = -\frac{v_1 h_{fe} R_C}{2h_{ie}}$$

and

$$v_{o2} = i_c R_C = \frac{v_1 h_{fe} R_C}{2h_{ie}}$$

The balanced output voltage (difference of two collector voltages) will be,

$$v_o = v_{o2} - v_{o1} = \frac{v_1 h_{fe} R_C}{2h_{ie}} - \left(-\frac{v_1 h_{fe} R_C}{2h_{ie}} \right)$$

$$v_o = \frac{v_1 h_{fe} R_C}{h_{ie}}$$

Therefore, the voltage gain will be,

$$A_v = \frac{v_o}{v_1} = \frac{h_{fe} R_C}{h_{ie}} \quad (14-33)$$

This voltage gain is exactly same as given by Equ.(14-24). Thus the analysis of other configurations of differential amplifier using hybrid modeling will give the same result as we have already found.

14.6 Common Mode Gain

The target of a differential amplifier is to amplify a signal applied to any one of the input terminals or the difference of two signals applied to two input terminals. But sometimes identical signals are induced, mainly due to noise, to both input terminals. This signal is called **common-mode signal**, v_{cm} . An ideal differential amplifier does not amplify the common-mode signal, but a practical differential amplifier provides little amplification to this common-mode signals. The gain of a differential amplifier to common-mode signal is called **common-mode gain**, A_{cm} . To determine the common-mode gain, let us apply a common-mode signal to both inputs of a differential amplifier as shown in Fig.14-15.

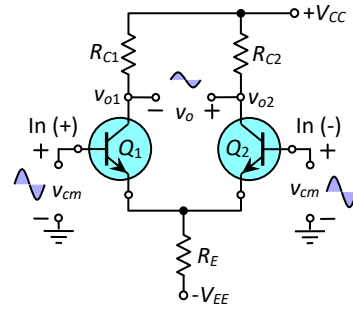


Fig.14-15: Single-input unbalanced output differential amplifier

The hybrid equivalent circuit of this amplifier is shown in Fig.14-16. As the transistors are perfectly matched, and the input signals are identical, we can assume $i_{b1} = i_{b2} = i_b$.

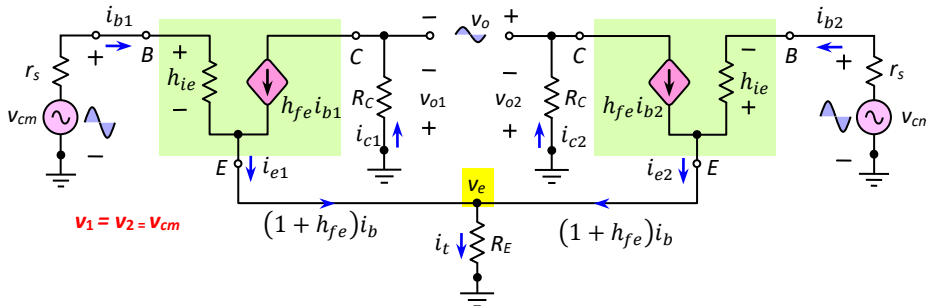


Fig.14-16: Equivalent circuit of differential amplifier for common mode signal

From the Fig.14-14, the tail current will be,

$$i_t = i_{e1} + i_{e2} = (1 + h_{fe})i_b + (1 + h_{fe})i_b = 2(1 + h_{fe})i_b \quad (14-34)$$

The value of the base current can be calculated as,

$$i_b = \frac{v_{cm} - v_e}{h_{ie}} = \frac{v_{cm} - i_e R_E}{h_{ie}} = \frac{v_{cm} - 2(1 + h_{fe})i_b R_E}{h_{ie}}$$

$$\text{or,} \quad i_b h_{ie} = v_{cm} - 2(1 + h_{fe})i_b R_E$$

$$\text{or,} \quad i_b [h_{ie} + 2(1 + h_{fe})R_E] = v_{cm}$$

$$\therefore i_b = \frac{v_{cm}}{h_{ie} + 2(1 + h_{fe})R_E} \quad (14-35)$$

So, the collector currents of the transistors will be,

$$i_{c1} = i_{c2} = i_c = h_{fe} i_b$$

Putting the value of i_b from Equ.(14-35),

$$i_c = \frac{h_{fe} v_{cm}}{h_{ie} + 2(1 + h_{fe})R_E}$$

So the magnitude of the output voltages will be,

$$v_{o1} = v_{o2} = i_c R_C = \frac{h_{fe} v_{cm} R_C}{h_{ie} + 2(1 + h_{fe})R_E}$$

Therefore, common-mode voltage gain (unbalanced) will be,

$$A_{cm} = \frac{v_{o1}}{v_{cm}} = \frac{v_{o2}}{v_{cm}} = \frac{h_{fe} R_C}{h_{ie} + 2(1 + h_{fe})R_E}$$

$$\boxed{A_{cm} = \frac{h_{fe} R_C}{h_{ie} + 2(1 + h_{fe})R_E}} \quad (14-36)$$

$$\because h_{ie} \ll 2(1 + h_{fe})R_E \quad \therefore A_{cm} \approx \frac{h_{fe} R_C}{2(1 + h_{fe})R_E}$$

Again as $h_{fe} \gg 1$,

$$\therefore A_{cm} \approx \frac{h_{fe} R_C}{2h_{fe} R_E} = \frac{R_C}{2R_E}$$

Thus the common-mode gain of the differential amplifier is,

$$\therefore \boxed{A_{cm} = \frac{R_C}{2R_E}} \quad (14-37)$$

From this expression, it is clear that, the gain will decrease with the increase in R_E and decrease in R_C .

14.7 Common-Mode Rejection Ratio (CMRR)

CMRR is a measure of how effectively a differential amplifier can eliminate the signals applied to both inputs called *common-mode signal*. Mainly the noises are the common-mode signals. It is

desired that an amplifier will amplify the applied signal (differential signal) and will reject the noise. An amplifier can do this successfully if it has very large value of CMRR. An ideal differential amplifier has $CMRR = \infty$. CMRR is defined as,

$$CMRR = \frac{A_{dm}}{A_{cm}} \quad (14-38)$$

Where, A_{dm} is the differential voltage gain and A_{cm} is the common-mode voltage gain. The value of A_{cm} is given by Equ.(14-37). The value of A_{dm} is given by Equ.(14-29) as,

$$A_{dm} = A_v = \frac{h_{fe}R_C}{2h_{ie}}$$

$$\because h_{ie} \approx r_e h_{fe} \quad A_{dm} \approx \frac{h_{fe}R_C}{2h_{fe}r_e} = \frac{R_C}{2r_e} \quad (14-39)$$

Using Equ.(14-37) and Equ.(14-39) into Equ.(14-38), we can calculate the value of CMRR as,

So, the value of CMRR of the BJT differential amplifier will be

$$CMRR = \frac{A_{dm}}{A_{cm}} \approx \frac{R_C}{2r_e} / \frac{R_C}{2R_E} = \frac{R_E}{r_e}$$

$$\therefore \quad \boxed{CMRR = \frac{R_E}{r_e}} \quad (14-40)$$

From this equation, we find that larger value of R_E will give the higher (better) CMRR. But practically we cannot increase the value of R_E because it reduces the differential voltage gain (high R_E results in low R_C for a fixed value of V_{CC} and I_C). That's why instead of using a single resistance (R_E), a current source is used which is shown in Fig.14-20.

14.8 Constant Current Source

The source that gives constant current irrespective of the load is called *constant current source*. Constant current sources can be designed using FET or BJT and are used in IC fabrication. Fig.14-17(a) shows a constant current source circuit using BJT. The load will be connected to the collector of the transistor.

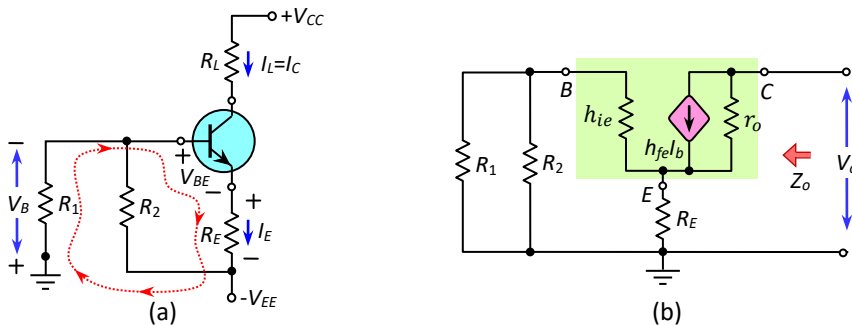


Fig.14-17: (a) Constant current source, (b) Its AC equivalent circuit

The base voltage will be set by the voltage divider network consisting of R_1 and R_2 . The value of base voltage will be,

$$V_B = \frac{R_1}{R_1 + R_2} V_{EE} \quad (14-41)$$

Applying KVL in the loop shown in Fig.14-17(a) we get,

$$\begin{aligned} -V_B - V_{BE} - I_E R_E + V_{EE} &= 0 \\ I_E &= \frac{V_{EE} - V_B - V_{BE}}{R_E} \approx I_C \end{aligned} \quad (14-42)$$

Using Equ.(14-41) and Equ.(14-42) we can calculate the value of I_E which is also approximately equal to I_C . From these equations, it is clear that I_C is independent of R_L and h_{fe} . Unless the transistor is saturated, the collector current will remain constant irrespective of the value of load resistance (R_L). For a fixed value of supply voltage, the amount of current can be set by changing the value of R_E , R_1 , and R_2 .

From the equivalent circuit of Fig.14-17(b), we can easily calculate the output impedance of the constant current source. The output impedance will be,

$$Z_o = (R_1 \parallel R_2 + h_{ie}) \parallel R_E + r_o \quad (14-43)$$

As $r_o \gg (R_1 \parallel R_2 + h_{ie}) \parallel R_E$, we can write

$$Z_o \approx r_o \quad (14-44)$$

From Equ.(14-44) we find that the output impedance of the constant current circuit is very large. This resistance will work as the internal resistance of the constant current source. **For current source, the higher the internal resistance the better is the current source.**

Example 13-3

The circuit of Fig.14-18 is a constant current source. Calculate the value of the constant current, and also determine the range of R_L for which the circuit will work as a constant current source.

Solution:

Using Equ.(14-41),

$$V_B = \frac{R_1(-V_{EE})}{R_1 + R_2} = \frac{1 \text{ k}(-12 \text{ V})}{1 \text{ k} + 10 \text{ k}} = -1.1 \text{ V}$$

Using Equ.(14-42),

$$I_E \approx I_C = \frac{12 \text{ V} - 1.1 \text{ V} - 0.7 \text{ V}}{2 \text{ k}} \approx 5.1 \text{ mA [Ans.]}$$

The minimum value of R_L may be zero, that is, if R_L is shorted, the same amount of current will flow. For the maximum value of R_L the transistor will be saturated. The transistor will be saturated when $V_{CE} = 0.2 \text{ V}$. Now applying KVL at the output loop, we get,

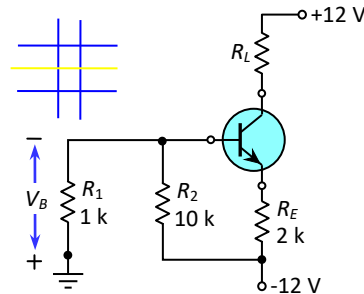


Fig.14-18: Constant current source for Example 13-3

$$V_{CE} = V_{CC} + V_{EE} - I_C(R_{L(max)} + R_E)$$

$$\text{or,} \quad 0.2 \text{ V} = 12 \text{ V} + 12 \text{ V} - 5.1 \text{ mA} (R_{L(max)} + 2 \text{ k})$$

$$\text{or,} \quad 5.1 R_{L(max)} = 24 \text{ V} - 10.2 \text{ V} - 0.2 \text{ V} = 13.6 \text{ V}$$

$$\therefore \quad R_{L(max)} = \frac{13.6 \text{ V}}{5.1 \text{ mA}} \approx 2.67 \text{ k [Ans.]}$$

Comments: This circuit will work as a constant current source for $R_L = 0$ to 2.67 k .

14.8.1 Constant Current Source using Zener Diode

The constant current source discussed above has a problem. The load current depends on the supply voltage, V_{EE} [Equ.(14-42)]. If the supply voltage V_{EE} changes the load current will also change. To make the load current independent of the supply voltage Zener diode can be used. Fig.14-19 shows a BJT constant current source using Zener diode.

Using KVL in the circuit of Fig.14-19 (loop containing V_Z), we can write

$$-V_B - V_Z + V_{EE} = 0$$

$$\text{or,} \quad V_B = V_{EE} - V_Z \quad (14-45)$$

Again using KVL in the circuit of Fig.14- (loop containing V_{BE}), we can write

$$-V_B - V_{BE} - I_E R_E + V_{EE} = 0$$

$$\text{or,} \quad I_E R_E = V_{EE} - V_B - V_{BE}$$

Putting the value of V_B from Equ.(14-45) into the above equation,

$$I_E R_E = V_{EE} - V_{EE} + V_Z - V_{BE}$$

$$\therefore \quad I_E = \frac{V_Z - V_{BE}}{R_E} \quad (14-46)$$

$$\therefore \quad I_C = I_L \approx \frac{V_Z - V_{BE}}{R_E} \quad (14-47)$$

From Equ.(14-47), we find that the load current is independent of supply voltage.

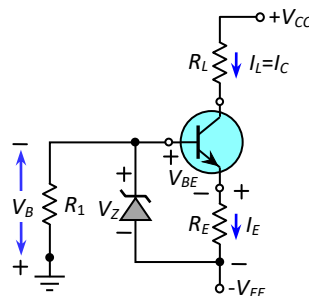


Fig.14-19: Constant current source with Zener diode

Example 13-4

For circuit of Fig.14-20, estimate the value of CMRR with and without the constant current source.

Solution:

At first we will calculate the value of I_{C3} .

Using Equ.(14-41),

$$V_B = \frac{R_1(-V_{EE})}{R_1 + R_2} = \frac{1 \text{ k}(-12 \text{ V})}{1 \text{ k} + 8.2 \text{ k}} = -1.3 \text{ V}$$

Using Equ.(14-42),

$$I_E \approx I_{C3} = \frac{12 \text{ V} - 1.3 \text{ V} - 0.7 \text{ V}}{3 \text{ k}} \approx 3.33 \text{ mA}$$

Emitter current of Q_1 and Q_2 will be,

$$I_{E1} = I_{E2} = \frac{I_{C3}}{2} = \frac{3.33 \text{ mA}}{2} \approx 1.67 \text{ mA}$$

Emitter resistance of Q_1 and Q_2 will be,

$$r_e = \frac{26 \text{ mV}}{1.67 \text{ mA}} = 15.57 \Omega$$

Without Current Source:

If the constant current source is not used, a resistor (R_E) has to be used in place of the constant current source. The voltage drop across R_E would be $12 \text{ V} - 0.7 \text{ V} = 11.3 \text{ V}$. Therefore, the value of R_E would be [H_E_atten],

$$R_E = \frac{11.3 \text{ V}}{3.33 \text{ mA}} \approx 3.4 \text{ k}$$

Using Equ.(14-40),

$$\text{CMRR} = \frac{R_E}{r_e} = \frac{3400 \Omega}{15.57 \Omega} \approx 218 \text{ [Ans.]}$$

With Current Source:

The internal resistance of the constant current source will be,

$$Z_o \approx r_o = 200 \text{ k}$$

This resistance will be the effective emitter resistance of the differential amplifier. Therefore, the CMRR using the current source will be,

$$\text{CMRR} = \frac{R_E}{r_e} = \frac{Z_o}{r_e} = \frac{200 \text{ k}}{15.57 \Omega} = 12845 \text{ [Ans.]}$$

Comments: Due to the constant current source, CMRR has been increased from 218 to 12845.

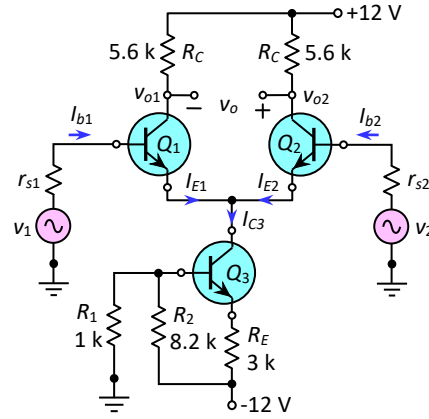


Fig.14-20: Differential amplifier with constant current source

14.9 Current Mirror Circuit

A **current mirror** is a circuit designed to copy a current through one active device, by controlling the current in another active device or a circuit, keeping the output current constant regardless of loading. In integrated circuit a number of current sources are used. As shown in Fig.14-20, if we set the emitter biasing current of a transistor using a constant current source, then the collector current and the emitter current will be independent of h_{fe} , R_B , and temperature. Only one requirement is- the biasing current should be constant. It should not change with the change of supply voltage and temperature. It is possible to design a very stable constant current source with complicated circuitry. Therefore, instead of designing constant current source for each amplifier stage of an IC, a single high quality constant current source is fabricated and it is replicated using current mirror circuit. So the replicated current source has the same characteristics as the reference current source. This means that the replicated current source is as stable as the reference current source. Thus, replicated current sources can be used to bias the amplifiers.

Current mirror is an active circuit that senses the reference current and generates copies of the reference current.

Current mirror circuits generally consist of two main transistors, although other devices such as FETs can be used. Some current mirror circuits may use more than two transistors to improve the level of performance. Fig.14-21(a) shows the basic construction principle of a current mirror circuit. A reference voltage (V_{ref}) is made using a current-to-voltage converter, and this voltage is applied to the B-E junction of a transistor. Fig.14-21(b) shows the circuit diagram of current mirror circuit using two perfectly matched transistors Q_1 , and Q_2 . The base and collector terminals of Q_1 are shorted together. So this transistor will work as a diode, with the collector working as the anode and the emitter working as the cathode. When the reference current (I_{ref}) flows through this diode-connected transistor, a voltage $V_{BE(ref)}$ is developed, which is proportional to I_{ref} . Thus, this transistor converts the reference current into reference voltage. In IC fabrication, this type of arrangement is used, because transistor can be easily made than a diode. The reference voltage is then applied to the B-E junction of the second transistor Q_2 . As the transistors are fully matched, the same amount of current flows through the collector of Q_2 , irrespective of the load R_L . In this way, I_{ref} is replicated in the collector of Q_2 .

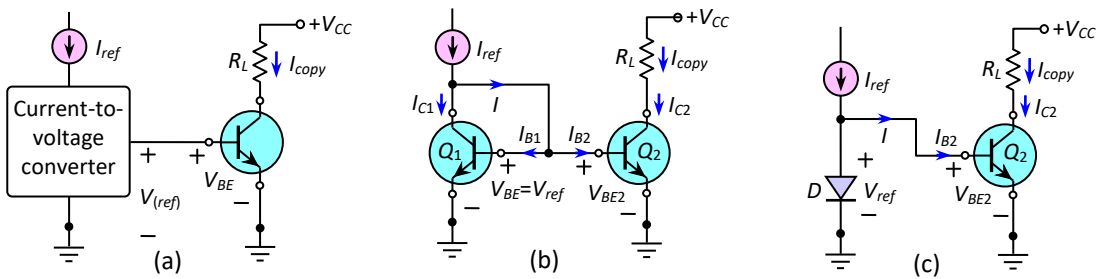


Fig.14-21: Current mirror circuit

For circuits of Fig.14-21(b) and (c), we will assume that the values of h_{fe} of these transistors are so large that the current flowing through the bases can be neglected, that is $I_B \approx 0$. When the reference current flows through the diode-connected transistor, a reference voltage is developed which can be calculated using standard diode current expression as,

$$I_{ref} = I_{C1} \approx I_{s1} e^{\left(\frac{V_{ref}}{V_T}\right)} \quad (14-48)$$

$$V_{ref} = V_T \ln\left(\frac{I_{ref}}{I_{s1}}\right) \quad (14-49)$$

Here, I_{s1} is the reverse saturation current of the B-E junction of Q_1 .

When this reference voltage is applied to Q_2 another collector current will be developed, which is given by

$$I_{copy} = I_{C2} \approx I_{s2} e^{\left(\frac{V_{ref}}{V_T}\right)} \quad (14-50)$$

Here, I_{s2} is the reverse saturation current of the B-E junction of Q_2 .

Dividing Equ.(14-50) by Equ.(14-48), we get,

$$\frac{I_{copy}}{I_{ref}} = \frac{I_{s2}}{I_{s1}}$$

$$I_{copy} = \frac{I_{s2}}{I_{s1}} \times I_{ref} \quad (14-51)$$

If the transistors are fully matched, $I_{s1} = I_{s2}$. Therefore,

$$I_{copy} = I_{ref} \quad (14-52)$$

Equ.(14-52) shows that for fully matched transistor, the replicated current (I_{copy}) will be equal to the reference current I_{ref} . This replicated current will have all properties of I_{ref} - like the temperature stability of I_{ref} . The manufacturer can fabricate Q_2 with different value of I_{s2} by controlling the area of the B-E junction. In that case I_{copy} can be any multiple of I_{ref} .

Fig.14-22 shows how multiple copies of I_{ref} can be produced. Here, the reference voltage developed across B-E junction of Q_1 is applied to the B-E junctions of three transistors in parallel (for simplicity, parallel connections of the transistors are shown in this way). So every transistor will produce a copy of the reference current.

Here the problem is the total base current will be high ($I = 4 \times I_C / \beta$). So I_C will no longer be equal to I_{ref} . Therefore, the replicated currents will be a little bit lower than I_{ref} . This problem can be overcome using another transistor as shown in Fig.14-23. Now, the value of I will be reduced to $4 \times I_C / \beta^2$.

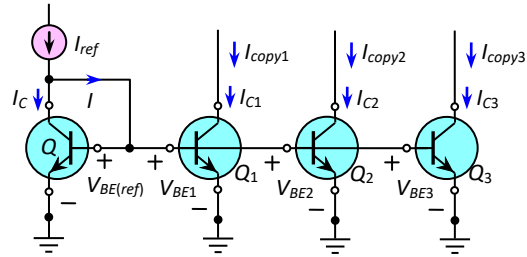


Fig.14-22: Production of multiple copies of reference current

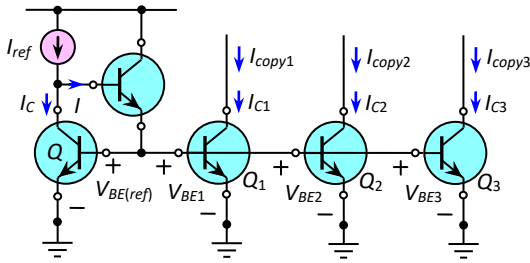


Fig.14-24: Base current compensated current mirror circuit

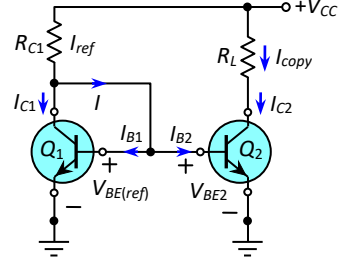


Fig.14-24: Constant current source

A current mirror circuit is also designed as in Fig.14-24. Here the current flowing through R_{C1} will be equal to the circuit flowing through R_L , that is $I_{C1} = I_{C2} = I_{copy}$.

14.10 Active Load for Differential Amplifier

Active loads are essentially transistor current sources used in place of the resistive loads in the differential amplifier circuits to increase differential-mode gain. For unbalanced output, the gain of a differential amplifier is,

$$A_{dm} = \frac{h_{fe}R_C}{2h_{ie}}$$

For balanced output, the gain of a differential amplifier is,

$$A_{dm} = \frac{h_{fe}R_C}{h_{ie}}$$

From these equations, it is clear that the **larger the R_C the greater is the differential voltage gain**. But for a fixed supply voltage, too large R_C will saturate the transistors. If the transistors are saturated, the circuit will no longer work as amplifier. To overcome this problem, active load (current mirror) circuits are used in place of R_C .

Fig.14-25 shows a differential amplifier with an active load. Here, Q_1 and Q_1 from the differential amplifier and Q_3 & Q_4 form the active load. Actually the active load is a current mirror circuit that works as a constant current source. As the tail current of the differential amplifier is I_T , the emitter current of Q_1 and Q_1 will be $I_T/2$. Therefore, we can assume that the collector current of Q_1 and Q_1 will also be approximately $I_T/2$. Same amount of current will flow through Q_3 and Q_4 . The internal output resistance of Q_3 and Q_4 will work as the collector resistance of the differential amplifier. As the internal output resistance of the active load is very high, hence the gain of the differential amplifier will also be very high.

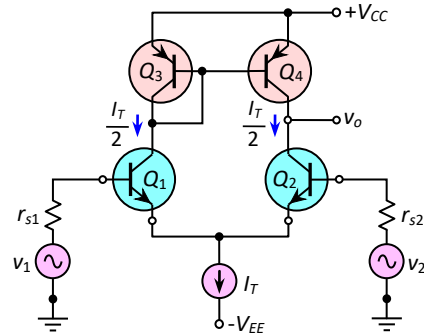


Fig.14-25: Differential amplifier with active load

14.11 Internal Architecture of Op-Amp

An operational amplifier is a high gain direct-coupled multistage differential amplifier followed by a level shifter and a push-pull complementary-symmetry power amplifier at the output stage. Fig.14-26 shows the block diagram of a typical operational amplifier which consists of four stages.

Input Stage

This is the first stage of the operational amplifier. This stage is usually a differential amplifier, which operates in dual-input and balanced output mode. The differential amplifier of the first stage provides most of the voltage gain along with the input resistance for the op-amp. The inputs denoted with '+' & '-' are the noninverting and inverting inputs of the op-amp, respectively. If the AC (or DC) signal is applied to the noninverting input, amplified signal with the same polarity is produced at the output of op-amp. On the other hand, if the AC (or DC) signal is applied to the inverting input terminal, amplified signal with the opposite polarity (or 180° phase shift) is produced at the output of op-amp.

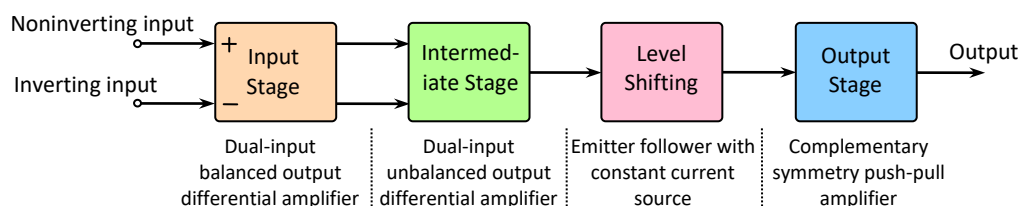


Fig.14-26: Block diagram of typical op-amp

Intermediate Stage

The intermediate stage is usually another differential amplifier that is operated in dual-input unbalanced output mode. The balanced output of the first stage is applied to this intermediate stage for further amplification of the signal. This stage provides additional gain of the op-amp and DC voltage stabilization.

Level Shifter

To amplify DC signal direct coupling is used between the input stage and the intermediate stages of op-amp. So, the voltage at the output of intermediate stage has a DC level above ground potential. Therefore, the DC level at its output must be shifted down to 0 V with respect to the ground. For this purpose, the level shifting stage is used where usually an emitter follower with the constant current source is employed.

Output Stage

This is the final stage of an op-amp, primarily designed to provide the power needed to drive external loads. The level shifted signal, from the third stage, is given to the output stage. This stage is usually a push-pull complementary-symmetry emitter follower amplifier. This stage mainly amplifies the current level of the signal. It has low output resistance and its voltage gain is approximately unity.

14.12 Simplified Schematic Diagram of Op-Amp (741)

The 741 op-amp IC is a monolithic integrated circuit, comprising of a general purpose Operational Amplifier. It was first manufactured by Fairchild semiconductors in the year 1963^[1]. The number 741 indicates that this operational amplifier IC has 7 functional pins, 4 pins capable of taking input and 1 output pin. Nowadays, a number of manufacturers fabricate op-amp ICs. Some of them are mentioned below:

Manufacturer	IC number	Manufacturer	IC number
Fairchild	μ A741	RCA	CA3741
National Semiconductor	LM741	Texas Instrument	SN5741
Motorola	MC1741	Signetics	N5741

This IC can be manufactured in three forms, and those are in 8-pin SOIC package, 8-pin Dual-in-line package, and in TO5-8 metal can form as shown in Fig.14-27.

The schematic diagram of the most common op-amp (741 IC) is shown in Fig.14-28. It consists of 24 transistors, 11 resistors and 1 capacitor all packaged in an 8-pin IC. Inverting and noninverting inputs are connected to two NPN transistors, Q_1 & Q_2 , respectively. Both the transistors work as NPN emitter followers, with their outputs fed to a pair of PNP transistors Q_3 & Q_4 , configured to function as common-base amplifiers. This configuration isolates both the inputs and prevents possible signal feedback that might occur. The emitter follower provides high input impedance of op-amp while the common-base amplifier provides voltage gain and good frequency response.

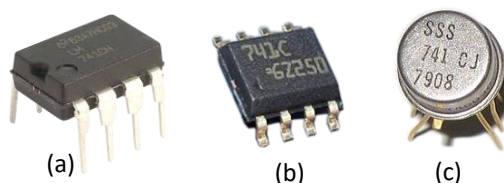


Fig.14-27: Op-amp IC package: (a) 8-pin DIP, (b) 8-pin SOIC, and (c) TO5-8 metal can

Transistor pairs Q_8 , Q_9 and Q_{12} , Q_{13} are arranged to form two current mirror circuits. The transistors Q_8 and Q_{12} are controlling transistors which set the emitter base voltage of the other transistor in the corresponding pair. This voltage is controlled precisely down to fractions of millivolts so as to allow only the required amount of current to flow. The first current mirror formed by Q_8 & Q_9 works as the active load for the input stage and the second current mirror formed by Q_{12} & Q_{13} are coupled to the output circuit and second stage amplifier.

A third current mirror formed by the transistors Q_{10} & Q_{11} acts as a high-impedance connection between the input circuit and negative of the power supply. It provides a reference voltage without loading the input circuit and sets the slight base bias current required by the PNP transistors at the input common-base amplifier circuit (Q_3 & Q_4).

Transistors Q_5 , Q_6 & Q_7 form a base compensated current mirror circuit that provides the tail current for the input stage. Transistor Q_7 , with 50 k resistance at the emitter, is connected to the base of Q_5 . An emitter follower consisting of Q_{16} and R_7 is connected to the collector of Q_6 for

loading Q_5 and Q_6 symmetrically. The output from this emitter follower is further amplified by Q_{17} . Double emitter transistor (Q_{13}) provides biasing current for Q_{17} and output stage. Capacitor C_C provides negative feedback to compensate the effect of transistor's parasitic capacitance and improve the frequency response of the op-amp. For this capacitor, 741 is called an internally compensated op-amp. It employs the Miller compensation technique, employed by C_C . Specifically, this is a 30 pF capacitor which is connected in the negative-feedback path of the second stage that set the lower cut-off frequency to ≈ 5 Hz in the open-loop gain response.

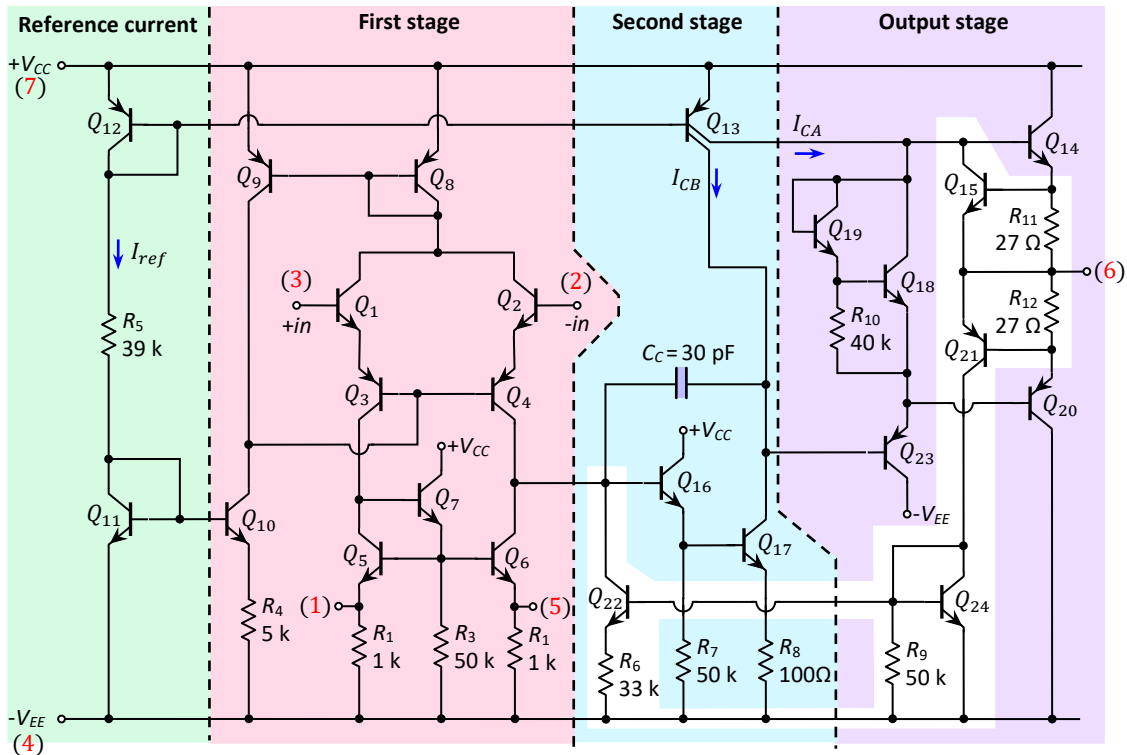


Fig.14-28: Equivalent internal schematic diagram of a typical op-amp 741 (the white shaded portion is for short-circuit protection)

Transistors Q_{14} , & Q_{20} are configured to work as a class AB complementary-symmetry power amplifier. Transistors Q_{18} , Q_{19} & Q_{23} are used to bias this complementary-symmetry amplifier. Transistors Q_{15} , Q_{21} , Q_{24} , and Q_{22} and resistance R_{11} are used for short-circuit protection purpose. Normally these transistors remain OFF, but if the output terminal is shorted to either $+V_{CC}$ or $-V_{EE}$, these transistors bypass the current and protect the op-amp.

As shown in Fig.14-28, pin number (2) and (3)

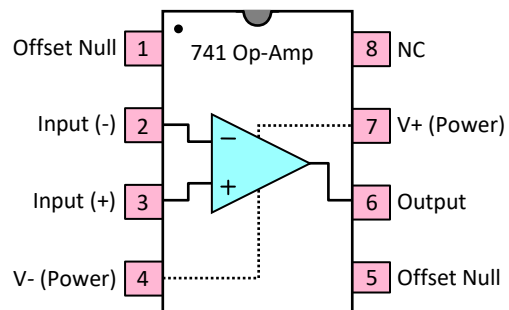


Fig.14-29: Pin diagram of op-amp 741

are the inverting and noninverting input terminal, respectively. Pin number (1) and (5) are for offset null inputs. Pin number (7) and (5) are positive and negative supply voltage. The output of this operational amplifier is taken from pin number (6). The pin-diagram of this IC is shown in Fig.14-29. This IC comes in an 8-pin package and pin number (8) is not used.

Example 13-5

DC Analysis of 741 Op-Amp:

Calculate the DC operating condition of different stages of the circuit of Fig.14-28. Assume: $\beta = 200$, $I_s = 10^{-14}$ A for all NPN transistors and $\beta = 50$, $I_s = 10^{-14}$ A for all PNP transistors, but for Q_{14} and Q_{20} , $I_s = 3 \times 10^{-14}$ A. In Q_{13} , 25% collector current goes to output section, and 75% collector current flows downward. Assume that the supply voltage is ± 15 V.

Solution:

Calculation of I_{ref} :

The function of Q_{12} and Q_{13} is to generate reference currents. This portion of the circuit is redrawn in Fig.14-30(a) with polarity of different voltage drops. Applying KVL in this circuit we get,

$$V_{CC} - V_{BE12} - R_5 \times I_{ref} - V_{BE11} + V_{BE11} = 0$$

$$\text{or,} \quad +15 \text{ V} - V_{BE12} - 39 \times I_{ref} - V_{BE11} + 15 \text{ V} = 0$$

$$\therefore I_{ref} = \frac{15 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} + 15 \text{ V}}{39 \text{ k}} \approx 0.733 \text{ mA}$$

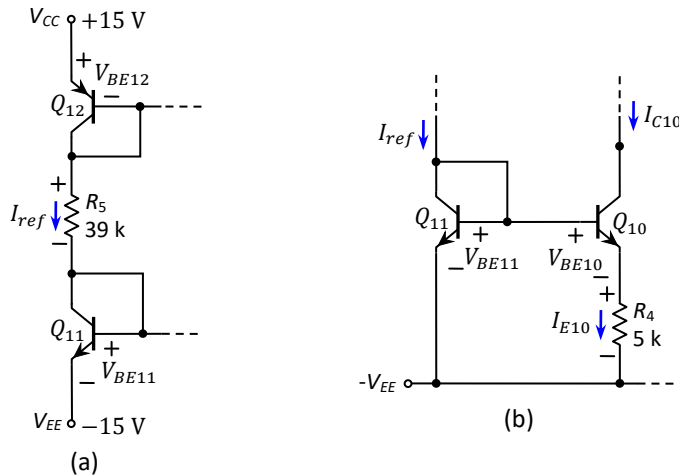


Fig.14-30: (a) Reference current section, (b) Current mirror circuit, and (c) Upper portion of first stage

Current mirror circuit (Q_{10}):

As shown in Fig.14-30(b), the base-emitter voltage of Q_{11} biases transistor Q_{10} . Using KVL, we get,

$$V_{BE11} = V_{BE10} + I_{E10}R_4$$

$$\therefore I_{E10} \approx I_{C10}$$

$$V_{BE11} = V_{BE10} + I_{C10}R_4$$

$$\text{or,} \quad V_T \ln \left(\frac{I_{ref}}{I_s} \right) = V_T \ln \left(\frac{I_{C10}}{I_s} \right) + I_{C10} R_4$$

$$\text{or,} \quad V_T \ln \left(\frac{I_{ref}}{I_s} \times \frac{I_s}{I_{C10}} \right) = V_T \ln \left(\frac{I_{ref}}{I_{C10}} \right) = I_{C10} R_4$$

$$\therefore (26 \times 10^{-3}) \ln \left(\frac{0.733 \text{ mA}}{I_{C10}} \right) = I_{C10} (5 \times 10^3 \Omega)$$

Solving this equation for I_{C10} we get, $I_{C10} = 19 \mu\text{A}$.

Input Stage:

The upper portion of the input stage is shown in Fig.14-30(c). As the base currents of the transistors are very small we can neglect them. So, current through Q_9 will be equal to I_{C10} , that is $I_{C9} = I_{C10} = 19 \mu\text{A}$. Transistors Q_9 and Q_8 form a current mirror circuit. Therefore, $I_{C8} = I_{C9} = 19 \mu\text{A}$. As Q_1 and Q_2 are matched transistors, $I_{C1} = I_{C2} = I_{C8}/2 = 19 \mu\text{A}/2 = 9.5 \mu\text{A}$. For the same reason, we can write $I_{C3} = I_{C4} = 9.5 \mu\text{A}$.

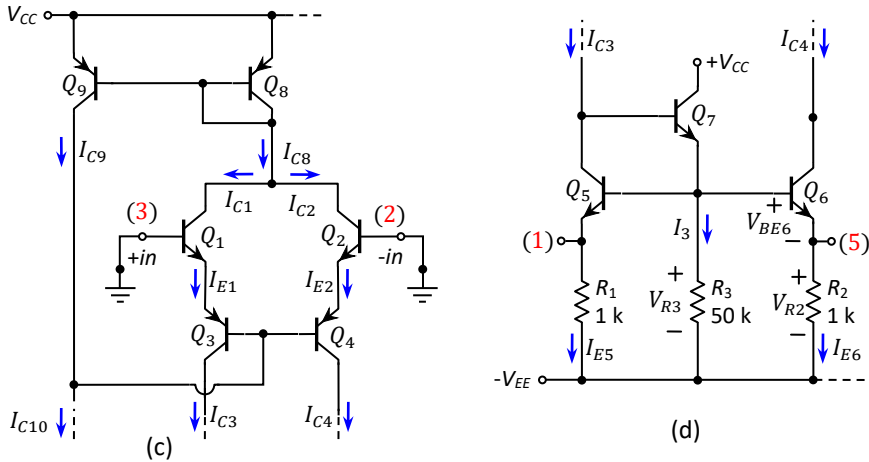


Fig.14-30: Input stage: (C) Upper portion, (d) Lower portion

The lower portion of the input stage is shown in Fig.14-30(d). For these transistors the base currents will also be very small and can be neglected. So, current through Q_5 will be equal to I_{C3} , that is $I_{E5} = I_{C5} = I_{C3} = 9.5 \mu\text{A}$. Similarly, $I_{E6} = 9.5 \mu\text{A}$.

From Fig.14-30(d), we find that the voltage across R_3 will be equal to the sum of V_{BE6} and voltage drop across R_2 , that is,

$$V_{R3} = V_{BE6} + V_{R2} = V_T \ln \left(\frac{I_{E6}}{I_s} \right) + V_{R2}$$

$$\text{or,} \quad I_3 R_3 = V_T \ln \left(\frac{I_{E6}}{I_s} \right) + I_{E6} R_2$$

It is given that $I_s = 10^{-14} \text{ A}$

$$\therefore I_3(50 \text{ k}) = (26 \text{ mV}) \ln \left(\frac{9.5 \text{ } \mu\text{A}}{10^{-14} \text{ A}} \right) + (9.5 \text{ } \mu\text{A})(1 \text{ k})$$

$$\therefore I_3 \approx 11 \text{ } \mu\text{A}$$

Again, as the base currents are negligible, we can write, $I_{C7} = I_{E7} = I_3 = 11 \text{ } \mu\text{A}$. Emitter voltage of transistors Q_5 and Q_6 will be,

$$V_{E5} = V_{E6} = -V_{EE} + I_{E5}R_1$$

$$\therefore V_{E5} = V_{E6} = -15 \text{ V} + (9.5 \text{ } \mu\text{A})(1 \text{ k}) \approx -15 \text{ V}$$

Second Stage:

The second stage of the op-amp along with its related biasing current mirror circuit is shown in Fig.14-30(e). Here, transistors Q_{12} and Q_{13} form a current mirror circuit. So, the emitter current of Q_{13} will be equal to I_{ref} or 0.733 mA . This current will be divided between the two collectors of Q_{13} according to given ratio. Thus, $I_{CB} = 0.75 \times 0.733 \text{ mA} = 0.55 \text{ mA}$ and $I_{CA} = 0.183 \text{ mA}$. Here, also we will neglect the base currents. So, the emitter current of Q_{17} will be approximately equal to I_{CB} , that is, $I_{E17} = I_{CB} = 0.55 \text{ mA}$.

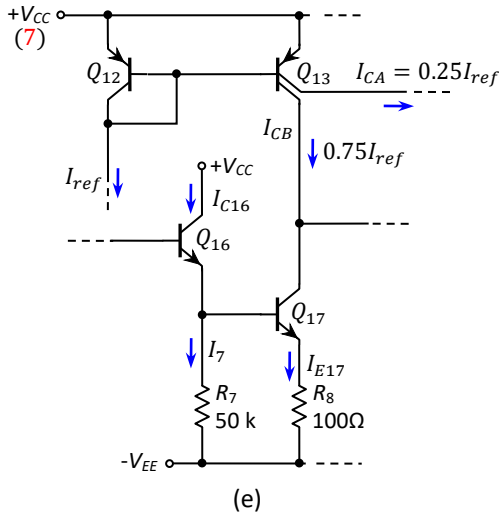


Fig.14-30: (e) Second stage of the op-amp

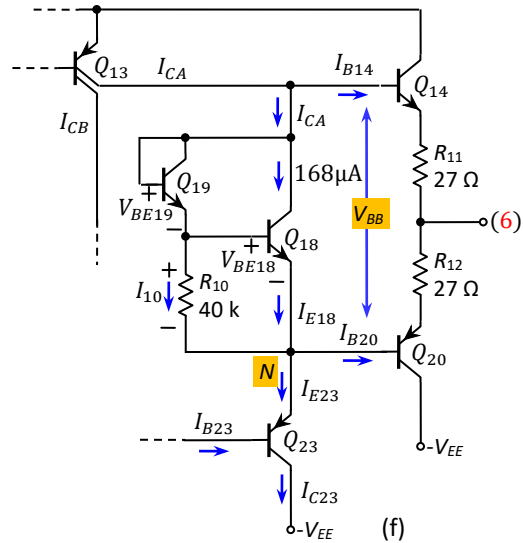


Fig.14-30: (f) Output stage of the op-amp

The value of I_7 can be calculated using the same method of I_3 . Therefore,

$$I_7 R_7 = V_T \ln \left(\frac{I_{E17}}{I_s} \right) + I_{E17} R_8$$

$$\text{or, } I_7(50 \text{ k}) = (26 \text{ mV}) \ln \left(\frac{0.55 \text{ mA}}{10^{-14} \text{ A}} \right) + (0.55 \text{ mA})(100 \text{ } \Omega)$$

$$\therefore I_7 \approx 14 \text{ } \mu\text{A}$$

The collector current of Q_{16} will be approximately I_7 , that is $I_{C16} = 14 \text{ } \mu\text{A}$.

Emitter voltage of transistors Q_{17} will be,

$$V_{E17} = -V_{EE} + I_{E17}R_{17}$$

$$\therefore V_{E17} = -15 \text{ V} + (0.55 \text{ mA})(100 \Omega) \approx -15 \text{ V}$$

Output Stage:

The output stage of the op-amp of Fig.14-28 is redrawn in Fig.14-30(f). The transistors used for the short-circuit protection purpose (shown in white shading) are not shown here. As the value of the base currents I_{B14} and I_{B20} are very small, we can neglect them as before. Therefore, $I_{E23} = I_{CA} = 183 \mu\text{A}$. The collector current of Q_{23} will be approximately $I_{C23} \approx 183 \mu\text{A}$. As this is a PNP transistor, it has $\beta = 50$. So the base current will be,

$$I_{B23} = \frac{I_{C23}}{\beta} = \frac{0.183 \text{ mA}}{50} \approx 3.67 \mu\text{A}$$

Let us assume that, $V_{BE18} = 0.6 \text{ V}$. The voltage across R_{10} will be equal to $V_{BE18} = 0.6 \text{ V}$. So,

$$I_{10} = \frac{V_{BE18}}{R_{10}} = \frac{0.6 \text{ V}}{40 \text{ k}} \approx 15 \mu\text{A}$$

Now applying KCL at point N ,

$$I_{10} + I_{E18} = I_{E23}$$

$$\text{or, } 15 \mu\text{A} + I_{E18} = 183 \mu\text{A}$$

$$\therefore I_{E18} = 168 \mu\text{A}$$

We have assumed $V_{BE18} = 0.6 \text{ V}$. Let us calculate this value to make a check,

$$V_{BE18} = V_T \ln \left(\frac{I_{E18}}{I_s} \right) = (26 \text{ mV}) \ln \left(\frac{168 \mu\text{A}}{10^{-14} \text{ A}} \right)$$

$$\therefore V_{BE18} = 0.612 \text{ V [almost } 0.6 \text{ V]}$$

Now if we neglect the base current of Q_{18} , we can write $I_{C18} \approx I_{E18} = 168 \mu\text{A}$.

$$I_{E19} = I_{10} + I_{B18}$$

$$\therefore I_{E19} = 15 \mu\text{A} + \frac{168 \mu\text{A}}{200} \approx 15.84 \mu\text{A}$$

For this transistor also we can neglect the base current and thus $I_{E19} \approx I_{C19}$. Therefore, the base-emitter voltage of Q_{19} can be calculated as,

$$V_{BE19} = V_T \ln \left(\frac{I_{E19}}{I_s} \right) = (26 \text{ mV}) \ln \left(\frac{15.84 \mu\text{A}}{10^{-14} \text{ A}} \right)$$

$$V_{BE19} = 0.55 \text{ V}$$

As the base and collector of transistor Q_{19} is short, the value of V_{BB} (voltage between the bases of Q_{14} and Q_{20}) will be,

$$V_{BB} = V_{BE19} + V_{BE18} = 0.55 \text{ V} + 0.612 \text{ V} = 1.162 \text{ V}$$

The voltage drops across R_{11} and R_{12} will be very small and we will neglect them. As Q_{14} and Q_{20} are matched transistors, V_{BB} will be equally divided between the B-E junction of these transistors.

$$V_{BE14} = V_{BE20} = \frac{1.162 \text{ V}}{2} \approx 0.58 \text{ V}$$

$$I_{E14} = I_{E20} = I_s e^{V_{BE}/V_T}$$

$$I_{E14} = I_{E20} = (3 \times 10^{-14}) e^{0.58 \text{ V}/26 \text{ mV}} \approx 146 \mu\text{A}$$

Comments: Although lengthy process, this DC analysis will help the students to get some idea about the operating currents of transistors at different stages.

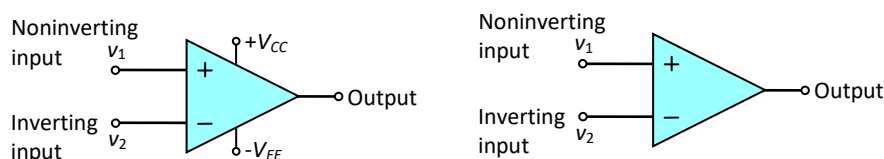


Fig.14-31: Electrical symbol of operational amplifier

14.13 Op-amp Symbol and Equivalent Circuit

In schematic diagram the operation amplifier is represented by a symbol as shown in Fig.14-31. The symbol of op-amp has two inputs: one is inverting and the other is noninverting input. The noninverting input is represented by '+' sign and the inverting input is represented by '-' sign. The symbol has one output. In some cases, power supply connection is shown in the symbol [Fig.14-31(a)] while in some other cases the power supply connection is not shown [Fig.14-31(b)]. An AC signal (or DC voltage) applied to noninverting input, produces an in-phase (or same polarity) signal at the output. On the other hand, an AC signal (or DC voltage) applied to inverting input, produces an out of phase (or opposite polarity) signal at the output.

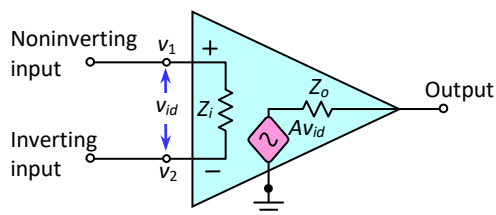


Fig.14-32: Equivalent circuit of operational amplifier

The equivalent circuit of an operational amplifier is shown in Fig.14-32 in the same way as we represented single-stage amplifier before. The input circuit is represented by a very large impedance, Z_i and the output circuit is represented by a dependant voltage source and a series output impedance, Z_o . The voltage level of the dependant voltage source is the product of the gain of the op-amp (A) and the difference of two voltages ($v_{id} = v_1 - v_2$) applied to the inputs, that is, Av_{id} .

14.14 Power Supply for Op-amps

Already we have studied that operational amplifier is constructed using a number of differential amplifiers, emitter follower, output amplifier etc. Differential amplifiers need both positive and

negative voltage sources (called *split power supply* or *dual-supply*) for their linear operation. This means that most op-amp ICs need both a positive supply and a negative supply of equal values. A few earlier ICs require unequal power supplies, and some ICs require only a positive supply. For example, the 702 op-amp requires unequal supplies, whereas the 324 requires only a positive supply. Some split-supply op-amp ICs can also be operated from a single supply voltage, provided that a special external biasing circuit is used for them.

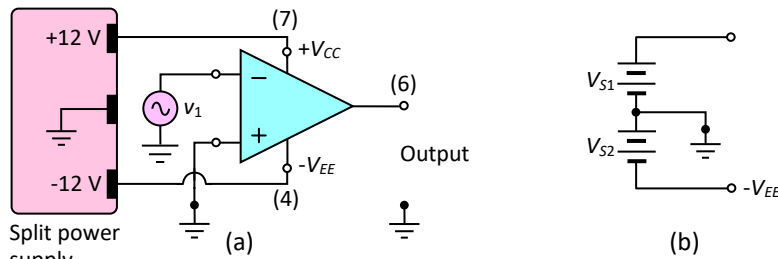


Fig.14-33: (a) Connection of split power supply, (b) Two power supplies used as split supply

Dual or split power supply can be constructed using centre-tapped transformer as discussed in [Chapter \[H_S_atten\]](#). The connection of this power supply with the op-amp (741) is shown in Fig.14-33(a). Pin (7) is connected to the positive terminal (+V_{CC}) and Pin (4) is connected to the negative terminal (-V_{EE}) of the supply. The output is taken from Pin (6) with respect to ground. The input signals have also to be connected with respect to the ground.

Instead of using a split-supply, we can connect two separate power supplies as shown in Fig.14-33(b). Sometimes, it may not be possible to use two supplies for a single circuit containing op-amp. In that case, we can build obtain (+V_{CC}) and (-V_{EE}) using a single power supply as shown in Fig.14-34. For the supply of Fig.14-34(a), the value of total resistance ($2R$) should be ≥ 10 k, so that it cannot draw much current from the supply source V_S , that is, to minimize power loss. Two capacitors are used for supply decoupling and their values should be 0.1 to 10 μ F. In Fig.14-34(b) Zener diodes are used to obtain symmetrical supply voltages. The value of series resistance (R_s) should be selected in such a way that it barely draws the current required for the op-amp and to operate the Zener diodes in the breakdown region. Due to the tolerance (variation) of standard resistors, the voltages of Fig.14-34(a) may not be equal.

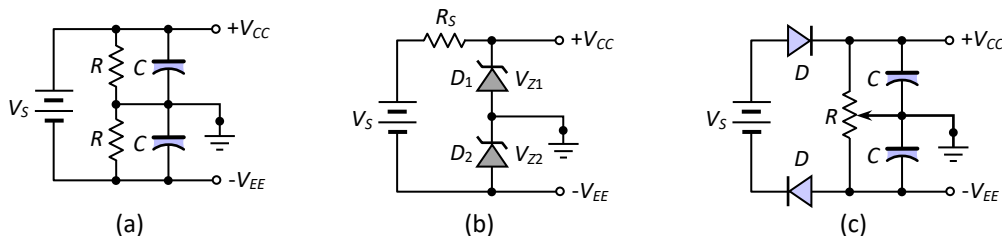


Fig.14-34: Conversion of single power supply into split power supply

So the circuit can be improved by adopting the ability to adjust the voltage levels as shown in Fig.14-34(c). Diodes are used to protect the IC in case of accidental reverse connection of source

(V_s). Note that these diodes could be used in the circuits of Fig.14-34(a) and Fig.14-34(b) for the same protection.

14.15 Op-amp Characteristics

An operational amplifier is fundamentally a voltage amplifying device designed to be used with external feedback components such as resistors and capacitors between its output and input terminals. These feedback components determine the resulting function or operation of the amplifier and by virtue of the different feedback configurations whether resistive, capacitive or both, the amplifier can perform a variety of different operations. Before going to understand the operation of an op-amp, we must learn about the op-amp characteristics. Some of the important characteristics of op-amp are described here.

14.15.1 Input Bias Current

In the inputs of an op-amp, while very high impedance, still allow some input current to flow. This is the input bias current (I_B) and it is in the range of nA in modern op-amps. I_B is the average of the currents that flow into the noninverting input terminal (I_{B1}), and into the inverting input terminal (I_{B2}) of the op-amp [see Fig.14-35]. Mathematically, it is given by

$$I_B = \frac{I_{B1} + I_{B2}}{2} \quad (14-53)$$

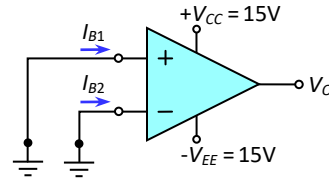


Fig.14-35: Bias current of operational amplifier

In fact, the two input bias currents I_{B1} and I_{B2} are the base currents of the two transistors of the first differential amplifier in the op-am. The typical value of I_B for $\mu A741C$ is 80 nA and the maximum value is 500 nA. Whereas, the maximum value of I_B for the precision op-amp OP07C is only 12 nA.

14.15.2 Input Offset Current

In ideal case, I_{B1} should be equal to I_{B2} . But in practical op-amp, due to the mismatch of the transistors (mainly V_{BE}), I_{B1} and I_{B2} may not be equal. The algebraic difference between I_{B1} and I_{B2} is called the input offset current and is denoted as I_{io} . Mathematically,

$$I_{io} = |I_{B1} - I_{B2}| \quad (14-54)$$

As shown in Fig.14-35, I_{B1} is the bias current into the noninverting input and I_{B2} is the bias current into the inverting input terminal.

The input offset current for $\mu A741C$ is typically 20 nA and maximum 200 nA. Whereas, the value of I_{io} for the precision op-amp OP07C is only 6 nA maximum.

14.15.3 Input Offset Voltage

The output of an operational amplifier should be zero when the voltage difference between the inverting and the noninverting inputs is zero ($v_{id} = |v_1 - v_2| = 0$), or when both inputs are grounded, i.e. $v_1 = v_2 = 0$. Due to the mismatch of the transistors in differential amplifier (mainly

different V_{BE}), practical op-amps have some input offset voltage. This input offset voltage is amplified by the op-amps and produce a large output voltage. This output offset voltage is an unwanted signal. To eliminate the output offset voltage, we can apply a voltage of opposite polarity at the input side as shown in Fig.14-36. Thus, the *input offset voltage* is defined as the voltage (V_{io}) that must be applied between the two input terminals of the op-amp to obtain zero volts at the output. V_{io} is symbolically represented by a voltage source that is in series with either the noninverting or inverting input terminal. It can be either negative or positive in polarity, varying from device to device of the same wafer lot.

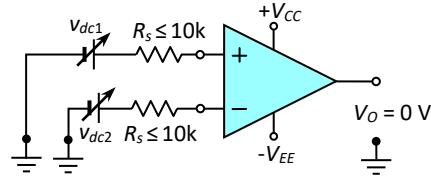


Fig.14-36: Offset voltage measurement technique of operational amplifier

The cause of input offset voltage is well known—it is due to the inherent mismatch of the input transistors and components during fabrication of the silicon die, and stresses placed on the die during the packaging process (minor contribution). These effects collectively produce a mismatch of the bias currents that flow through the input circuit, and primarily the input devices, resulting in a voltage difference at the input terminals of the op-amp. This voltage difference is amplified by the op-amp and large offset output voltage is developed.

For $\mu A741C$ the maximum value of V_{io} is 6 mV DC. Whereas, the value of V_{io} for the precision op-amp OP07C is 150 μV .

Although from Fig.14-36, we get the concept of input offset voltage, but using this circuit we cannot measure the value of V_{io} . Due to the high gain of the op-amp, if we try to adjust the level of input voltage, the op-amp output will be saturated, instead of being zero. Practically it can be measured by reducing the gain of the op-amp using negative feedback as shown in Fig.14-37. Here the gain of the overall circuit is 101 (will be discussed later). So the value of V_{io} will be,

$$V_{io} = \frac{V_{oo}}{101}$$

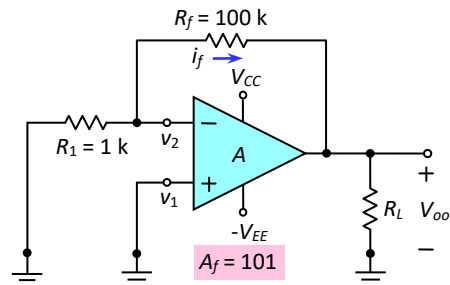


Fig.14-37: Offset voltage measuring circuit arrangement with gain = 101

For some op-amps there have mechanism for nulling these input mismatches. In $\mu A741C$ a 10 k to 100 k potentiometer is used as shown in Fig.14-37. By adjusting the potentiometer the output voltage is made zero for zero input voltages.

14.15.4 Output Offset Voltage

Output offset voltage is the voltage which will appear at the output even when there is no input (that is inputs are 0V). It is produced due to input offset voltage. The output offset voltage is nothing but the input offset voltage multiplied by the gain of the op-amp circuit configuration. For this reason most datasheets do not mention its value.

14.15.5 Differential Input Resistance

Differential input resistance, R_i is defined as the equivalent resistance that would be measured at either input terminal with the other terminal grounded. This means that the input resistance R_{i1} seen from the input signal source v_1 is determined with the signal source v_2 set at zero. Similarly, the input signal v_1 is set at zero to determine the input resistance R_{i2} seen from the input signal source v_2 . Generally, $R_i = R_{i1} = R_{i2}$. For the $\mu A741C$ op-amp, the input resistance is very high, typical value is $2\text{ M}\Omega$. Whereas, for the AD743 op-amp the input resistance is very high, typical value is $10^4\text{ M}\Omega$. As shown in Fig.14-38, this resistance is also represented by Z .

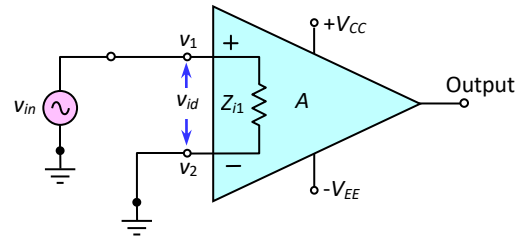


Fig.14-38: Equivalent input resistance seen in noninverting terminal.

14.15.6 Input Capacitance

Input capacitance, C_i is defined as the equivalent capacitance that would be measured at either input terminal with the other terminal shorted to ground. All op-amps exhibit a differential-mode input capacitance C_{dm} and a common-mode (with the inputs tied together) input capacitance C_{cm} . C_{dm} is the capacitance between two input terminals of the op-amp, but C_{cm} is the capacitance measured between the inputs (shorted together) and the ground. Fig.14-39 shows these capacitances. Though internal the capacitances are shown externally for better visualization. The capacitance measured from one terminal to ground will just half of C_{cm} , that is, $C_{cm1} = C_{cm2} = 0.5 C_{cm}$.

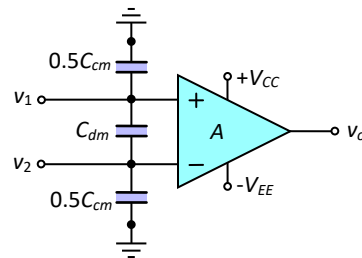


Fig.14-39: Equivalent input capacitance of op-amp

Generally, the values of these capacitances are not mentioned in the datasheets. The typical value of input capacitance for 741 IC is 1.4 pF which is the equivalent capacitance measured from any one of the input terminals by keeping the other terminal connected to ground.

14.15.7 Offset Voltage Adjustment Range

One of the most important features of the 741 op-amps family is *offset voltage null capacity*. 741 op-amp has two special pins (pin 1 and pin 5) marked as *offset null* for this purpose. As shown in Fig.14-40, a potentiometer ($10\text{ k}\Omega$ to $100\text{ k}\Omega$) is to be connected between offset null pins 1 and 5 and the wiper of the potentiometer should be connected to the negative supply $-V_{EE}$. By varying the potentiometer, the output offset voltage (output voltage without any input voltage) can be reduced to zero volts. The *offset voltage adjustment range* is the range of output voltage that can be adjusted (reduced to zero) by varying a $10\text{ k}\Omega$

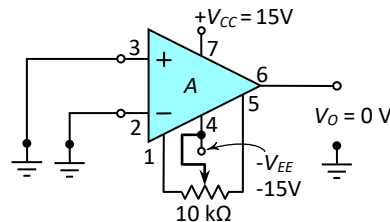


Fig.14-40: To nullify the offset voltage of 741 using potentiometer

potentiometer. For μAF741 op-amp, the value of offset voltage adjustment range is ± 15 mV. It means that if the output offset voltage is greater than ± 15 mV, it cannot be nullify using the potentiometer.

14.15.8 Input Voltage Range

When the same voltage is applied to both input terminals, the voltage is called a common-mode voltage, V_{cm} , and the op-amp is said to be operating in the common-mode configuration as shown in Fig.14-41. The maximum range of this common-mode voltage, for which the op-amp works properly (linearly) is called the *input voltage range*. For μAF741 op-amp, the value of the input voltage range is ± 13 V. This means that the common-mode voltage, applied to both input terminals, can be as high as +13 V or as low as -13 V without disturbing the proper functioning of the op-amp.

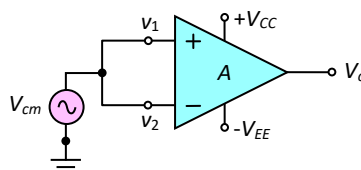


Fig.14-41: Common-mode signal applied to an op-amp

14.15.9 Common-Mode Rejection Ratio

We have already defined this term for differential amplifier. The same parameter is also used for the operational amplifiers. For op-amp, CMRR can be defined as the ratio of the differential voltage gain A_{dm} to the common-mode voltage gain A_{cm} , that is,

$$\text{CMRR} = \frac{A_{dm}}{A_{cm}} \quad (14-55)$$

The differential voltage gain A_{dm} is the open-loop voltage gain (A) of the op-amp which is given in the data sheet. The common-mode gain is not given in datasheet, but we can measure or calculate it [Example 13-6].

Generally the value of A_{cm} is very small and the value of $A_{dm} = A$ is very large. Therefore, the value of CMRR is also very large. Being a large value, CMRR is most often expressed in decibels (dB) as,

$$\text{CMRR(dB)} = 20 \log \left(\frac{A_{dm}}{A_{cm}} \right) \quad (14-56)$$

For the 741C, CMRR is 90 dB typically whereas, for the OP97F the typical value of CMRR is 132 dB. This means that OP97F op-amp has a better ability to reject common-mode voltages (which are mainly noise) than 741C.

Example 13-6

OP97 is a precision op-amp. The typical values of large-signal differential voltage gain (A_{dm}) and the CMRR is 2000 V/mV, and 132 dB, respectively. Calculate the value of common-mode voltage gain (A_{cm}) in absolute value and in dB.

Solution:

Here, $A_{dm} = 2000 \text{ V/mV} = 2000 \text{ V}/10^{-3} \text{ V} = 2 \times 10^6$

and, $\text{CMRR(dB)} = 132 \text{ dB}$

According to the definition of dB we can write,

$$132 \text{ dB} = 20 \log(\text{CMRR}) \text{ dB}$$

or, $\log(\text{CMRR}) = 132 / 20 = 6.6$

$\therefore \text{CMRR} = \log_{10}^{-1} 6.6 = 10^{6.6} \approx 3981072$

Using Equ.(14-55), $\text{CMRR} = \frac{A_{dm}}{A_{cm}}$

or, $A_{cm} = \frac{A_{dm}}{\text{CMRR}} = \frac{2 \times 10^6}{3981072} \approx 0.5 \text{ [Ans.]}$

or, $A_{cm}(\text{dB}) = 20 \log 0.5 \approx -6 \text{ dB [Ans.]}$

Comments: Negative value of $A_{cm}(\text{dB})$ means that the common mode gain is less than unity. So any noise induced in the input terminals will be attenuated by a factor 0.5.

14.15.10 Supply Voltage Sensitivity

Supply voltage sensitivity is also called **supply voltage rejection ratio** or **power supply rejection**. In datasheets it is denoted as k_{SVS} . Variations in V_{CC} and/or $-V_{EE}$ produce some change in the input offset voltage and thus, change the output voltage. The **supply voltage sensitivity** is defined as the change in input offset voltage (ΔV_{io}) to the change in supply voltage (ΔV_{CC} or ΔV_{EE}). Mathematically,

$$k_{SVS} = \frac{\Delta V_{io}}{\Delta V_{CC}} \quad (14-57)$$

Generally, k_{SVS} is measured in $\mu\text{V/V}$ unit. This value is sometimes represented in dB unit. That is,

$$k_{SVS}(\text{dB}) = 20 \log \left(\frac{\Delta V_{io}}{\Delta V_{CC}} \right) \quad (14-58)$$

For the 741C op-amp, the maximum value of k_{SVS} is $150 \mu\text{V/V}$ whereas, for OP07 the maximum value of k_{SVS} is $32 \mu\text{V/V}$.

14.15.11 Large Signal Differential Voltage Gain

Since the op-amp amplifies the difference voltage between two input terminals ($V_{id} = v_1 - v_2$), and the output voltage swing is very large (compared to small signal amplifier), the gain of an op-amp is called **large signal differential voltage gain** ($A_{dm} = A$). The definition of this parameter is same as the normal amplifier. That is,

$$A = \frac{\text{output voltage}}{\text{differential input voltage}} = \frac{V_o}{V_{id}} \quad (14-59)$$

The large-signal voltage gain of the 741C is $200 \text{ V/mV} = 0.2 \times 10^6$ typically, whereas, for OP07 the typical value of A is 2000 V/mV .

14.15.12 Peak Output Voltage

It is the range of output voltage for which the op-amp works linearly, that is, the input signal is amplified properly (without being saturated). As the op-amp can produce large signal, the value of *peak output voltage* is high. The output voltage swing is limited by the supply voltage. As, for a particular op-amp, there has a limit on supply voltage, the peak output voltage has a limited value as well. For example, 741C op-amp supports maximum $\pm 15\text{ V}$ as power supply. With this maximum power supply, and with a load resistance of $R_L > 2\text{ k}$, the peak output voltage will be $\pm 13\text{ V}$, generally 2 V lower than the supply voltage [see Fig.14-42].

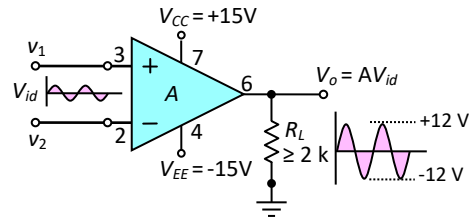


Fig.14-42: Peak output voltage of operational amplifier

14.15.13 Input Impedance

In datasheets input impedance is also mentioned as input resistance and is denoted as R_i or Z_i . It is the equivalent resistance that is measured between the inverting and the noninverting terminals of op-amp. As shown in Fig.14-43, Z_i is the equivalent resistance between the inverting and the noninverting terminals. This resistance works as the load resistance for the input signal (if not changed by external components). The higher the input impedance, the better is the op-amp. Typical values of input impedance range from $1\text{ M}\Omega$ to $10\text{ T}\Omega$. For example, the typical value of the input impedance of 741C is only $2\text{ M}\Omega$.

14.15.14 Output Impedance

In datasheets output impedance is also mentioned as output resistance and is denoted as R_o or Z_o . It is the equivalent resistance that is measured between the output terminal and ground. As shown in Fig.14-43, Z_o is the equivalent series resistor of the output dependant voltage source (Av_{id}). The typical value of the output impedance of 741C is only $75\text{ }\Omega$. The lower the output impedance, the better is the op-amp.

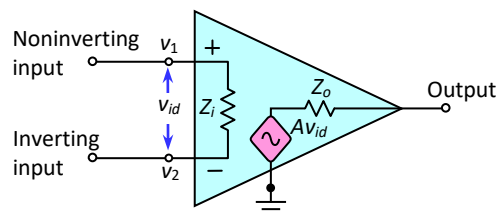


Fig.14-43: Equivalent input and output impedance of an operational amplifier

14.15.15 Output Short-Circuit Current

Op-amp manufacturers specify the value of output short-circuit current (I_{sc}). Nobody would deliberately short the output terminal of the op-amp to ground, but if such an event happens accidentally, then I_{sc} amount of current will flow through the short path. This high current may damage the op-amp if it does not have output short-circuit protection. However, in section 13-12, we found that the 741 family op-amps has in-built short-circuit protection circuitry. From the output short-circuit current we get some idea about the current capacity of the op-amp. For the 741C op-amp the short-circuit current $I_{sc} = 25\text{ mA}$.

14.15.16 Supply Current

Supply current, I_{CC} , is the current drawn by the op-amp from the power supply, that is, from $+V_{CC}$ to $-V_{EE}$ with the output terminal open ($I_o = 0$). From this supply current, we get some idea of power dissipation of the op-amp. For 741C op-amp, the value of supply current is $I_{CC} = 2.8$ mA maximum.

14.15.17 Power Consumption

Power consumption, P_c , is the amount of quiescent power that is consumed by the op-amp in order to operate properly with the input terminals connected to ground ($V_{id} = 0$) and with no output ($V_o = 0$). From this parameter, we get some idea about how much power is wasted by the op-amp in an electronic system. For 741C op-amp, the maximum value of power consumption is $P_D = 85$ mW.

14.15.18 Transient Response

The response of any network to a given input is composed of two parts: the **transient response** and **steady-state response**. The transient response is that portion of the complete response before the output attains some fixed value. This response is time dependant. After the transient response the system reaches to a fixed value that does not depend on time. Therefore, this time independent response is called **steady-state response**. **Rise time** and **overshoot** are two important parameters of transient response.

The time required by the output to go from 10% to 90% of its final value is called the **rise time** and is denoted by t_r . Conversely, the maximum amount of voltage by which the output deviates from the steady-state value is called **overshoot**. Overshoot is generally expressed as a percentage of the steady-state value.

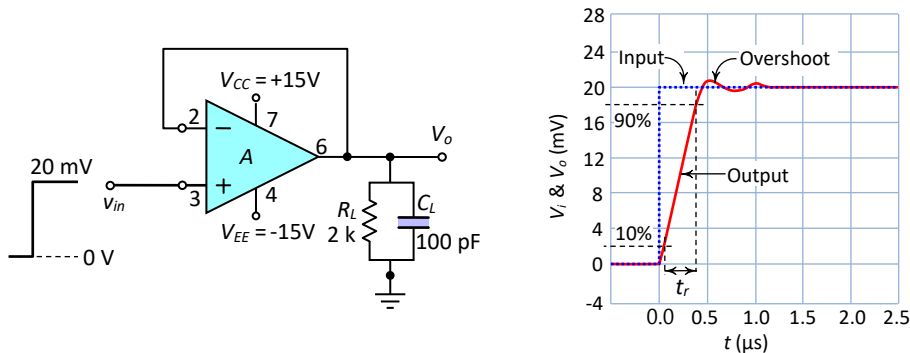


Fig.14-44: (a) Slew-rate measurement circuit, and (b) Transient response of operational amplifier

The transient response test circuit for the 741C is shown in Fig.14-44. The input is a step function with $V_{in} = 20$ mV DC. From this circuit arrangement, it is found that the rise time $t_r = 0.3\text{ }\mu\text{s}$ and the overshoot $OS = 5\%$ for the 741C op-amp. The transient response is one of the important considerations in selecting an op-amp for AC applications. In fact, the rise time is inversely proportional to the unity gain bandwidth of the op-amp. The smaller the value of rise time, the higher is the bandwidth.

14.15.19 Slew Rate

Slew rate (SR) is defined as the maximum rate of change of output voltage per unit time. Generally slew rate is measured in volt per microsecond ($V/\mu s$) unit, that is, the maximum change in output voltage in one microsecond. Mathematically,

$$SR = \frac{\text{max. change in output voltage}}{\text{change in time}} = \frac{\Delta V_o}{\Delta t} \Big|_{\text{max}} \quad V/\mu s = \frac{dV_o}{dt} \Big|_{\text{max}} \quad V/\mu s \quad (14-60)$$

This parameter expresses how fast the output voltage of an op-amp can change in response to input [see Fig.14-45]. To measure the slew rate a step input (or a square wave) is applied to the input of op-amp. The slew rate changes with change in voltage gain and is normally specified at unity gain ($A_v = 1$). Slew rate limits the maximum frequency that the op-amp can amplify. Thus, slew rate is one of the important factors in selecting, the op-amp for AC applications. The higher the value of slew rate, the better is the op-amp.

The typical value of the slew rate of 741C op-amp is only $0.5 V/\mu s$ which is very small. Whereas, for LM318-N op-amp the typical value of slew rate is as high as $70 V/\mu s$. The slew rate is further discussed in [Section 13.40 \[H_S_atten\]](#).

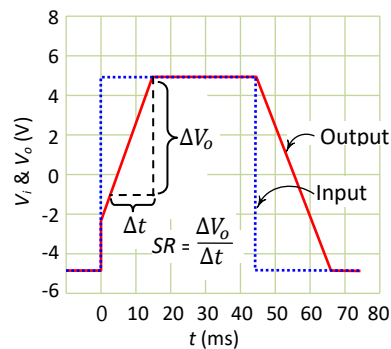


Fig.14-45: Slew rate limited response of operational amplifier

14.15.20 Gain-Bandwidth Product

The **gain-bandwidth product (GB)** is the product of the gain of an op-amp and its bandwidth at that gain. Generally the gain-bandwidth product is measured with unity gain. If the gain of an op-amp is set to 1 (using negative feedback), the value of the bandwidth will also be equal to the gain-bandwidth product. Although the datasheet of $\mu A741$ op-amp does not mention this value, but from the open-loop voltage gain versus frequency graph, this parameter can be determined easily. Fig.14-46 shows the open-loop voltage gain versus frequency graph for $\mu A741$. From this graph we find that the bandwidth is 1 MHz at the gain level 1 (0 dB). Therefore, the gain-bandwidth product of $\mu A741$ is 1 MHz. For any gain level this product remains constant (if the frequency response has only one pole frequency). There are many op-amp ICs with higher gain-bandwidth product. For example, THS4567 op-amp has 220 MHz gain-bandwidth product.

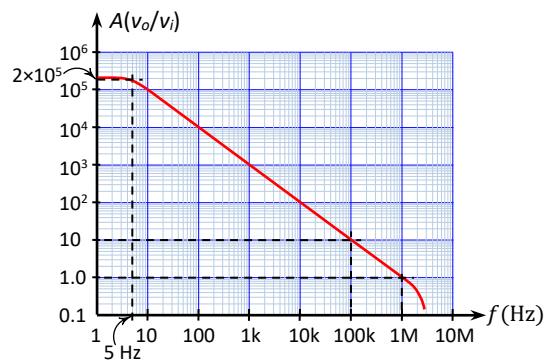


Fig.14-46: Frequency response of a typical 741 operational amplifier

14.15.21 Transfer Characteristics of Operational Amplifier

Already we have discussed that operational amplifier is a high gain voltage amplifier. It has a very large differential voltage gain (A) and a very small common-mode voltage gain. Op-amp mainly amplifies the difference of the voltages applied to its two inputs. Considering the output offset voltage zero, we can write,

$$v_o = Av_{id} = A(v_1 - v_2) \quad (14-61)$$

here, v_1 is the voltage applied to the noninverting input, and v_2 is the voltage applied to the inverting input as shown in Fig.14-47.

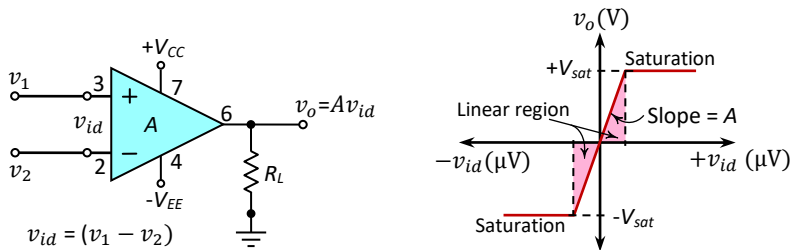


Fig.14-47: (a) Differential amplifier, and (b) Transfer characteristics of operational amplifier

Equ.(14-61) shows the linear relationship between the output (v_o) and the differential input v_{id} . By plotting this equation we will get the voltage transfer characteristics of the op-amp which is shown in Fig.14-47(b). From this graph, we find that the output voltage changes rapidly for very small change in differential input voltage (v_{id}) due to the high voltage gain of the op-amp. The output voltage cannot exceed the maximum swing of the op-amp which is imposed by the supply voltage or by the device itself. This means that the output voltage is directly proportional to the input voltage difference (v_{id}) only until it reaches the maximum limit, where the output gets saturated. The graph shown in Fig.14-47(b) is an ideal voltage transfer curve, because the output offset voltage has assumed to be zero. In practical op-amp circuit there may have output offset voltage. But for simplicity this output offset voltage is ignored unless specified.

Example 13-7

For the circuit of Fig.14-47(a) determine the minimum value of input voltage for which the output will be saturated. Assume $A = 2 \times 10^6$ and the maximum output voltage swing is ± 13 V.

Solution:

Using Equ.(14-61),

$$v_o = Av_{id}$$

or,

$$v_{id} = \frac{v_o}{A} = \frac{\pm 13 \text{ V}}{2 \times 10^6} = \pm 6.5 \mu\text{V} \text{ [Ans.]}$$

Comments: Due to the very high differential gain, the output saturates for a small value of input $\pm 6.5 \mu\text{V}$. Beyond $\pm 6.5 \mu\text{V}$, the op-amp will lose its linear property. For this reason, negative feedback is used to decrease the gain and adopt the advantages of negative feedback.

14.16 Characteristics of Ideal and Practical Op-Amp

Already we have discussed various parameters related with op-amps. Although we cannot fabricate an ideal op-amp, we can imagine an op-amp with ideal characteristics. Generally the characteristics of practical op-amp are compared with those of ideal op-amp. The following table gives a comparison between the ideal and practical operational amplifiers.

Table 13-: Characteristics of an ideal and practical op-amp

Parameters	Ideal Op-amp	Practical Op-amp
Open-loop gain (A)	Infinite	Very high (>10000)
Bandwidth (BW)	Infinite	High ($\approx 10 \text{ MHz}$)
CMRR	Infinite	High ($> 60 \text{ dB}$)
Input impedance (Z_{in})	Infinite	Very high ($>1 \text{ M}\Omega$)
Output impedance (Z_o)	Zero	Very low ($< 100 \Omega$)
Input bias currents (I_B)	Zero	Very low ($< 50 \text{ nA}$)
Input offset voltage (V_{io})	Zero	Low ($< 10 \text{ mV}$)
Input offset current (I_{io})	Zero	Low ($< 50 \text{ nA}$)
Slew rate (SR)	Infinite	High (few $\text{V}/\mu\text{s}$)
Drift	Zero	Low

14.17 Operational Amplifier Datasheet

Operational amplifier is such a versatile IC that it has many applications in electronics. Different applications need different types of op-amps. For proper selection of op-amp IC we must refer to their datasheets. To get some idea about the datasheet of op-amp, some portion of the datasheet of LM741 op-amp IC is reproduced below (courtesy of Fairchild Semiconductor, USA).



February 2007

LM741

Single Operational Amplifier

Features

- Short Circuit Protection
- Excellent Temperature Stability
- Internal Frequency Compensation
- High Input Voltage Range
- Null of Offset

Description

The LM741 series are general purpose operational amplifiers. It is intended for a wide range of analog applications. The high gain and wide range of operating voltage provide superior performance in integrator, summing amplifier, and general feedback applications.

8-DIP



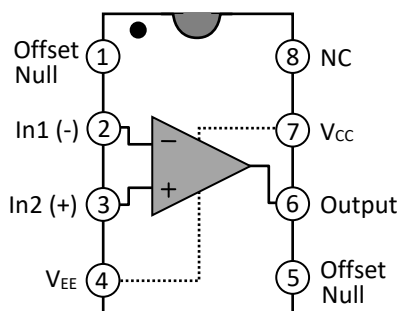
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8-SOP



1

Internal Block Diagram



Absolute Maximum Ratings

The “Absolute Maximum Ratings” are those values beyond which the safety of the device cannot be guaranteed. The device should not be operated at these limits. The parametric values defined in the Electrical Characteristics tables are not guaranteed at the absolute maximum ratings. $T_A = 25^\circ\text{C}$, unless otherwise specified.

Symbol	Parameter	Value	Unit
V_{CC}	Supply Voltage	± 18	V
$V_{I(DIFF)}$	Differential Input Voltage	30	V
V_I	Input Voltage	± 15	V
-	Output Short Circuit Duration	Indefinite	-
P_D	Power Dissipation	500	mW
T_{OPR}	Operating Temperature Range	$0 \sim +70$	$^\circ\text{C}$
T_{STG}	Storage Temperature Range	$-65 \sim +150$	$^\circ\text{C}$

Electrical Characteristics

($V_{CC} = 15\text{V}$, $V_{EE} = -15\text{V}$, $T_A = 25^\circ\text{C}$, unless otherwise specified)

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Input Offset Voltage	V_{IO}	$R_S \leq 10\text{k}\Omega$	-	1.0	6.0	mV
		$R_S \leq 50\text{k}\Omega$	-	-	-	
Input Offset Voltage Adjustment Range	$V_{IO(R)}$	$V_{CC} = \pm 20$	-	± 15	-	mV
Input Offset Current	I_{IO}	-	-	20	200	nA
Input Bias Current	I_{BIAS}	-	-	80	500	nA
Input Resistance (Note1)	R_I	$V_{CC} = \pm 20$	0.3	2.0	-	$\text{M}\Omega$
Input Voltage Range	$V_{I(R)}$	-	± 12	± 13	-	V
Large Signal Voltage Gain	G_V	$R_L \geq 2\text{k}\Omega$	$V_{CC} = \pm 20$ $V_{C(P-P)} = \pm 15$	-	-	V/mV
			$V_{CC} = \pm 15$ $V_{C(P-P)} = \pm 10$	20	200	
			-	-	-	
			-	-	-	
Output Short Circuit Current	I_{SC}	-	-	25	-	mA
Output Voltage Swing						
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq 10\text{k}\Omega$	$V_{CC} = \pm 20$	-	-	V
			$V_{CC} = \pm 15$	± 12	± 14	
		$R_L \geq 2\text{k}\Omega$	$V_{CC} = \pm 20$	-	-	
			$V_{CC} = \pm 15$	± 10	± 13	
Common Mode Rejection Ratio	CMRR	$R_S \leq 10\text{k}\Omega$, $V_{CM} = \pm 12$	70	90	-	dB
		$R_S \leq 50\text{k}\Omega$, $V_{CM} = \pm 12$	-	-	-	

Electrical Characteristics (Continued)(V_{CC} = 15V, V_{EE} = -15V, T_A = 25°C, unless otherwise specified)

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Power Supply Rejection Ratio	PSRR	R _S ≤ 50Ω, V _{CC} = ±15	-	-	-	dB
		R _S ≤ 10kΩ, V _{CM} = ±15	77	96	-	
Transient Response	Rise Time	Unity Gain	-	0.3	-	μs
	Overshoot		-	10	-	%
Slew Rate	SR	Unity Gain	-	0.5	-	V/μs
Supply Current	I _{CC}	R _L = ∞Ω	-	1.5	2.8	mA
Power Consumption	P _C	V _{CC} = ±15	-	-	-	mW
		V _{CC} = ±15	-	50	85	

Note:

1. Guaranteed by design.

Electrical Characteristics (Continued)(0°C ≤ T_A ≤ 70°C, V_{CC} = 15V, unless otherwise specified)The following specification apply over the range of 0°C ≤ T_A ≤ 70°C for the LM741C

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Input Offset Voltage	V _{IO}	R _S ≤ 50Ω	-	-	-	mV
		R _S ≤ 10kΩ	-	-	7.5	
Input Offset Current	I _{IO}	-	-	-	300	nA
Input Bias Current	I _{BIAS}	-	-	-	0.8	μA
Input Voltage Range	V _{I(R)}	-	±12	±13	-	V
Output Voltage Swing	V _{O(P-P)}	R _L ≥ 10kΩ	V _{CC} = ±20	-	-	V
			V _{CC} = ±15	±12	±14	
		R _L ≥ 2kΩ	V _{CC} = ±20	-	-	
			V _{CC} = ±15	±10	±13	
Output Short Circuit Current	I _{SC}	-	10	-	40	mA
Common Mode Rejection Ratio	CMRR	R _S ≤ 10kΩ, V _{CM} = ±12	70	90	-	dB
		R _S ≤ 50kΩ, V _{CM} = ±12	-	-	-	
Power Supply Rejection Ratio	PSRR	R _S ≤ 50Ω, V _{CC} = ±15	-	-	-	dB
		R _S ≤ 10kΩ, V _{CM} = ±15	77	96	-	
Large Signal Voltage Gain	G _V	R _L ≥ 2kΩ V _{CC} = ±15 V _{C(P-P)} = ±10	15	-	-	V/mV

14.18 Open-loop Applications of Op-amps

Open-loop means that there is no feedback from the output to input. The gain of an amplifier without feedback is called **open-loop gain**. So, when an op-amp is used in a circuit without feedback, the open-loop gain (A) of that op-amp will be effective. In open-loop condition op-amps can be used in four ways. These are:

- Differential amplifier
- Noninverting amplifier
- Inverting amplifier
- Voltage comparator

14.18.1 Open-Loop Differential Amplifier

An open-loop differential amplifier using op-amp is shown in Fig.14-48. Input signal v_{s1} is applied to the noninverting input (denoted by '+' sign) and input signal v_{s2} is applied to the inverting input (denoted by '-' sign). The op-amp will amplify the difference between v_1 and v_2 . That is, the op-amp will amplify the differential input $v_{id} = v_1 - v_2$. The output of the circuit can be expressed using Equ.(14-61),

$$v_o = Av_{id} = A(v_1 - v_2) \quad (14-61)$$

Since, the op-amp amplifies the difference of two signals, this configuration is called *differential amplifier*. Operational amplifier can amplify both DC and AC signals. Therefore, v_1 and v_2 can be either DC voltages or AC signals. For both types of signals, if $v_1 > v_2$, the output will be in phase with the input (output increases with the increase in v_{id}). But if $v_2 > v_1$, the output will be 180° out of phase with the input (output decreases with the increase in v_{id}).

As the open-loop input impedance of op-amp is very large, very small amount of current will flow through the input terminals. On the other hand, the internal resistances r_{s1} and r_{s2} of the sources are very small compared to the op-amp input impedance. So voltage drops across these resistances can be neglected. In that case, $v_1 = v_{s1}$, and $v_2 = v_{s2}$. Then we can write,

$$v_o = Av_{id} \approx A(v_{s1} - v_{s2}) \quad (14-62)$$

The open-loop gain of an op-amp is very large. Therefore, to operate the op-amp in its linear range the differential voltage, v_{id} , should be very small (few μV). Otherwise, the op-amp will be saturated and the signal will be distorted [illustrated in Example 13-8]. In case of AC operation, the input signals v_{s1} and v_{s2} must be of same frequency and same phase.

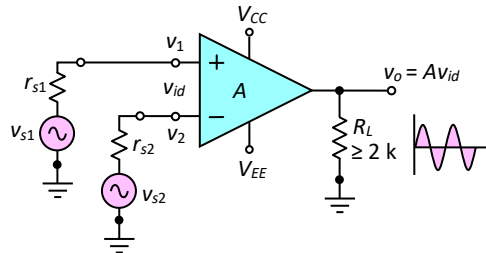


Fig.14-48: Open-loop differential amplifier

Example 13-8

For the differential amplifier of Fig.14-48, assume that the open-loop gain $A = 2 \times 10^5$, $v_{s1} = 50 \mu\text{V}$ (RMS), $v_{s2} = 10 \mu\text{V}$ (RMS) and the maximum output voltage swing is $\pm 13 \text{ V}$. Neglecting the voltage dropped across the source resistance and assuming both the input signals have same frequency and same phase, determine the RMS output voltage and draw the waveform. Repeat the same process considering $v_{s1} = 80 \mu\text{V}$ (RMS).

Solution:

For the first case, $v_{s1} = 50 \mu\text{V}$ (RMS), and $v_{s2} = 10 \mu\text{V}$ (RMS). Therefore,

Using Equ.(14-61), $v_o \approx A(v_{s1} - v_{s2}) = 2 \times 10^5(50 \mu\text{V} - 10 \mu\text{V})$

or, $v_o \approx 2 \times 10^5(40 \mu\text{V}) = 8 \text{ V}$ (RMS) [Ans.]

or, $v_{o(\text{peak})} = 8 \text{ V (RMS)} \times \sqrt{2} \approx 11.3 \text{ V (peak)}$

For the second case, $v_{s1} = 80 \mu\text{V}$ (RMS).

Using Equ.(14-61),

$$v_o \approx A(v_{s1} - v_{s2}) = 2 \times 10^5(80 \mu\text{V} - 10 \mu\text{V})$$

or, $v_o \approx 2 \times 10^5(70 \mu\text{V}) = 14 \text{ V}$ (RMS) [Ans.]

or, $v_{o(\text{peak})} = 14 \text{ V (RMS)} \times \sqrt{2} \approx 19.8 \text{ V (peak)}$

For the first case, the op-amp will produce sinusoidal output, but for the second case the output will be clipped off at the maximum output swing ($\pm 13 \text{ V}$). The waveforms are shown in Fig.14-49.

Comments: Due to the large open-loop gain, the op-amp has been saturated for a small differential voltage ($v_{id} = 70 \mu\text{V}$).

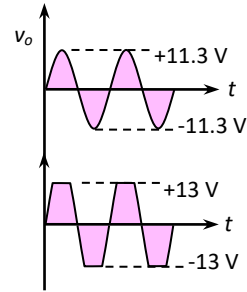


Fig.14-49: Output waveforms for Example 13-8

14.18.2 Open-Loop Noninverting Amplifier

The circuit diagram of an open-loop inverting amplifier is shown in Fig.14-50. In this configuration, only one signal (v_{s1}) is applied to the noninverting input terminal. The inverting terminal is connected to zero, so that $v_2 = 0$. If we neglect the voltage dropped across the source resistance, we can write, $v_1 = v_{s1}$. Since, $v_2 = 0$, the differential voltage will be,

$$\begin{aligned} v_{id} &= v_1 - v_2 = v_{s1} - 0 \\ &= v_{s1} \end{aligned}$$

Using Equ.(14-61) we can write,

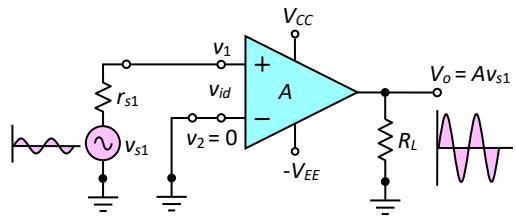


Fig.14-50: Open-loop noninverting amplifier using op-amp

$$v_o = Av_{id} = v_{s1}A \quad (14-63)$$

The output signal will be the product of the open-loop gain (A) and the input signal, v_{s1} . Here, the output is in phase with the input. If v_{s1} is a sinusoidal input signal, the output will be of same phase, or if v_{s1} is a DC voltage the output will be of same polarity.

14.18.3 Open-Loop Inverting Amplifier

The circuit diagram of an open-loop noninverting amplifier is shown in Fig.14-51. This amplifier is just opposite to the open-loop noninverting amplifier. Here also, only one signal (v_{s2}) is applied to the inverting input terminal. The noninverting terminal is connected to ground, so that $v_1 = 0$. If we neglect the voltage dropped across the source resistance, we can write, $v_2 = v_{s2}$. Since, $v_1 = 0$, the differential voltage will be,

$$v_{id} = v_1 - v_2 = 0 - v_{s2} = -v_{s2}$$

Using Equ.(14-61) we can write,

$$v_o = Av_{id} = -v_{s2}A \quad (14-64)$$

The output signal will be the product of the open-loop gain (A) and the input signal, v_{s2} . The negative sign indicates 180° phase reversal. If v_{s2} is a sinusoidal signal, the output will have 180° phase shift, or if v_{s2} is a DC voltage the output will be of opposite polarity.

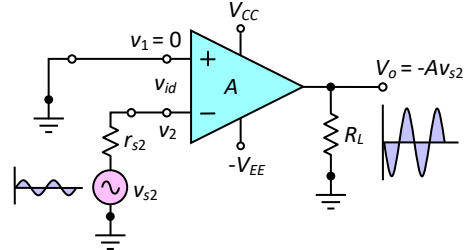


Fig.14-51: Open-loop inverting amplifier using op-amp

Example 13-9

Calculate and draw the output voltage for the inverting amplifier of Fig.14-51. If

- $v_{s1} = 5 \mu\text{V} \sin(\omega t)$
- $v_{s1} = 5 \text{ mV} \sin(\omega t)$

Assume the open-loop gain of the op-amp is 2×10^5 , supply voltage is $\pm 15 \text{ V}$, and neglect the voltage drop across the source internal resistance.

Solution:

For case a):

Using $v_o = -v_{s2}A$
 Equ.(14-64),
 $= -2 \times 10^5 \times 5 \mu\text{V} \sin(\omega t)$

or, $v_o = -1 \text{ V} \sin(\omega t)$ [Ans.]

For case a):

Using Equ.(14-64), $v_o = -v_{s2}A = -2 \times 10^5 \times 5 \text{ mV} \sin(\omega t)$

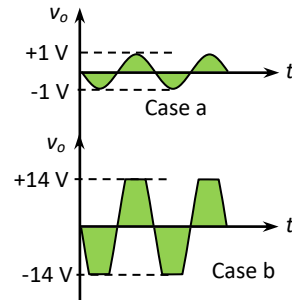


Fig.14-52: Output waveforms of Example 13-19

or, $v_o = -1000 \text{ V} \sin(\omega t)$ [Ans.]

As the supply voltage is $\pm 15 \text{ V}$, the maximum output swing may be $\pm 14 \text{ V}$. So for the second case the op-amp will be saturated. The two waveforms are shown in Fig.14-52.

Comment: The maximum output swing of an op-amp will be 1 V lower than the supply voltage (unless otherwise specified).

14.18.4 Open-Loop Voltage comparator

A voltage comparator is a circuit that compares an input voltage with another voltage called the **reference voltage**. The voltage comparator may be either inverting or noninverting. Fig.14-53 shows a non-inverting open-loop voltage comparator. Here, the reference voltage is a constant voltage with which the input sinusoidal voltage, v_i is compared. The output of a comparator will be always the maximum saturation voltage either positive or negative. As the open-loop gain of an op-amp is very high, the output will be saturated for a very small differential input voltage [as explain in

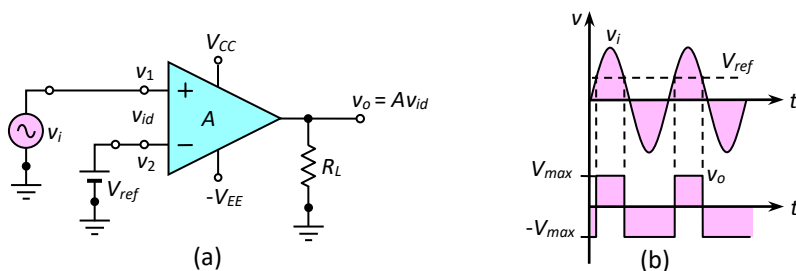


Fig.14-53: (a) Noninverting voltage comparator, (b) Output waveform

Example 13-8, and 13-9]. Let us represent this differential voltage as $\pm V_{th}$ (its value is few micro volts). The op-amp works as a linear amplifier if $v_{id} \leq V_{th}$. However, if $v_{id} > V_{th}$ the output of the op-amp will be saturated. Therefore, for the comparator of Fig.14-53, if $v_{i1} > V_{ref} + V_{th}$ the op-amp will be saturated and will produce the maximum positive voltage ($+V_{max}$). Similarly, if $v_i < V_{ref} - V_{th}$ the op-amp will produce the maximum negative voltage ($-V_{max}$). If the reference voltage and the input voltage are very large compared to $\pm V_{th}$, this threshold voltage can be neglected. Then we can write, if $v_i > V_{ref}$, then, $v_o = +V_{max}$ or if $v_i < V_{ref}$, then, $v_o = -V_{max}$. The output of the comparator, for a positive reference voltage, and a sinusoidal input voltage (with peak value $> V_{ref}$) is shown in Fig.14-53(b). The input voltage can be either DC voltage or AC voltage.

14.19 Op-Amp with Negative Feedback

Already we have studied four circuits of op-amp without using feedback, that is, all of them were open-loop circuits. The open-loop-amplifiers can amplify a very small input signal. In Example 13-8, we have seen that if the differential voltage (v_{id}) is $70 \mu\text{V}$, the output voltage is clipped off. This clipping occurs due to the very high open-loop gain of the amplifier. Moreover, in open-loop condition, the bandwidth of op-amps is very small (few Hz), and the gain is not stable, that is, the

gain changes with temperature, power supply and some other reasons. Therefore, to increase the bandwidth and gain stability, negative feedback is used in op-amp amplifiers. The open-loop gain of op-amps is so large that, even using negative feedback, the closed-loop amplifiers provide sufficient gain for signal amplification. Sacrificing some gain, the closed-loop amplifiers adopt all the advantages of negative feedback (described in Chapter 12). However, for the comparator circuits, positive feedback is used. Some of the closed-loop op-amp circuits are described in the following topics.

14.20 Closed-Loop Noninverting Amplifier

Fig.14-54 shows the circuit diagram of a **closed-loop noninverting amplifier** using op-amp. In this circuit, voltage series feedback is used. The feedback network is constructed using the resistors R_1 and R_f . As the input signal is applied to the noninverting terminal and negative feedback is deployed, this circuit is called **closed-loop noninverting amplifier**. As shown in the circuit, the full output voltage is applied to the feedback network, and the feedback voltage (v_f) is connected in series with the input voltage, v_i . Therefore, it is a voltage series feedback. Moreover, v_f reduces the overall input voltage ($v_{id} = v_i - v_f$), that is, v_f is subtracted from v_i . So the type of the feedback is negative feedback.

The open-loop gain of the op-amp is,

$$A = \frac{v_o}{v_{id}} \quad (14-65)$$

As the input impedance of op-amp is very large, using voltage divider rule we can calculate the feedback voltage as,

$$v_f = \frac{R_1 v_o}{R_1 + R_f}$$

or,

$$\frac{v_f}{v_o} = \frac{R_1}{R_1 + R_f} = \beta$$

$$\therefore \quad \boxed{\beta = \frac{R_1}{R_1 + R_f}} \quad (14-66)$$

$$\therefore \quad v_f = \frac{R_1 v_o}{R_1 + R_f} = \beta v_o \quad (14-67)$$

14.20.1 Closed-loop Voltage Gain

As this is a noninverting amplifier, the output voltage will be in phase with the input signal. So for the positive half-cycle of input, the output will be also positive. The polarity of voltages in the circuit is shown for the positive half-cycle. Now, using KVL, we can write,

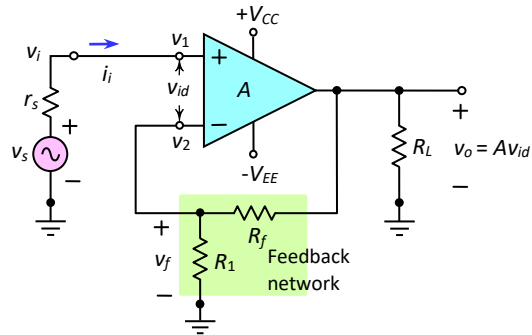


Fig.14-54: Closed-loop noninverting amplifier

$$v_s - i_i r_s - v_{id} - v_f = 0$$

As the input impedance of op-amp is very large, the input current, i_i will be very small. Moreover, the internal source resistance (r_s) is also very small. So, we can neglect the voltage drop across r_s , that is $i_i r_s = 0$ and $v_s = v_i$. Therefore,

$$v_{id} = v_i - v_f$$

The output of the op-amp will be just the product of the differential voltage, v_{id} , and the open-loop gain, A . Thus,

$$v_o = A v_{id} = A(v_i - v_f)$$

Putting the value of v_f from Equ.(14-67),

$$v_o = A(v_i - \beta v_o)$$

or,

$$v_o + A\beta v_o = A v_i$$

or,

$$v_o(1 + A\beta) = A v_i$$

$\therefore$

$$\frac{v_o}{v_i} = \frac{A}{1 + A\beta} \quad (14-68)$$

This the gain of the amplifier with feedback and is denoted as A_f . So,

$$\boxed{A_f = \frac{A}{1 + A\beta}} \quad (14-69)$$

As the loop-gain, $A\beta$ is very larger than 1, we can write,

$$A_f = \frac{A}{1 + A\beta} \approx \frac{A}{A\beta} = \frac{1}{\beta}$$

$$\boxed{A_f = \frac{1}{\beta}} \quad (14-70)$$

Putting the value of β from Equ.(14-66) into the above equation we get,

$$A_f = \frac{1}{R_1/(R_1 + R_f)} = \frac{R_1 + R_f}{R_1}$$

$\therefore$

$$\boxed{A_f = 1 + \frac{R_f}{R_1}} \quad (14-71)$$

From this expression, we find that the closed-loop gain depends only on the feedback network not on the open-loop gain (A) of the op-amp. So the closed-loop gain is very stable.

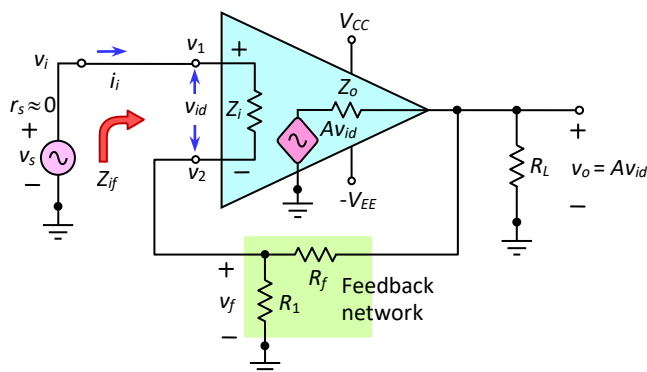


Fig.14-55: Input impedance of closed-loop noninverting amplifier

14.20.2 Closed-loop Input Impedance

In Fig.14-55, the noninverting amplifier is redrawn using the equivalent circuit of op-amp. Here, Z_i is the open-loop input impedance of the op-amp, that is, the input impedance without feedback. Its value will be,

$$Z_i = \frac{v_{id}}{i_i} \quad (14-72)$$

Z_{if} is the closed-loop input impedance or the input impedance with feedback. Its value will be,

$$Z_{if} = \frac{v_i}{i_i} = \frac{v_s}{i_i}$$

Using Equ.(14-72),

$$Z_{if} = \frac{v_i}{i_i} = \frac{v_i}{v_{id}/Z_i} = \frac{v_i Z_i}{v_{id}}$$

Using Equ.(14-68),

$$Z_{if} = \frac{Z_i v_o (1 + A\beta)/A}{v_{id}} = \frac{Z_i v_o (1 + A\beta)}{v_{id} A}$$

From Equ.(14-65), we get $v_{id} = v_o/A$. Putting this value into the above equation,

$$Z_{if} = \frac{Z_i v_o (1 + A\beta)}{A(v_o/A)}$$

$\therefore$

$$\boxed{Z_{if} = Z_i (1 + A\beta)} \quad (14-73)$$

14.20.3 Closed-loop Output Impedance

To determine the closed-loop output impedance, that is, the output impedance with feedback, we have to short the input source. Then we have to apply an output voltage, say v_x , and determine the output current for this voltage. From the output voltage and current we can determine the output impedance. The circuit arrangement is shown in Fig.14-56.

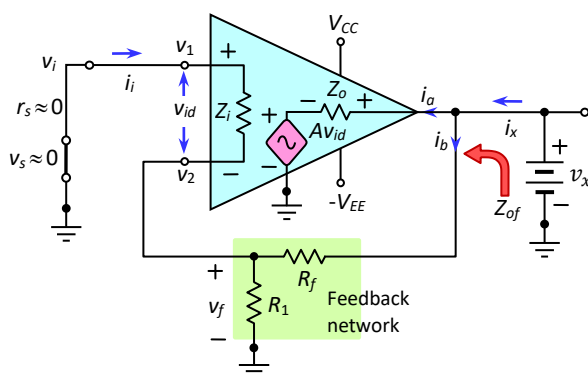


Fig.14-56: Output impedance of closed-loop noninverting amplifier

From the feedback network, we can write, $v_f = \beta v_x$.

Applying KVL in the input circuit, we can write,

$$v_{id} = -v_f = -\beta v_x \quad (14-74)$$

Now, applying KVL at the output circuit, we can write,

$$v_x - i_a Z_o - A v_{id} = 0$$

We will assume that the resistance of the feedback network is so large that, $i_b \approx 0$, so $i_a \approx i_x$. Putting this value into the above equation, we find,

$$v_x - i_x Z_o - A v_{id} = 0$$

Putting the value of v_{id} from Equ.(14-74) into the above equation, we get,

$$v_x + A\beta v_x = i_x Z_o$$

or,

$$v_x(1 + A\beta) = i_x Z_o$$

or,

$$\frac{v_x}{i_x} = \frac{Z_o}{1 + A\beta}$$

v_x/i_x is nothing but the output impedance with feedback, that is,

$$Z_{of} = \frac{Z_o}{1 + A\beta} \quad (14-75)$$

14.20.4 Closed-Loop Bandwidth

As described in Chapter X [H_S_atten], bandwidth is the range of frequencies that is properly amplified by an amplifier. This is the difference between the higher cut-off frequency (f_H) and the lower cut-off frequency (f_L). Being a direct-coupled amplifier, op-amp can amplify DC voltage. So the lower cut-off frequency of op-amp is zero. Then, the higher cut-off frequency will be equal to the bandwidth. One of the main drawbacks of op-amp open-loop-amplifier is very small bandwidth. When negative feedback is used, the bandwidth increases. Manufacturer's datasheet provides either the gain-bandwidth product or the open-loop frequency response graph. Fig.14-57 shows the open-loop gain vs. frequency curve for a typical 741 op-amp. As shown in this figure, for the open-loop gain of $A = 2 \times 10^5$ the bandwidth is only 5 Hz. So, the gain-bandwidth product is 1 MHz. On the other extreme, for the unity gain ($A_f = 1$), the bandwidth is 1 MHz. Here also the gain-bandwidth product is 1 MHz. It is called unity-gain-bandwidth (UGB) and sometimes is given in the datasheets. Moreover, from the figure it is clear that the gain-bandwidth product is constant. This is the property of all frequency response that has only a single break frequency (the portion of the curve up to the unity gain cut-off frequency). If UGB is given in datasheets, the bandwidth at any closed-loop gain can be calculated. As the gain-bandwidth product is constant we can write,

$$UGB = A \times f_o \quad (14-76)$$

Here, A = open-loop voltage gain and

f_o = open-loop higher cut-off frequency

At any other closed-loop gain,

$$UGB = A_f \times f_f \quad (14-77)$$

Here, A_f = closed-loop voltage gain and

f_f = closed-loop higher cut-off frequency

From Equ.(14-76) and Equ.(14-77),

$$A \times f_o = A_f \times f_f$$

But using Equ.(14-69), we can write,

$$A \times f_o = \frac{A}{1 + \beta A} \times f_f$$

or,

$$f_o = \frac{1}{1 + \beta A} \times f_f$$

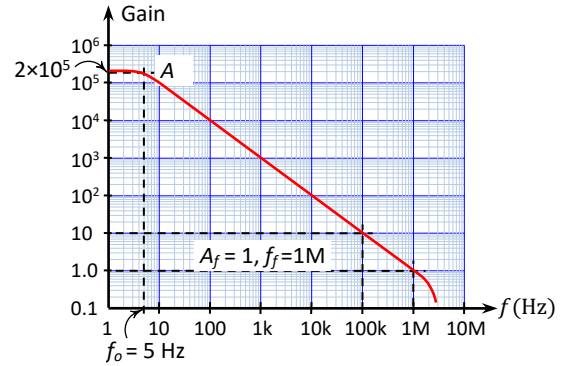


Fig.14-57: Frequency response of op-amp

$$\therefore \boxed{f_f = f_o(1 + \beta A)} \quad (14-78)$$

From this equation, we find that the closed-loop bandwidth has been increased by a factor of $(1 + \beta A)$.

Example 13-10

For the non-inverting amplifier of Fig.14-54, assume $A = 2 \times 10^5$, $f_o = 5 \text{ Hz}$, $Z_i = 1 \text{ M}\Omega$, $Z_o = 75 \Omega$, $R_f = 510 \text{ k}$, and $R_1 = 10 \text{ k}$. Determine closed-loop gain, closed-loop input impedance, closed-loop output impedance, and bandwidth.

Solution:

Using Equ.(14-66),
$$\beta = \frac{R_1}{R_1 + R_F} = \frac{10 \text{ k}}{10 \text{ k} + 510 \text{ k}} \approx 0.0192$$

Closed-loop gain:

Using Equ.(14-71),
$$A_f = 1 + \frac{R_F}{R_1} = 1 + \frac{510 \text{ k}}{10 \text{ k}} = 52 \text{ [Ans.]}$$

Closed-loop input impedance:

Using Equ.(14-73),
$$Z_{if} = Z_i(1 + A\beta) = 1 \text{ M}\Omega(1 + 2 \times 10^5 \times 0.0192)$$

$$\therefore Z_{if} \approx 3.8 \text{ G}\Omega \text{ [Ans.]}$$

Closed-loop output impedance:

Using Equ.(14-75),
$$Z_{of} = \frac{Z_o}{(1 + A\beta)} = \frac{75}{(1 + 2 \times 10^5 \times 0.0192)}$$

$$\therefore Z_{of} \approx 0.02 \Omega \text{ [Ans.]}$$

Closed-loop bandwidth:

Using Equ.(14-77),
$$f_f = f_o(1 + \beta A) = 5 \text{ Hz}(1 + 2 \times 10^5 \times 0.0192)$$

$$\therefore f_f \approx 19.2 \text{ kHz [Ans.]}$$

Comments: Due to the negative feedback, the gain has been decreased, but other parameters have been improved.

14.21 Closed-Loop Inverting Amplifier

Fig.14-58 shows the circuit diagram of a closed-loop inverting amplifier using op-amp. In this circuit voltage shunt feedback is applied. Here also, the feedback network is constructed using the

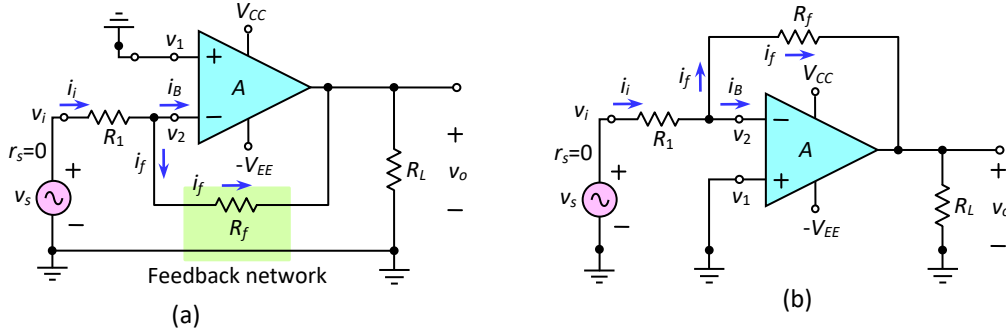


Fig.14-58: Closed-loop inverting amplifier: (a) Circuit with feedback concept, (b) Circuit with amplifier concept

resistors R_1 and R_f . As the input signal is applied to the inverting terminal and negative feedback is used, this circuit is called *closed-loop inverting amplifier* or simply *inverting amplifier*. As shown in the circuit, the full output voltage is applied to the feedback network, and the feedback current (i_f) is connected in parallel with the input signal. Therefore, it is a voltage shunt feedback. Moreover, i_f will be subtracted from i_i , that is, the overall current to the amplifier will be reduced. So, the type of the feedback is negative feedback. The input signal will be amplified by the op-amp and the output will be 180° out of phase with respect to the input.

14.21.1 Closed-loop Voltage Gain

Using KCL at the input node v_2 , we can write,

$$i_i = i_B + i_f$$

Since the input impedance of op-amp (Z_i) is very large, we can neglect i_B . Therefore,

$$i_i \approx i_f$$

or,

$$\frac{v_i - v_2}{R_1} = \frac{v_2 - v_o}{R_f} \quad (14-79)$$

From Equ.(14-61), we get,

$$v_1 - v_2 = \frac{v_o}{A}$$

But for this circuit, $v_1 = 0$. So,

$$v_2 = -\frac{v_o}{A} \quad (14-80)$$

Putting this value of v_2 into Equ.(14-79) we get,

$$\frac{v_i + v_o/A}{R_1} = \frac{-v_o/A - v_o}{R_f}$$

or,

$$v_i R_f + \frac{R_f v_o}{A} = -\frac{R_1 v_o}{A} - R_1 v_o$$

$$\text{or,} \quad v_i R_f A + R_f v_o = -R_1 v_o - R_1 v_o A$$

$$\text{or,} \quad v_i R_f A = -v_o (R_f + R_1 + R_1 A)$$

$$\text{or,} \quad \frac{v_o}{v_i} = -\frac{R_f A}{R_f + R_1 + R_1 A}$$

$$\text{or,} \quad \frac{v_o}{v_i} = -\frac{R_f A}{R_f + R_1 + R_1 A}$$

But, v_o/v_i is the closed-loop voltage gain of the inverting amplifier. So,

$$\boxed{A_f = -\frac{R_f A}{R_f + R_1 + R_1 A}} \quad (14-81)$$

As the open-loop gain, A , is very large (ideally infinity), so $R_1 A \gg R_f + R_1$. Therefore, the above equation can be approximate as,

$$A_f = -\frac{R_f A}{R_1 A} = -\frac{R_f}{R_1}$$

That is the approximate gain of the inverting amplifier is,

$$\boxed{A_f = -\frac{R_f}{R_1}} \quad (14-82)$$

We can represent the gain in terms of feedback theory. To do that, let us divide Equ.(14-81) by $(R_f + R_1)$,

$$A_f = -\frac{R_f A / (R_f + R_1)}{1 + R_1 A / (R_f + R_1)}$$

$$\text{or,} \quad \boxed{A_f = -\frac{KA}{1 + A\beta}} \quad (14-83)$$

where,

$$\boxed{K = \frac{R_f}{R_f + R_1}} \quad (14-84)$$

and

$$\boxed{\beta = \frac{R_1}{R_f + R_1}} \quad (14-85)$$

Although this circuit uses voltage shunt feedback, the feedback factor is calculated as $\beta = v_f/v_o = R_1/(R_f + R_1)$.

It is also possible to derive Equ.(14-82) from Equ.(14-83). Let us check. As $A\beta \gg 1$, Equ.(14-83) can be written as,

$$A_f \approx -\frac{KA}{A\beta} = -\frac{K}{\beta}$$

Putting the value of K and β from Equ.(14-84) and Equ.(14-85) into the above equation,

$$A_f = -\frac{R_f}{R_f + R_1} \times \frac{R_f + R_1}{R_1} = -\frac{R_f}{R_1}$$

This expression is same as Equ.(14-82).

14.21.2 Simplified Calculation of Closed-loop Voltage Gain

As A is very large, ideally infinity, from Equ.(14-80), we get,

$$v_2 = -\frac{v_o}{A} \approx 0 \quad (14-86)$$

Putting this value into Equ.(14-79),

$$\frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_f}$$

or,

$$\frac{v_i}{R_1} = \frac{-v_o}{R_f}$$

$\therefore$

$$A_f = \frac{v_o}{v_i} = -\frac{R_f}{R_1}$$

14.21.3 Closed-Loop Input Impedance

From Fig.14-58(b) we can write,

$$i_i = \frac{v_i - v_2}{R_1}$$

But from Equ.(14-86), we get $v_2 = 0$. Putting this value of v_2 into the above equation,

$$i_i = \frac{v_i - 0}{R_1}$$

or,

$$\frac{v_i}{i_i} = R_1$$

But, v_i/i_i is the input impedance with feedback. So,

$$Z_{if} = R_1 \quad (14-87)$$

14.21.4 Alternate Calculation of Z_{if}

We can redraw the inverting amplifier of Fig.14-58 as in Fig13-59. Here, instead of op-amp symbol, its equivalent circuit is used, and resistor R_f has been split using Miller's theorem. From this figure, we can easily calculate the value of Z_{if} as,

$$Z_{if} = R_1 + \frac{R_f}{1 + A} \parallel Z_i$$

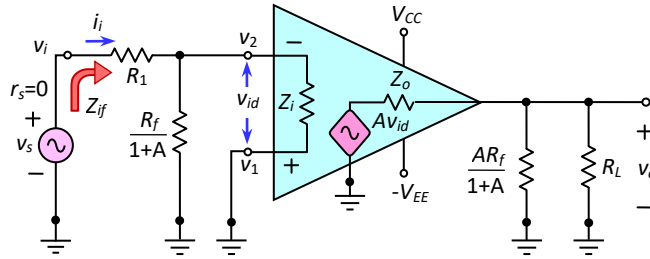


Fig.14-59: Inverting amplifier with feedback resistance split using Miller's theorem

The above equation gives the exact value of Z_{if} . But as A is very large, we can assume $R_f/(1+A) \approx 0$. Therefore,

$$Z_{if} \approx R_1$$

The input impedance is lower than that of the noninverting amplifier. Generally, current mixing feedback (current shunt or voltage shunt feedbacks) decreases the input impedance, and this decrease in input impedance is good for the amplifier.

14.21.5 Closed-loop Output Impedance

To determine the closed-loop output impedance, we have to short the input source and apply an output voltage, say v_x , and determine the output current for this voltage. From the output voltage and current we can determine the output impedance. The arrangement is shown in Fig.14-60. Carefully observing this circuit, we find that this circuit is exactly same as that of the

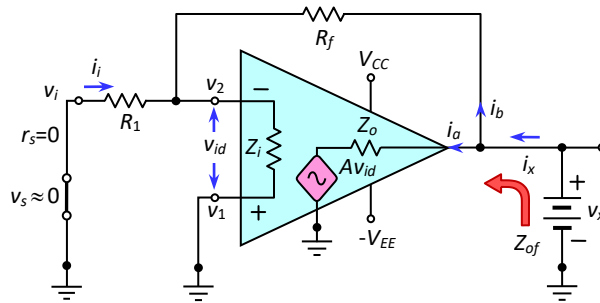


Fig.14-60: Output impedance of closed-loop noninverting amplifier

noninverting amplifier (Fig.14-56). In both circuits, R_f is connected between the inverting terminal and output, and R_1 is connected from inverting terminal to ground. So the output impedance will be exactly same as that of noninverting amplifier. That is,

$$Z_{of} = \frac{Z_o}{1 + A\beta} \quad (14-88)$$

14.21.6 Closed-loop Bandwidth of Inverting Amplifier

As already discussed, the gain-bandwidth product of a single break frequency op-amp (like 741) is always constant. If f_o is the open-loop bandwidth, A is the open-loop gain, and UGB is the unity gain bandwidth, then we can write,

$$f_o \times A = \text{UGB} \times 1$$

$$\text{or,} \quad f_o = \frac{\text{UGB}}{A} \quad (14-89)$$

According to the theory of negative feedback, the closed-loop bandwidth is given by,

$$f_f = f_o(1 + A\beta) \quad (14-90)$$

Putting the value of f_o from Equ.(14-89), we get,

$$f_f = \frac{\text{UGB}}{A}(1 + A\beta)$$

$$\text{or,} \quad f_f = \text{UGB} \times K \left(\frac{1 + A\beta}{AK} \right)$$

$$\text{or,} \quad f_f = \frac{\text{UGB} \times K}{AK/(1 + A\beta)}$$

Using Equ.(14-83) into the above equation, we get

$$\boxed{f_f = \frac{\text{UGB} \times K}{A_f}} \quad (14-91)$$

where, $A_f = KA/(1 + A\beta)$, and $K = R_f/(R_f + R_1)$.

Equ.(14-90) can be used to determine the closed-loop bandwidth if f_o is given, or Equ.(14-91) can be used if UGB is given.

Example 13-11

For the inverting amplifier of Fig.14-58, assume $A = 2 \times 10^5$, $\text{UGB} = 1 \text{ MHz}$, $Z_i = 1 \text{ M}\Omega$, $Z_o = 75 \Omega$, $R_f = 470 \text{ k}$, and $R_1 = 10 \text{ k}$. Determine closed-loop gain, closed-loop input impedance, closed-loop output impedance, and bandwidth.

Solution:

$$\text{Using Equ.(14-85),} \quad \beta = \frac{R_1}{R_1 + R_f} = \frac{10 \text{ k}}{10 \text{ k} + 470 \text{ k}} \approx 0.021$$

Closed-loop gain:

$$\text{Using Equ.(14-82),} \quad A_f = -\frac{R_f}{R_1} = -\frac{470 \text{ k}}{10 \text{ k}} = -47 \text{ [Ans.]}$$

Closed-loop input impedance:

$$\text{Using Equ.(14-87),} \quad Z_{if} = R_1 = 10 \text{ k [Ans.]}$$

Closed-loop output impedance:

Using Equ.(14-88),
$$Z_{of} = \frac{Z_o}{(1 + A\beta)} = \frac{75}{(1 + 2 \times 10^5 \times 0.021)}$$

$\therefore Z_{of} \approx 0.018 \Omega$ [Ans.]

Closed-loop bandwidth:

Using Equ.(14-91),
$$f_f = \frac{UGB \times K}{A_f}$$

where,
$$K = \frac{R_f}{R_f + R_1} = \frac{470 \text{ k}}{470 \text{ k} + 10 \text{ k}} \approx 0.979$$

$\therefore f_f = \frac{1 \text{ MHz} \times 0.979}{47} \approx 20.8 \text{ k}$ [Ans.]

Comments: Inverting amplifier gives less gain, lower input impedance and lower bandwidth compared to noninverting amplifier.

14.22 Virtual Ground in Inverting Amplifier

As shown in the inverting amplifier of Fig.14-58, the noninverting terminal is directly connected to ground, and the signal is applied to the inverting terminal via resistor R_1 . It can be proved that as long as the circuit works linearly, the potential at the inverting terminal is *zero*. We also know that the potential of *ground terminal* is also zero. For this reason, we can consider the inverting terminal as *ground*. However, a *true ground* can source or sink infinite amount of current. But, though the potential of the inverting terminal is zero, it cannot source or sink large current, due to the high input impedance of op-amp. For this reason the inverting terminal of the closed-loop inverting amplifier is called *virtual ground*. In Fig.14-61, the virtual ground point is shown by the shaded area.

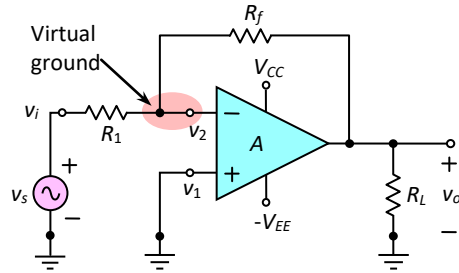


Fig.14-61: Virtual ground in inverting amplifier

To prove that the potential of the inverting terminal is zero, let us start with the basic expression of op-amp.

Using Equ.(14-61),

$$v_o = Av_{id} = A(v_1 - v_2)$$

$\therefore v_1 - v_2 = \frac{v_o}{A}$

As, the open-loop gain of an op-amp (A) is very large (ideally infinity), we can write,

$$v_1 - v_2 = \frac{v_o}{A} \approx 0$$

$$\therefore v_2 = v_1$$

But, $v_1 = 0$, because the noninverting terminal is directly connected to ground. Therefore,

$$v_2 = v_1 = 0$$

In this circuit the noninverting terminal is *true ground* and the inverting terminal is *virtually ground*. If the negative feedback is removed from this circuit, it will not work in linear region then, the inverting terminal will no longer be in virtual ground.

14.23 Offset Voltage due to Bias Current

We have already defined the input bias current as,

$$\text{Equ.(14-53), } I_B = \frac{I_{B1} + I_{B2}}{2}$$

Although very small (few nA), the bias current may produce some output offset voltage, even though there is no offset voltage (V_{io}) due to mismatch of transistors. To understand this, let us consider the inverting amplifier with the signal source shorted to ground (Fig.14-62). As $v_i = 0$, the voltage of the output dependant source will be also zero. Moreover the output impedance of op-amp is very small. So we can assume that the output terminal is shorted to ground. Thus, the bias current I_{B2} will flow through $R_1 \parallel R_f$. So, the voltage at the inverting terminal (v_2) will be,

$$v_2 = -I_{B2}(R_1 \parallel R_f) = -\frac{R_1 R_f}{R_1 + R_f} I_{B2} \quad (14-92)$$

As the bias current flows from ground to inverting terminal, v_2 will be negative.

However, $v_1 = 0$, because the noninverting terminal is directly connected to ground. Thus a differential voltage is developed at the inputs of the op-amp ($v_1 = 0, v_2 < 0$). This difference in the input will be amplified by the amplifier and will produce an output offset voltage, v_{oI_B} .

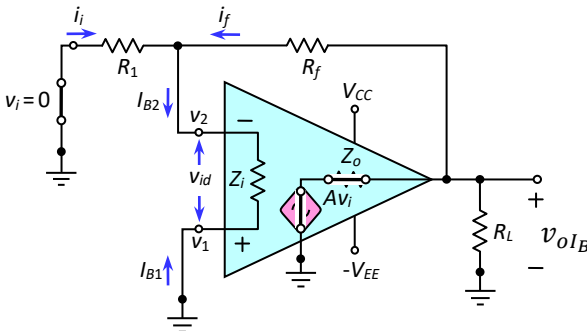


Fig.14-63: Bias current produces offset voltage in op-amp circuit

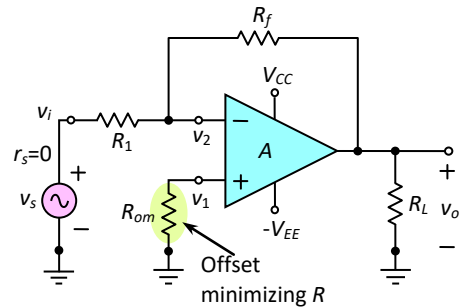


Fig.14-63: Inverting amplifier with bias offset minimizing resistance

To determine the output offset voltage due to this biasing current, let us apply KCL at the inverting terminal of the op-amp.

$$i_i + i_f = I_{B2}$$

$$\text{or, } \frac{0 - v_2}{R_1} + \frac{v_{oI_B} - v_2}{R_f} = \frac{v_2}{Z_i}$$

$$\text{or, } \frac{v_{oI_B}}{R_f} = \frac{v_2}{R_1} + \frac{v_2}{R_f} + \frac{v_2}{Z_i}$$

$$\text{or, } \frac{v_{oI_B}}{R_f} = v_2 \left(\frac{1}{R_1} + \frac{1}{R_f} + \frac{1}{Z_i} \right)$$

As Z_i is very large, we can consider $1/Z_i = 0$. Therefore,

$$\frac{v_{oI_B}}{R_f} = v_2 \left(\frac{1}{R_1} + \frac{1}{R_f} \right) = v_2 \frac{R_1 + R_f}{R_1 R_f}$$

Putting the value of v_2 from Equ.(14-92) into the above equation, we get,

$$\frac{v_{oI_B}}{R_f} = -\frac{R_1 R_f}{R_1 + R_f} I_{B2} \frac{R_1 + R_f}{R_1 R_f}$$

$$\boxed{v_{oI_B} = -I_{B2} R_f}$$

To overcome this offset voltage, we have to ensure same amount of voltage at the noninverting terminal. We can use a *offset minimizing resistor* (R_{om}) in the noninverting terminal. The value of R_{om} should be equal to the resistance through which I_{B2} flows. That is,

$$v_1 = -R_{om} I_{B1} = v_2 = -\frac{R_1 R_f}{R_1 + R_f} I_{B2}$$

As $I_{B1} = I_{B2}$, the above equation gives,

$$\boxed{R_{om} = \frac{R_1 R_f}{R_1 + R_f}} \quad (14-93)$$

An inverting amplifier with bias current offset minimizing resistance (R_{om}) is shown in Fig.14-63.

14.24 Voltage Follower

The **voltage follower** is a unity-gain closed-loop noninverting amplifier. As the output is equal to the input or the output follows the input, it is called so. Although it is similar to discrete emitter-follower, the op-amp voltage follower is preferred because it has larger input impedance, lower

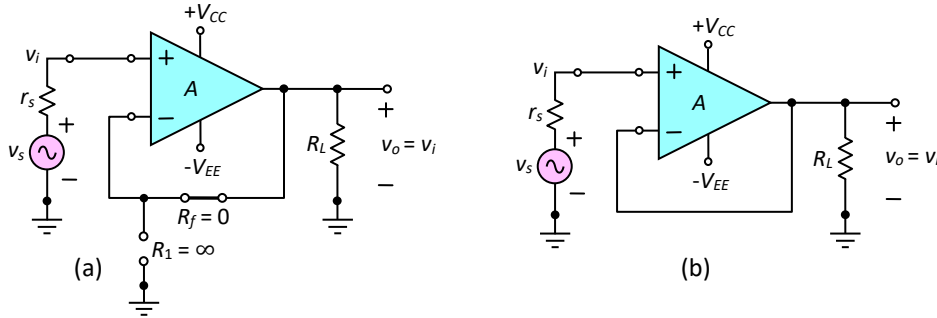


Fig.14-64: Circuit diagram of voltage follower: (a) Noninverting amplifier with $R_1 = \infty$, and $R_f = 0$, (b) Actual voltage follower

output impedance and the output is exactly equal to the input. Due to its very high input impedance and very low output impedance, voltage follower is used for isolation purpose. For this reason it also called **buffer amplifier**. We know that the gain of a closed-loop noninverting amplifier is,

$$\text{Equ. (14-71),} \quad A_f = 1 + \frac{R_f}{R_1}$$

If we set, $R_f = 0$, and $R_1 = \infty$, then the gain of the closed-loop noninverting amplifier will be,

$$A_f = 1 + \frac{0}{\infty} = 1$$

Considering $R_f = 0$ (short), and $R_1 = \infty$ (open) we can redraw the circuit of Fig.14-54 as in Fig.14-64,

Therefore, to use an op-amp as voltage follower, the signal has to be applied to the noninverting terminal, and the inverting terminal has to be shorted to output [Fig.14-64(b)].

Since the voltage follower is a special case of closed-loop noninverting amplifier, all the expressions derived in Section 13.20 will be applicable here. Therefore, the feedback factor can be calculated using Equ.(14-66) as,

$$\beta = \frac{R_1}{R_1 + R_f} = \frac{R_1}{R_1 + 0} = 1 \quad (14-94)$$

The closed-loop voltage gain using Equ.(14-69) will be,

$$A_f = \frac{A}{1 + A\beta} = \frac{A}{1 + A} \approx 1 \quad (14-95)$$

The closed-loop input impedance can be calculated using Equ.(14-73),

$$Z_{if} = Z_i(1 + A\beta) = Z_i(1 + A) \quad (14-96)$$

The closed-loop output impedance can be calculated using Equ.(14-75),

$$Z_{of} = \frac{Z_o}{1 + A\beta} = \frac{Z_o}{1 + A} \quad (14-97)$$

Example 13-12

For the voltage follower of Fig.14-64, assume $A = 2 \times 10^5$, $f_o = 5 \text{ Hz}$, $Z_i = 1 \text{ M}\Omega$, and $Z_o = 75 \Omega$. Determine A_f , Z_{if} , Z_{of} and bandwidth.

Solution:

Closed-loop gain:

Using Equ.(14-95),
$$A_f = \frac{A}{1 + A} = \frac{2 \times 10^5}{1 + 2 \times 10^5} = 0.999995 \approx 1 \text{ [Ans.]}$$

Closed-loop input impedance:

Using Equ.(14-96),
$$Z_{if} = Z_i(1 + A) = 1 \text{ M}\Omega(1 + 2 \times 10^5)$$

$$\therefore Z_{if} \approx 200 \text{ G}\Omega \text{ [Ans.]}$$

Closed-loop output impedance:

Using Equ.(14-97),
$$Z_{of} = \frac{Z_o}{(1 + A)} = \frac{75}{(1 + 2 \times 10^5)}$$

$$\therefore Z_{of} \approx 0.38 \text{ m}\Omega \text{ [Ans.]}$$

Closed-loop bandwidth:

Using Equ.(14-90),
$$f_f = f_o(1 + A) = 5 \text{ Hz}(1 + 2 \times 10^5)$$

$$\therefore f_f \approx 1 \text{ MHz [Ans.]}$$

Comments: The input impedance, and bandwidth are very large and the output impedance is very small. All these properties are essential for a good quality voltage follower or buffer amplifier.

14.25 Differential Amplifier Using Op-Amps

We have already studied differential amplifier using BJTs. We have learnt that, the differential amplifier amplifies the difference between to input signals either DC or AC. We have also studied the open-loop differential amplifier using op-amp [Section 13.18.1] [\[H_S_atten\]](#). Here, we will study the closed-loop differential amplifiers using op-amp.

If we measure the voltage difference between any two points of a circuit other than ground, it is called **differential voltage**. On the other hand, if a voltage is measured at any point of a circuit with respect to ground, it is called **single-ended voltage**. Sometimes it is required to convert a differential voltage into a single-ended voltage. For example, suppose we want to convert the voltage difference between point x and y into digital signal as shown in Fig.14-65. We have an ADC that has only one single-ended analog input. If we connect x terminal to the input of ADC and the y terminal to ground, then the circuit inside the black box will be shorted to ground, which is not permitted. In this case, we have to use a differential amplifier to convert the differential voltage (v_{xy}) into single-ended voltage and then apply to the ADC [shown in Fig.14-65(b)].

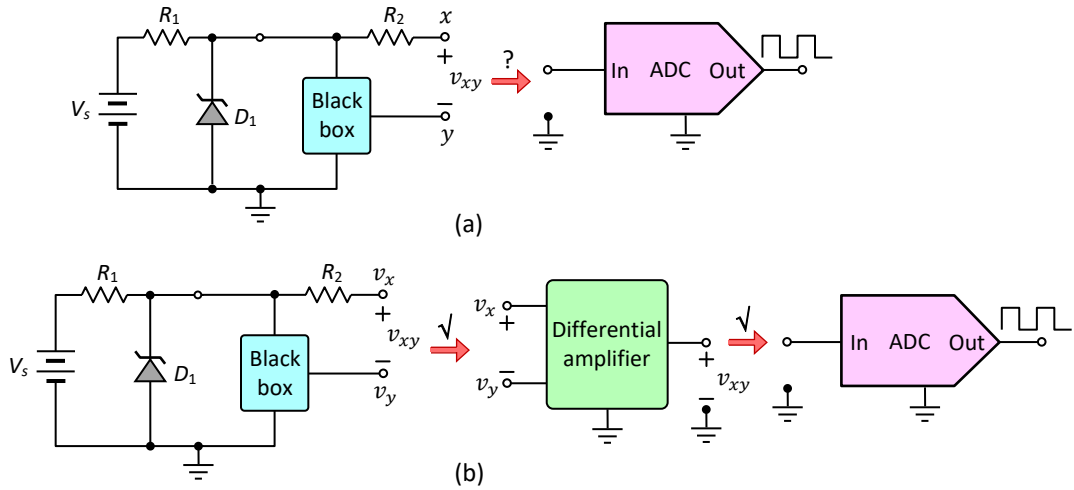


Fig.14-65: Necessity of differential amplifier: (a) A single-ended input ADC does not support v_{xy} , (b) A differential amplifier convert v_{xy} to single ended output for the ADC

The circuit diagram of a closed-loop differential amplifier is shown in Fig.14-66. Actually this circuit is the combination of inverting and noninverting amplifier. That is, when v_x is set to zero (shorted to ground) the circuit becomes a noninverting amplifier, whereas the circuit becomes an inverting amplifier when v_y is reduced to zero.

Voltage Gain:

As the circuit has two input voltage, we will use superposition theorem for its analysis. First, let us consider the effect of v_x by shorting v_y . When v_y is shorted to ground, the circuit works as an inverting amplifier with input signal v_x . Therefore, the output voltage will be [Using Equ.(14-82)],

$$v_{ox} = -\frac{R_f}{R_1} v_x$$

Similarly, when v_x is shorted to ground, the circuit works as a noninverting amplifier with input signal v_1 . Therefore, the output voltage will be [Using Equ.(14-71)],

$$v_{oy} = \left(1 + \frac{R_f}{R_1}\right) v_1 \quad (14-98)$$

The value of v_1 can be calculated using voltage divider rule, as

$$v_1 = \left(\frac{R_3}{R_2 + R_3}\right) v_y$$

Putting this value of v_1 into Equ.(14-98), we get,

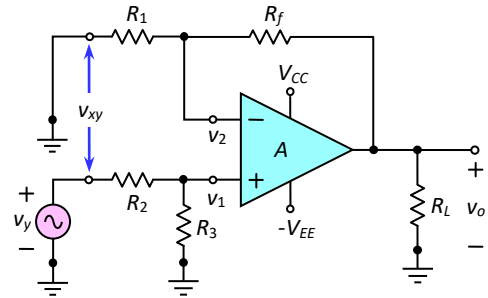


Fig.14-66: Op-amp closed-loop differential amplifier

$$v_{oy} = \left(\frac{R_1 + R_f}{R_1} \right) \left(\frac{R_3}{R_2 + R_3} \right) v_y$$

To make the circuit symmetrical, generally it is considered that $R_1 = R_2$, and $R_f = R_3$. Therefore, the above equation can be written as,

$$v_{oy} = \left(\frac{R_1 + R_f}{R_1} \right) \left(\frac{R_f}{R_1 + R_f} \right) v_y = \frac{R_f}{R_1} v_y$$

According to the superposition theorem, the net output voltage will be the algebraic sum of v_{ox} and v_{oy} , that is,

$$v_o = v_x + v_y = -\frac{R_f}{R_1} v_x + \frac{R_f}{R_1} v_y$$

or,

$$v_o = -\frac{R_f}{R_1} (v_x - v_y) \quad (14-99)$$

From this equation, it is clear that, the output is equal to the difference of the voltages v_x and v_y multiplied by a constant, that is, the circuit amplifies the difference between v_x and v_y (or v_{xy}). The gain of this amplifier will be,

$$A_d = \frac{v_o}{v_x - v_y} = \frac{v_o}{v_{xy}} = -\frac{R_f}{R_1} \quad (14-100)$$

From this expression, we find that the differential amplifier has a gain exactly same as the gain of inverting amplifier.

If we set $R_f = R_1$, and connect the higher potential as v_y and the lower potential as v_x , then the output will be exactly equal to the input voltage difference, but will be converted into single-ended voltage (voltage with respect to ground).

Input Impedance

In a differential amplifier, every input source will see an impedance, that is, two input impedance in two terminals. To determine the input impedance of any terminal the other terminal has to be shorted to ground (or replaced by its internal source resistance). If v_y is shorted to ground, the circuit becomes an inverting amplifier, so the input impedance seen by v_x will be,

Using Equ.(14-87),

$$Z_{ix} \approx R_1 \quad (14-101)$$

Similarly, if v_x is shorted to ground, the circuit becomes a noninverting amplifier, so the input impedance seen by v_y will be (considering Z_i of the op-amp very high),

$$Z_{iy} \approx R_2 + R_3 = R_1 + R_3 \quad (14-102)$$

From Equ.(14-101), and Equ.(14-102), it is clear that the input impedances are low and not equal. This problem can be solved by using two op-amps in the differential amplifier circuit.

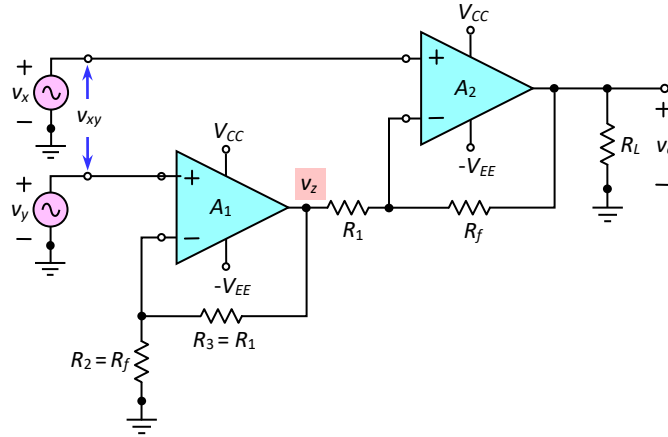


Fig.14-67: Differential amplifier with two op-amps

14.26 Differential Amplifier with two Op-Amps

Fig.14-67 shows a differential amplifier using two op-amps. This amplifier eliminates the problem of low input impedance which is present in the differential amplifier with one op-amp. If we carefully observe this circuit, we find that it consists of a differential amplifier (using op-amp A_2), and a noninverting amplifier (using op-amp A_1). The differential amplifier amplifies the difference of v_x , and v_z whereas the noninverting amplifier amplifies the signal v_y . Here, the gain of the noninverting amplifier is $(1 + R_3/R_2)$. So the output voltage (v_z) of this amplifier will be,

$$v_z = \left(1 + \frac{R_3}{R_2}\right) v_y \quad (14-103)$$

The output voltage of the differential amplifier can be calculated using superposition theorem. If we short v_z terminal to ground (i.e., $v_z = 0$), this amplifier will work as a noninverting amplifier with the input voltage v_x connected to noninvertign terminal. The output voltage will be,

$$v'_o = \left(1 + \frac{R_f}{R_1}\right) v_x$$

If we short v_x source (i.e., $v_x = 0$), this amplifier (A_2) will work as an inverting amplifier with the input voltage v_z . So, the output voltage will be,

$$v''_o = -\frac{R_f}{R_1} v_z$$

Putting the value of v_z from Equ.(14-103) into the above equation we get,

$$v''_o = -\frac{R_f}{R_1} \left(1 + \frac{R_3}{R_2}\right) v_y = -\left(\frac{R_f}{R_1} + \frac{R_f}{R_1} \times \frac{R_3}{R_2}\right) v_y$$

According the superposition theorem, the net output voltage of the circuit will be,

$$v_o = v'_o + v''_o = \left(1 + \frac{R_f}{R_1}\right) v_x - \left(\frac{R_f}{R_1} + \frac{R_f}{R_1} \times \frac{R_3}{R_2}\right) v_y$$

Generally, in this type of amplifier we set $R_2 = R_f$, and $R_3 = R_1$. Putting these values into the above equation,

$$v_o = \left(1 + \frac{R_f}{R_1}\right)v_x - \left(\frac{R_f}{R_1} + \frac{R_f}{R_1} \times \frac{R_1}{R_f}\right)v_y$$

or,

$$v_o = \left(1 + \frac{R_f}{R_1}\right)v_x - \left(\frac{R_f}{R_1} + 1\right)v_y$$

$\therefore$

$$v_o = \left(1 + \frac{R_f}{R_1}\right)(v_x - v_y) \quad (14-104)$$

From this expression, we find that the output is proportional to the difference of two voltages v_x , and v_y . The differential voltage gain of this amplifier will be,

$$A_d = \frac{v_o}{v_{xy}} = \frac{v_o}{v_x - v_y} = 1 + \frac{R_f}{R_1} \quad (14-105)$$

Input Impedance

The input impedance seen by the source v_y is the input impedance of the inverting amplifier with negative feedback consisting of R_3 and R_2 . Therefore, using Equ.(14-73),

$$Z_{iy} = Z_i(1 + \beta A) \quad (14-106)$$

where, A is the open-loop gain of op-amp A_1 and β is the feedback factor and its value is $R_2/(R_2 + R_3) = R_f/(R_f + R_1)$ [$\because R_2 = R_f, R_3 = R_1$].

Similarly, the input impedance seen by the source v_x is the input impedance of the inverting amplifier with negative feedback consisting of R_1 and R_f . Therefore, using the same equation,

$$Z_{ix} = Z_i(1 + \beta A) \quad (14-107)$$

where, A is the open-loop gain of op-amp A_1 and β is the feedback factor with value $R_1/(R_1 + R_f)$.

The impedance seen by the two sources will be very large but will not be identical though identical op-amps are used, as the expressions for β are different.

Example 13-13

For the differential amplifier of Fig.14-67, assume $R_1 = R_3 = 1 \text{ k}$, $R_f = R_2 = 10 \text{ k}$ and the op-amps have $Z_i = 1 \text{ M}\Omega$, $Z_o = 75 \Omega$, and $A = 2 \times 10^5$. Calculate the differential voltage gain, input impedance seen by the sources.

Solution:

Using Equ.(14-105),

$$A_d = \frac{v_o}{v_{xy}} = 1 + \frac{R_f}{R_1} = 1 + \frac{10 \text{ k}}{1 \text{ k}} = 11 \text{ [Ans.]}$$

Using Equ.(14-106),

$$Z_{iy} = Z_i(1 + \beta A)$$

where,

$$\beta = \frac{R_2}{R_2 + R_3} = \frac{10 \text{ k}}{10 \text{ k} + 1 \text{ k}} \approx 0.91$$

$$\therefore Z_{iy} = 1 \text{ M}\Omega[(1 + 0.91 \times (2 \times 10^5))] \approx 182 \text{ G}\Omega [\text{Ans.}]$$

$$\text{Using Equ.(14-), } Z_{ix} = Z_i(1 + \beta A)$$

$$\text{where, } \beta = \frac{R_1}{R_1 + R_f} = \frac{1 \text{ k}}{1 \text{ k} + 10 \text{ k}} \approx 0.091$$

$$\therefore Z_{ix} = 1 \text{ M}\Omega[(1 + 0.091 \times (2 \times 10^5))] \approx 18.2 \text{ G}\Omega [\text{Ans.}]$$

Comments: The input impedances are very large and the impedance seen by v_y will be always R_f/R_1 times the impedance seen by v_x .

14.27 Current-to-Voltage Converter

Most of the electronic circuits can process voltage signals, not the current signals. But there are some sensors that produce output current proportional to the physical parameter. For example, a photo-detector produces an output current proportional to the input light. The sensitivity of the photo-detector will depend on the load resistance to which it is connected. For a fixed value of light input, it will provide most sensitivity (maximum current will be produced) when the sensor is shorted. But short-circuit will produce zero output voltage, and zero output voltage cannot be processed further. To solve this problem, we can use a current-to-voltage converter. A current-to-voltage converter can easily be made by setting $R_1 = 0$ of the inverting amplifier as shown in Fig.14-68.

Using KCL at the input node v_2 , we can write,

$$i_i = i_B + i_f$$

Since the input impedance of op-amp (Z_i) is very large, we can neglect i_B . Therefore,

$$\begin{aligned} i_i &\approx i_f \\ \therefore i_i &= \frac{v_2 - v_o}{R_f} \end{aligned} \quad (14-108)$$

From Equ.(14-61), we get,

$$v_1 - v_2 = \frac{v_o}{A}$$

But for this circuit, $v_1 = 0$. So,

$$v_2 = -\frac{v_o}{A}$$

As A is very large, we can approximately assume,

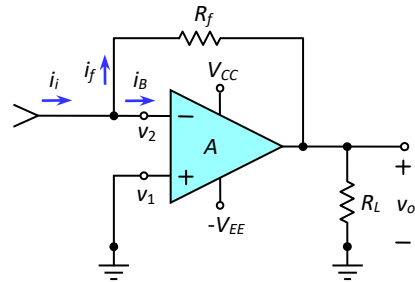


Fig.14-68: Current-to-voltage converter

$$v_2 = -\frac{v_o}{A} \approx 0$$

This assumption is also correct due to the virtual grounding of the inverting terminal.

Putting this value of v_2 into Equ.(14-108) we get,

$$i_i = \frac{0 - v_o}{R_f}$$

$$\therefore v_o = -i_i R_f$$

This expression reveals that the output voltage is directly proportional to input current i_i and out of phase with the input. That is, if the input current increases, the output voltage will decrease. If needed, we can eliminate this phase inversion by adding another inverting amplifier at the output of this current-to-voltage converter.

Input Impedance

We have already calculated the input impedance of an inverting amplifier which is given by Equ.(14-87),

$$Z_{if} \approx R_1$$

But in this circuit as $R_1 = 0$, the input impedance seen by the current source will be,

$$Z_{if} \approx R_1 = 0$$

Thus, the current source will be shorted by this converter which will maximize the output current of the current source. But the final output of the converter is a voltage which is proportional to the input current.

14.28 Summing Amplifier

Summing amplifier is basically an op-amp circuit that can combine a number of input signals to a single output, that is, the weighted sum of the applied inputs. Both the inverting and noninverting amplifiers can be used as summing amplifier. The inverting summing amplifier is shown in Fig.14-69. This circuit is also called *summing inverter* or even a *voltage adder*. This circuit has three input terminals (a , b , and c), so it can sum three signals. But we can add as many inputs as we need.

Applying KCL at the inverting terminal of this circuit we get,

$$i_a + i_b + i_c - i_f - i_B = 0$$

As the input impedance of the op-amp is very large, we can neglect i_B . Therefore,

$$i_a + i_b + i_c = i_f$$

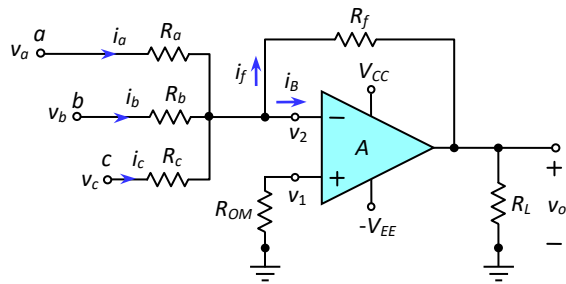


Fig.14-69: Inverting summing amplifier

$$\therefore \frac{v_a - v_2}{R_a} + \frac{v_b - v_2}{R_b} + \frac{v_c - v_2}{R_c} = \frac{v_2 - v_o}{R_f}$$

We know that the inverting terminal is a virtual ground as described before, that is, $v_2 = 0$. Putting this value of v_2 into the above equation, we get,

$$\frac{v_a}{R_a} + \frac{v_b}{R_b} + \frac{v_c}{R_c} = -\frac{v_o}{R_f}$$

$$\therefore \boxed{v_o = -\left(\frac{R_f}{R_a}v_a + \frac{R_f}{R_b}v_b + \frac{R_f}{R_c}v_c\right)} \quad (14-109)$$

If we use identical input resistors, that $R_a = R_b = R_c = R$, then the above equation can be written as,

$$\boxed{v_o = -\frac{R_f}{R}(v_a + v_b + v_c)} \quad (14-110)$$

Again if we set $R_f = R$, then

$$\boxed{v_o = -(v_a + v_b + v_c)} \quad (14-111)$$

From this expression, we find that the output of this circuit is the inverting sum of the input voltages. That is the circuit of Fig.14-69 works as a summing amplifier. If we need amplified sum of the signals, then we can use Equ.(14-110) and changing the ratio R_f/R , we can multiply the summing output by any value. Summing amplifier has many applications. Some of them are discussed below.

14.28.1 Scaling or Weighted Amplifier

The Summing Amplifier is a very flexible circuit indeed, enabling us to effectively *add* or *sum* (hence its name) together several individual input signals. If the inputs resistors R_a, R_b, R_c , and the feedback resistor R_f are of equal values, the circuit will work as *unity gain inverting adder*. However, if the input resistors are of different values a *scaling* or *weighted summing amplifier* is produced which will output a weighted sum of the input signals. Then the output expression of this circuit will be,

$$\boxed{v_o = -R_f\left(\frac{v_a}{R_a} + \frac{v_b}{R_b} + \frac{v_c}{R_c}\right)} \quad (14-112)$$

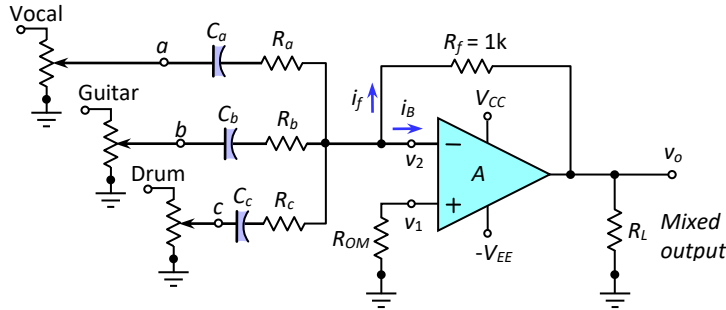


Fig.14-70: Summing amplifier as audio mixer

If we want any input signal to contribute more to the output, then, the resistance corresponding to that input signal has to be set lower compared to other resistances. For example, if we set $R_b = R_c = 10\text{ k}$, and $R_f = R_a = 2\text{ k}$, then the contribution of signal v_a will be 5 times that of signals v_b and v_c .

This weighted summing amplifier can be used as *audio mixer*. The arrangement is shown in Fig.14-70. By adjusting the potentiometer, it is possible to increase or decrease the audio signal coming from any instrument.

This weighted summing amplifier can also be used as *digital-to-analog converter*. The arrangement is shown in Fig.14-71. The resistance of one input should be just double the input resistance of previous input. The expression of the output voltage of this circuit will be [Using Equ.(14-112)],

$$v_o = -R_f \left(v_a + \frac{v_b}{2\text{ k}} + \frac{v_c}{4\text{ k}} \right)$$

Let us use $R_f = 1\text{ k}$. So the above equation reduces to,

$$v_o = -(v_a + 0.5 v_b + 0.25 v_c) \quad (14-113)$$

Although, the above equation gives the analog value of 3-bit digital input, we can increase the number of bits by adding as many inputs as we need. The value of input signals will be the logic '1' = 5 V or logic '0' = 0 V. The step size and the full-scale output voltage of this circuit can be changed by changing the value of R_f . The accuracy of the output voltage depends on the accuracy of the resistances and the voltage of the logic levels.

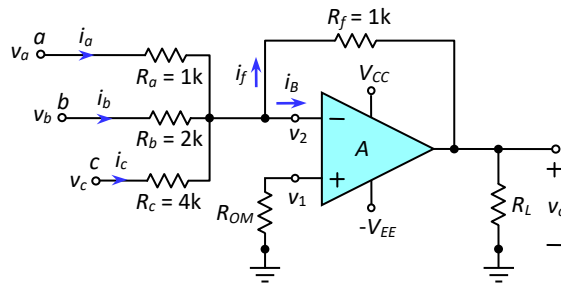


Fig.14-71: Summing amplifier as DAC

Example 13-14

An 8-input DAC is designed as the circuit of Fig.14-71 with $R_a = 1\text{ k}$, and $R_f = 1\text{ k}$. Calculate the value of the resistance of last (least significant) input R_h . Also calculate the step-size, and full-scale output voltage for this DAC. Assume, logic '1' = 5 V or logic '0' = 0 V.

Solution:

The value of the resistance of one input will be just double of the previous input. As $R_a = 1 \text{ k}$, $R_b = 2 \times 1 \text{ k} = 2 \text{ k}$. In this way, the resistance of the last input will be,

$$R_h = 2^{8-1} \times 1 \text{ k} = 128 \text{ k [Ans.]}$$

The expression for the output voltage of this DAC can be found using Equ.(14-112) as,

$$v_o = -\left(\frac{v_a}{1} + \frac{v_b}{2} + \frac{v_c}{4} + \frac{v_d}{8} + \frac{v_e}{16} + \frac{v_f}{32} + \frac{v_g}{64} + \frac{v_h}{128}\right)$$

The step-size is the minimum change in output voltage that will be equal to the output for digital input 00000001. Given, logic '1' = 5 V and logic '0' = 0 V. Thus the step-size will be,

$$v_{o(st)} = -\left(\frac{0}{1} + \frac{0}{2} + \frac{0}{4} + \frac{0}{8} + \frac{0}{16} + \frac{0}{32} + \frac{0}{64} + \frac{5 \text{ V}}{128}\right)$$

$$\therefore v_{o(st)} \approx -39.1 \text{ mV [Ans.]}$$

The full-scale output voltage will be found for the maximum digital input which is 11111111.

$$v_{o(fs)} = -\left(\frac{5 \text{ V}}{1} + \frac{5 \text{ V}}{2} + \frac{5 \text{ V}}{4} + \frac{5 \text{ V}}{8} + \frac{5 \text{ V}}{16} + \frac{5 \text{ V}}{32} + \frac{5 \text{ V}}{64} + \frac{5 \text{ V}}{128}\right)$$

$$\therefore v_{o(fs)} \approx -9.97 \text{ V [Ans.]}$$

The full-scale output can also be calculated using the following formula,

$$v_{o(fs)} = v_{o(st)} \times (2^8 - 1) \approx -9.97 \text{ V}$$

Comments: For proper functioning of this DAC the supply voltage should be 1 V greater than maximum output. Here, we can use $\pm 12 \text{ V}$ or above.

14.28.2 Average Circuit

The summing amplifier of Fig.14-69 can be used as an averaging circuit, that is, the output voltage will be equal to the average of all the input signals. To do that we have to use input resistors with equal values, that is $R_a = R_b = R_c = R$ and the gain of the amplifier should be 1 over the number of inputs ($1/n$). For example, if there are n number of inputs, the value of R should n times the value of the feedback resistor, i.e., $R_a = R_b = R_c = R = nR_f$. Using $n = 3$, Equ.(14-112) can be written as,

$$v_o = -\frac{R_f}{R}(v_a + v_b + v_c) = -\frac{R_f}{3R_f}(v_a + v_b + v_c)$$

$$\text{or, } v_o = -\frac{1}{3}(v_a + v_b + v_c) \quad (14-114)$$

For n number of inputs, the above equation can be rewritten as,

$$v_o = -\frac{1}{n}(v_a + v_b + v_c + \dots + v_n) \quad (14-115)$$

Equ.(14-114) and Equ.(14-115) shows that the output is the average of the input voltages.

14.29 Noninverting Summing Amplifier

We have seen that the summing amplifier of Fig.14-69 gives inverted output (180° phase shift). But using noninverting summing amplifier we can avoid the polarity inversion. A three-input noninverting summing amplifier is shown in Fig.14-72. Here the input signals v_a , v_b , and v_c are applied to the noninverting input via input resistances R_a , R_b , and R_c . Whereas, the negative feedback is implemented to the inverting terminal using R_1 and R_f .

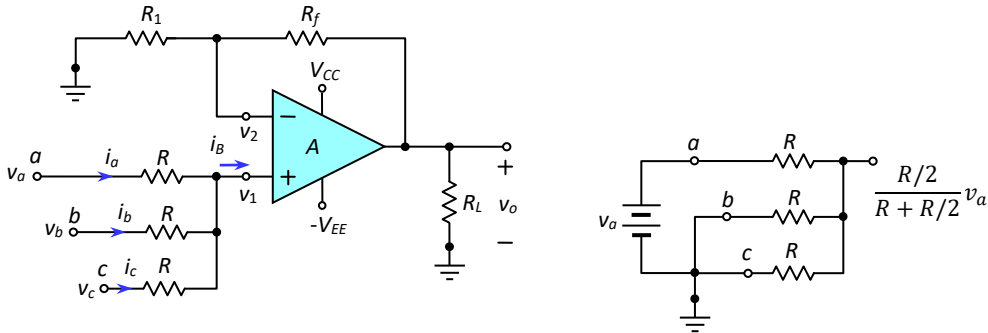


Fig.14-72: (a) 3-input noninverting summing amplifier, and (b) Equivalent circuit of input side for v_a

To find the expression of output voltage, first we have to determine the value of v_1 using superposition theorem. Let us consider v_a by shorting v_b and v_c to ground [Fig.14-72(b)]. The contribution of v_a to v_1 will be

$$\frac{R/2}{R + R/2} v_a$$

The contributions of other sources will be same, as all the input resistances are of same value. Therefore, we can write the value of v_1 as,

$$v_1 = \frac{R/2}{R + R/2} v_a + \frac{R/2}{R + R/2} v_b + \frac{R/2}{R + R/2} v_c$$

$$\text{or,} \quad v_1 = \frac{v_a}{3} + \frac{v_b}{3} + \frac{v_c}{3} = \frac{v_a + v_b + v_c}{3} \quad (14-116)$$

Now, we can assume the summing amplifier as a noninverting amplifier that has an input signal v_1 . So, the output of this amplifier can be calculated using Equ.(14-71) as,

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_1$$

Putting the value of v_1 from Equ.(14-116) into the above equation we get,

$$\boxed{v_o = \left(1 + \frac{R_f}{R_1}\right) \left(\frac{v_a + v_b + v_c}{3}\right)} \quad (14-117)$$

From this expression, it is clear that the output of a noninverting amplifier is the sum of individual input multiplied by a constant. If we set $R_f = 2R_1$, the above equation reduces to

$$v_o = \left(1 + \frac{2R_1}{R_1}\right) \left(\frac{v_a + v_b + v_c}{3}\right)$$

$$\boxed{v_o = (v_a + v_b + v_c)} \quad (14-118)$$

From this expression we find that the output is just the sum of three input voltages.

14.30 Instrumentation Amplifier

An instrumentation amplifier is used to amplify very low-level signals, rejecting noise and interference signals. Examples can be heartbeats, blood pressure, temperature, earthquakes signals and so on. Therefore, the instrumentation amplifier must have some essential characteristics which are given below:

- Inputs to the instrumentation amplifiers will have very low signal energy.
- Therefore, the instrumentation amplifier should have high and accurate gain.
- The gain should be easily adjustable using a single control (resistor).
- It must have high input impedance and low output impedance to prevent loading.
- The instrumentation amplifier should have high CMRR since the transducer output will usually contain common mode signals such as noise when transmitted over long wires.
- It must also have a high slew rate to handle sharp rise times of events and provide a maximum undistorted output voltage swing.

An instrumentation amplifier using three op-amps is shown in Fig.14-73. The op-amps A_1 and A_2 are arranged as buffer amplifiers and op-amp A_3 is a difference amplifier like the circuit of Fig.14-66. The first two-stages provide very high input impedances for the instrumentation amplifier. The last stage amplifies the difference of the input signals v_x and v_y . Analysis of this circuit will show that the gain can be adjusted by varying a single resistor R_{gn} .

From the basic expression of op-amp, we can write,

Equ.(14-61),
$$v_o = (v_1 - v_2)A$$

or,
$$v_1 - v_2 = \frac{v_o}{A}$$

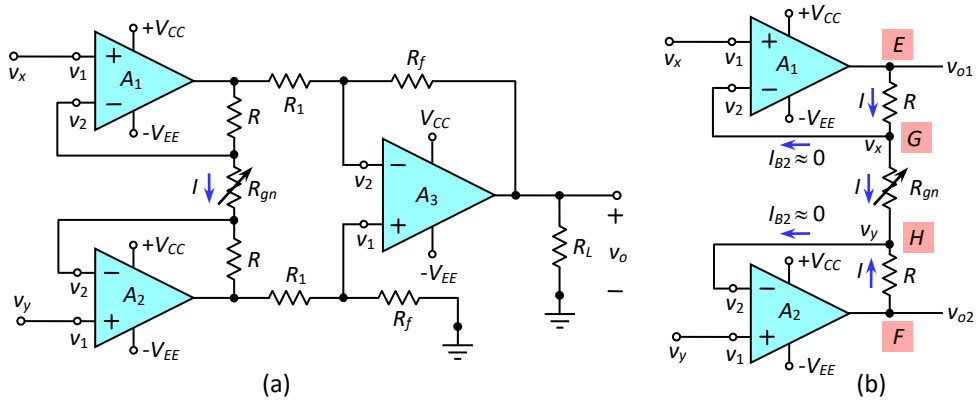


Fig.14-73: (a) Instrumentation amplifier using op-amp, and (b)

As the value of A is very large (ideally infinity), we can reasonably consider $v_o/A \approx 0$. Therefore,

$$v_1 - v_2 \approx 0$$

or,

$$v_1 \approx v_2$$

$\therefore v_1 = v_x, \therefore$

$$v_2 = v_x$$

That is, the voltage at point G [shown in Fig.14-73(b)] is v_x . Similarly, we can find that the voltage at point H is v_y .

Let the current through R_g be I . Using Ohm's law, we can write,

$$I = \frac{v_x - v_y}{R_{gn}} \quad (14-119)$$

Due to the large input impedance, the current to the feedback paths of op-amps A_1 and A_2 can be neglected. Therefore, current flowing through the entire path between points E and F remains same. Let the outputs of op-amp A_1 and A_2 be v_{o1} and v_{o2} , respectively. Therefore,

$$I = \frac{v_{o1} - v_{o2}}{R_{gn} + 2R} \quad (14-120)$$

Combining Equ.(14-119) and Equ.(14-120), we can write,

$$\frac{v_x - v_y}{R_{gn}} = \frac{v_{o1} - v_{o2}}{R_{gn} + 2R}$$

or,

$$(v_{o1} - v_{o2}) = \frac{(v_x - v_y)(R_{gn} + 2R)}{R_{gn}}$$

The output of the differential amplifier consisting of op-amp A_3 can be calculated using Equ.(14-99) as,

$$v_o = -\frac{R_f}{R_1}(v_{o1} - v_{o2})$$

$$\text{or,} \quad v_o = -\frac{R_f (R_{gn} + 2R)}{R_1 R_{gn}} (v_x - v_y)$$

$$\text{or,} \quad \frac{v_o}{v_x - v_y} = -\frac{R_f (R_{gn} + 2R)}{R_1 R_{gn}}$$

$$\therefore \quad \boxed{A_d = -\frac{R_f (R_{gn} + 2R)}{R_1 R_{gn}}} \quad (14-121)$$

From this expression of output voltage, we find that the gain depends on R_{gn} . Therefore, changing the value of R_{gn} , we can change the gain of the instrumentation amplifier.

Example 13-15

An instrumentation amplifier like that of Fig.14-73 is designed with $R_1 = R = 1 \text{ k}$, $R_f = 100 \text{ k}$, and potentiometer in place of R_{gn} . If R_{gn} can be varied from 1 k to 100 k , calculate the lowest and highest gain of the instrumentation amplifier.

Solution:

The lowest gain will be found for highest value of $R_{gn} = 100 \text{ k}$.

$$\text{Using Equ.(14-121),} \quad A_{d(l)} = -\frac{R_f (R_{gn} + 2R)}{R_1 R_{gn}} = -\frac{100 \text{ k} (100 \text{ k} + 2 \times 1 \text{ k})}{1 \text{ k} \cdot 100 \text{ k}}$$

$$\therefore \quad A_{d(l)} = 102 \text{ [Ans.]}$$

The highest gain will be found for lowest value of $R_{gn} = 1 \text{ k}$.

$$\text{Using Equ.(14-121),} \quad A_{d(h)} = -\frac{R_f (R_{gn} + 2R)}{R_1 R_{gn}} = -\frac{100 \text{ k} (1 \text{ k} + 2 \times 1 \text{ k})}{1 \text{ k} \cdot 1 \text{ k}}$$

$$\therefore \quad A_{d(h)} = 300 \text{ [Ans.]}$$

Comments: Selecting suitable components value, we can achieve any range of gain.

14.31 Op-Amp Integrator Circuit

An integrator is a circuit in which the output voltage is the integral of the input voltage. That is, an integrating circuit produces an output voltage which is proportional to the area (amplitude multiplied by time) contained under the waveform. An inverting amplifier can easily be converted into an integrator replacing the feedback resistor (R_f) by a capacitor as shown in Fig.14-74.

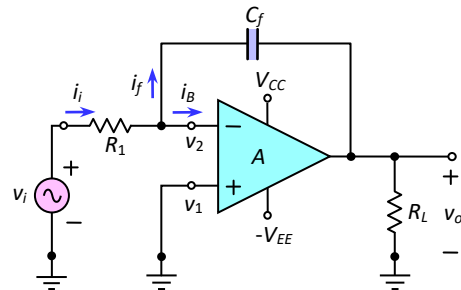


Fig.14-74: Basic integrator circuit using op-amp

To derive the expression for the output voltage, we will apply KCL at the inverting terminal (v_2) of the circuit,

$$i_i = I_B + i_f$$

Since, the input impedance of op-amp is very large, I_B will be very small. Neglecting I_B , we get,

$$i_i \approx i_f \quad (14-122)$$

We know that, the current flowing through a capacitor is given by,

$$i_c = C \frac{dv_c}{dt}$$

From the circuit of Fig.14-74, we can write,

$$i_f = C_f \frac{d(v_2 - v_o)}{dt} \quad (14-123)$$

The value of input current will be,

$$i_i = \frac{v_i - v_2}{R_1} \quad (14-124)$$

Putting the value of i_f and i_i from Equ.(14-123) and Equ.(14-124) into Equ.(14-122) we get,

$$\frac{v_i - v_2}{R_1} = C_f \frac{d(v_2 - v_o)}{dt}$$

or,
$$\frac{v_i - v_2}{R_1} = C_f \frac{d(v_2 - v_o)}{dt}$$

However, as described before, the inverting terminal of this amplifier will be a virtual ground. So, $v_2 = 0$. Therefore, the above equation reduces to,

$$\frac{v_i}{R_1} = -C_f \frac{dv_o}{dt}$$

To find the value of v_o , we have to integrate both sides of the above equation with respect to time.

$$\int_0^t \frac{v_i}{R_1} dt = \int_0^t \left(-C_f \frac{dv_o}{dt} \right) dt$$

or,
$$\frac{1}{R_1} \int_0^t v_i dt = -C_f v_o + C_t$$

or,
$$v_o = -\frac{1}{R_1 C_f} \int_0^t v_i dt + C_t$$

Here, C_t is the integral constant, and its value is the initial output voltage, that is, the value of output voltage at $t = 0$.

∴
$$v_o = -\frac{1}{R_1 C_f} \int_0^t v_i dt + v_o|_{t=0} \quad (14-125)$$

Equ.(14-125) shows that the output (v_o) of the circuit is proportional to the integration of the input voltage with a phase inversion (due to minus sign).

The op-amp integrator lends itself to a variety of applications, ranging from integrating-type digital-to-analog converters, voltage-to-frequency converters, double-integrator-loop filters etc.

This basic integrator suffers from some limitations. If there is no input signal applied to this integrator, that is, if $v_i = 0$, the circuit will work as an open-loop-amplifier. The feedback capacitor, C_f , will work as open circuit ($X_{C_f} = \infty$) for the input offset voltage, V_{io} . This input offset voltage will be amplified by the open-loop gain of the op-amp, and will produce a large error voltage at the output. Another limitation is the output is inversely proportional to the time constant, $R_1 C_f$. To overcome these limitations, a practical integrator is made by connecting a resistor, R_f in parallel with C_f as shown in Fig.14-75(a). This R_f limits the low frequency gain and minimizes the output error voltage.

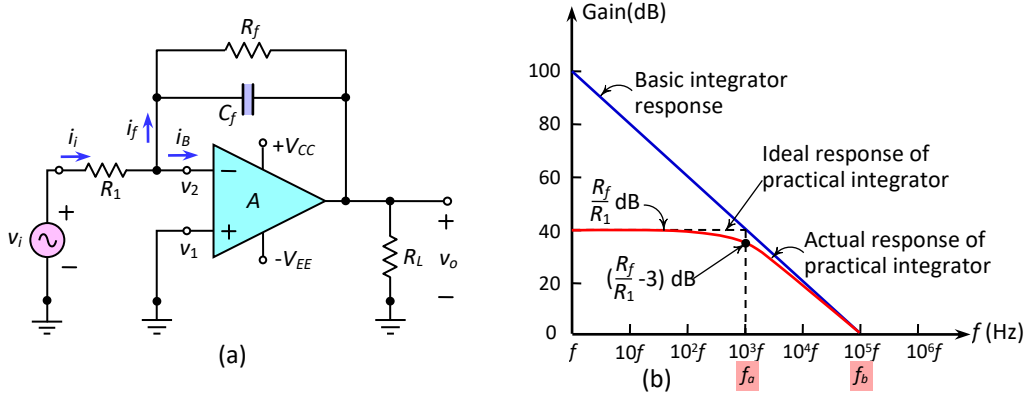


Fig.14-75: (a) Practical integrator, and (b) Frequency response of integrator

The frequency response of the basic integrator and the practical integrator is shown in Fig.14-75(b). In this response, f_a , called *gain limiting frequency*, is the frequency below which the gain of the integrator will be limited to R_f/R_1 . At this frequency the reactance of C_f and the resistance R_f will be equal. That is,

$$\frac{1}{2\pi f_a C_f} = R_f$$

$\therefore$

$$f_a = \frac{1}{2\pi R_f C_f}$$

(14-126)

As shown in Fig.14-75(b), for the frequencies greater than f_a , the gain decreases at a rate of 20 dB/decade. In this range, $X_{C_f} \ll R_f$, that is, X_{C_f} becomes effective in the circuit. So the circuit works as an integrator. On the other hand, for the frequencies lower than f_a , $X_{C_f} \gg R_f$, that is, R_f will be effective in the circuit. In this frequency range, the circuit works as an amplifier the gain remains constant at R_f/R_1 .

Another frequency, f_b is the frequency at which the gain is 0 dB or 1. At this frequency, $X_{C_f} \ll R_f$, and $X_{C_f} = R_1$. Therefore,

$$\frac{1}{2\pi f_b C_f} = R_1$$

$$\therefore \quad \boxed{f_b = \frac{1}{2\pi R_1 C_f}} \quad (14-127)$$

For a practical integrator, f_b should be greater than f_a and within the range of frequencies from f_a to f_b the circuit works as an integrator.

A workable integrator can be designed by following the subsequent steps:

- i) Set f_a equal to the lowest frequency of the input signal. Then select a suitable value of $C_f \leq 1 \mu\text{F}$, and calculate the value of R_f using Equ.(14-127).
- ii) Set $f_b = 20f_a$, and calculate R_1 using Equ.(14-126).

Example 13-16

Suppose an integrator is so designed that its time constant is $R_1 C_f = 1 \text{ s}$. Draw the input and output waveforms for

- i) A 4 V peak-to-peak square wave with the frequency of 50 Hz.
- ii) A 4 V peak-to-peak sinusoidal wave with the frequency of 50 Hz.

Assume that the initial output of the integrator is 0.

Solution:

i) Square wave:

The input square wave is drawn in Fig.14-76(a). Time period of the square wave will be $1/50 \text{ s} = 20 \text{ ms}$ and the peak value will be $4/2 = 2 \text{ V}$. For this square wave, the output of the integrator will be an inverted triangular wave. For the positive half-cycle (10 ms), the output of the integrator will decrease from 0 V (initial output) and the minimum value will be:

$$\text{Using Equ.(14-125),} \quad v_o = -\frac{1}{R_1 C_f} \int_0^t v_i dt + v_o|_{t=0} = -\frac{1}{1 \text{ s}} v_i t + 0$$

Putting $t = T/2 = 10 \text{ ms} = 10 \times 10^{-3} \text{ s}$, and $R_1 C_f = 1 \text{ s}$

$$v_o = \left(-\frac{1}{1 \text{ s}}\right) 2 \text{ V} \times (10 \times 10^{-3} \text{ s}) = -0.02 \text{ V}$$

For the negative half-cycle (10 ms), the output of the integrator will start from -0.02 V and the maximum value will be

$$v_o = -\frac{1}{R_1 C_f} \int_0^t v_i dt + v_o|_{t=0} = -\frac{1}{1 \text{ s}} v_i t - 0.02 \text{ V}$$

$$\therefore \quad v_o = \left(-\frac{1}{1 \text{ s}}\right) (-2 \text{ V}) \times 10 \times 10^{-3} \text{ s} - 0.02 \text{ V} = 0$$

The slope of the triangular wave will be $\pm 0.02 \text{ V}/10 \text{ ms} = \pm 2 \text{ V/s}$.

The output waveform is shown in Fig.14-76(b).

ii) Sinusoidal wave:

As the peak-to-peak value is 4 V, the amplitude of the sinusoidal will be 2 V. Therefore, the sinusoidal wave can be represented as,

$$v_i = 2 \sin(\omega t) = 2 \sin(2\pi f t) = 2 \sin(2\pi \times 50 t)$$

$$\text{or,} \quad v_i = 2 \sin(100\pi t)$$

$$\text{Using Equ.(14-125),} \quad v_o = -\frac{1}{1 \text{ s}} \int_0^t 2 \sin(100\pi t) dt + v_o|_{t=0}$$

$$\text{or,} \quad v_o = -\left[\frac{-2 \cos(100\pi t)}{100\pi} \right]_0^t + 0 \quad \because v_o|_{t=0} = 0$$

$$\text{or,} \quad v_o = \frac{2}{100\pi} [\cos(100\pi t) - 1]$$

$$\therefore v_o = \frac{1}{50\pi} \cos(100\pi t) - \frac{1}{50\pi}$$

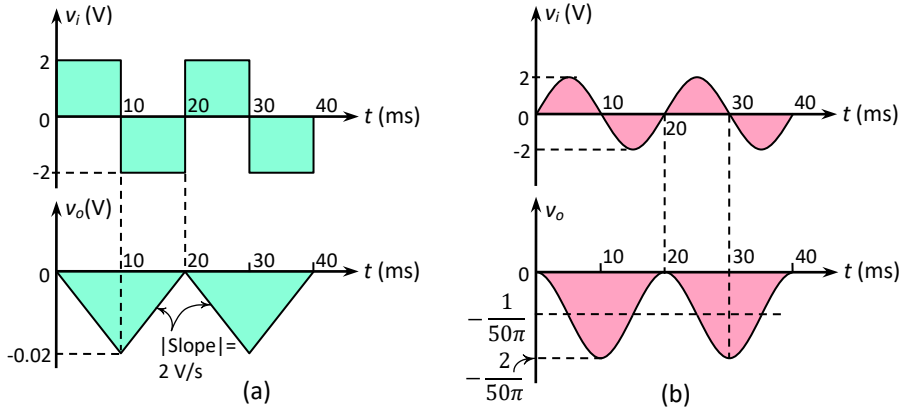


Fig.14-76: Waveforms of Example 13-16: (a) Input and output for square wave, and (b) Input and output for sinusoidal wave

The output waveform is shown in Fig.14-76(b). To understand this output, let us calculate some points of this waveform. When $t = 0$,

$$\therefore v_o = \frac{1}{50\pi} \cos(100\pi \times 0) - \frac{1}{50\pi} = 0$$

When $t = 10 \text{ ms}$,

$$v_o = \frac{1}{50\pi} \cos(100\pi \times 10 \text{ ms}) - \frac{1}{50\pi}$$

$$\text{or,} \quad v_o = \frac{1}{50\pi} \cos(\pi \times 1000 \text{ ms}) - \frac{1}{50\pi}$$

$$\text{or,} \quad v_o = \frac{1}{50\pi} \cos(\pi \times 1 \text{ s}) - \frac{1}{50\pi}$$

or,

$$v_o = \frac{1}{50\pi}(-1) - \frac{1}{50\pi} = -\frac{2}{50\pi}$$

Comments: The output is always negative due to the negative sign in Equ.(14-).

14.32 Op-Amp Differentiator Circuit

A **differentiator** is a circuit in which the output voltage is the derivative of the input voltage. That is, a differentiator circuit produces an output voltage which is proportional to the time rate of change of input voltage. An inverting amplifier can easily be converted into a differentiator replacing the input resistor (R_1) by a capacitor as shown in Fig.14-77.

The expression for the output voltage can be derived by applying KVL at the inverting terminal (node v_2) of the circuit:

$$i_i = I_B + i_f$$

Since, the input impedance of op-amp is very large, I_B will be very small. Neglecting I_B , we get,

$$i_i \approx i_f$$

or,

$$C_1 \frac{d(v_i - v_2)}{dt} = \frac{v_2 - v_o}{R_f}$$

Here also $v_2 = 0$, due to the virtual grounding of the inverting terminal. Therefore,

$$C_1 \frac{dv_i}{dt} = -\frac{v_o}{R_f}$$

or,

$$v_o = -R_f C_1 \frac{dv_i}{dt} \quad (14-128)$$

From this expression, it is clear that the output is directly proportional to the derivative of the input signal with phase reversal (minus sign).

The basic differentiator of Fig.14-77 has some limitations. The gain of the circuit, R_f/X_{C1} , increases with the increase in frequency at a rate of 20 dB/decade which makes the circuit unstable. As the input impedance X_{C1} decreases with frequency, and hence increases the gain, the circuit will be very susceptible to high frequency noise.

To avoid these limitations, an additional series resistor (R_1) and a parallel capacitor (C_f) are used in practical differentiator as shown in Fig.14-78(a). The frequency response of the basic and practical differentiator is shown in Fig.14-78(b). The component values should be chosen in such a way that, at lower frequencies, $X_{C1} \gg R_1$, and $R_f \ll X_{Cf}$. So in the input branch, the higher reactance, X_{C1} , and in the feedback branch the lower resistance, R_f , will be effective. Therefore, the circuit will

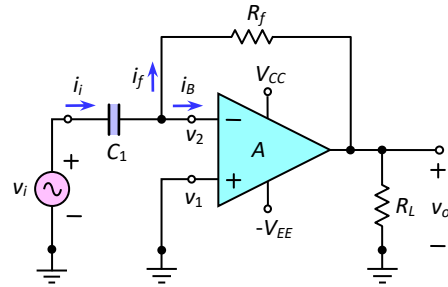


Fig.14-77: Basic differentiator circuit using op-amp

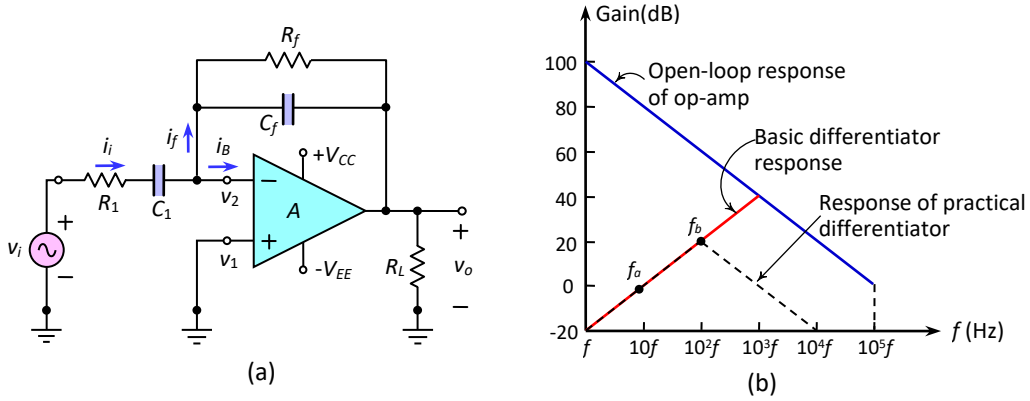


Fig.14-78: (a) Practical differentiator, and (b) Frequency response of differentiator

work as a differentiator. But, at some higher frequencies (say f_b); $X_{C1} \ll R_1$ and $R_f \gg X_{Cf}$. So in the input branch R_1 and in the feedback branch C_f will be effective. Therefore, the circuit will work as an integrator. Thus we find that, the circuit works as a differentiator if the signal frequency is less than f_b . Up to frequency f_b the gain increases at a rate of 20 dB/decade and after f_b the gain decreases at a rate of 20 dB/decade. In the differentiator range, if frequency is increased gradually X_{C1} will decrease. At a particular frequency (say f_a), X_{C1} will be equal to R_f and hence the gain of the circuit will be 1 or 0 dB. The value of this zero dB frequency can be calculated by considering

$$X_{C1} = R_f$$

or,

$$\frac{1}{2\pi f_a C_1} = R_f$$

$\therefore$

$$f_a = \frac{1}{2\pi R_f C_1} \quad (14-129)$$

The frequency (f_b) beyond which the circuit starts to work as an integrator must satisfy the conditions: $X_{C1} = R_1$, and $X_{Cf} = R_f$. Therefore, the value of this cut-off frequency can be calculated using either of the conditions:

$$X_{C1} = R_1$$

or,

$$\frac{1}{2\pi f_b C_1} = R_1$$

$\therefore$

$$f_b = \frac{1}{2\pi R_1 C_1} \quad (14-130)$$

Using the other condition,

$$X_{Cf} = R_f$$

or,

$$\frac{1}{2\pi f_b C_f} = R_f$$

$$\therefore \quad \boxed{f_b = \frac{1}{2\pi R_f C_f}} \quad (14-131)$$

Another limiting frequency of this circuit is f_c , which is the **unity-gain frequency** of the op-amp. The relation among these three frequencies is,

$$f_a < f_b < f_c \quad (14-132)$$

For satisfactory performance of the circuit as differentiator, the range of the frequency of the input signal should be f_a to f_b . From Equ.(14-130) and Equ.(14-131) it is found that the two time constants $R_1 C_1$, and $R_f C_f$ should be equal.

A workable differentiator can be designed by following the steps given below:

- iii) Set f_a equal to the lowest frequency of the input signal. Then select a suitable value for $C_1 \leq 1 \mu\text{F}$, and calculate the value of R_f using Equ.(14-129).
- iv) Set $f_b = 20f_a$, and calculate R_1 using Equ.(14-130), and C_f using Equ.(14-131).

Example 13-17

Suppose a differentiator is designed with time constant $R_1 C_f = 1 \text{ s}$. Draw the input and output waveforms for

- iii) A 4 V peak-to-peak triangular wave with the frequency of 50 Hz.
- iv) A 4 V peak-to-peak sinusoidal wave with the frequency of 50 Hz.

Solution:

i) Triangular wave:

The input triangular wave is drawn in Fig.14-79. Time period of the triangular wave will be $1/50 \text{ s} = 20 \text{ ms}$. The output of the differentiator will be the rate of change of input voltage (or slope of the input). The value of the rate of change of input voltage is $\pm 4 \text{ V}/10 \text{ ms} = \pm 400 \text{ V/s}$. For the positive slope, the output will be

$$\text{Using Equ.(14-128),} \quad v_o = -R_f C_1 \frac{dv_i}{dt} = (-1 \text{ s}) 400 \text{ V/s} = -400 \text{ V}$$

For the negative slope, the output will be

$$v_o = -R_f C_1 \frac{dv_i}{dt} = (-1 \text{ s})(-400 \text{ V/s}) = 400 \text{ V}$$

The output square wave is shown in Fig.14-79(b).

ii) Sinusoidal wave:

As the peak-to-peak value is 4 V, the amplitude of the sinusoidal wave will be 2 V. Therefore, the sinusoidal wave can be represented as

$$v_i = 2 \sin(\omega t) = 2 \sin(2\pi f t) = 2 \sin(2\pi \times 50 t)$$

$$\text{or,} \quad v_i = 2 \sin(100\pi t)$$

As $R_1 C_f = 1 \text{ s}$, the output will be,

Using Equ.(14-128),
$$v_o = -(1\text{ s}) \frac{d}{dt} [2 \sin(100\pi t) \text{ V}]$$

or,
$$v_o = (-2\text{ s}) [\cos(100\pi t)] (100\pi) \text{ V/s}$$

$\therefore v_o = -200\pi \cos(100\pi t) \text{ V}$

The output waveform will be an inverted *cosine* wave with amplitude 200π as shown in Fig.14-79(b).

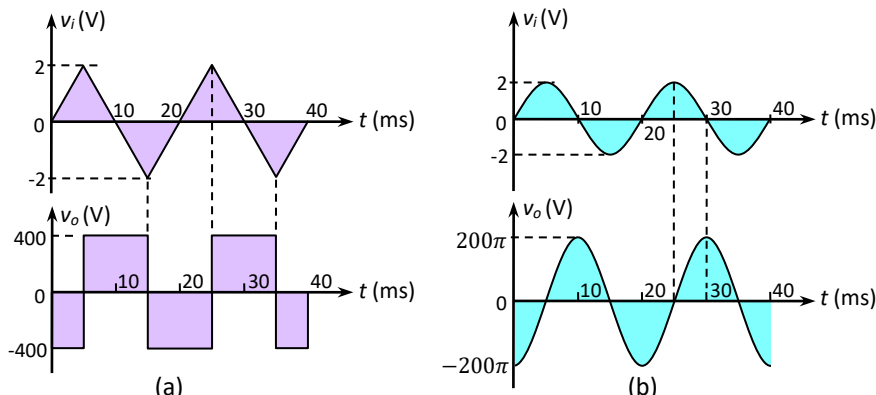


Fig.14-79: Waveforms of Example 13-17: (a) Input and output for triangular wave, and (b) Input and output for sinusoidal wave

Comments: The output is negative for positive slope due to the negative sign in Equ.(14-128) and vice versa.

14.33 Logarithmic Amplifier

A **logarithmic amplifier** is a circuit in which the output voltage is proportional to the logarithm of the input voltage. This amplifier can be constructed using either diodes or transistors.

14.33.1 Logarithmic Amplifier using Diode

We can easily convert an inverting amplifier into a logarithmic amplifier using a diode in place of the feedback resistor as shown in Fig.14-80.

By applying KCL at the inverting terminal we can write,

$$i_i = I_B + i_f$$

Since, the input impedance of op-amp is very large, I_B will be very small. Neglecting I_B , we get,

$$i_i \approx i_f = i_D \quad (14-133)$$

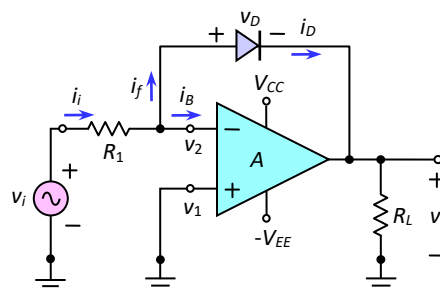


Fig.14-80: Logarithmic amplifier using diode

The diode current is given by,

$$i_D = i_s \left(e^{\frac{v_D}{nV_T}} - 1 \right) \quad (14-134)$$

where,

v_D = forward bias voltage across the diode

n = ideality factor. For ideal diode $n = 1$, and for practical diode its value may be 1 to 2.

$V_T = \frac{KT}{q}$ = thermal voltage. At room temperature $V_T \approx 26$ mV

K = Boltzmann Constant ($= 1.38 \times 10^{-23}$ J/Kelvin)

T = Temperature of the diode

q = Charge on an electron ($= 1.6 \times 10^{-19}$ C)

As $e^{\frac{v_D}{nV_T}} \gg 1$, the diode current of Equ.(14-134) can be approximate as,

$$i_D \approx i_s e^{\frac{v_D}{nV_T}}$$

Putting the value of i_D from Equ.(14-133) into the above equation we get,

$$i_i = i_s e^{\frac{v_D}{nV_T}}$$

or,

$$\frac{v_i - v_2}{R_1} = i_s e^{\frac{v_2 - v_o}{nV_T}}$$

Due to the virtual ground, $v_2 = 0$. Therefore, the above equation reduces to,

$$\frac{v_i}{R_1} = i_s e^{\frac{-v_o}{nV_T}}$$

or,

$$\frac{v_i}{R_1 i_s} = e^{\frac{-v_o}{nV_T}}$$

or,

$$\ln \left(\frac{v_i}{R_1 i_s} \right) = \frac{-v_o}{nV_T}$$

$\therefore$

$$v_o = -nV_T \ln \left(\frac{v_i}{R_1 i_s} \right) \quad (14-135)$$

From this expression, it is clear that the output voltage is proportional to the logarithm of the input voltage. Although the circuit gives natural log ($\ln$), we can find $\log_{10}$, by using proper scaling factor as,

$$v_{o(\log_{10})} = -nV_T \ln \left(\frac{v_i}{R_1 i_s} \right) \times (0.4343) \quad (14-136)$$

In case of ideal diode, the value of n is 1. But for practical diode the value of $n > 1$. The higher value of n reduces the dynamic range of the logarithmic amplifier.

This problem can be overcome by using BJT instead of diode. The logarithmic amplifier using a transistor is shown in Fig13-81. The base of the transistor is directly connected to ground and the collector is connected to inverting terminal of op-amp. As we know, the inverting terminal of op-amp is virtually ground. Thus both the base and the collector of the transistor are connected to zero potential. So, we can consider that the base of the transistor is shorted to the collector terminal. Hence, only the B-E junction will be effective in the feedback path, i.e., the transistor will work as a diode. The emitter current of this transistor will be,

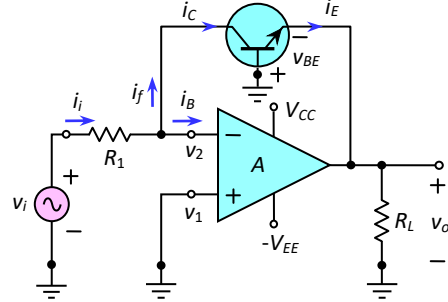


Fig.14-81: Logarithmic amplifier using BJT

$$i_E = I_s \left(e^{\frac{v_{BE}}{V_T}} - 1 \right) \approx I_s e^{\frac{v_{BE}}{V_T}} \approx i_C = i_f$$

Using the same analysis of diode logarithmic amplifier, we can find that,

$$v_o = -V_T \ln \left(\frac{v_i}{R_1 I_s} \right) \quad (14-137)$$

The output is proportional to the logarithm of input and does not depend on diode ideality factor. However, both the circuits suffer from the temperature dependency.

Example 13-18

Determine the output voltage for the circuit of Fig.14-81, if $v_i = 0.5 \text{ V}$, $R_1 = 50 \text{ k}$, $I_s = 30 \text{ nA}$. Also determine the output for inputs of 1.0 V and 2 V. Assume, $V_T \approx 26 \text{ mV}$.

Solution:

For $v_i = 0.5 \text{ V}$:

Using Equ.(14-137),

$$v_o = -V_T \ln \left(\frac{v_i}{R_1 I_s} \right)$$

or,

$$v_o = -26 \text{ mV} \times \ln \left(\frac{0.5 \text{ V}}{50 \text{ k} \times 30 \text{ nA}} \right)$$

$\therefore$

$$v_o = -151 \text{ mV [Ans.]}$$

For $v_i = 1 \text{ V}$:

Using Equ.(14-137),

$$v_o = -V_T \ln \left(\frac{v_i}{R_1 I_s} \right)$$

or,

$$v_o = -26 \text{ mV} \times \ln \left(\frac{1 \text{ V}}{50 \text{ k} \times 30 \text{ nA}} \right)$$

$$\therefore v_o = -169 \text{ mV [Ans.]}$$

For $v_i = 2 \text{ V}$:

$$\text{Using Equ.(14-137),} \quad v_o = -V_T \ln \left(\frac{v_i}{R_1 I_s} \right)$$

$$\text{or,} \quad v_o = -26 \text{ mV} \times \ln \left(\frac{2 \text{ V}}{50 \text{ k} \times 30 \text{ nA}} \right)$$

$$\therefore v_o = -187 \text{ mV [Ans.]}$$

Comments: For each doubling of the input signal, the output signal increases by a constant value of 18 mV. This is the nature of a log amplifier.

14.33.2 Temperature Compensated Logarithmic Amplifier

A temperature compensated logarithmic amplifier is shown in Fig.14-82. In this circuit the diodes D_1 and D_2 are matched diodes.

If we carefully observe the circuit, we find that the first circuit with op-amp A_1 is a logarithmic amplifier using diode. This circuit is same as that of Fig.14-80. The output voltage of his circuit

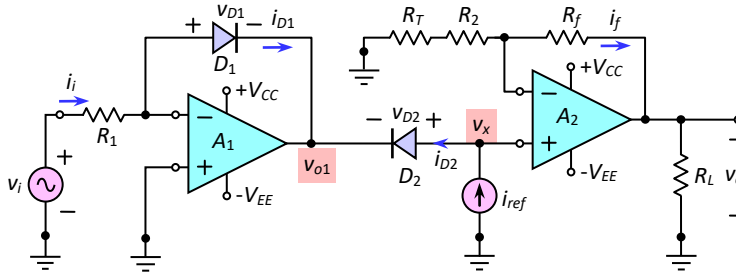


Fig.14-82: Temperature compensated logarithmic amplifier

can be found using Equ.(14-135) as,

$$v_{o1} = -nV_T \ln \left(\frac{v_i}{R_1 i_s} \right) \quad (14-138)$$

As the input impedance of op-amp is very large, we can assume $i_{D2} = i_{ref}$. So, for diode D_2 , we can write,

$$i_{D2} = i_{ref} = i_s e^{\frac{v_{D2}}{nV_T}}$$

$$\text{or,} \quad \frac{i_{ref}}{i_s} = e^{\frac{v_{D2}}{nV_T}}$$

$$\therefore v_{D2} = nV_T \ln \left(\frac{i_{ref}}{i_s} \right) \quad (14-139)$$

This input voltage will be amplified by the noninverting amplifier consisting of op-amp A_2 . So the overall output voltage will be,

$$v_o = \left(1 + \frac{R_f}{R_T + R_1 + R_f}\right) v_x$$

$$\text{or,} \quad v_o = -nV_T \left(1 + \frac{R_f}{R_T + R_1 + R_f}\right) \ln \left(\frac{v_i}{R_1 i_{ref}}\right)$$

$$\therefore \quad \boxed{v_o = K_1 \ln \left(\frac{v_i}{v_{ref}}\right)} \quad (14-140)$$

where, K_1 , and v_{ref} are constant and given by

$$K_1 = -nV_T \left(1 + \frac{R_f}{R_T + R_1}\right) \quad (14-141)$$

$$v_{ref} = R_1 i_{ref} \quad (14-142)$$

From Equ.(14-141), we find that the output voltage is independent of reverse saturation current but not the thermal voltage V_T and K_1 depends on V_T . As V_T increases with temperature, the output voltage will also increase. To eliminate this temperature dependency, the thermal resistance R_T has been used. With the increase in V_T (due to temperature), R_T will also increase, that will decrease the gain of the noninverting amplifier. Thus, the output will be almost independent temperature.

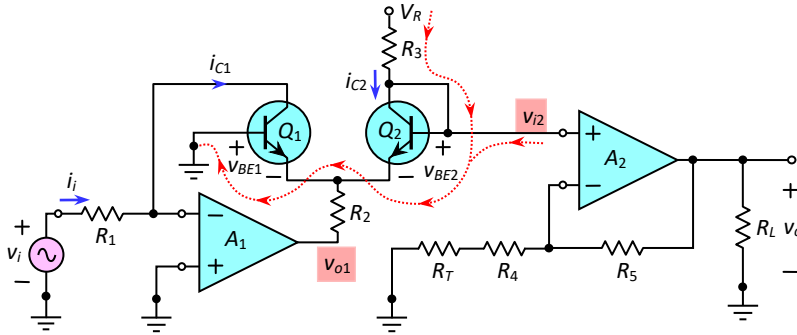


Fig.14-83: Temperature compensated logarithmic amplifier using BJT

14.33.3 Temperature Compensated Logarithmic Amplifier with BJT

A temperature compensated logarithmic amplifier using two op-amps and two transistors are shown in Fig.14-83. It provides wide dynamic range amplification for input voltage then the logarithmic amplifier using diodes.

As the noninverting terminal of op-amp A_1 is grounded, the inverting terminal will also be virtually grounded. Therefore,

$$i_i = \frac{v_i}{R_1} \approx i_{C1} \quad (14-143)$$

Now, applying KVL, starting from V_R and going up to the base of Q_1 we can write,

$$V_R - i_{C2}R_3 - v_{BE2} + v_{BE1} = 0$$

In this circuit transistors Q_1 , and Q_2 are matched transistors. So $I_{s1} = I_{s2} = I_s$, and $v_{BE2} = v_{BE1}$. Thus the above equation reduces to,

$$V_R - i_{C2}R_3 = 0$$

$$\therefore i_{C2} = \frac{V_R}{R_3} \quad (14-144)$$

For transistor Q_1 we can write,

$$i_{C1} = I_s \left(e^{\frac{v_{BE1}}{V_T}} - 1 \right)$$

$$\text{Using Equ.(14-144),} \quad \frac{v_i}{R_1} \approx I_s e^{\frac{v_{BE1}}{V_T}} [\text{neglecting } 1]$$

$$\text{or,} \quad \ln \left(\frac{v_i}{R_1 I_s} \right) = \frac{v_{BE1}}{V_T}$$

$$\therefore v_{BE1} = V_T \ln \left(\frac{v_i}{R_1 I_s} \right) \quad (14-145)$$

Similarly, for transistor Q_2 we can find

$$v_{BE2} = V_T \ln \left(\frac{V_R}{R_3 I_s} \right) \quad (14-146)$$

Applying KVL, from v_{i2} going through two B-E junctions of the transistors we get,

$$v_{i2} - v_{BE2} + v_{BE1} = 0$$

$$\text{or,} \quad v_{i2} = v_{BE2} - v_{BE1}$$

Putting the value of v_{BE1} , and v_{BE2} from Equ.(14-146) and Equ.(14-147),

$$v_{i2} = V_T \ln \left(\frac{V_R}{R_3 I_s} \right) - V_T \ln \left(\frac{v_i}{R_1 I_s} \right)$$

$$\text{or,} \quad v_{i2} = V_T \ln \left(\frac{V_R}{R_3 I_s} \times \frac{R_1 I_s}{v_i} \right)$$

$$\text{or,} \quad v_{i2} = V_T \ln \left(\frac{V_R R_1}{v_i R_3} \right)$$

$$\therefore v_{i2} = -V_T \ln \left(\frac{v_i R_3}{V_R R_1} \right) \quad (14-147)$$

v_{i2} is the input of noninverting amplifier consisting of op-amp A_2 . So the overall output will be,

$$v_o = v_{i2} \left(1 + \frac{R_5}{R_T + R_4} \right) = -V_T \left(1 + \frac{R_5}{R_T + R_4} \right) \ln \left(\frac{v_i R_3}{V_R R_1} \right) \quad (14-148)$$

From Equ.(14-149), we find that the output voltage is proportional to the natural log (ln) of the input voltage. Moreover, the output is independent of reverse saturation current and the variation of V_T (due to temperature) is compensated by thermal resistance R_T . Therefore, this circuit will work as a better logarithmic amplifier.

14.34 Antilog Amplifier

Antilog amplifier or **anti-logarithmic amplifier** is an electronic circuit that produces output proportional to the anti-logarithm of the input. Antilog amplifier does the exactly opposite to a log amplifier. Like the logarithmic amplifiers, antilog amplifiers can also be designed using either diodes or transistors. In fact, a logarithmic amplifier can be converted into an antilog amplifier by interchanging the position of resistor and diode or transistor.

14.34.1 Antilog Amplifier using Diode

An antilog amplifier is shown in Fig.14-84. Compare this circuit with the log amplifier of Fig.14-80. The positions of the diode and the resistor have been interchanged. In this circuit, the input voltage V_i is applied to the inverting terminal of op-amp through a diode D_1 . The noninverting terminal is directly connected to ground. So the inverting terminal will be virtually ground, that is $v_2 = 0$.

By applying KCL at the inverting terminal we can write,

$$I_i = I_B + I_f$$

Since, the input impedance of the op-amp is very large, I_B will be very small. Neglecting I_B , we get,

$$I_i \approx I_f$$

or,

$$I_s \left(e^{\frac{v_D}{nV_T}} - 1 \right) = \frac{v_2 - V_o}{R_f}$$

Neglecting 1,

$$I_s e^{\frac{(v_i - v_2)}{nV_T}} = \frac{v_2 - V_o}{R_f}$$

$\because v_2 = 0$

$$I_s e^{\frac{v_i}{nV_T}} = \frac{-V_o}{R_f}$$

$\therefore$

$$\boxed{V_o = -R_f I_s e^{\frac{v_i}{nV_T}}}$$

(14-149)

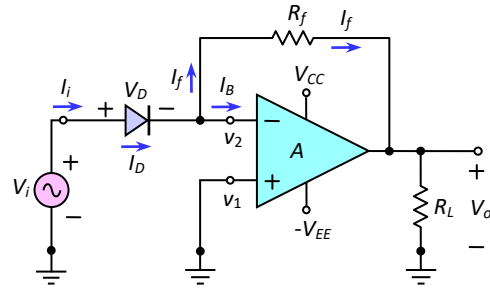


Fig.14-84: Antilog amplifier using diode

From this equation, we find that the output voltage is proportional to the exponential function (e -base) of input voltage. The exponential function is nothing but the antilog and thus the circuit works as an antilog amplifier. The output is temperature dependant as it contains I_s and V_T . Moreover, the output depends on the diode ideality factor n . The output can be made independent of n by using a BJT instead of diode.

14.34.2 Antilog Amplifier using BJT

An antilog amplifier using a PNP transistor is shown in Fig.14-85. Using KVL, we can easily find that,

$$V_{BE} = V_i$$

The emitter current of the transistor can be calculated as,

$$I_E = I_s \left(e^{\frac{V_{BE}}{V_T}} - 1 \right)$$

or,

$$I_E = I_s e^{\frac{V_i}{V_T}} \approx I_C$$

Applying KCL at the inverting terminal we can write,

$$I_C = I_B + I_f$$

Since, the input impedance of the op-amp is very large, I_B will be very small. Neglecting I_B , we get,

$$I_C \approx I_f$$

or,

$$I_s e^{\frac{V_i}{V_T}} = \frac{v_2 - V_o}{R_f}$$

Due to the virtual ground $v_2 = 0$. Therefore,

$$I_s e^{\frac{V_i}{V_T}} = \frac{-V_o}{R_f}$$

$\therefore$

$$\boxed{V_o = -R_f I_s e^{\frac{V_i}{V_T}}} \quad (14-150)$$

The output voltage is proportional to the exponential function of input voltage. Although the output is independent of ideality factor (n), it will dependant on temperature.

Antilog amplifiers along with log amplifiers are used to perform analogue computations (division, multiplication, square-root etc.) on the input signals. Hence these amplifiers are used in analog computers. They are also used for data compression and inflation.

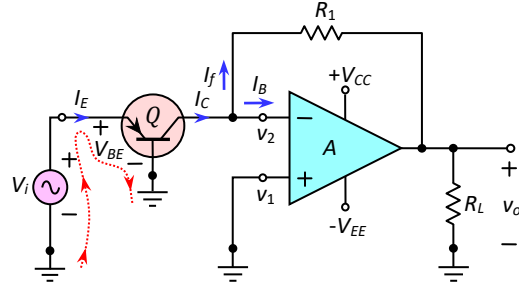


Fig.14-85: Antilog amplifier using transistor

14.35 Analog Multiplier Circuit

An **analog multiplier** is a circuit that produces an output voltage or current that is proportional to the product of two or more independent input voltages or currents. In addition to multiplying and dividing, multipliers can perform squaring, square-rooting and modulation functions. An analog multiplier can be constructed using log amplifiers, summing amplifier and an antilog amplifier. To multiply two analog signals two log amplifiers have to be used and similarly, to multiply n -number of signals n -number of log amplifiers have to be used. A two-input multiplier is shown in Fig.14-86. It consists of two log amplifiers, a two-input summing amplifier and an antilog amplifier.

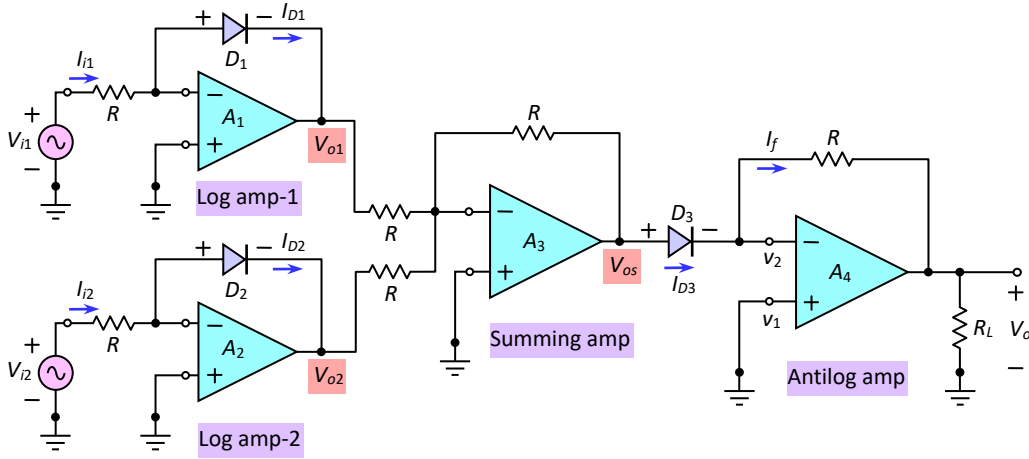


Fig.14-86: Analog multiplier circuit

The outputs of the log amplifiers are given by Equ.(14-135). We will assume all diodes are identical and all resistors are of same value. Therefore,

$$V_{o1} = -nV_T \ln \left(\frac{V_{i1}}{RI_s} \right) \quad (14-151)$$

$$V_{o2} = -nV_T \ln \left(\frac{V_{i2}}{RI_s} \right) \quad (14-152)$$

The output of the summing amplifier can be found using Equ.(14-109),

$$V_{os} = -\frac{R}{R}(V_{o1} + V_{o2}) = -(V_{o1} + V_{o2})$$

Putting the value of V_{o1} , and V_{o2} from Equ.(14-152) and Equ.(14-153) into the above equation,

$$V_{os} = - \left[-nV_T \ln \left(\frac{V_{i1}}{RI_s} \right) - nV_T \ln \left(\frac{V_{i2}}{RI_s} \right) \right]$$

or,

$$V_{os} = nV_T \ln \left(\frac{V_{i1}}{RI_s} \right) + nV_T \ln \left(\frac{V_{i2}}{RI_s} \right)$$

or,

$$V_{os} = nV_T \left[\ln \left(\frac{V_{i1}}{RI_s} \right) + \ln \left(\frac{V_{i2}}{RI_s} \right) \right] \quad (14-153)$$

$$\therefore V_{os} = nV_T \ln \left(\frac{V_{i1} \times V_{i2}}{R^2 I_s^2} \right) \quad (14-154)$$

Now this V_{os} will be the input for the antilog amplifier. Using Equ.(14-150) the output of the antilog amplifier can be calculated.

$$V_o = -RI_s e^{\frac{1}{nV_T} \times nV_T \ln \left(\frac{V_{i1} \times V_{i2}}{R^2 I_s^2} \right)}$$

$$\text{or,} \quad V_o = -RI_s e^{\ln \left(\frac{V_{i1} \times V_{i2}}{R^2 I_s^2} \right)}$$

$$\text{or,} \quad V_o = -RI_s \left(\frac{V_{i1} \times V_{i2}}{R^2 I_s^2} \right)$$

$$\therefore \quad \boxed{V_o = -\frac{V_{i1} \times V_{i2}}{RI_s}} \quad (14-155)$$

That is, the output is proportional to the product of the two input signals V_{i1} , and V_{i2} .

14.36 Analog Divider Circuit

An analog divider circuit can easily be constructed using a difference amplifier instead of the summing amplifier of Fig.14-86. A two-input divider is shown in Fig.14-87. It consists of two log amplifiers, a two-input difference amplifier and an antilog amplifier.

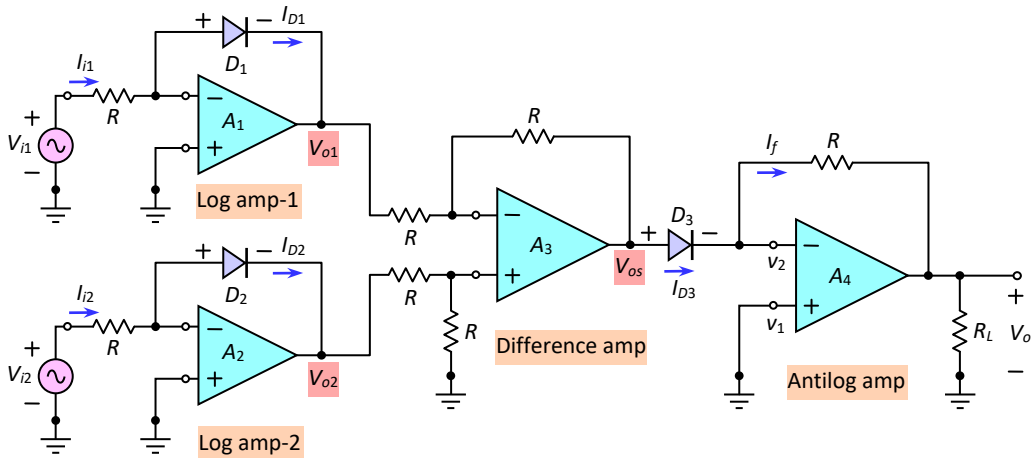


Fig.14-87: Analog divider circuit

The analysis of this circuit will be exactly similar to that of multiplier circuit. Only the difference will be in Equ.(14-154). As this circuit uses a difference amplifier, the value of V_{os} will be,

$$V_{os} = nV_T \left[\ln \left(\frac{V_{i1}}{RI_s} \right) - \ln \left(\frac{V_{i2}}{RI_s} \right) \right]$$

$$\text{or,} \quad V_{os} = nV_T \ln \left(\frac{V_{i1}}{V_{i2}} \right)$$

Using Equ.(14-150) the output of the antilog amplifier can be calculated as,

$$V_o = -RI_s e^{\frac{1}{nV_T} \times nV_T \ln\left(\frac{V_{i1}}{V_{i2}}\right)}$$

or,

$$V_o = -RI_s e^{\ln\left(\frac{V_{i1}}{V_{i2}}\right)}$$

∴

$$\boxed{V_o = -RI_s \left(\frac{V_{i1}}{V_{i2}}\right)} \quad (14-156)$$

Thus, we find that the output of this circuit is proportional to the division of two input signals.

14.37 Precision Rectifier

A rectifier is a circuit that converts AC voltage (and current) into DC voltage (and current). In Chapter 2, we have studied half-wave and full-wave rectifiers using PN junction diodes. Normal rectifier diodes have a threshold voltage V_T . For silicon diode the value of $V_T = 0.7$ V and for germanium diode this value is $V_T = 0.3$ V. Therefore, normal rectifiers drop V_T amount of voltage from the input signal. Moreover, the normal rectifiers cannot rectify very small signals (less than 0.7 V). In electronics, we need to rectify small signals with values less than 0.7 V, e.g., in demodulation circuit. Therefore, we need precision rectifiers. Precision rectifiers can be constructed using op-amp and normal diodes.

14.37.1 Precision Half-Wave Rectifier

A **half-wave precision rectifier** is shown in Fig.14-88. This circuit is also called a *super diode*. It consists of an op-amp and a normal silicon diode (D).

This precision rectifier will give the positive half-cycles at the output and the negative half-cycles will be clipped off. For the negative half-cycles, the output of the op-amp will be negative, and hence, the diode will be open as shown in Fig.14-89(b). So no output is found for negative half-cycles. For the positive half-cycles the output of op-amp will be positive, so, the diode will work as a short-circuit as shown in Fig.14-89(a). With the diode short, the op-amp is nothing but a voltage follower. Therefore, the full positive half-cycle of input will be found at the output.

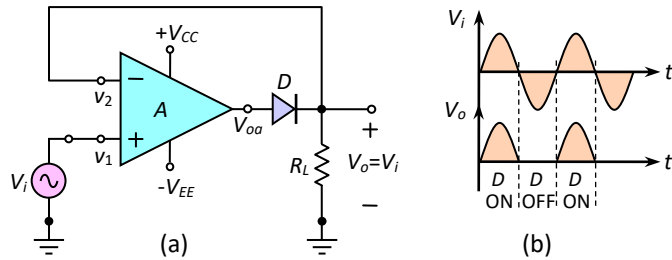


Fig.14-88: (a) Precision half-wave rectifier, and (b) Output waveforms of this circuit

Now, we will explain how this rectifier overcomes the 0.7 V drop of the input signal. Let us consider a very small positive input voltage for which the diode is open. With the diode open, no output will be found and op-amp circuit will work as an open-loop noninverting amplifier. So long as the value of op-amp's output $V_{oa} < V_T$ the diode will remain open and no output will be developed. But as soon as $V_{oa} = V_T$ the diode will be short, the op-amp will work again as a voltage follower, and

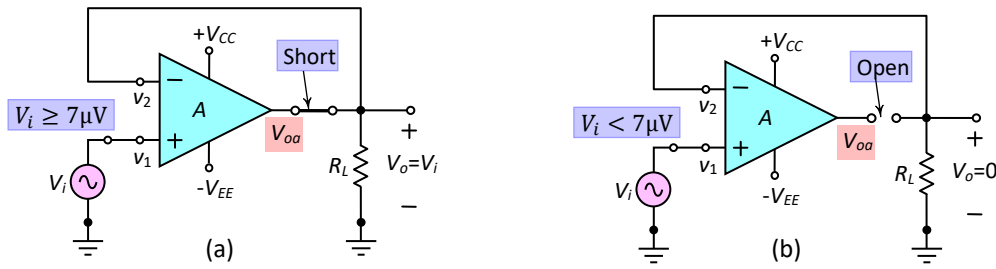


Fig.14-89: Precision half-wave rectifier: (a) $V_i \geq 0.7 \mu\text{V}$, and (b) $V_i < 0.7 \mu\text{V}$

output will be found. When the op-amp works as noninverting amplifier the input voltage will be amplified by the open-loop gain, A . Therefore, the value of input voltage required to produce $V_{oa} = V_T$ will be,

$$V_i = \frac{V_T}{A}$$

A typical value of $A = 10^5$. So,

$$V_i = \frac{V_T}{A} = \frac{0.7 \text{ V}}{10^5} = 7 \mu\text{V} \quad (14-157)$$

Thus, if the value of input voltage is less than $7 \mu\text{V}$, output will not be found. But, for positive input voltage $V_i \geq 0.7 \mu\text{V}$, the output will be exactly same as input. Therefore, we can say that using op-amp, the threshold voltage of a normal diode has been reduced to only $0.7 \mu\text{V}$.

One limitation of this half-wave precision rectifier is: it suffers from speed. As for the negative half-cycle, the circuit is converted into open-loop noninverting amplifier, the op-amp gets saturated to negative voltage. Suppose that the op-amp is in negative saturation and that a

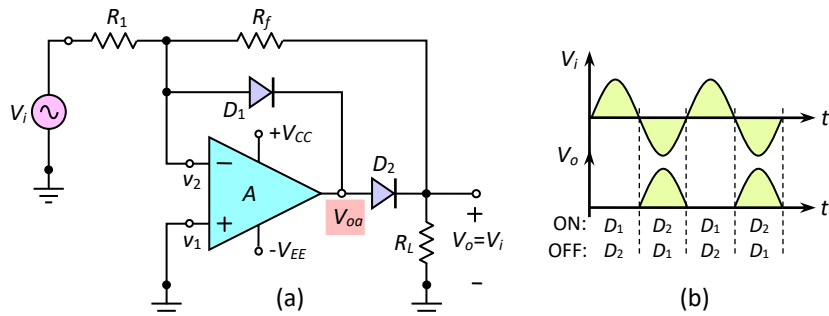


Fig.14-90: (a) Precision half-wave rectifier, and (b) Output waveforms of this circuit

quick positive input pulse occurs. In order to track this, the op-amp must climb out of negative saturation first. Using a 741 op-amp with $\pm 15 \text{ V}$ supplies, it will take about $26 \mu\text{s}$ to go from negative saturation (-13 V) to zero. If the aforementioned pulse is only $20 \mu\text{s}$ wide, the circuit doesn't have enough time to produce that pulse. The input pulse will have gone negative again, before the op-amp has a chance to climb out of its saturation. If the positive pulse were a bit longer, say $50 \mu\text{s}$, the op-amp would be able to track a portion of it. This problem can be overcome by avoiding the saturation of the op-amp during negative half-cycles. Such a half-wave precision rectifier is shown in Fig.14-90.

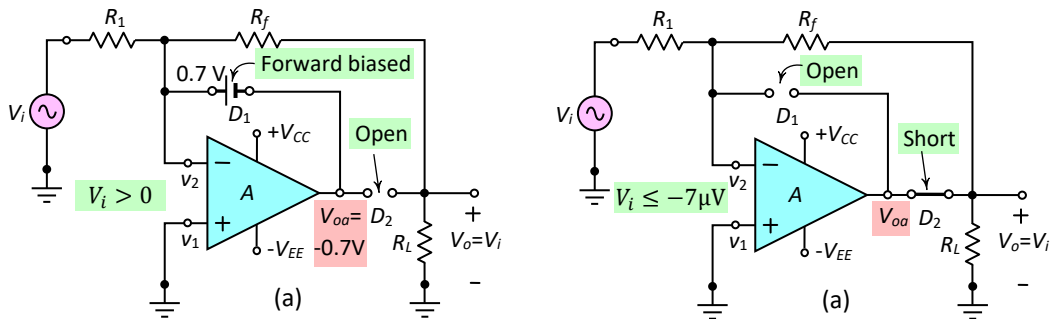


Fig.14-91: Precision half-wave rectifier: (a) $V_i \geq 0.7 \mu\text{V}$, and (b) $V_i < 0.7 \mu\text{V}$

Unlike the previous circuit, here the input signal, V_i , is applied to the inverting terminal through resistor R_1 . For the positive half-cycle, the output of the op-amp (V_{oa}) will be negative, because the op-amp works as an inverting amplifier. Due to this negative voltage, D_2 will be open and D_1 will be forward biased with a voltage drop of 0.7 as shown in Fig.14-91(a). As D_2 is open, no output will be found. All of the input voltage (V_i) will be dropped across R_1 resistor. Moreover, the output voltage of the op-amp (V_{oa}) will be only -0.7 V which is the voltage dropped across D_1 . This small output voltage restricts the op-amp from getting saturated.

For the negative half-cycles, V_{oa} will be positive, so, diode D_2 will be short but diode D_1 will be open. Therefore, the circuit will work as an inverting amplifier with gain R_f/R_1 (generally set to unity). With D_2 short and $R_f/R_1 = 1$, the negative half-cycles will be inverted and appear at the output. Here also the maximum voltage drop of the input signal will be only $-7 \mu\text{V}$. As the op-amp does not get saturated, its speed will be higher than the precision rectifier of Fig.14-88.

14.37.2 Precision Full-Wave Rectifier

In a precision full-wave rectifier, for both the half-cycles output is produced & in one direction only (either positive or negative). The diagram of Fig.14-92 shows a precision full-wave rectifier for

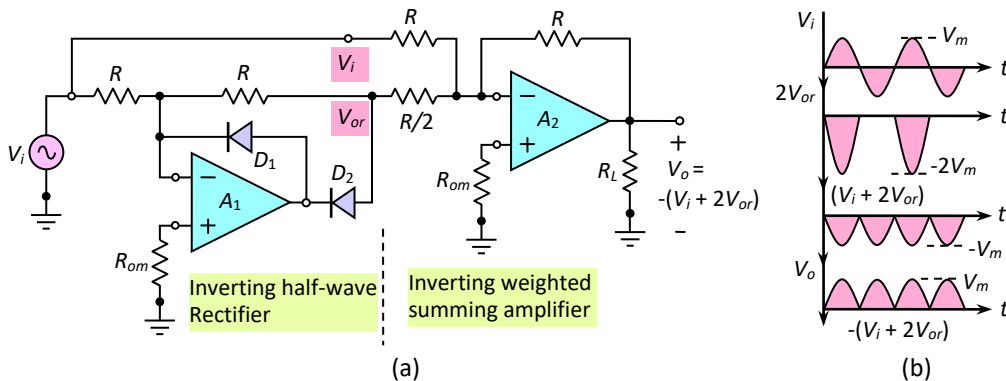


Fig.14-92: (a) Precision full-wave rectifier, and (b) Output waveforms of this circuit

positive output. It is also called an *absolute value circuit*, because the output is always positive. So we get absolute value of input signal.

This precision rectifier is convenient in that it only requires two op-amps and all resistors are of the same value (except one $R/2$). The offset minimizing resistors, R_{om} , are optional. This rectifier is comprised of two circuits: an inverting half-wave rectifier and an inverting weighted summing amplifier.

Look at the diodes of the rectifier circuit. They are connected in opposite polarity than the diodes in Fig.14-90. So, this rectifier will pass the positive half-cycles of the input, but in inverted form (that is negative half-cycles). The operation process is same as that of Fig.14-90. Now, look at the inverting weighted summing amplifier. It will sum up the input signal, V_i and the output of the rectifier, V_{or} . For V_i input, the scaling factor is $R/R = 1$, but for V_{or} the scaling factor is $R/(R/2) = 2$. Being an inverting summing amplifier, its output will be, $V_o = -(V_i + 2V_{or})$. As shown in Fig.14-92(b), $2V_{or}$ has only negative half-cycles with peak value of $-2V_m$. But V_i has both positive and negative half-cycles with peak value $\pm V_m$. When they are sum up and inverted by the summing amplifier, all positive half-cycles will be found with peak value V_m .

14.37.3 Another Fullwave Precision Rectifier (5 figures 3 equations)

In this circuit, both the op-amps work in inverting mode. So, during positive half cycle of the input, output of the op-amp-1, v_{o1} , will be negative. Therefore, diode D_2 will be forward biased, and diode D_1 will be reverse biased. So, the circuit of Fig.14-127 can be redrawn as in Fig.14-128. As discussed earlier, the diode will be forward biased for a very small input voltage ($\approx 0.7 \mu V$). Both the op-amp circuits will work in linear region. So the inverting input of op-amp-1 will be virtually ground. Considering this virtual ground, we can further simplify the circuit as in Fig.14-129.

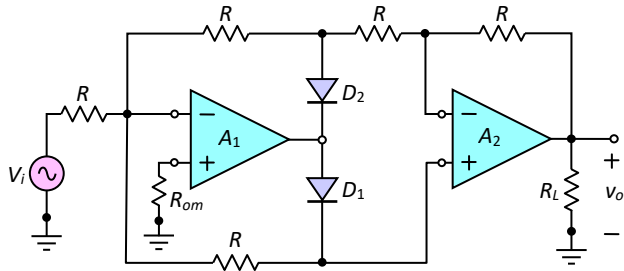


Fig.14-93: Full-wave precision rectifier

From this circuit, it is clear that both the op-amps work as inverting amplifier. The voltage gain of these circuits will be,

$$A_{v1} = A_{v2} = -\frac{R}{R} = -1$$

Therefore, the output of the first circuit will be,

$$v_{o1} = v_i \times A_{v1} = -v_i$$

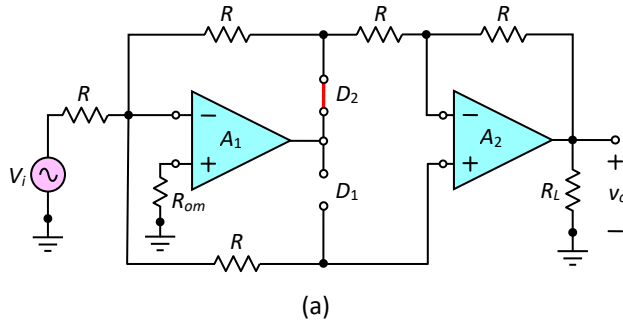


Fig.14-94: State of the precision full-wave rectifier for +ve half-cycle

This output will work as the input of second amplifier. So, the output of the second amplifier circuit will be,

$$v_{o2} = v_o = (-v_i) \times A_{v1} = v_i$$

From this expression, we find that for a positive half-cycle in the input terminal, a similar positive half-cycle will be developed at the output ($v_o = v_{o2}$).

During negative half-cycle of the input, diode D_2 will be reverse biased, and diode D_1 will be

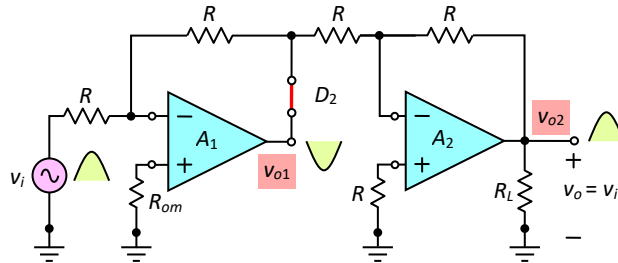


Fig.14-95: The circuit of Fig.14-128 is redrawn here with of effect of virtual ground

forward biased. So, the circuit of Fig. 13-127 can be redrawn as in Fig. 13-130. Now applying KCL at the inverting terminal of the op-amp A_1 , we get,

$$i_i + i_1 + i_f = i_b$$

But the input current into the inverting terminal (i_b) of the op-amp will be negligibly small. So,

$$\frac{v_i - v_2}{R} + \frac{v_{o1} - v_2}{R} + \frac{v_{o1} - v_2}{2R} = 0 \quad (14-158)$$

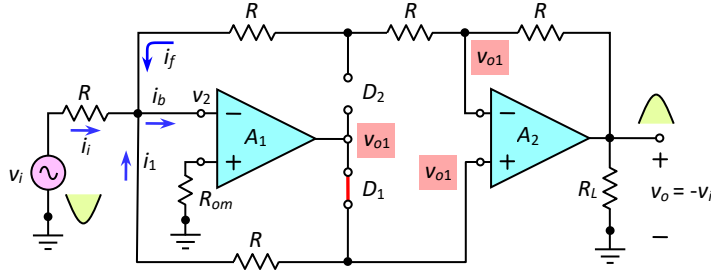


Fig.14-96: State of the precision full-wave rectifier for negative half-cycle

As the inverting terminal is virtual ground, we can consider, $v_2 = 0$. Thus, Equ.13-228 will be,

$$\frac{v_i}{R} + \frac{v_{o1}}{R} + \frac{v_{o1}}{2R} = 0$$

or,

$$\frac{v_{o1}}{R} + \frac{v_{o1}}{2R} = -\frac{v_i}{R}$$

or,

$$2v_{o1} + v_{o1} = -2v_i \quad \text{Multiplying both sides by } 2R$$

or,

$$3v_{o1} = -2v_i$$

or,

$$v_{o1} = -\frac{2v_i}{3} \quad (14-159)$$

Considering the virtually ground of inverting terminal of the op-amp-1, we can redraw the second op-amp circuit as in Fig.14-131. Now, it is clear that this is a noninverting amplifier. So the output of this amplifier will be,

$$v_{o2} = \left(1 + \frac{R}{2R}\right) v_{o1} \quad (14-160)$$

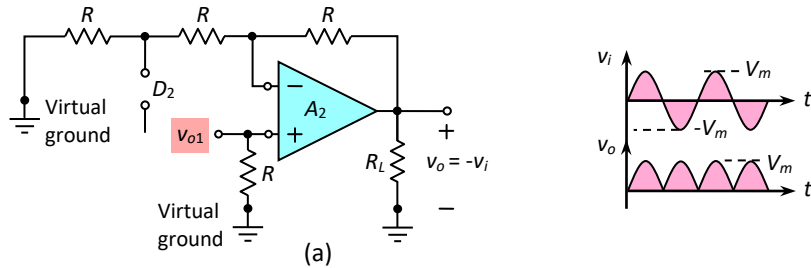


Fig.14-97: The circuit of Fig.14-130 is redrawn here with the effect of virtual ground

Putting the value of v_{o1} into Equ.(14-260), we get

$$v_{o2} = \left(1 + \frac{R}{2R}\right) \left(-\frac{2v_i}{3}\right) = \left(\frac{3R}{2R}\right) \left(-\frac{2v_i}{3}\right)$$

∴

$$v_{o2} = -v_i$$

From this expression, we find that for a negative half-cycle at the input terminal, a negative half-cycle of same magnitude will appear at the output ($v_o = v_{o2}$).

14.38 Capacitor-Coupled AC Amplifiers

The inverting and noninverting amplifiers discussed in Section 13.20 and 13.21, can amplify both DC signals and AC signals as there is no coupling capacitor. In electronic systems, we sometimes need pure AC amplifier that will amplify only the AC signal and block the DC signals (levels). For example, in an audio receiver system that consists of a number of stages, because of thermal drift, component tolerances, and variations, a DC level is produced. To prevent the amplification of this DC level, coupling capacitors must be used between the stages. Fig.14-98 shows inverting and

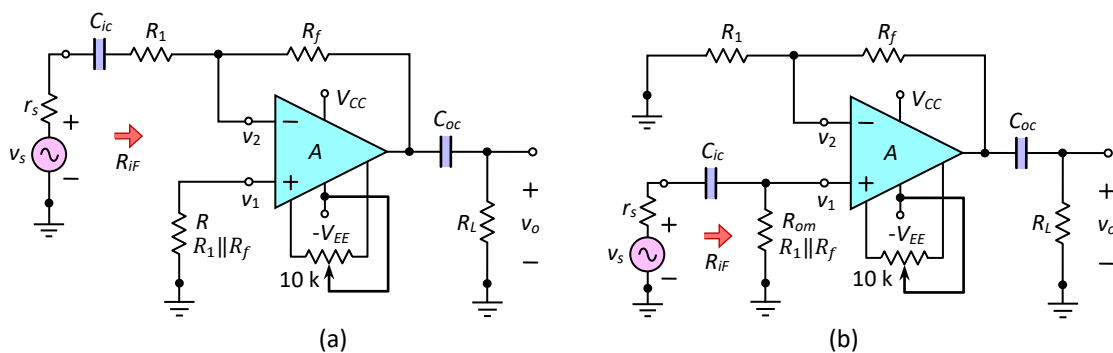


Fig.14-98: Capacitor-coupled AC amplifiers: (a) Inverting, and (b) Noninverting

noninverting amplifiers with input and output coupling capacitors. The coupling capacitors not only block the DC voltage but also set the lower cut-off frequency of the amplifiers as,

$$C_{ic} = \frac{1}{2\pi f_L(r_s + R_{iF})} \quad (14-161)$$

$$C_{oc} = \frac{1}{2\pi f_L(Z_o + R_L)} \quad (14-162)$$

For a particular lower cut-off frequency (f_L), we have to use either of the above two expressions which produces the larger value of capacitance. The other capacitor should be 10 times the value found by the equation. For example, if $C_{ic} > C_{oc}$, then C_{ic} should be calculated using Equ.(14-161) and C_{oc} should be equal to $10/2\pi f_L(Z_o + R_L)$. But, if $C_{oc} > C_{ic}$, then C_{oc} should be calculated using Equ.(14-162) and C_{ic} should be equal to $10/2\pi f_L(r_s + R_{iF})$.

Depending on the closed-loop gain (A_f) and the gain-bandwidth product of the op-amp, a higher cut-off frequency (f_H) will also be set. The calculation process is same as described before. So, the bandwidth of the AC amplifier will be,

$$BW = f_H - f_L$$

In the mid-band frequency range, all the capacitors will be almost short. So all the expressions derived previously for these circuits will work properly. The gain of the inverting amplifier will be,

$$A_f \approx -\frac{R_f}{R_1}$$

and the gain of the noninverting amplifier will be,

$$A_f \approx 1 + \frac{R_f}{R_1}$$

14.39 AC Amplifiers with Single-Supply

Most of the op-amp circuits, discussed so far, have used split-supply voltage ($+V_{CC}$, and $-V_{EE}$). But, it is often desirable to avoid split-supplies, especially with small battery powered applications. One option in such a case is to use a single-supply op amp. However, these devices will clip the output if the input attempts to go negative, making them unsuitable for AC applications. Clipping can be avoided by applying a DC voltage level to the noninverting terminal as shown in Fig.14-99 and Fig.14-100. Generally the value of the inserted DC level is $+V_{CC}/2$, so that the positive swings are equal to the negative swing. As shown in Fig.14-99 and Fig.14-100, the DC level will be exactly $V_{CC}/2$,

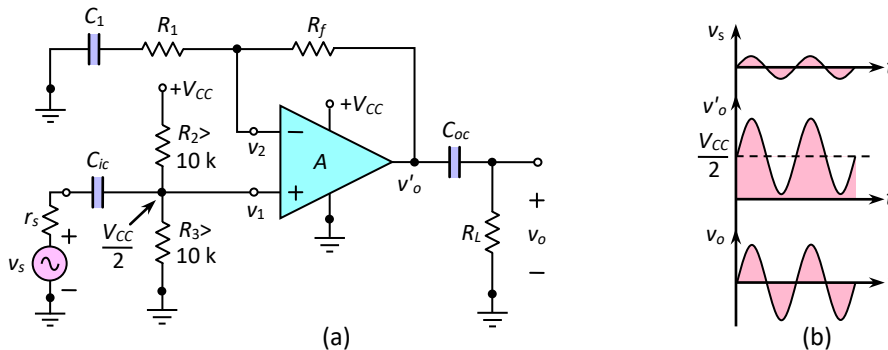


Fig.14-99: AC amplifier with single-supply: (a) Noninverting amplifier, and (b) Output waveforms

if the resistances of the voltage divider network are equal, that is, $R_2 = R_3$. With the input DC voltage in place, when an input signal goes negative, the voltage applied to the input of the op-amp will dip below the DC voltage, but will not go below ground (provided the bias voltage is large enough, and provided the input signal is not too large). To block the DC voltage from the source and the load, input and output coupling capacitors are used. In both the circuits, for DC input ($f = 0\text{ Hz}$), the input capacitors (C_i and C_1) will work as open circuit. So, the amplifiers will work as voltage followers. Therefore, the output will be exactly the same as the DC voltage level added in the noninverting terminal, that is, $v_o = V_{CC}/2$. For the AC input signal, the output will fluctuate about this DC level.

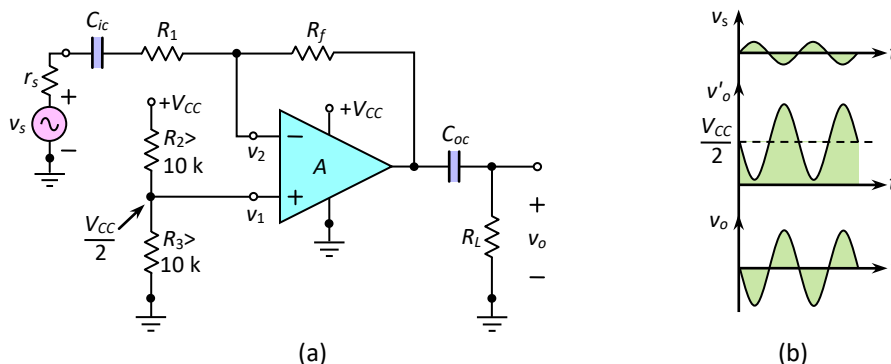


Fig.14-100: AC amplifier with single-supply: (a) Inverting amplifier, and (b) Output waveforms

The reactance of the capacitors used in the circuits should be very small for the range of operating frequencies. The lower cut-off frequency of these amplifiers will be set by these capacitors. The values of the capacitors can be calculated using the same method as described in Section 13-38.

14.40 Effect of Slew-Rate on AC Signals

The **unity-gain bandwidth** is the small-signal high-frequency limitation on the use of op-amp. *Slew-rate*, on the other hand, is related with large-signal application. Already we have defined the slew rate of op-amp. This is the maximum rate of change of output voltage and is measured in $\mu\text{V/s}$. This limitation on the maximum rate of change of output voltage is imposed by the internal compensating capacitor C_c that is connected in the feedback path of the second-stage amplifier of op-amp [see Fig.14-28]. The rate of change of output depends on the rate of change of voltage across this capacitor as,

$$\frac{dv_c}{dt} = \frac{I}{C_c}$$

where, I will be the bias current of the transistor across which the compensating capacitor is connected. The typical value of slew-rate of 741 op-amp is only $0.5\text{ V}/\mu\text{s}$.

Although, slew-rate affects the large signal input, it also limits the maximum frequency of AC signal that can be properly produced at the output of an op-amp. To find the slew-rate limited maximum frequency, let us consider an amplifier [Fig.14-101] with closed-loop gain of $A_f = 1$, as the slew-rate is measured for unity gain. Let us apply a high frequency sinusoidal input signal with large peak-to-peak value to this amplifier which is given by,

$$v_i = V_p \sin(\omega t)$$

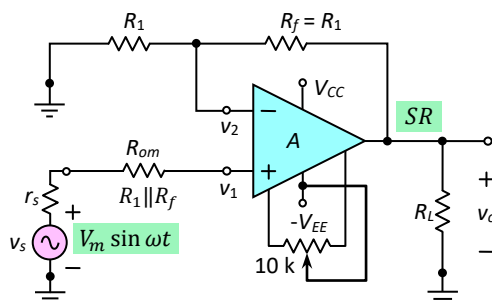


Fig.14-101: Unity gain noninverting amplifier

Here, the value of V_p is so selected that the amplifier will never be saturated. The output will be equal to this input voltage, that is,

$$v_o = V_p \sin(\omega t)$$

Differentiating this value, we will get the rate of change of output voltage as,

$$\frac{dv_o}{dt} = \omega V_p \cos(\omega t)$$

The maximum value of $\cos(\omega t)$ is 1. Therefore, the maximum rate of change of output will be,

$$\left. \frac{dv_o}{dt} \right|_{max} = \omega_m A_v V_p \quad (14-163)$$

Now, the operational amplifier will properly produce this signal at the output if the slew-rate of the op-amp (SR) is greater than or equal to this maximum rate of change. Therefore, we can write,

$$SR = \omega_m V_p \text{ V/s}$$

or,

$$SR = \frac{2\pi f_m V_p}{10^6} \text{ V}/\mu\text{s}$$

$\therefore$

$$\boxed{f_m = \frac{10^6 SR}{2\pi V_p}} \quad (14-164)$$

where,

SR = slew rate of the op-amp in $\text{V}/\mu\text{s}$

f_m = maximum frequency of the input signal

V_p = peak value of the signal

Using this equation, we can calculate the slew-rate limited maximum frequency, peak value of the output if all other parameters are known. If the gain of the amplifier is more than unity we can use the above equation putting $V_p = A_f V_{pi}$, where, A_f is the gain of the amplifier and V_{pi} is the peak value of the input sinusoidal signal, that is,

$$\boxed{f_m = \frac{10^6 SR}{2\pi A_f V_{pi}}} \quad (14-165)$$

But there is no grantee that the above equation will work properly, because the slew-rate of op-amp, measured in unity gain, may change for higher gain.

Example 13-19

A closed-loop noninverting amplifier is designed with $A_f = 100$ using 741 operational amplifier that provides slew-rate of $0.5 \text{ V}/\mu\text{s}$ at this gain. The input signal is $v_i = (10 \text{ mV}) \sin \omega t$. Assuming that the amplifier does never saturate, determine the maximum frequency of the input signal and the output peak-to-peak value. Draw the output waveform if the signal frequency is 100 kHz considering slew-rate of ∞ and $0.5 \text{ V}/\mu\text{s}$.

Solution:

Using Equ.(14-165),

$$f_m = \frac{10^6 SR_{op}}{2\pi A_f V_{pi}} = \frac{10^6 (0.5)}{2\pi (100) (10 \text{ mV})} \approx 79.6 \text{ kHz [Ans]}$$

The output peak-to-peak value will be,

$$V_{o(pp)} = 2A_f V_{pi} = 2(100)(10 \text{ mV}) = 2 \text{ V [Ans.]}$$

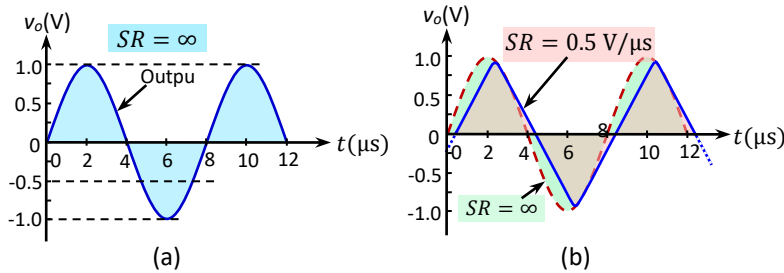


Fig.14-102: Output waveforms of Example 13-19: (a) For infinite slew rate, and (b) For slew rate = 0.5 V/ μ s

For 125 kHz signal the time period will be $1/125 \text{ kHz} = 8 \mu\text{s}$. If the op-amp has ∞ slew rate the 2 V peak-to-peak sinusoidal output will be produced properly. The waveform is shown in Fig.14-102(a). However, if the op-amp has 0.5 V/ μ s slew-rate, the output will not follow the shape of sinusoidal waveform rather increase and decrease linearly with a rate of 0.5 V/ μ s. The output waveform will be like a triangular wave as shown in Fig.14-102(b).

Comments: Due to higher frequency than supported by slew-rate, the output has been turned into a triangular wave.

14.41 Comparator Circuits Using Op-Amps

Already we have discussed about comparator circuit in Section 13.18.4. A voltage comparator is a circuit that compares an input voltage with a *reference voltage*. The output of a comparator may be either positive saturation voltage ($+V_{max}$) or negative saturation voltage ($-V_{max}$) depending on the input voltage. A comparator has two trip levels (or points). For input voltage greater than a particular voltage level called **higher trip level** (HTL) or *upper trip level* (UTL), the comparator's output goes to one saturation level (either positive or negative), and for input voltage lower than a particular voltage level called **lower trip level** (LTL), the comparator's output goes to the opposite saturation level. If the output goes to positive saturation for higher input voltage and negative saturation for lower input voltage, the comparator is called **noninverting comparator**, whereas, if the output goes to negative saturation for higher input voltage and positive saturation for lower input voltage, the comparator is called **inverting comparator**.

The comparator of the Section 13.18.4 is redrawn in Fig.14-103.

The op-amp will work as a linear amplifier if $v_{id} = (v_i - V_{ref}) \leq V_{th}$ [here, V_{th} is a threshold voltage and its value will be V_{max}/A]. However, if $v_{id} > V_{th}$ the output of the op-amp will be saturated.

Therefore, for the comparator of Fig.14-103, if $v_i > V_{ref} + V_{th}$ the op-amp will be saturated and will produce the maximum positive voltage ($+V_{max}$).

Similarly, if $v_i < V_{ref} - V_{th}$ the op-amp will produce the maximum negative voltage ($-V_{max}$). If the value of open-loop gain of the op-amp is $A = 10^5$ and the saturation voltage is ± 15 V, the value of the threshold voltage will be $15 \text{ V} / 10^5 = 0.15 \text{ mV}$. Generally, the reference voltage and the input voltage are very large compared to V_{th} . So, the threshold voltage can be neglected. Then we can say, if $v_i > V_{ref}$, then, $v_o = V_{max}$ and if $v_i < V_{ref}$, then, $v_o = -V_{max}$. The output of the comparator, for a positive reference voltage, and a sinusoidal input voltage (with peak value $> V_{ref}$) is shown in Fig.14-103(b). Here, the value of HTL = $(V_{ref} + V_{th})$, and the value of lower trip level is LTL = $(V_{ref} - V_{th})$. The difference between HTL, and LTL is only 0.3 mV. If we neglect V_{th} , then HTL $\approx$ LTL.

The comparator with HTL very close to LTL, may suffer from mistriggering by noise, or by some other reasons (e.g. loading effect). To understand this let us consider a lead-acid battery charging system as shown in Fig.14-105. It is desired that the comparator will stop the charging as soon as the battery voltage rises to 14.5 V.

Assume that the battery has an internal resistance of 0.1Ω , and the charging current is 10 A. The battery charging is ON/OFF by turning the switch (S) ON/OFF by an inverting comparator with HTL = LTL = 14.5 V. The voltage drop across the internal resistance of the battery will be $I_{ch} \times r_s = 10 \text{ A} \times 0.1 \Omega = 1 \text{ V}$. When the battery voltage $V_B = 13.5 \text{ V}$, the comparator will sense it as $13.5 \text{ V} + 1 \text{ V} = 14.5 \text{ V}$. So the comparator's output will go LOW and the switch will be OFF.

Thus, the charging of the battery will stop. But, as soon as charging stops, the voltage drop across r_s will be zero. So, the terminal voltage of the battery will fall to 13.5 V which is less than LTL. Therefore, the comparator will be HIGH, and charging will start again. Thus, the switch will be continuously ON/OFF which may damage the switch and overcharge the battery.

This problem could have been overcome, if the LTL of the comparator were set to less than 13.5 V. In that case, once the charging is OFF, the comparator senses the battery voltage 13.5 V, but as $LTL < 13.5 \text{ V}$, the comparator would have not gone to HIGH [H_E_atten]. The comparators that has different values of LTL and HTL is called **Schmitt trigger** circuit.

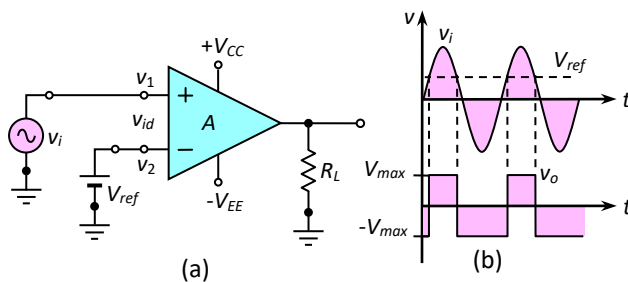


Fig.14-103: Output waveforms of Example 13-19: (a) For infinite slew rate, and (b) For slew rate = $0.5 \text{ V}/\mu\text{s}$

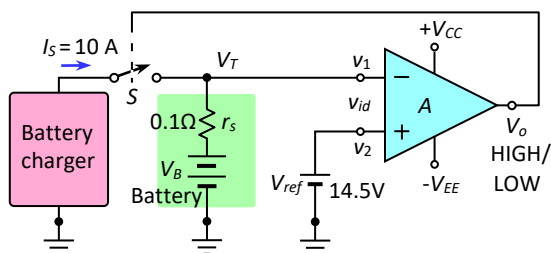


Fig.14-104: Control system of battery charging

14.42 Comparator with Hysteresis

Comparators can be designed with different values of LTL, and UTL. The difference between UTL, and LTL is called *hysteresis* as the term used in magnetic hysteresis. In its most general case, *hysteresis* is a property that means a device (or circuit) behaves differently when its input is increasing from the way it behaves when its input is decreasing. A comparator with hysteresis is called **Schmitt trigger circuit**. The Schmitt trigger was named after its inventor, Otto Schmitt. Otto Schmitt invented the Schmitt trigger circuit in 1937, which he originally named the **thermionic trigger**—but Schmitt's inventions had such a profound impact on electronics that everyone called it the **Schmitt trigger** in his honor^[1] [H_R_atten]. In some applications (like the battery charging system), hysteresis is a desired characteristic, because it prevents the comparator from switching back and forth in response to loading effect or noise in the input. Hysteresis can be introduced in a comparator using positive feedback. Like normal comparators, Schmitt trigger circuit can be inverting and noninverting types.

14.42.1 Inverting Schmitt Trigger Circuit

Fig.14-105 shows how hysteresis can be introduced into a comparator. Here, the input is connected to the inverting terminal and reference voltage is applied to the noninverting terminal. Unlike the amplifiers, here feedback is applied to the noninverting terminal using a voltage divider network consisting of R_1 and R_f which ensures positive feedback. Thus, this

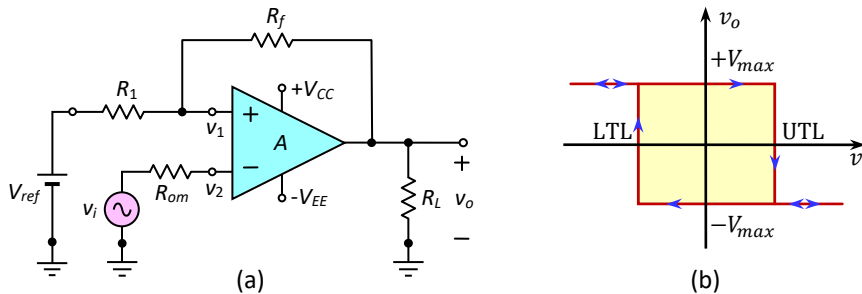


Fig.14-105: (a) Inverting Schmitt trigger circuit, and (b) Transfer characteristics of this circuit

circuit is an inverting Schmitt trigger circuit or an inverting comparator. When the input (v_i) is greater than the UTL, the output of the comparator will go LOW ($-V_{max}$). But, when the input (v_i) is less than the LTL, the output of the comparator will go HIGH ($+V_{max}$). The value of $\pm V_{max}$ will be generally 1 V less than the supply voltage $\pm V_{CC}$. The transfer characteristic of this comparator is shown in Fig.14-105(b). From this characteristic, it is clear that, the output switches to $+V_{max}$ when v_i falls below LTL, but will not switch to $-V_{max}$ unless v_i rises past the upper trigger level, UTL. For values $LTL < v_i < UTL$, the output of the circuit will not change, that is, the op-amp will retain its previous state.

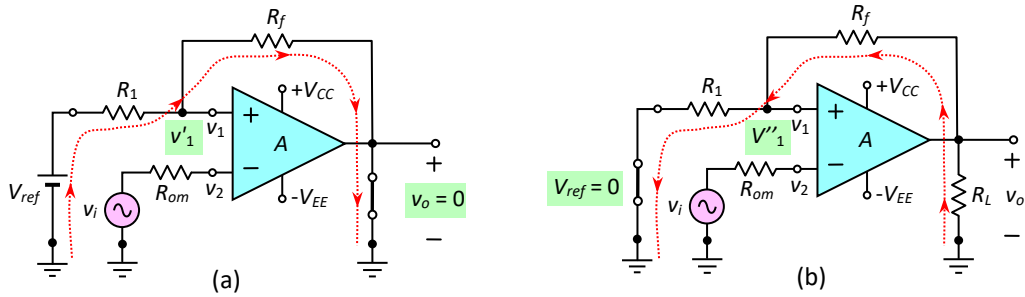


Fig.14-106: (a) Circuit showing the effect of V_{ref} only, and (b) Circuit showing the effect of v_o only

The value of LTL, and UTL can be calculated using Superposition theorem. First, let us consider that the output terminal of the op-amp is shorted to ground (V_{ref} present but $v_o = 0$) as shown in Fig.14-106(a). The contribution of V_{ref} to the noninverting terminal can be calculated using voltage divider rule,

$$v'_1 = \frac{R_f V_{ref}}{R_1 + R_f} \quad (14-166)$$

Next, let us consider that the reference voltage is shorted ($V_{ref} = 0$ but v_o present) as shown in Fig.14-106(b). Then, the contribution of v_o to the noninverting terminal will be,

$$v''_1 = \frac{R_1 v_o}{R_1 + R_f} \quad (14-167)$$

Therefore, according to the Superposition theorem, the net voltage at the noninverting terminal of the output will be,

$$v_1 = v'_1 + v''_1 = \frac{R_f V_{ref}}{R_1 + R_f} + \frac{R_1 v_o}{R_1 + R_f} \quad (14-168)$$

Let us consider that the output of the comparator is at HIGH state ($v_o = +V_{max}$). Therefore, the voltage at the noninverting terminal will be,

$$v_1 = \frac{R_f V_{ref}}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (+V_{max}) \quad (14-169)$$

So, to switch the comparator to LOW state, $v_i (= v_2)$ must be greater than or equal to v_1 , where the value of v_1 is given by the above expression [the threshold value, V_{th} of the comparator has been neglected. Due to positive feedback V_{th} will be even smaller then before]. This value of v_i will be the UTL, that is,

$$\boxed{UTL = \frac{R_f V_{ref}}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (+V_{max})} \quad (14-170)$$

Now, suppose that the comparator has switched to LOW state. So, $v_o = -v_{max}$. Putting this value into Equ.(14-170) we get,

$$v_1 = \frac{R_f V_{ref}}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (-V_{max}) \quad (14-171)$$

The comparator will stay in the LOW state, as long as $v_i > v_1$. To switch the comparator to HIGH state again, v_i should be less than or equal to v_1 which is given by the Equ.(14-172). This value of v_1 will be the LTL of the comparator, that is,

$$\text{LTL} = \frac{R_f V_{ref}}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (-V_{max}) \quad (14-172)$$

The value of hysteresis of this comparator will be,

$$\text{Hysteresis} = \text{UTL} - \text{LTL} = \frac{2R_1 V_{max}}{R_1 + R_f} \quad (14-173)$$

Therefore, by changing the value of V_{ref} , R_f , and R_1 Schmitt trigger circuit can be designed with any value of UTL, LTL and hysteresis. [See the following Example.]

Example 13-20

For the inverting comparator of Fig.14-105(a), assume $V_{ref} = 5 \text{ V}$, $R_1 = 7.5 \text{ k}$, $R_f = 10 \text{ k}$, and the maximum output voltage is $\pm V_{max} = \pm 12 \text{ V}$.

- i) Calculate the value of UTL, LTL, and hysteresis. Draw the transfer characteristic curve and output waveform for a sinusoidal input of peak value 15 V.
- ii) Repeat (i) considering $V_{ref} = 0 \text{ V}$.
- iii) Repeat (i) considering $V_{ref} = 0 \text{ V}$, and $-V_{max} = 0 \text{ V}$.

Solution:

i)

Using Equ.(14-171),
$$\text{UTL} = \frac{10 \text{ k} \times 5 \text{ V}}{7.5 \text{ k} + 10 \text{ k}} + \frac{7.5 \text{ k} \times 12 \text{ V}}{7.5 \text{ k} + 10 \text{ k}} \approx 8 \text{ V [Ans.]}$$

Using Equ.(14-173),
$$\text{LTL} = \frac{10 \text{ k} \times 5 \text{ V}}{7.5 \text{ k} + 10 \text{ k}} + \frac{7.5 \text{ k} \times (-12 \text{ V})}{7.5 \text{ k} + 10 \text{ k}} \approx -2.3 \text{ V [Ans.]}$$

Using Equ.(14-174),
$$\text{Hysteresis} = \frac{2R_1 V_{max}}{R_1 + R_f} = \frac{2 \times 7.5 \text{ k} \times 12 \text{ V}}{7.5 \text{ k} + 10 \text{ k}} \approx 10.3 \text{ V [Ans.]}$$

The transfer characteristic and the output waveforms are shown in Fig.14-107(a), (b).

ii)

Using Equ.(14-170),
$$\text{UTL} = \frac{7.5 \text{ k} \times 12 \text{ V}}{7.5 \text{ k} + 10 \text{ k}} \approx 5.14 \text{ V}$$

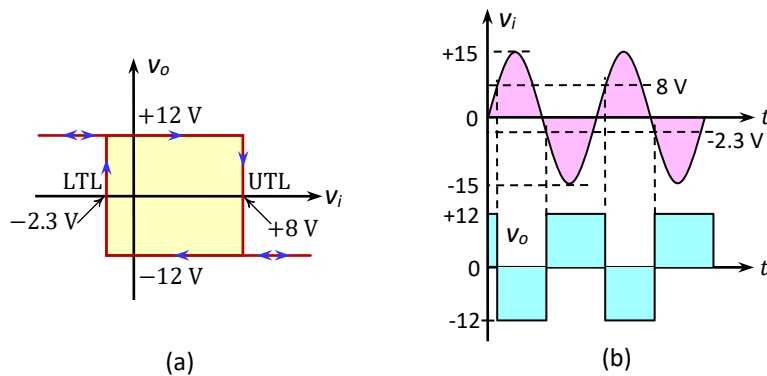


Fig.14-107: (a) Transfer characteristics of Example 13-20(i), and
(b) Input and output waveforms of Example 13-20(i)

Using Equ.(14-172),
$$\text{LTL} = \frac{7.5\text{ k} \times (-12\text{ V})}{7.5\text{ k} + 10\text{ k}} \approx -5.14\text{ V}$$

Using Equ.(14-173),
$$\text{Hysteresis} = \frac{2R_1 V_{\max}}{R_1 + R_f} = \frac{2 \times 7.5\text{ k} \times 12\text{ V}}{7.5\text{ k} + 10\text{ k}} \approx 10.3\text{ V}$$

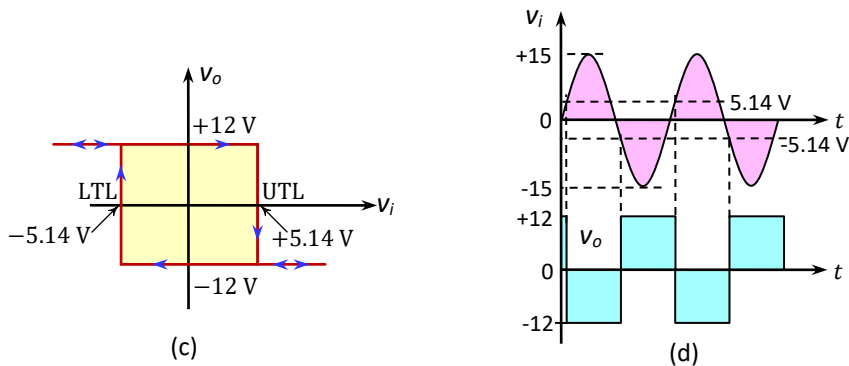


Fig.14-107: (c) Transfer characteristics of Example 13-20(ii), and
(d) Input and output waveforms of Example 13-20(ii)

The transfer characteristic and the output waveforms are shown in Fig.14-107(c), (d).
iii)

Using Equ.(14-171),
$$\text{UTL} = \frac{7.5\text{ k} \times 12\text{ V}}{7.5\text{ k} + 10\text{ k}} \approx 5.14\text{ V}$$

Using Equ.(14-173),
$$\text{LTL} = \frac{7.5\text{ k} \times (0\text{ V})}{7.5\text{ k} + 10\text{ k}} = 0\text{ V}$$

Using Equ.(14-174),
$$\text{Hysteresis} = \text{UTL} - \text{LTL} = 5.14\text{ V}$$

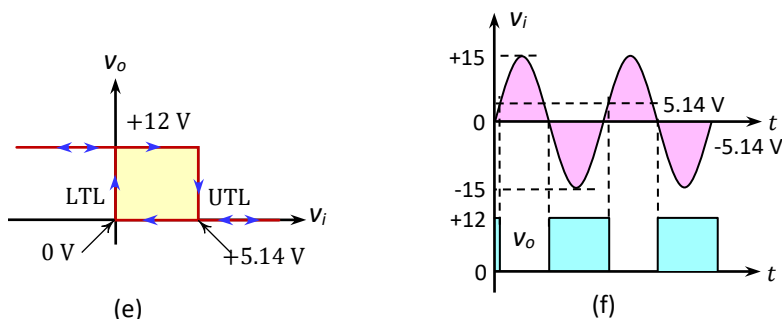


Fig.14-102: (e) Transfer characteristics of Example 13-20(iii), and
(f) Input and output waveforms of Example 13-20(iii)

The transfer characteristic and the output waveforms are shown in Fig.14-107(e), (f).

Comments: The amount of hysteresis does not depend on reference voltage but the trip levels depend on it.

14.43 Noninverting Schmitt Trigger Circuit

Fig.14-108(a) shows a noninverting Schmitt trigger circuit. Here, the input signal is connected to the noninverting terminal and the reference voltage is applied to the inverting terminal. The positive feedback is applied to the noninverting terminal using a voltage divider network, R_1 and R_f as before. Thus, this circuit is a noninverting Schmitt trigger circuit or a noninverting comparator. When the input (v_i) is greater than UTL, the comparator will go HIGH ($+V_{max}$). When the input (v_i) is less than the LTL, the output will go LOW ($-V_{max}$). The transfer characteristic of this comparator is shown in Fig.14-108(b). From this characteristic, it is clear that, the output switches to $-V_{max}$ when v_i falls below LTL, and the output switches to $+V_{max}$ when v_i rises above UTL. For values $LTL < v_i < UTL$, the output will not change, that is, the op-amp will retain its previous state.

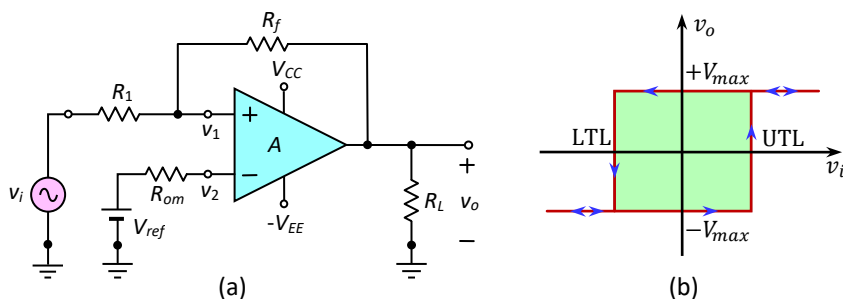


Fig.14-108: (a) Non-inverting Schmitt trigger circuit, and (b) Transfer characteristics of this circuit

Using Superposition theorem we can find the value of v_1 in the same way as described for inverting comparator. The value of voltage at the inverting terminal can be found using Equ.(14-169) only replacing the reference voltage V_{ref} by the input voltage v_i . Therefore,

$$v_1 = \frac{R_f v_i}{R_1 + R_f} + \frac{R_1 v_o}{R_1 + R_f} \quad (14-174)$$

Now let us consider that the output of the comparator is at HIGH state ($v_o = +V_{max}$). Therefore, the voltage at the noninverting terminal will be,

$$v_1 = \frac{R_f v_i}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (+V_{max}) \quad (14-175)$$

As $v_2 = V_{ref}$, to toggle to comparator from the HIGH state to the LOW state ($v_o = -V_{max}$), the value of v_i has to be decreased to such a value that $v_1 = V_{ref}$. Putting this value of v_1 into the above equation, we get,

$$V_{ref} = \frac{R_f v_i}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (+V_{max})$$

or,

$$v_i = \frac{R_1 + R_f}{R_f} V_{ref} - \frac{R_1}{R_f} (+V_{max})$$

The value of v_i given by the above equation, is the LTL. Therefore,

$$\boxed{\text{LTL} = \frac{R_1 + R_f}{R_f} V_{ref} - \frac{R_1}{R_f} (+V_{max})} \quad (14-176)$$

Once the comparator has been toggled to LOW state, the value of v_1 will be,

$$v_1 = \frac{R_f v_i}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (-V_{max})$$

To toggle to comparator from the LOW state to the HIGH state again, the value of v_i has to be increased to such a value that $v_1 = V_{ref}$. Putting this value of v_1 into the above equation, we get,

$$V_{ref} = \frac{R_f v_i}{R_1 + R_f} + \frac{R_1}{R_1 + R_f} (-V_{max})$$

or,

$$v_i = \frac{R_1 + R_f}{R_f} V_{ref} + \frac{R_1}{R_f} (+V_{max})$$

This value of v_i will be the UTL. Therefore,

$$\boxed{\text{UTL} = \frac{R_1 + R_f}{R_f} V_{ref} + \frac{R_1}{R_f} (+V_{max})} \quad (14-177)$$

The value of hysteresis of this noninverting comparator will be,

$$\boxed{\text{Hysteresis} = \text{UTL} - \text{LTL} = \frac{2R_1 V_{max}}{R_f}} \quad (14-178)$$

Like the inverting comparator, by changing the value of V_{ref} , R_f , and R_1 Schmitt trigger circuit can be designed with any value of UTL, LTL and hysteresis. [See the following Example.]

Example 13-21

For the noninverting comparator of Fig.14-108(a), assume $V_{ref} = 2\text{ V}$, $R_1 = 7.5\text{ k}$, $R_f = 51\text{ k}$, and the maximum output voltage is $\pm V_{max} = \pm 12\text{ V}$. Calculate the value of UTL, LTL, and hysteresis. Draw the transfer characteristic curve and output waveform for a sinusoidal input of peak value 6 V .

Solution:

Using Equ.(14-177),
$$UTL = \frac{7.5\text{ k} + 51\text{ k}}{51\text{ k}}(+2\text{ V}) + \frac{7.5\text{ k}}{51\text{ k}}(+12\text{ V})$$

$\therefore UTL \approx 2.3\text{ V} + 1.8 = 4.1\text{ V [Ans.]}$

Using Equ.(14-176),
$$LTL = \frac{7.5\text{ k} + 51\text{ k}}{51\text{ k}}(+2\text{ V}) - \frac{7.5\text{ k}}{51\text{ k}}(+12\text{ V})$$

$\therefore LTL \approx 2.3\text{ V} - 1.8 = 0.5\text{ V [Ans.]}$

Using Equ.(14-178),
$$\text{Hysteresis} = UTL - LTL = 4.1\text{ V} - 0.5\text{ V} = 3.5\text{ V [Ans.]}$$

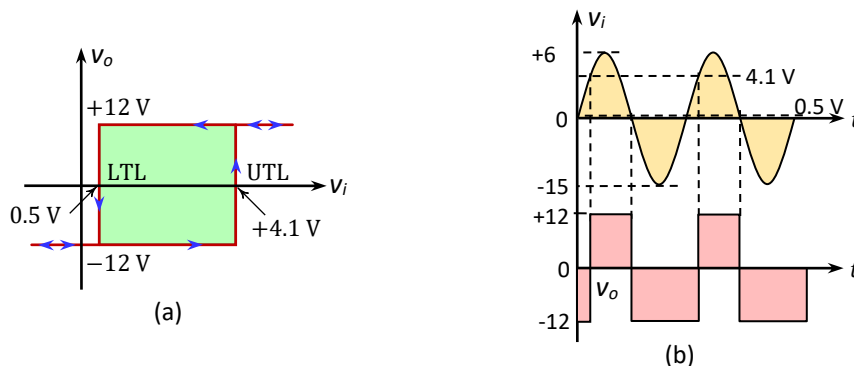


Fig.14-109: (a) Transfer characteristics of Example 13-21(iii), and (b) Input and output waveforms of Example 13-21(iii)

The transfer characteristic and the output waveforms are shown in Fig.14-109(a) and (b).

Comments: UTL, and LTL are more affected by V_{ref} , than in inverting comparator. Here the value of hysteresis is higher than that of inverting comparator for the same values of resistances.

14.44 Application of Comparators

Comparators are mainly used in controlling system. The output of the comparator is digital and analog. Nowadays, most of the controlling systems are digital (made of digital logic gates, or microcontrollers or even microprocessors). This digital circuits can easily employ the voltage comparators. Although not all some important applications of comparators are mentioned below:

- In wave-shaping circuit (convert non-square signal into square wave)
- To generate Pulse Width Modulated signal
- To build zero-crossing detector

- In charge controller (to ON/OFF the charging and discharging of the battery)
- They are used to implement multivibrators
- To construct Day/Night sensitive light controlling systems
- To implement ON/OFF switching with time delay
- To build Analog to Digital Converter (ADC) circuits
- Temperature, humidity, liquid flow controlling and so on

14.44.1 Over-Temperature Controlling Circuit

Fig.14-110 shows an op-amp comparator used in a precision over-temperature sensing and controlling system. The comparator always sample the temperature of the place of interest and when the temperature reaches a certain critical value its output goes LOW. The circuit consists of a Wheatstone bridge, an op-amp comparator, a transistor and a relay. The Wheatstone bridge has been used to increase sensitivity and the transistor has been used to ON/OFF (energize/de-energize) the relay. The temperature generating device is connected to the relay and works when the relay is ON.

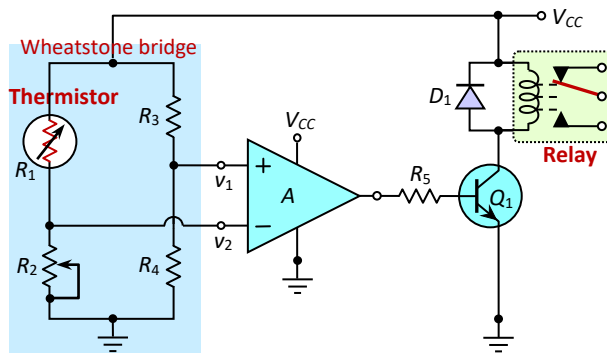


Fig.14-110: Over temperature controlling circuit

Here, the comparator always observes when the bridge goes to balanced condition. One leg of the bridge contains a thermistor (R_1), which is a temperature-sensing resistor with a negative temperature coefficient (its resistance decreases with the decrease of temperature increases). The potentiometer (R_2) is used to set the level of the critical temperature. Actually its value is set equal to the resistance of the thermistor at the critical temperature. Generally the value of R_3 is set equal to the value of R_4 . At normal temperatures (less than critical), R_1 is greater than R_2 , so v_1 will be greater than v_2 thus creating an unbalanced condition that drives the op-amp to its HIGH saturated output level and keeps transistor Q_1 ON. So the heating device will produce heat continuously.

However, as the temperature increases, the resistance of the thermistor decreases. When the temperature reaches the critical value, R_1 becomes equal to R_2 , and the bridge becomes balanced (since $R_3 = R_4$). At this point the op-amp switches to its LOW saturated output level, turning Q_1 OFF. The relay will be de-energized and the heating device will stop generating heat.

Moreover, we can change the polarity of the comparator to produce a HIGH output when the bridge balanced. Then, the relay can be used to activate an alarm or initiate an appropriate response to the over-temperature condition.

Here, no feedback is used, so the relay may, sometimes, make chattering sound. In that case we have to an Schmitt trigger comparator, instead of this simple comparator.

14.45 Active Filter

Electric filters are the circuits that have the ability to pass a particular frequency band (called **pass-band**) through it, while rejecting the other frequencies (called **stop-band**) outside the pass-band. Filter circuits basically exhibit the property of frequency selectivity. An ideal filter has constant gain for all frequencies in the pass-band and zero gain for all frequencies in the stop-band. The most commonly used filters are:

- Low-pass filter
- High-pass filter
- Band-pass filter
- Band-reject filter
- All-pass filter

Fig.14-111 shows the ideal frequency response for the first four filters, along with typical responses for practical filters of the same types. In each case, cut-off frequencies are shown to be those frequencies at which the response is 3 dB down from its maximum value in the pass band. Filters are widely used to extract desired frequency components from complex signals, and/or reject undesired ones, such as noise.

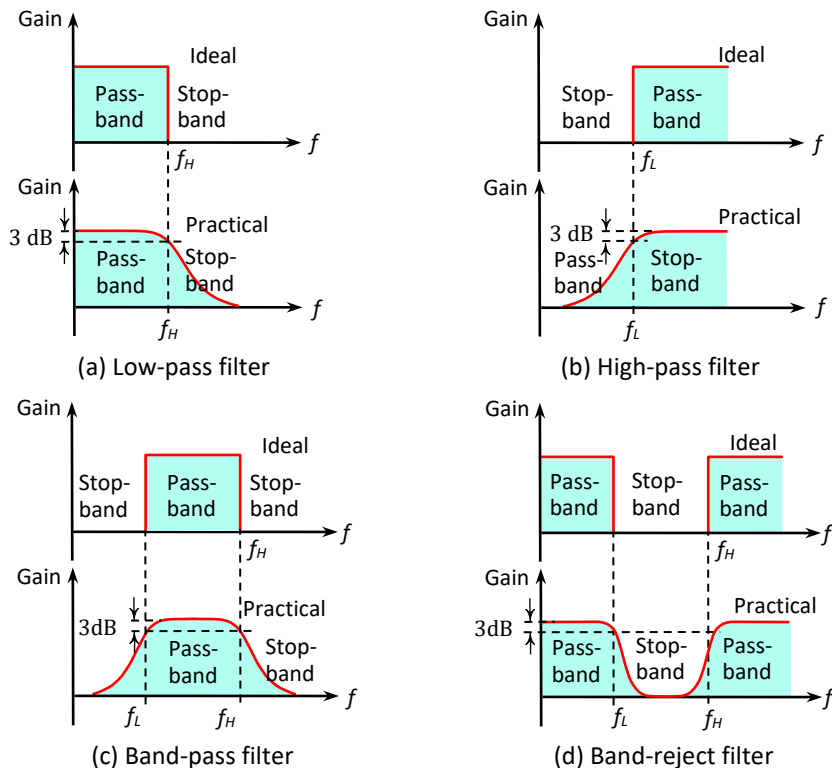


Fig.14-111: Ideal and practical responses of different filters

Filters can be classified in a number of ways:

- According to the components used: Passive or active filters

- According to the type of signals: Analog or digital filters
- According to operating frequency: Audio (AF) or radio (RF) filters
- According to order: 1st order, 2nd order etc.
- According to design techniques: Butterworth, Chebyshev or Cauer (or elliptic) filters

The major difference between **active** and **passive filter** is that an active filter uses active components like transistor or op-amp for the filtering of electronic signals. On the other hand, a passive filter uses passive components like resistor, inductor and capacitor to select signals of a particular band. Another major difference between the two is that an active filter needs an external source of power for its operation. While no external source is needed in case of passive filters.

Analog filters are designed to process analog signals, while **digital filters** are designed to process analog signals using digital techniques. Analog filters are circuits made of analog components such as resistors, capacitors, inductors, and op amps. Digital filters are often embedded in a chip that operates on digital signals, such as an MCU (microcontroller unit), microprocessor, or DSP (digital signal processor). Analog filters are fairly simple but increase in complexity with its order. Digital filters can be more precise in filtering, but the signal must be digital. Examples of digital filters are: IIR filters (Infinite Impulse Response filters), FIR filters (Finite Impulse Response filters) etc.

According to the operating frequency range, the filters may be classified as **audio frequency** (AF) or **radio-frequency** (RF) filters. Audio frequency filters work with the range of frequencies below 20 kHz while, radio-frequency filters work in the range of frequencies around 20 kHz to about 300 GHz.

Filters are also classified by their **order**, an integer number, also called the number of *poles*. For example, a filter that has only one pole is called a **first order filter**; a filter that has two poles is called a **second order filter** and so on. In general, the higher the order of a filter, the more closes its response to the ideal filter [H_E_atten]. With the increase in order the circuits become more and more complex.

Filters are also classified according to their design type that affects their response characteristics. For example, Butterworth filters, Chebyshev filters, Cauer (Elliptic) filters, Bessel filters, etc. The responses of these filters are shown in Fig.14-112.

The first, and probably the best-known filter approximation is the **Butterworth filter** that has maximally-flat response. It exhibits a nearly flat pass-band with no ripple. The roll-off is smooth and monotonic, with a low-pass or high-pass roll-off rate of 20 dB/decade or 6 dB/octave for every pole. Thus, a 3rd-order Butterworth low-pass filter will have an attenuation rate of 60 dB for every factor of ten increase in frequency beyond the cut-off frequency. It has a reasonably good phase response.

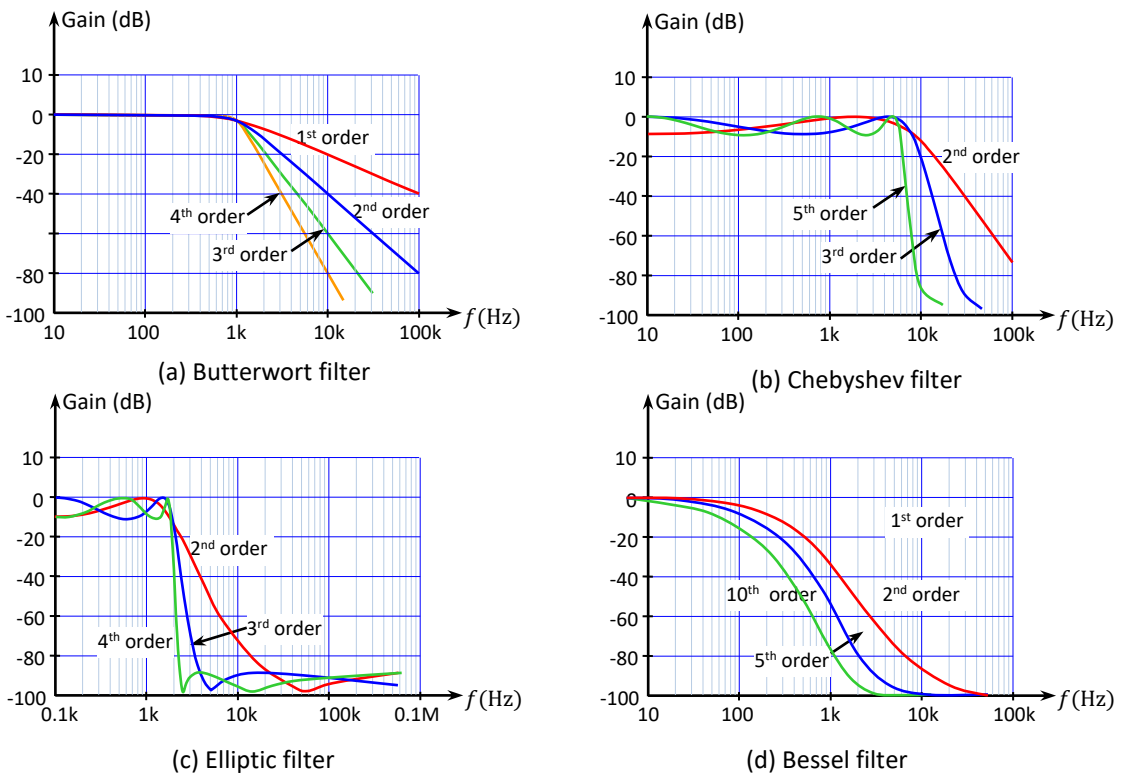


Fig.14-112: Approximate responses of some common filters

The **Chebyshev response** is a mathematical strategy for achieving a faster roll-off by allowing ripple in the frequency response. As the ripple increases (bad for a filter), the roll-off becomes sharper (good for a filter). The Chebyshev response is an optimal trade-off between these two parameters. Chebyshev filters where the ripple is only allowed in the pass-band are called type-1 filters. Chebyshev filters that have ripple only in the stop-band are called type-2 filters, but are seldom used. Chebyshev filters have a poor phase response.

The cut-off slope of a Cauer (elliptic) filter is steeper than that of a Butterworth, Chebyshev, or Bessel filters, but the amplitude response has ripple in both the pass-band and the stop-band, and the phase response is very nonlinear. However, if the primary concern is to pass frequencies within a certain frequency band and reject frequencies outside that band, regardless of phase shifts or ringing, the elliptic response will perform that function with the lowest-order filter.

The **Bessel filters** have a flat pass-band and a monotonic stop-band similar to those of the Butterworth filters. For the same filter order, however, the roll-off in the transition region is much less with a Bessel filter than with a Butterworth filter.

14.45.1 Comparison of Active and Passive Filters

Active filters have some advantages and some disadvantages over passive filters. A comparison between active and passive filters is given in the following table.

Parameters	Active Filters	Passive Filters
------------	----------------	-----------------

Components	Active components like op-amp, transistor, etc. along with passive components	Only passive components like resistors, capacitors, inductors etc.
Cost	High	Comparatively low
Circuit complexity	More complex	Less complex
Q-factor	High	Very low
External power	Required	Not required
Sensitivity	More sensitive	Comparatively less sensitive
Gain	Gain may be greater than 1	Gain is always less than 1
Input impedance	Very high	Comparatively low
Output impedance	Very low	Comparatively high
Frequency	Below 1 MHz	Above 1 MHz
Insertion loss	No insertion loss	High insertion loss

14.45.2 First-Order Low-Pass Butterworth Filter

Fig.14-113 shows a first-order low-pass Butterworth filter that uses an RC network for filtering purpose. Here the op-amp is used as a noninverting amplifier. So the RC network is connected

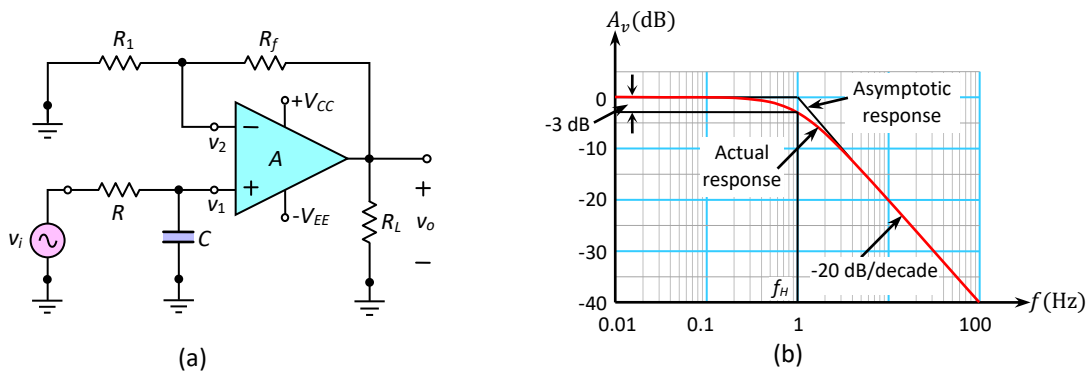


Fig.14-113: First-order low-pass filter: (a) Circuit diagram, and (b) Normalized frequency response

to noninverting input and negative feedback is applied to inverting terminal using R_1 and R_f resistors. Being a noninverting amplifier its gain will be $(1 + R_f/R_1)$. The operation process of this filter is exactly same as the passive low-pass filter. Here the op-amp will provide the gain (if required), high input impedance and low output impedance. For lower frequencies, the reactance of the capacitor (C) will be high, so most of the input voltage will be developed across the capacitor which will be amplified by the op-amp. On the other hand, of higher frequencies, the reactance of the capacitor (C) will be low, so the voltage developed across the capacitor will be low. Therefore, the output of the amplified will also be low.

The value of the voltage at the noninverting terminal (v_1) can be calculated using voltage divider rule as,

$$v_1 = \frac{-jX_C}{R - jX_C} v_i$$

$$\text{or, } v_1 = \frac{1}{R/(-jX_C) + 1} v_i$$

$$\text{or, } v_1 = \frac{v_i}{1 + j2\pi fRC} \quad (14-179)$$

The output of the circuit (noninverting amplifier) will be,

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_1$$

Putting the value of v_1 from Equ.(14-179) into the above equation we get,

$$v_o = \left(1 + \frac{R_f}{R_1}\right) \frac{v_i}{1 + j2\pi fRC}$$

Therefore, the gain of the circuit will be,

$$A_{vf} = \frac{v_o}{v_i} = \left(1 + \frac{R_f}{R_1}\right) \frac{1}{1 + j2\pi fRC}$$

$$\text{or, } A_{vf} = \frac{v_o}{v_i} = \left(1 + \frac{R_f}{R_1}\right) \frac{1}{1 + jf/(1/2\pi RC)}$$

$$\therefore \boxed{A_{vf} = \frac{v_o}{v_i} = \frac{A_v}{1 + j(f/f_H)}} \quad (14-180)$$

where, $A_{vf} = v_o/v_i$ = gain of the circuit as a function of frequency

$A_v = \left(1 + \frac{R_f}{R_1}\right)$ = maximum gain or gain in the pass-band

f_H = higher cut-off frequency of the filter, which is given by

$$\boxed{f_H = \frac{1}{2\pi RC}} \quad (14-181)$$

The magnitude of the gain can be calculated from Equ.(14-180) as,

$$\boxed{|A_{vf}| = \left|\frac{v_o}{v_i}\right| = \frac{A_v}{\sqrt{1 + (f/f_H)^2}}} \quad (14-182)$$

At the higher cut-off frequency ($f = f_H$) the magnitude of gain of the circuit will be,

$$|A_{vf}| = \frac{A_v}{\sqrt{1 + (f/f_H)^2}} = \frac{A_v}{\sqrt{1 + (1)^2}} = \frac{A_v}{\sqrt{2}} \quad (14-183)$$

The phase angle of output signal will be,

$$\phi = -\tan^{-1}\left(\frac{f}{f_H}\right) = -\tan^{-1}(2\pi fRC) \quad (14-184)$$

The operation of the low-pass filter can also be explain from Equ.(14-182). For signals with very low frequencies, that is $f \ll f_H$, the gain of the circuit will be $A_{vf} = A_v$, which is the maximum gain or the pass-band gain. If $f = f_H$, the gain of the filter will be [Equ.(14-183)] $A_{vf} = A_v/\sqrt{2} = 0.707A_v$. That is at the higher cut-off frequency the gain of the filter will be $1/\sqrt{2}$ times the maximum gain. For very high frequencies, ($f \gg f_H$) Equ.(14-182) gives $A_{vf} \ll A_v$. In summery we find, the low-pass filter has a constant gain A_v form 0 Hz up to the higher cut-off frequency, f_H . At f_H the gain is $0.707A_v$, and after f_H the gain decreases at a constant rate of 20 dB/decade or 6 dB/octave. The normalized response characteristic of this filter is shown in Fig.14-113(b). Here, the parameter in x -axis has been divided by the cut-off frequency (f_H) and the parameter in the y -axis have been divided by the maximum gain. The frequency f_H is called the cut-off frequency because at this frequency the gain is 3 dB down from the maximum gain. This frequency is also called **−3 dB frequency, break frequency, corner frequency or half-power frequency**.

The design criteria of a low-pass filter are to determine the value of R and C for a given value of higher cut-off frequency. At first we have to select a capacitor with less than or equal to $1 \mu\text{F}$, then putting the value of C and f_H into Equ.(14-181), the value of R has to be determined. For better performance, we can use Mylar or tantalum capacitors. By proper selection of R_1 and R_f we can set any pass-band gain of the filter.

Example 13-22

Design a first-order low-pass filter with the higher cut-off frequency of 5 kHz and the pass-band gain of 6.

Solution:

Let us use $C = 0.01 \mu\text{F}$. Now putting the value of C and $f_H = 5 \text{ kHz}$ into Equ.(14-181), we can calculate the value of R .

$$\begin{aligned} \text{Equ.(14-181)} \quad f_H &= \frac{1}{2\pi RC} \\ \text{or,} \quad R &= \frac{1}{2\pi f_H C} = \frac{1}{2\pi(5 \text{ kHz})(0.01 \mu\text{F})} \\ \therefore R &\approx 3.2 \text{ k [use standard 3.3 k] [Ans.]} \end{aligned}$$

Let us select $R_f = 10 \text{ k}$. Using the expression for the pass-band gain,

$$\begin{aligned} 6 &= \left(1 + \frac{10 \text{ k}}{R_1}\right) \\ \therefore R_1 &= 2 \text{ k [Ans.]} \end{aligned}$$

Comments: We should not select such a small C that its value is comparable with the parasitic capacitance (few pF) of the op-amp.

14.45.3 Second-Order Low-Pass Butterworth Filter

Second-order (or two-pole) filters consist of two RC filter sections connected together to provide a -40 dB/decade roll-off rate. The circuit diagram of second order filter is shown in Fig.14-114. This arrangement is also referred to as VCVS filter, because the op-amp is used as a *Voltage Controlled Voltage Source* amplifier. If the gain of the amplifier is unity (a buffer amplifier), then the filter is called a *Sallen-Key filter* according to its inventors^[2][H_R_atten]. Second-order filters are very important because combining this filter with the first order filters, higher order filter circuits can be designed. The filter design is based around a non-inverting op-amp configuration so the filter's gain, A_{vf} , can be set any convenient value. Also the op-amp has high input impedance which means that it can be easily cascaded with other active filters to give more higher-order filters.

At low frequencies, both capacitors appear to be open and the circuit will work as a noninverting amplifier with gain $(1 + R_f/R_3)$. As the frequency increases, the reactance of C_2 decreases so the noninverting input (v_1) voltage also decreases. At the same time, capacitor C_1 is feeding back a signal that is in phase with the input signal. Since the feedback signal adds to the source signal, the feedback is positive. As a result, the decrease in the noninverting input voltage caused by C_2 is not as large as it would be without the positive feedback. The larger the value of C_1 with respect to C_2 , the more is the positive feedback; this is equivalent to increasing the Q of the circuit. If C_1 is large enough to make Q greater than 0.707, peaking appears in the frequency response.

The normalized frequency response of this filter [Fig.14-114(b)] is fixed by the RC networks and is generally identical to that of the first order type. The main difference between a 1st and 2nd order low-pass filter is that the stop band roll-off is twice the 1st order filter, that is, at 40 dB/decade or 12 dB/octave as the operating frequency increases above the cut-off frequency f_H . The cut off frequency f_H for the filter is now decided by R_1 , C_1 , R_2 and C_2 . The gain of the filter is set by the resistance R_3 and R_f .

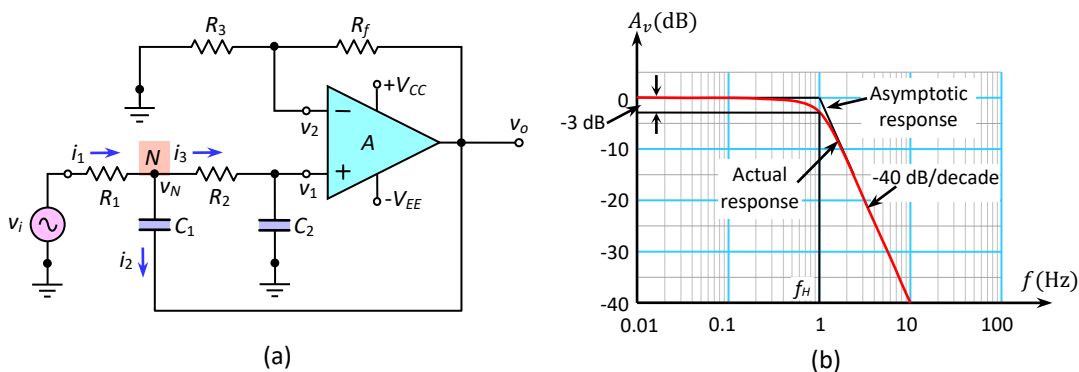


Fig.14-114: Second-order low-pass filter: (a) Circuit diagram, and (b) Normalized frequency

Let us represent this maximum gain of the amplifier by A_f , that is,

$$A_f = \left(1 + \frac{R_f}{R_3}\right) \quad (14-185)$$

and let, $s = j\omega$.

Applying KCL at node N , we get,

$$i_1 = i_2 + i_3$$

$$\text{or, } \frac{v_i - v_N}{R_1} = \frac{v_N - v_o}{X_{C1}} + \frac{v_N - v_1}{R_2} \quad (14-186)$$

Using voltage divider rule we can find the value of v_1 as,

$$v_1 = \frac{X_{C2}}{R_2 + X_{C2}} v_N$$

$$\text{or, } v_N = \left(1 + \frac{R_2}{X_{C2}}\right) v_1 \quad (14-187)$$

The output voltage of the circuit will be,

$$v_o = \left(1 + \frac{R_f}{R_3}\right) v_1 = A_f v_1$$

$$\therefore v_1 = \frac{v_o}{A_f} \quad (14-188)$$

Putting this value of v_1 into Equ.(14-187),

$$v_N = \left(1 + \frac{R_2}{X_{C2}}\right) v_o / A_f$$

$$\therefore X_{C2} = 1/sC_2, \quad v_N = \frac{(1 + sR_2C_2)v_o}{A_f} \quad (14-189)$$

Now, putting the value of v_N and v_1 from Equ.(14-189) and Equ.(14-188) into Equ.(14-186),

$$\frac{v_i - (1 + sR_2C_2)v_o/A_f}{R_1} = \frac{(1 + sR_2C_2)v_o/A_f - v_o}{X_{C1}} + \frac{(1 + sR_2C_2)v_o/A_f - v_o/A_f}{R_2}$$

$$\text{or, } \frac{v_i}{R_1} = \frac{(1 + sR_2C_2)v_o}{A_f R_1} + \frac{(1 + sR_2C_2)v_o/K - v_o}{1/sC_1} + \frac{sC_2 v_o}{A_f}$$

$$\text{or, } \frac{v_i}{R_1} = v_o \left[\frac{(1 + sR_2C_2)}{A_f R_1} + \frac{(sC_1 + s^2 R_1 R_2 C_1 C_2)}{A_f} - sC_1 + \frac{sC_2}{A_f} \right]$$

$$\text{or, } v_i = v_o \left[\frac{(1 + sR_2C_2)}{A_f} + \frac{(sC_1 R_1 + s^2 R_1 R_2 C_1 C_2)}{A_f} - sR_1 C_1 + \frac{sR_1 C_2}{A_f} \right]$$

$$\text{or, } v_i = v_o \left(\frac{1 + sR_2C_2 + sC_1 R_1 + s^2 R_1 R_2 C_1 C_2 - sR_1 C_1 K + sR_1 C_2}{A_f} \right)$$

$$\text{or, } \frac{v_o}{v_i} = \frac{A_f}{s^2 (R_1 R_2 C_1 C_2) + s[R_2 C_2 + C_1 R_1 (1 - K) + R_1 C_2] + 1}$$

$$\text{or, } \frac{v_o}{v_i} = \frac{A_f / R_1 R_2 C_1 C_2}{s^2 + s[R_2 C_2 + C_1 R_1 (1 - A_f) + R_1 C_2] / R_1 R_2 C_1 C_2 + 1 / R_1 R_2 C_1 C_2} \quad (14-190)$$

$$\therefore \frac{v_o}{v_i} = \frac{K\omega_H^2}{s^2 + s\omega_H/Q + \omega_H^2} \quad (14-191)$$

Where, ω_H is the higher cut-off frequency and is given by,

$$\omega_H = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \quad (14-192)$$

and Q is the quality factor which is given by,

$$Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{R_2 C_2 + C_1 R_1 (1 - A_f) + R_1 C_2} \quad (14-193)$$

The quality factor Q determines the peaking effect of the response around the cut-off frequency as shown in Fig.14-115.

For design simplicity we generally assume, $R_1 = R_2 = R$, and $C_1 = C_2 = C$. Therefore, Equ.(14-192) reduces to,

$$\omega_H = \frac{1}{RC} \quad (14-194)$$

$$\text{or, } f_H = \frac{1}{2\pi RC} \quad (14-195)$$

Similarly, Equ.(14-193) reduces to

$$Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{R_2 C_2 + C_1 R_1 (1 - A_f) + R_1 C_2} = \frac{RC}{RC + RC(1 - A_f) + RC}$$

$$\text{or, } Q = \frac{1}{3 - A_f} \quad (14-196)$$

Therefore, the gain of a second-order filter should less than 3. Otherwise the filter will be unstable. As shown in Fig.(14-109), to get the Butterworth response the quality factor should be $Q = 0.707$. Putting this value into Equ.(14-196), we get,

$$0.707 = \frac{1}{3 - A_f}$$

$$\text{or, } A_f = 1.586 \quad (14-197)$$

Putting the value of A_f from Equ.(14-185),

$$1 + \frac{R_f}{R_3} = 1.586$$

$$\text{or, } \frac{R_f}{R_3} = 0.586$$

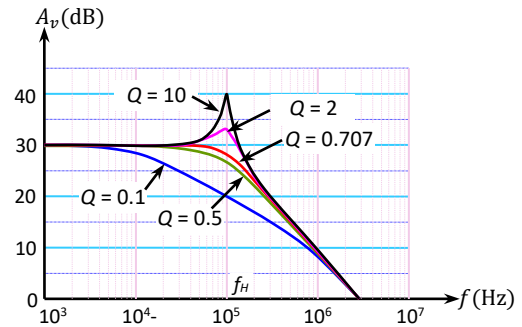


Fig.14-115: Variation of response of second-order low-pass with the quality factor Q

or,

$$R_f = 0.586R_3 \quad (14-198)$$

Being a 2nd order filter, the magnitude of the gain of Equ.(14-190) can be represented in terms of cut-off frequency as [derivation is not shown here],

$$|A_{vf}| = \left| \frac{v_o}{v_i} \right| = \frac{A_f}{\sqrt{1 + (f/f_H)^4}} = \frac{1.586}{\sqrt{1 + (f/f_H)^4}} \quad (14-199)$$

The phase relation is given by

$$\phi = \quad (14-200)$$

The design process of the second-order low-pass filter will be same as that of first-order filter. Here, the only difference is that the gain of the amplifier should be 1.586 to design a Butterworth filter [see Example 13-].

Example 13-23

Design a second-order Butterworth low-pass filter with the cut-off frequency of 5 kHz. Also, show the frequency response of the filter in a graph.

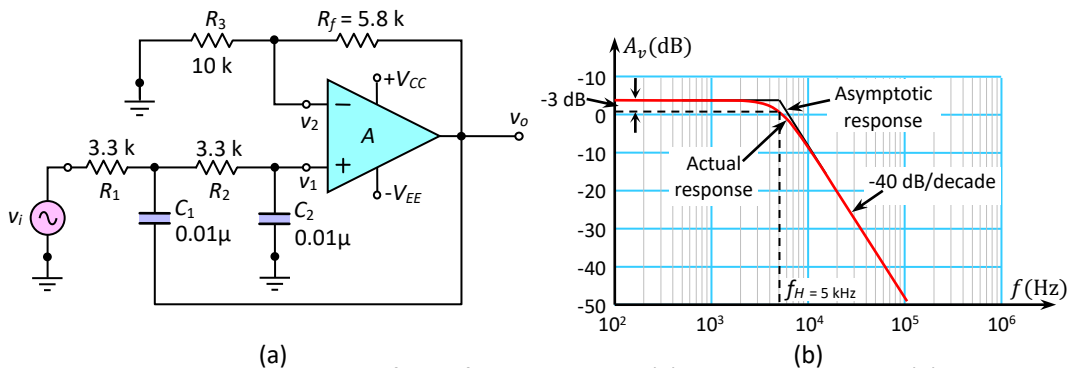


Fig.14-116: Second-order low-pass filter of Example 13-23: (a) Circuit diagram, and (b) Frequency response

Solution:

Let us use $C_1 = C_2 = C = 0.01 \mu\text{F}$. Now putting the value of C and $f_H = 5 \text{ kHz}$ into Equ.(14-195), we get,

$$R = \frac{1}{2\pi f_H C} = \frac{1}{2\pi(5 \text{ kHz})(0.01 \mu\text{F})}$$

$$\therefore R \approx 3.2 \text{ k [use standard 3.3 k] [Ans.]}$$

Let us select $R_3 = 10 \text{ k}$. Using Equ.(14-197),

$$R_f = 0.586R_3 = 0.586 \times 10 \text{ k}$$

$$\therefore R_f = 5.8 \text{ k} [\text{use standard } 5.6 \text{ k}] [\text{Ans.}]$$

The frequency response of the filter is drawn using Equ.(14-198) and is shown in Fig.14-116.

Comments: The values of the frequency selecting components (R and C) are same as in the first-order filter design of Example 13-22 as the cut-off frequency is same.

14.45.4 First-Order High-pass Butterworth Filter

High-pass filters are often formed simply by interchanging the position of the frequency selecting components (R and C). Thus, a first-order high-pass Butterworth filter can be formed from the filter circuit of Fig.14-113 by interchanging the position of R and C . The circuit arrangement is shown in Fig.14-117.

The value of the voltage at the noninverting terminal (v_1) can be calculated using voltage divider rule as,

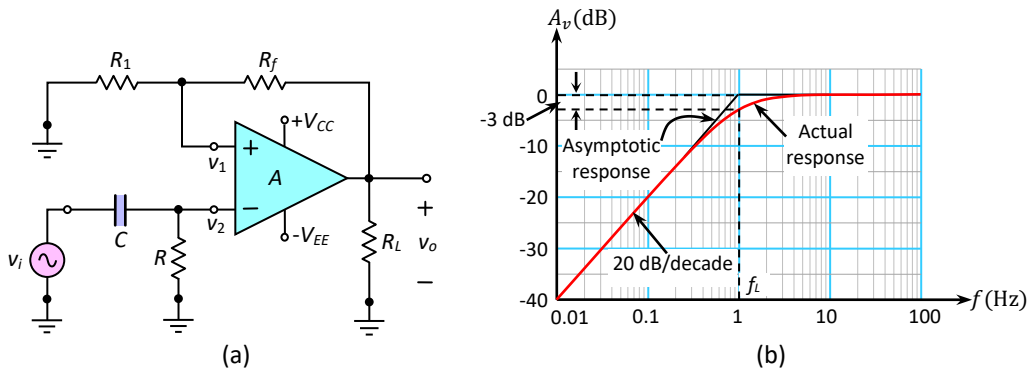


Fig.14-117: First-order high-pass filter: (a) Circuit diagram, and (b) Normalized frequency response

$$v_1 = \frac{R}{R - jX_C} v_i$$

or,

$$v_1 = \frac{R}{R + 1/j(2\pi fC)} v_i$$

or,

$$v_1 = \frac{j2\pi fRC}{1 + j2\pi fRC} v_i \quad (14-201)$$

The output of the circuit (noninverting amplifier) will be,

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Putting the value of v_1 from Equ.(14-201) into the above equation we get,

$$v_o = \left(1 + \frac{R_f}{R_1}\right) \frac{j2\pi fRC}{1 + j2\pi fRC} v_i$$

Therefore, the gain of the circuit will be,

$$A_{vf} = \frac{v_o}{v_i} = \left(1 + \frac{R_f}{R_1}\right) \frac{j2\pi fRC}{1 + j2\pi fRC}$$

$$\text{or, } A_{vf} = \frac{v_o}{v_i} = \left(1 + \frac{R_f}{R_1}\right) \frac{jf/(1/2\pi RC)}{1 + jf/(1/2\pi RC)}$$

$$\text{or, } A_{vf} = \frac{v_o}{v_i} = \frac{A_v j(f/f_L)}{1 + j(f/f_L)}$$

where, $A_{vf} = v_o/v_i$ = gain of the circuit as a function of frequency

$A_v = \left(1 + \frac{R_f}{R_1}\right)$ = maximum gain or gain in the pass-band

$f_L = \frac{1}{2\pi RC}$ = lower cut-off frequency of the filter

The magnitude of the gain will be,

$$|A_{vf}| = \left| \frac{v_o}{v_i} \right| = \frac{A_v(f/f_L)}{\sqrt{1 + (f/f_L)^2}} \quad (14-202)$$

and the phase angle will be,

$$\phi = \tan^{-1}\left(\frac{f}{f_L}\right) = \tan^{-1}(2\pi fRC) \quad (14-203)$$

The operation of the high-pass filter can be explain from Equ.(14-202). For signals with very high frequencies, that is $f \gg f_L$, the gain of the circuit will be $A_{vf} \approx A_v$, which is the maximum gain or the pass-band gain. If $f = f_L$, the gain of the filter will be [from Equ.(14-202)] $A_{vf} = A_v/\sqrt{2} = 0.707A_v$. That is, at the lower cut-off frequency the gain of the filter will be $1/\sqrt{2}$ times the maximum gain. For the very low frequencies, ($f \ll f_L$) Equ.(14-202) gives $A_{vf} \ll A_v$. In summery we find, the high-pass filter has a constant gain A_v form f_L up to higher frequency. At f_L the gain is $0.707A_v$, and for frequencies lower than f_L the gain decreases at a constant rate of 20 dB/decade or 6 dB/octave. The response characteristic of this filter is shown in Fig.14-117(b). The frequency f_L is called the cut-off frequency and is given by,

$$f_L = \frac{1}{2\pi RC} \quad (14-204)$$

The design process of a high-pass filter is exactly same as that of low-pass filter.

14.45.5 Second-Order High-Pass Butterworth Filter

A second-order high-pass filter can be constructed from a second-order low-pass filter simply by interchanging the position of the frequency selecting resistors and capacitors. The circuit diagram of a second-order high-pass filter is shown in Fig.14-118. The filter will pass all frequencies greater than a lower cut-off frequency (f_L), and all frequencies below f_L will be attenuated with a rate of 40 dB/decade. The expression for the cut-off frequency will be same as second-order low-pass filter, that is,

$$f_L = \frac{1}{2\pi RC} \quad (14-205)$$

To achieve Butterworth response, the value of the gain will be,

$$A_f = 1.586 \quad (14-206)$$

Being a 2nd order filter, the magnitude of the gain of Equ.(14-) can be represented in terms of cut-off frequency as,

$$|A_{vf}| = \left| \frac{v_o}{v_i} \right| = \frac{A_f}{\sqrt{1 + (f_L/f)^4}} = \frac{1.586}{\sqrt{1 + (f_L/f)^4}} \quad (14-207)$$

The frequency response of this filter is shown in the Fig.14-117 of Example 13-24.

Example 13-24

Design a second-order Butterworth high-pass filter with the cut-off frequency of 10 kHz. Also, show the frequency response of the filter in a Bode plot.

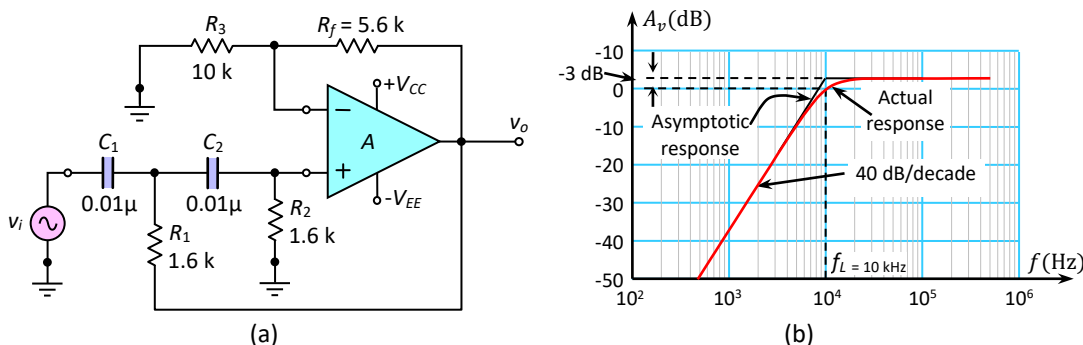


Fig.14-118: Second-order high-pass filter for Example 13-24: (a) Circuit diagram, and (b) Frequency response

Solution:

Let us use $C_1 = C_2 = C = 0.01 \mu\text{F}$. Now putting the value of C and $f_L = 10 \text{ kHz}$ into Equ.(14-205), we get,

$$R = \frac{1}{2\pi f_L C} = \frac{1}{2\pi(10 \text{ kHz})(0.01 \mu\text{F})}$$

$$\therefore R \approx 1.6 \text{ k [Ans.]}$$

Let us select $R_3 = 10 \text{ k}$. Using Equ.(14-203),

$$R_f = 0.586 R_3 = 0.586 \times 10 \text{ k}$$

$$\therefore R_f = 5.8 \text{ k [use standard } 5.6 \text{ k] [Ans.]}$$

The frequency response of the filter is drawn using Equ.(14-204) and is shown in Fig.14-118.

Comments: The design process is exactly same as that of Example 13-23.

14.45.6 Higher-Order Filters

From the discussion on filters given so far, it may be concluded that in the stop-band the gain of the filter changes at the rate of 20 dB/decade for first-order filters and 40 dB/decade for second-order filters. This means that as the order of the filter is increased, the actual stop-band response of the filter approaches its ideal stop-band characteristics. In general, an N -order filter produces $N \times 20$ dB/decade or $N \times 6$ dB/octave attenuation in the stop-band.

Higher-order filters, such as third, fourth, fifth etc. are built simply by using the first and second-order filters. The simplest way to build a third-order low-pass filter is by cascading a first order filter with a second-order. Similarly a fourth-order low-pass filter can be formed by cascading two second-order low-pass filters. Although there is no limit to the order of the filter that can be formed, as the order of the filter increases, so does its size. Also the accuracy declines, in that the difference between the actual stop-band response and the theoretical stop-band response increases with increase in the order of the filter.

14.45.7 Third-Order Low-Pass Butterworth Filter

Fig.14-119 shows a third-order low-pass Butterworth filter. The voltage-gain of the first section is optional, it can be set whatever is required. The voltage-gain of the second section, however

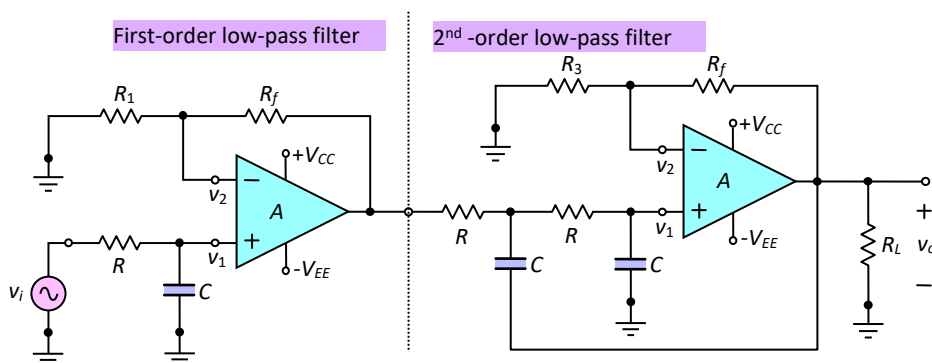


Fig.14-119: Third-order low-pass filter cascading a first-order and a second-order filter

affects the flatness of the overall response. If closed-loop gain is kept 1.586, then the overall gain will be down 6 dB (3 dB for each section) at the cut-off frequency. By slightly increasing the voltage gain of the second section, cumulative loss of voltage gain is compensated. By using an advanced mathematical derivation (out of scope of this book), it can be proved that an $A_f = 2$ is the critical value required for a maximally flat response. For the circuit of Fig.14-119, it is required that $R_f = R_1$ and hence $A_f = 2$. Considering all the frequency selecting components identical, the value of the cut-off frequency will be,

$$f_H = \frac{1}{2\pi RC} \quad (14-208)$$

At cut-off frequency, the overall voltage gain is down 3 dB. Above the cut-off frequency, the voltage gain drops at a rate of 60 dB/decade, or equivalent to 18 dB/octave.

14.45.8 Fourth-Order Low-Pass Butterworth Filter

A fourth-order low-pass Butterworth filter is illustrated in Fig.14-120. It is formed by cascading two second-order low-pass filters. If $A_f = 1.586$ is used for both sections, the voltage gain will be down 6 dB at the cut-off frequency. By using different gain for each section, we can adopt a compromise that produces a maximally flat response. An advanced derivation shows that we need to use $A_f = 1.152$ for the first section and $A_f = 2.235$ for the second section.

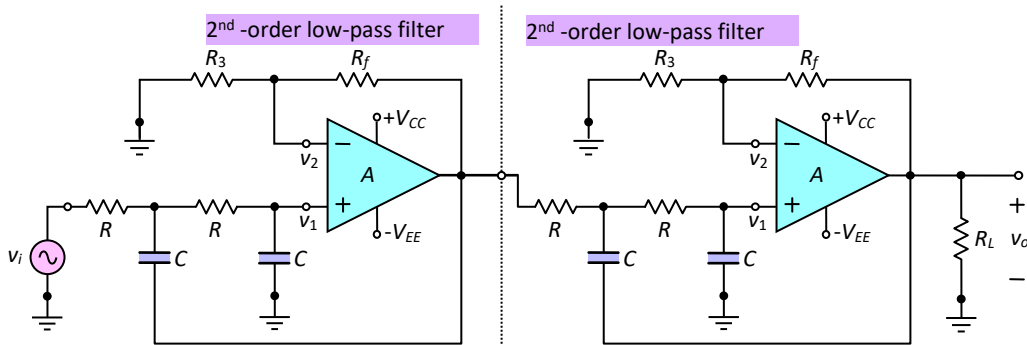


Fig.14-120: Fourth-order low-pass filter cascading two second-order filters

Also, the overall filter gain is equal to the product of the individual voltage gains of the filter sections that is, $1.152 \times 2.235 = 2.575$. Here also, considering all the frequency selecting components identical, the cut-off frequency will be,

$$f_H = \frac{1}{2\pi RC} \quad (14-209)$$

At cut-off frequency, the overall voltage gain is down 3 dB. Above the cut-off frequency, the voltage gain drops at a rate of 80 dB/decade, or equivalent to 24 dB/octave.

Like the higher-order low-pass filters, higher-order high-pass filters are formed by simply interchanging the positions of the frequency determining resistors and capacitors in the corresponding low-pass filters.

14.45.9 Band-Pass Filters

As the name implies, a band-pass filter passes a band of frequencies between a lower (f_L) and a higher (f_H) cut-off frequencies, and rejects any other frequencies outside this pass-band. There are two types of band-pass filters:

- i. wide band-pass filters and
- ii. narrow band-pass filters

Unfortunately, there is no set dividing line between the two. However, a band-pass filter is defined as a wide band-pass if its figure of merit or quality factor Q is less than 10 while the band-pass filters with $Q \geq 10$ are called the narrow band-pass filters. The Q -factor is a measure of selectivity, meaning that the higher the value of Q the more selective is the filter, or the narrower is the bandwidth (BW). The relationship between Q , 3 dB bandwidth, and the centre frequency f_c is given by the following equations:

$$f_c = \sqrt{f_L f_H} \quad (14-210)$$

$$Q = \frac{f_c}{BW} = \frac{f_c}{f_H - f_L} \quad (14-211)$$

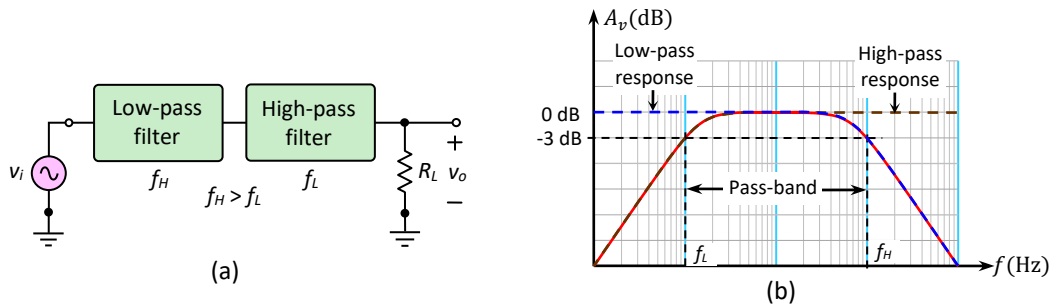


Fig.14-121: Wide band-pass filter: (a) Block diagram, and (b) Frequency response

14.45.10 Wide Band-pass Filters

Wide band-pass filters are designed by simply cascading a low-pass and a high-pass filter. The lower cut-off frequency (f_L) of the band-pass filter will be set by the high-pass filter and the higher cut-off frequency (f_H) of the filter will be set by the low-pass filter. There should have an overlap in the response of the low-pass and high-pass filters. That is, the higher cut-off frequency of low-pass filter should be larger than the lower cut-off frequency of the high-pass filter ($f_H > f_L$). The roll-off rate will be exactly same as the roll-off of the high-pass and low-pass filters. The overall (total) gain of the band-pass filter will be product of the individual gain of the component filters. The magnitude of gain of the band-pass filter can be found by multiplying the Equ.(14-183) and Equ.(14-199),

$$\left| \frac{v_o}{v_i} \right| = \frac{A_{ft}(f/f_L)}{\sqrt{[1 + (f/f_L)^2][1 + (f/f_H)^2]}} \quad (14-212)$$

where,

A_{ft} = total gain of the band-pass filter

f_L = lower cut-off frequency of the band-pass filter, and

f_H = higher cut-off frequency of the band-pass filter.

The design process is illustrated in Example 13-.

Example 13-25

Design a band-pass filter with $f_L = 1$ kHz, $f_H = 10$ kHz, and a pass-band gain $A_{ft} = 6$. Draw the frequency response of the filter.

Solution:

Using Equ.(14-207), $f_c = \sqrt{f_L f_H} = \sqrt{1 \text{ kHz} \times 10 \text{ kHz}} = 3.16 \text{ kHz}$

Using Equ.(14-208),
$$Q = \frac{f_c}{f_H - f_L} = \frac{3.16 \text{ kHz}}{10 \text{ kHz} - 1 \text{ kHz}} \approx 0.35 < 1$$

So, we will design the filter using a low-pass section with gain of 2 and a high-pass section with gain of 3.

Low-pass section:

Let us use $C = 0.01 \mu\text{F}$. Now putting the value of C and $f_H = 10 \text{ kHz}$ into Equ.(14-), we get,

Using Equ.(14-179),
$$R = \frac{1}{2\pi f_H C} = \frac{1}{2\pi(10 \text{ kHz})(0.01 \mu\text{F})}$$
$$R \approx 1.6 \text{ k}$$

Let us select $R_f = 10 \text{ k}$. Using the expression for the pass-band gain,

$$2 = \left(1 + \frac{10 \text{ k}}{R_1} \right)$$

$\therefore R_1 = 10 \text{ k}$

High-pass section:

Let us use $C = 0.01 \mu\text{F}$. Now putting the value of C and $f_L = 1 \text{ kHz}$ into Equ.(14-), we get,

Using Equ.(14-201),
$$R = \frac{1}{2\pi f_L C} = \frac{1}{2\pi(1 \text{ kHz})(0.01 \mu\text{F})}$$
$$R \approx 16 \text{ k}$$

Let us select $R_f = 10 \text{ k}$. Using the expression for the pass-band gain,

$$3 = \left(1 + \frac{10 \text{ k}}{R_1} \right)$$

$\therefore R_1 = 5 \text{ k}$ [use standard 5.1 k]

The design circuit is shown in Fig.14-112(a). The frequency response can be plotted using Equ.(14-199) as,

$$\left| \frac{v_o}{v_i} \right| = \frac{6 \times (f/1 \text{ k})}{\sqrt{[1 + (f/1 \text{ k})^2][1 + (f/10 \text{ k})^2]}}$$

The graph of this expression has been plotted in MS Excel and is shown in Fig.14-122(b).

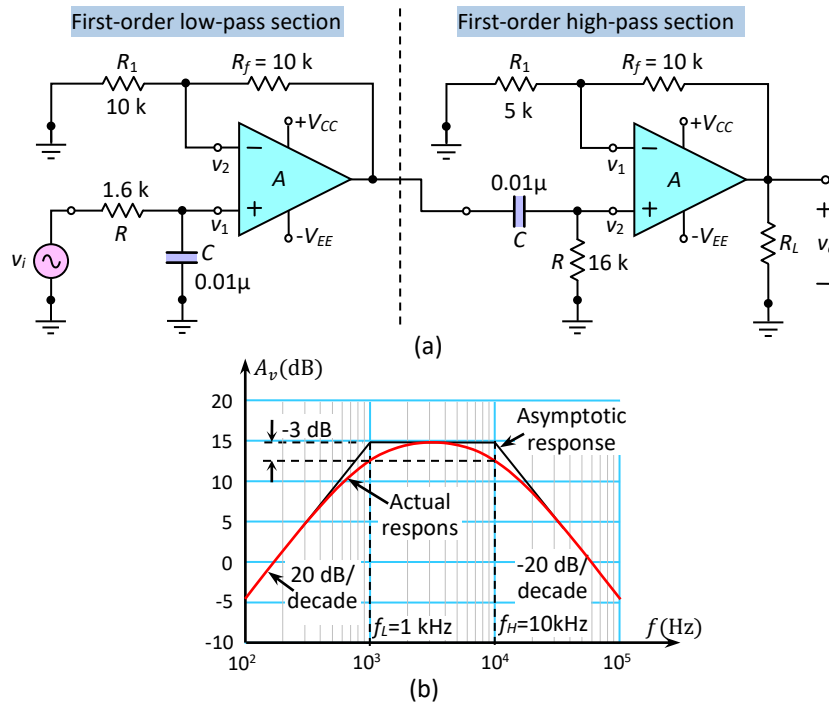


Fig.14-122: Wide band-pass filter of Example 13-25: (a) Circuit diagram, and (b) Frequency response

Comments: It is also possible to design the filter using two second-order filters or even more higher-order filters.

14.45.11 Narrow Band-Pass Filter

A narrow band-pass filter employing multiple feedback is depicted in Fig.14-123. This filter employs only one op-amp. In comparison to all filters discussed so far, this filter has some unique features that are given below.

- It has two feedback paths, and for this reason it is called a multiple-feedback filter.
- The op-amp is used in the inverting mode.

At lower frequency region, the reactance of C_f is so large that it can be ignored. In this range, the reactance of C_1 decreases with the increase in frequency and hence the gain of the circuit increases. In the high frequency region, C_1 is almost short and the reactance of C_f gradually decreases with the increase in frequency, and hence the gain of the circuit decreases. That is, R_f and C_1 provide the high-pass response, and R_1 and C_f provide the low-pass response. The frequency response of the narrow band-pass filter is shown in Fig.14-123(b).

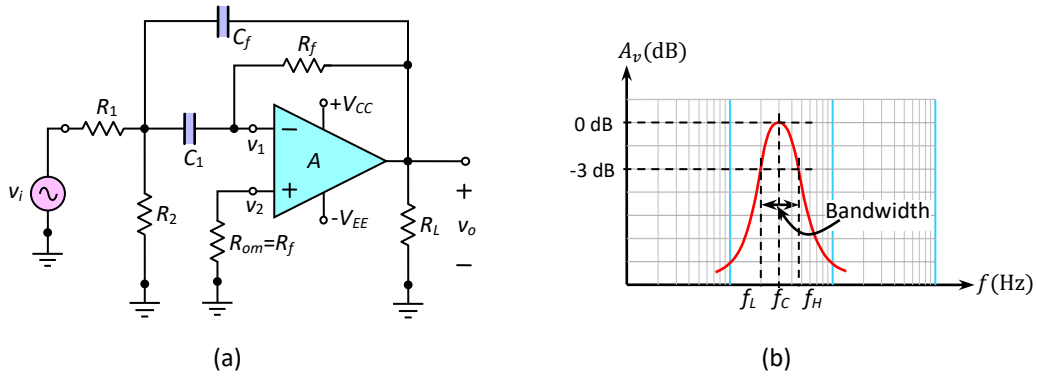


Fig.14-123: Narrow band-pass filter: (a) Circuit diagram, and (b) Frequency response

To derive an expression for the centre frequency, let us find the equivalent resistance seen by C_f . This value will be $R_1 \parallel R_2$. Therefore, the expression for the centre frequency will be (using the concept of two poles),

$$f_c = \frac{1}{2\pi\sqrt{(R_1 \parallel R_2)C_f R_f C_1}} \quad (14-213)$$

For simplification of design calculations it is assumed that $C_1 = C_f = C$.

$$f_c = \frac{1}{2\pi\sqrt{(R_1 \parallel R_2)R_f C^2}} = \frac{1}{2\pi C\sqrt{(R_1 \parallel R_2)R_f}}$$

$$\therefore f_c = \frac{1}{2\pi C} \sqrt{\frac{R_1 + R_2}{R_1 R_2 R_f}} \quad (14-214)$$

The value of the quality factor can be calculated as,

$$Q = \frac{f_c}{BW} = \frac{f_c}{f_H - f_L} \quad (14-215)$$

Generally, the narrow band-pass filter is designed for specific values of centre frequency, f_c and quality factor Q , or f_c and bandwidth BW . The circuit components are determined from the following relationships:

$$R_1 = \frac{Q}{2\pi f_c C A_f} \quad (14-216)$$

$$R_2 = \frac{Q}{2\pi f_c C (2Q^2 - A_f)} \quad (14-217)$$

$$R_f = \frac{Q}{\pi f_c C} \quad (14-218)$$

From the Equ.(14-213) and Equ.(14-215) the value of Q can be found as,

$$Q = 2\pi f_c C A_f R_1$$

and,

$$Q = \pi f_c C R_f$$

$\therefore$

$$2\pi f_c C A_f R_1 = \pi f_c C R_f$$

or,

$$2A_f R_1 = R_f$$

$\therefore$

$$A_f = \frac{R_f}{2R_1} \quad (14-219)$$

In order for the denominator of the Equ.(14-214) to be positive, $A_f < 2Q^2$, which impose a limitation on the gain.

Example 13-26

Design a narrow band-pass filter with $Q = 5$, $f_c = 5$ kHz, and $A_f = 10$.

Solution:

For simplicity let us assume that $C_1 = C_2 = C = 0.01 \mu\text{F}$.

Using Equ.(14-213),
$$R_1 = \frac{Q}{2\pi f_c C A_f} = \frac{5}{2\pi(5 \text{ kHz})(0.01 \mu\text{F})10} \approx 1.6 \text{ k [Ans.]}$$

Using Equ.(14-214),
$$R_2 = \frac{5}{2\pi(5 \text{ kHz})(0.01 \mu\text{F})(2 \times 5^2 - 10)} \approx 398 \Omega \text{ [use } 390\Omega\text{] [Ans.]}$$

Using Equ.(14-215),
$$R_f = \frac{Q}{\pi f_c C} = \frac{5}{\pi(5 \text{ kHz})(0.01 \mu\text{F})} \approx 6.4 \text{ k [Ans.]}$$

Comments: We can verify the value of gain using Equ.(14-216), that is, $R_f/2R_1 \approx 10$.

14.45.12 Band-Reject Filters

The band-reject filter is also called *band-stop* or *band-elimination filter*. As the name implies, a band-reject filter blocks (or rejects) a band of frequencies between a lower (f_L) and a higher (f_H) cut-off frequencies, and passes any other frequencies outside this stop-band. Like the band-pass filters, band-reject filters may also be of two types:

- i. Wide band-reject filters and
- ii. narrow band-reject filters

They are differentiated depending of the value of Q as same as the band-pass filters. The narrow band-reject filter is also called *notch filter*.

14.45.13 Wide Band-Reject Filter

A band-reject filter can be implemented by connecting a low-pass and a high-pass filter at the inputs of a summing amplifier as shown in Fig.14-124. As shown in Fig.14-124(b), the lower cut-off frequency (f_L) of the high-pass filter must be greater than the higher cut-off frequency (f_H) of the low-pass filter. In fact, the band of frequencies between f_H and f_L will be rejected by the filter. Generally, the gain of the two filters should be same.

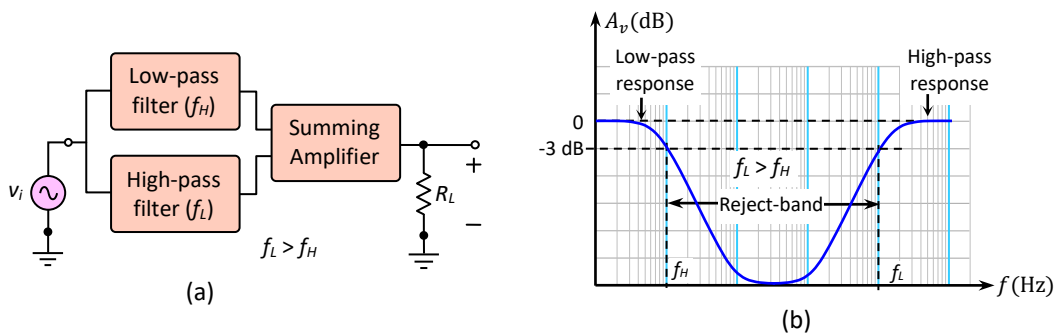


Fig.14-124: Wide band-reject filter: (a) Block diagram, and (b) Frequency response

Example 13-27

Design a band-reject filter that can eliminate the range of frequencies 1 kHz to 5 kHz. The pass-band gain of the filter should be 10.

Solution:

The centre frequency of the filter will be $f_c = \sqrt{f_L f_H} = \sqrt{5 \text{ kHz} \times 1 \text{ kHz}} = 2.236 \text{ kHz}$.

The quality factor will be,

$$\text{Using Equ.(14-212),} \quad Q = \frac{f_c}{f_L - f_H} = \frac{2.236 \text{ kHz}}{5 \text{ kHz} - 1 \text{ kHz}} \approx 0.6$$

As the Q is very low, we can implement the filter as wide band-reject filter. The higher cut-off frequency of the low-pass filter should be $f_H = 1 \text{ kHz}$, and the lower cut-off frequency of the high-pass filter should be $f_L = 5 \text{ kHz}$,

Here, we will design the filters with gain of 3 and the summing amplifier with a gain of 2. So the overall gain will be $3 \times 2 = 6$.

Low-pass section:

Let us use $C = 0.01 \mu\text{F}$. Now putting the value of C and $f_H = 1 \text{ kHz}$ into Equ.(14-179), we get,

$$R = \frac{1}{2\pi f_H C} = \frac{1}{2\pi(1 \text{ kHz})(0.01 \mu\text{F})}$$

$$\therefore R \approx 16 \text{ k}$$

Let us select $R_f = 10 \text{ k}$. Using the expression for the pass-band gain,

$$A_v = \left(1 + \frac{R_f}{R_1}\right) \text{ or, } 3 = \left(1 + \frac{10 \text{ k}}{R_1}\right)$$

$$\therefore R_1 = 5 \text{ k}$$

High-pass section:

Let us use $C = 0.01 \mu\text{F}$. Now putting the value of C and $f_L = 5 \text{ kHz}$ into Equ.(14-201), we get,

$$R = \frac{1}{2\pi f_L C} = \frac{1}{2\pi(5 \text{ kHz})(0.01 \mu\text{F})}$$

$$\therefore R \approx 3.2 \text{ k} [\text{use standard } 3.3 \text{ k}]$$

Again considering $R_f = 10 \text{ k}$,

$$3 = \left(1 + \frac{10 \text{ k}}{R_1}\right)$$

$$\therefore R_1 = 5 \text{ k} [\text{use standard } 5.1 \text{ k}]$$

Summing amplifier:

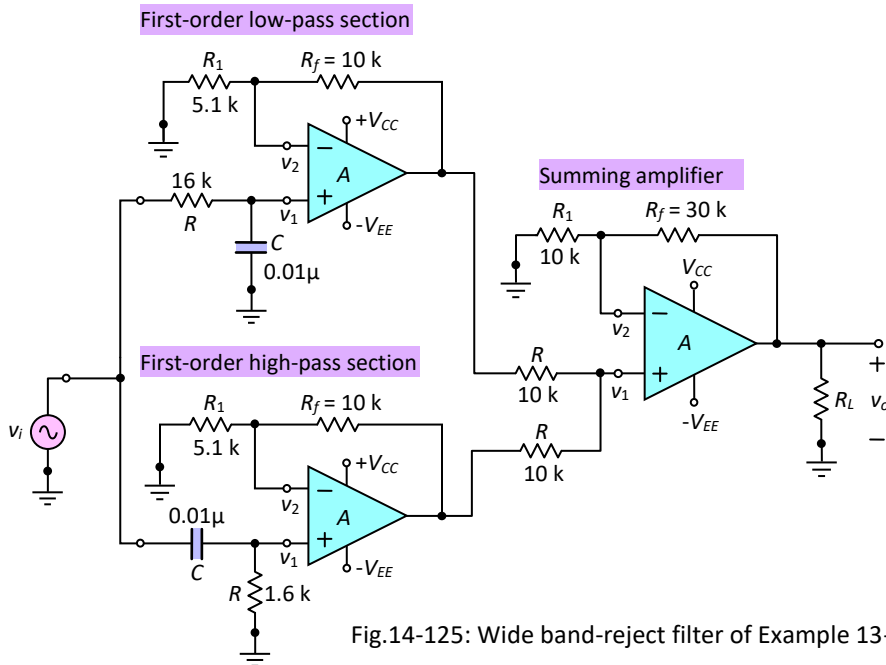


Fig.14-125: Wide band-reject filter of Example 13-27

From Equ.(14-117),

$$v_o = \left(1 + \frac{R_f}{R_1}\right) \left(\frac{v_a + v_b}{2}\right)$$

or,

$$\frac{v_o}{v_a + v_b} = 2 = \left(1 + \frac{R_f}{R_1}\right) \left(\frac{1}{2}\right)$$

Let $R_1 = 10 \text{ k}$,

$$4 = \left(1 + \frac{R_f}{10 \text{ k}}\right)$$

$\therefore$

$$R_f = 30 \text{ k}$$

$\therefore$

The design circuit is shown in Fig.14-125.

Comments: It is also possible to design the filter using two second-order filters or even more higher-order filters.

14.45.14 Narrow Band-Reject Filter

This filter is also called a **notch filter**. It is commonly used to eliminate a single frequency such as 50 Hz power line frequency hum. The most widely used notch filter is the twin-T network as shown in Fig.14-126(a). The frequency response of the active notch filter is shown in Fig.14-126(b).

Notch filters are most commonly used in communications and biomedical instruments for eliminating the undesired frequencies. A mathematical analysis of this circuit shows that it acts as a lead-lag circuit with a phase angle, shown in Fig.14-(b). There is a frequency f_c at which the phase shift is equal to 0° . In Fig.14-125(b), the voltage gain is equal to 1 (0 dB) at low and high frequencies. In between, there is a frequency f_c at which voltage gain drops to almost zero. Thus the filter notches out, or blocks frequencies near f_c . The frequency at which maximum attenuation occurs is called the **notch-out frequency** or simply **notch frequency** and is given by,

$$\therefore f_N = \frac{1}{2\pi RC} \quad (14-220)$$

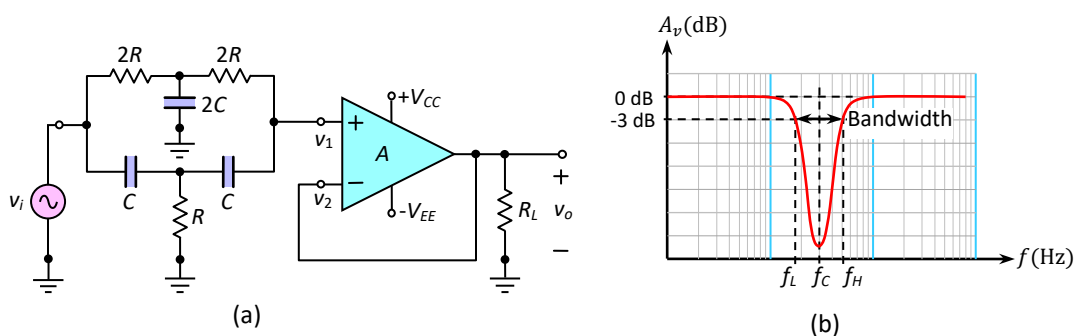


Fig.14-126: Narrow band-reject filter: (a) Circuit diagram, and (b) Frequency response

14.46 All-pass Filters

The amplitude response of the filters, discussed so far, is of primary concern. Another type of filter that leaves the amplitude of the signal unchanged, but introduces phase shift is called an **all-pass filter**. The purpose of this filter is to add phase shift (delay) to the response of the circuit. The gain of an all-pass filter is unity for all frequencies. The phase response, however, changes from 0° to 360° (for a 2-pole filter) as the frequency is swept from 0 to infinity. When signals are transmitted over transmission lines, such as telephone/co-axial wires, they undergo changes in phase. To compensate for these phase changes, all-pass filters are used. Other uses of an all-pass filter is to provide phase equalization, typically in pulse circuits, single side-band suppressed carrier (SSB-SC) modulation circuits etc. The all-pass filters are also called **delay equalizers** or **phase correctors**.

14.46.1 First-Order All-pass

A first-order all-pass filter with lagging phase shift is shown in Fig.14-127(a). The circuit has a phase shift that goes from 0° at 0 Hz to -180° at high frequency. It is -90° at $\omega = 1/RC$. The resistor may be made variable to allow adjustment of the delay (phase shift) at a particular frequency. The output of this filter can be determined using superposition theorem.

When the noninverting input is grounded, keeping v_i connected to inverting terminal, as shown in Fig.14-128(a), the circuit will work just as an inverting amplifier. So the output will be,

$$v'_o = \left(-\frac{R_f}{R_1}\right) v_i \quad (14-221)$$

When the inverting input is grounded, keeping v_i connected to noninverting terminal, as shown in Fig.14-128(b), the circuit will work as a low-pass filter. The low-pass filter has been analyzed in previous section. The output can be calculated using Equ.(14-178),

$$v''_o = \left(1 + \frac{R_f}{R_1}\right) \frac{v_i}{1 + j2\pi fRC} \quad (14-222)$$

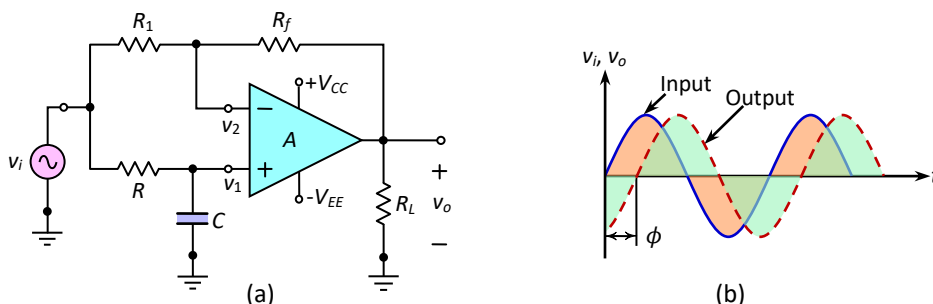


Fig.14-127: First-order all-pass filter: (a) Circuit diagram, and (b) Phase change response

According to the superposition theorem the overall output will be,

$$v_o = v'_o + v''_o = \left(-\frac{R_f}{R_1}\right) v_i + \left(1 + \frac{R_f}{R_1}\right) \frac{v_i}{1 + j2\pi fRC}$$

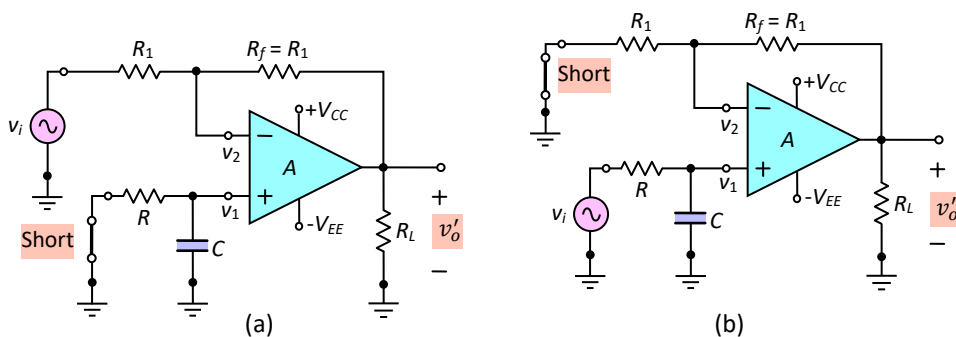


Fig.14-128: First-order all-pass filter: (a) Working as inverting amplifier, and (b) Working as noninverting amplifier

Let us assume, $R_f = R_1$. Therefore,

$$v_o = -v_i + \frac{2v_i}{1 + j2\pi fRC}$$

So, the gain of the circuit will be,

$$A_{vf} = \frac{v_o}{v_i} = -1 + \frac{2}{1 + j2\pi fRC} = \frac{-1 - j2\pi fRC + 2}{1 + j2\pi fRC}$$

$$A_{vf} = \frac{1 - j2\pi fRC}{1 + j2\pi fRC} \quad (14-223)$$

The magnitude of this gain will be,

$$|A_{vf}| = \frac{\sqrt{1^2 + (2\pi fRC)^2}}{\sqrt{1^2 + (2\pi fRC)^2}} \quad (14-224)$$

$$\boxed{|A_{vf}| = 1} \quad (14-225)$$

As the magnitude of the gain is unity, the output will be exactly same as the input signal. The gain expression [Equ.(14-220)] can be represented in polar form as,

$$|A_{vf}| = 1 \angle \frac{\tan^{-1}(-2\pi fRC/1)}{\tan^{-1}(2\pi fRC/1)}$$

$$\text{or,} \quad |A_{vf}| = 1 \angle [-\tan^{-1}(2\pi fRC) - \tan^{-1}(2\pi fRC)]$$

$$\therefore \quad |A_{vf}| = 1 \angle -2\tan^{-1}(2\pi fRC)$$

That is, the output will lag the input by an angle given by,

$$\boxed{\phi = -2\tan^{-1}(2\pi fRC)} \quad (14-226)$$

The range of phase shift of this filter will be -180° (for $f = \infty$) to 0° (for $f = 0$).

The all-pass filter that produces leading phase shift is shown in Fig. 13-123. Here, the position of the R and C has been interchanged.

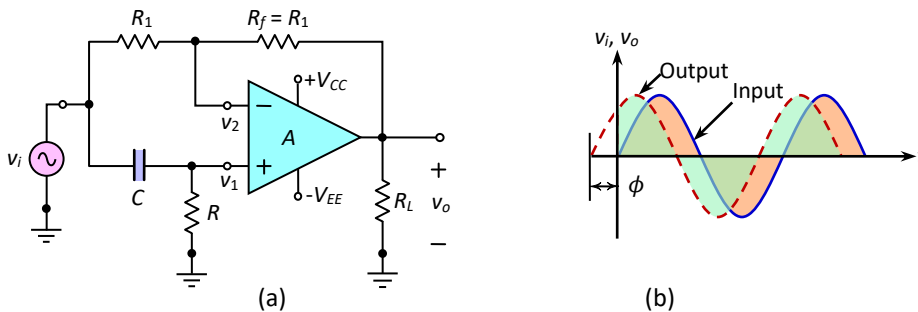


Fig.14-129: First-order all-pass filter: (a) Circuit diagram, and (b) Phase change response

This circuit can also be analyzed in the same way as lagging all-pass filter. The value of the gain will be same as before, that is, $|A_{vf}| = 1$, and the phase angle will be,

$$\phi = 180^\circ - 2\tan^{-1}(2\pi fRC) \quad (14-227)$$

The range of phase shift of this filter will be 180° (for $f = 0$) to 0° (for $f = \infty$).

14.47 Switched Capacitor Filters

Switched capacitor filters (SCFs) are active IC filters made of op-amps, capacitors, and transistor switches. Also known as **analog sampled data filters** or **commutating filters**, these devices are usually implemented with MOS or CMOS circuits. The primary advantage of SCFs is that they provide a way to make tuned or selective circuits in an IC without the use of discrete inductors, capacitors, or resistors. All components are fully integrated on a single chip, making external discrete components unnecessary. The secret to the SCF is that all resistors are replaced by capacitors that are switched by MOSFET switches. Resistors are more difficult to make in IC form and take up far more space on the chip than transistors and capacitors. With switched capacitors, it is possible to make complex active filters on a single chip. Other advantages are select ability of filter type, full adjustability of the cut-off or center frequency, and full adjustability of bandwidth. One filter circuit can be used for many different applications and can be set to a wide range of frequencies and bandwidths.

The circuit in Fig.14-130 consists of two MOSFETs and a capacitor (C). A two-phase clock provides complementary but nonoverlapping gate pulses (ϕ_1 and ϕ_2). When a clock pulse is HIGH, the corresponding transistor turns ON; when the gate pulse is LOW, the transistor is OFF.

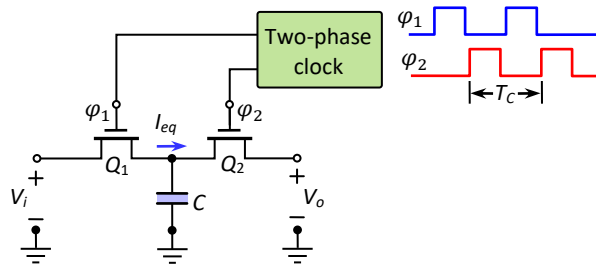


Fig.14-130: Simulation of resistance using switched capacitor system

When ϕ_1 goes HIGH, Q_1 turns on and capacitor C charges up to V_i . When ϕ_2 goes HIGH, Q_2 turns ON and capacitor C discharges to V_o (assuming $V_i > V_o$). The amount of charge transferred during this process is $Q = C(V_i - V_o)$ and the transfer occurs during one clock period T_c . The equivalent current is then,

$$I_{eq} = \frac{Q}{T_c} = \frac{C(V_i - V_o)}{T_c} = Cf_c(V_i - V_o)$$

The average current can also be calculated considering an average effective resistance (R_{eq}) as,

$$I_{eq} = \frac{V_i - V_o}{R_{eq}}$$

Combining these two equations we get,

$$C f_C (V_i - V_o) = \frac{V_i - V_o}{R_{eq}}$$

∴

$$R_{eq} = \frac{1}{C f_C} \quad (14-228)$$

Therefore, the equivalent resistance is inversely proportional to the switching frequency f_C . Using this technique, we can simulate an equivalent resistance by alternately charging and discharging a capacitor between two voltage levels. A large equivalent resistance can be simulated by using a small capacitance and an appropriate clock frequency. The circuit in Fig.14-130 is therefore called a **switched-capacitor circuit**.

14.47.1 Example of Switched Capacitor Filter

Any filters can be implemented in a monolithic IC using switched capacitor technique. Fig.14-131 shows a first-order low-pass filter. The transfer function of this filter will be,

$$|A_{vf}| = \left| \frac{v_o}{v_i} \right| = \frac{R_f}{R_1} \frac{1}{\sqrt{1 + (f/f_H)^2}} \quad (14-229)$$

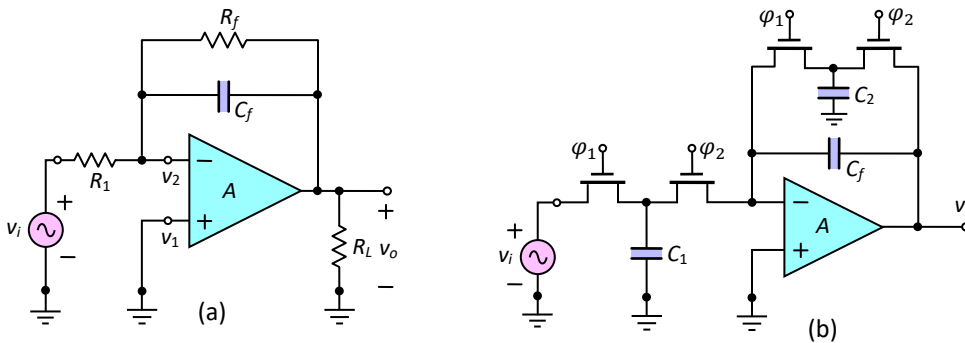


Fig.14-131: (a) First-order low-pass filter, and (b) Same filter designed using switched capacitor

and the cut-off frequency is

$$f_H = \frac{1}{2\pi R_f C_f} \quad (14-230)$$

If a 10 kHz cut-off frequency is required and if $C_f = 100$ pF, then the R_f resistance required is approximately 159 k. In addition, if a gain of -5 is desired, then resistance R_1 must be 31.8 k. The equivalent switched-capacitor filter is shown in Fig.14-131(b). Here the required resistances have been simulated using switched capacitor system. Let the frequency of the two-phase clock be f_C .

Therefore, the value of the resistances will be, $R_{feq} = 1/(f_c C_2)$ and $R_{1eq} = 1/(f_c C_1)$. The magnitude of gain is still given by Equ.(14-229). Then, the expression for the voltage gain will be,

$$|A_{vf}| = \left| \frac{v_o}{v_i} \right| = \frac{1/(f_c C_2)}{1/(f_c C_1)} \frac{1}{\sqrt{1 + (f/f_H)^2}}$$

$$\therefore |A_{vf}| = \left| \frac{v_o}{v_i} \right| = \frac{C_1}{C_2} \frac{1}{\sqrt{1 + (f/f_H)^2}}$$

The low-frequency gain is $-C_1/C_2$, which is just the ratio of two capacitances, and higher cut-off frequency will be,

$$\boxed{f_H = \frac{f_c C_2}{2\pi C_f}} \quad (14-231)$$

which is also proportional to the ratio of two capacitances. This means that switched-capacitor filter characteristics can be precisely controlled simply by controlling the values of capacitances and the switching frequency.

14.48 Sample Questions

1. What is transistor differential amplifier? Write some important features of a transistor differential amplifier.
2. Write the names of four configurations in which a differential amplifier can be used.
3. Describe the operation process of a transistor differential amplifier. Why one input of a transistor differential amplifier is called inverting input and the other is non-inverting input?
4. Draw the circuit diagram of a transistor differential amplifier and calculate its DC operating point.
5. Draw the circuit diagram of a transistor differential amplifier and calculate its tail current.
6. Draw the circuit diagram of a BJT differential amplifier and explain how this amplifier amplifies the difference to two input signals.
7. Draw the circuit diagram of a dual input balanced output BJT differential amplifier and derive the expression for the voltage gain, input impedance and output impedance.
8. Draw the circuit diagram of a single input balanced output BJT differential amplifier and derive the expression for the voltage gain, input impedance and output impedance.
9. Draw the circuit diagram of a single input unbalanced output BJT differential amplifier and derive the expression for the voltage gain, input impedance and output impedance.
10. Draw the circuit diagram of a dual input unbalanced output BJT differential amplifier and derive the expression for the voltage gain, input impedance and output impedance.
11. Draw the hybrid equivalent circuit of a single-input balanced output differential amplifier and derive the expression for the voltage gain.
12. What is common mode voltage gain and differential voltage gain? Why do we prefer low common mode voltage gain?
13. What is Common-Mode Rejection Ratio (CMRR)? Draw the equivalent circuit of a BJT differential amplifier and from this equivalent circuit derive the expression for CMRR.

14. What is constant current source? With a suitable diagram explain how a BJT can be used as a constant current source.
15. Why constant circuit source is used in differential amplifiers?
16. Draw the circuit diagram of a constant current source using BJT and Zener diode and derive the expression for the load (collector) current.
17. What is current mirror circuit? Draw a current mirror circuit using BJT and explain.
18. With suitable circuit diagram explain how multiple copies of reference current can be produced.
19. Draw the circuit diagram of a base current compensated current mirror current and explain how the base current is compensated.
20. What is active load? Draw a differential amplifier with active load and explain.
21. Draw the block diagram of the internal architecture of an operational amplifier. Explain the function of different units.
22. Draw the simplified schematic circuit diagram of the operational amplifier 741 and explain the function of different sections.
23. Draw the pin-diagram of 741 operational amplifier.
24. Draw the electrical symbol of an operational amplifier and explain.
25. Draw the equivalent circuit of an operational amplifier and explain.
26. What is split power supply? Explain how single polarity power supply can be converted into a split power supply.
27. Give very short description on the following terms of operational amplifier:
 - i) Bias current, ii) Input offset current iii) Output short circuit current
 - iv) Input offset current
28. Explain the following terms in context of op-amp:
 - i) Input offset voltage ii) Output offset voltage iii) Input voltage adjustment range
 - iv) Input voltage range v) Supply voltage sensitivity vi) Peak output voltage
29. Describe the following terms in the context of op-amp:
 - v) Differential input resistance vi) Input capacitance x) Large signal differential voltage gain xii) Input impedance xiii) Output impedance Power consumption
30. What is CMRR of an Op-Amp? Explain how CMRR can be calculated in dB scale.
31. What do you mean by steady-state response and transient response? Draw the transient response graph of a typical operation amplifier and explain rise time and overshoot.
32. What is slew rate? Draw the slew-rate limited response of an operational amplifier and explain how to calculate the value of slew rate.
33. What is gain-bandwidth product? Draw the frequency response of 741 op-amp and determine its gain-bandwidth product.
34. Draw the transfer characteristics of an op-amp and describe.
35. Write the ideal characteristics of an operational amplifier.
36. Write the characteristics of a practical operational amplifier.
37. What are the meanings of the following symbols as per the datasheet given in section 13.17?

i) T_{OPR} ii) T_{STG} iii) $V_{IO(R)}$ iv) G_V v) $PSRR$ vi) I_{CC} vii) P_C

38. What do you mean by open-loop and closed-loop gain of an amplifier- explain with figures.
39. Draw the circuit diagram of an open-loop differential amplifier and explain. What are the main limitations of this amplifier?
40. Draw the circuit diagram of an open-loop noninverting amplifier and explain. Mention the limitations of this amplifier.
41. Draw the circuit diagram of an open-loop inverting amplifier and explain. Mention the limitations of this amplifier.
42. What is open-loop voltage comparator? Draw the circuit diagram of an open-loop voltage comparator using op-amp and explain its operation procedure.
43. Why we use negative feedback in amplifier? Explain with diagram how negative feedback can be employed in op-amp.
44. Draw the circuit diagram of a closed-loop noninverting amplifier and prove that the closed-loop voltage gain is $A_f = \frac{A}{1+A\beta}$.
45. Draw the circuit diagram of a closed-loop noninverting amplifier and prove that the closed-loop voltage gain is independent of the open-loop gain A .
46. Draw the circuit diagram of a closed-loop noninverting amplifier and prove that the closed-loop voltage gain is $A_f = 1 + \frac{R_f}{R_1}$.
47. Draw the circuit diagram of a closed-loop noninverting amplifier and derive the expression for the voltage gain, input impedance and output impedance.
48. prove that the closed-loop voltage gain is $A_f = 1 + \frac{R_f}{R_1}$.
49. Draw the frequency response of an op-amp and calculate the closed loop bandwidth.
- 50. Inverting**
51. Draw the circuit diagram of a closed-loop inverting amplifier and prove that the closed-loop voltage gain is $A_f = -\frac{KA}{1+A\beta}$.
52. Draw the circuit diagram of a closed-loop inverting amplifier and prove that the closed-loop voltage gain is independent of the open-loop gain A .
53. Draw the circuit diagram of a closed-loop inverting amplifier and prove that the closed-loop voltage gain is $A_f = -\frac{R_f}{R_1}$.
54. Draw the circuit diagram of a closed-loop noninverting amplifier and derive the expression for the voltage gain, input impedance and output impedance.
55. Draw the frequency response of an op-amp and calculate the closed loop bandwidth for an inverting amplifier.
56. What is virtual ground? Prove that the inverting terminal of the closed-loop inverting amplifier is a virtual ground.
57. What do you mean by linear operation of an amplifier? Will the inverting terminal of the inverting amplifier will be virtual ground if the amplifier does not work in nonlinear range?
58. How input offset voltage is produced due to the input bias current? Explain how to minimize this offset voltage using R_{om} .

59. What is voltage follower? Explain with suitable figure how a noninverting amplifier can be used as a voltage follower.
60. What is single-ended voltage and differential voltage? Draw the circuit diagram of a differential amplifier and derive the expression for the voltage gain and input impedance.
61. Draw the circuit diagram of a differential amplifier using two op-amps. What is the advantage of using the second op-amp?
62. Draw the circuit diagram of a differential amplifier using two op-amps and derive the expression for the voltage gain and input impedance.
63. What is the advantage of using the second op-amp?
64. Why do we need current to voltage converter? Explain how an inverting amplifier can be used a current to voltage converter.
65. What is summing amplifier? Draw the circuit diagram of a four-input summing amplifier and prove that the output is the sum of the inputs.
66. Explain how a summing amplifier can be used as a scaling or weighted amplifier.
67. Explain how a summing amplifier can be used as an averaging circuit.
68. What is digital-to-analog converter (DAC)? Can we build a DAC using summing amplifier? If so draw the circuit and explain how?
69. What is binary-to-decimal converter (BDC)? Can we build a BDC using summing amplifier? If so draw the circuit and explain how?
70. Draw the circuit diagram of a three-input noninverting summing amplifier and derive the expression for the output voltage.
71. Why an instrument amplifier is different from other amplifiers? Draw the circuit diagram of an instrument amplifier and prove that the differential gain of this amplifier can be adjusted by changing the value of a single resistor (R_{gn}).
72. What is an integrator circuit? Draw the circuit diagram of an op-amp integrator and find the expression for the voltage.
73. Draw the circuit diagram of an op-amp integrator and prove that the output voltage is equal the integration of the input voltage.
74. What are the limitations of the basic integrator? Explain how these limitations can be overcome.
75. Draw the circuit diagram of a practical integrator and explain its operation with its frequency response graphs.
76. different.
77. What is a differentiator? Draw the circuit diagram of an op-amp differentiator and find the expression for the output voltage.
78. Draw the circuit diagram of an op-amp differentiator and prove that the output is equal the first derivative of the input.
79. What are the limitations of the basic differentiator? Explain how these limitations can be overcome.
80. Draw the circuit diagram of a practical differentiator and explain its operation with its frequency response graphs.

81. What is a logarithmic amplifier? Draw the circuit diagram of a logarithmic amplifier using a diode with op-amp and prove the output voltage is proportional to the logarithm of the input voltage.
82. Explain how the logarithmic circuit of question 81 can be used to determine the 10-base log.
83. Draw the circuit diagram of a logarithmic amplifier using a BJT with an op-amp and prove the output voltage is proportional to the logarithm of the input.
84. Draw the circuit diagram of a temperature compensated logarithmic amplifier using diodes and explain how this circuit compensates the effect of temperature.
85. Draw the circuit diagram of a temperature compensated logarithmic amplifier using bipolar junction transistors and explain how this circuit compensates the effect of temperature.
86. What is an antilog amplifier? Draw the circuit diagram of an antilog amplifier using diode and op-amp and prove the output voltage is proportional to the antilog of the input voltage.
87. Draw the circuit diagram of an antilog amplifier using a BJT and an op-amp and prove the output voltage is proportional to the antilog of the input.
88. What is an analog multiplier circuit? Prove that the output is proportional to the product of the input signals.
89. What is the precision rectifier? Why do we need a precision rectifier?
90. Draw the circuit diagram of a half-wave precision rectifier using a single diode and op-amp and describe its operation. What is the limitation of this circuit?
91. Explain how the limitation of the precision rectifier of question no. 90 can be removed using a second diode?
92. Draw the circuit diagram of a full-wave precision rectifier using diodes and op-amp and describe its operation.
93. Why do we need an AC amplifier? Draw the circuit diagram of a capacitor coupled AC amplifier with op-amp and explain its operation.
94. Explain how can we calculate the value of input and output coupling capacitors.
95. Normally a dual (split) power supply is used in op-amp amplifiers. But sometimes we need to use a single power supply in these amplifiers. Explain how we can implement an AC noninverting amplifier using a single power supply.
96. Explain how we can implement an AC inverting amplifier using a single-mode power supply.
97. Slew rate is the maximum rate of change of output voltage of an op-amp. Definitely it has an effect on the high frequency square wave. Is there any effect of slew rate on the sine wave – explain your answer.
98. The maximum frequency an op-amp can properly amplify is limited by its slew rate. Derive the expression for that slew rate limited maximum frequency.
99. Prove that the maximum slew rate limited frequency of an op-amp is $f_m = \frac{10^6 SR}{2\pi A_f V_{pi}}$
100. What is the problem with the open-loop comparator? Explain with an example.
101. What is a Schmitt trigger circuit? Explain how the problem mentioned in question 100 is overcome in a Schmitt trigger circuit.
102. Draw the circuit diagram of an inverting Schmitt trigger circuit and explain its operation.
103. Draw the circuit diagram of a noninverting Schmitt trigger circuit and derive the expression for the upper trip level (HTL) and lower trip level (LTL).

104. Draw the circuit diagram of a noninverting Schmitt trigger circuit and explain its operation.
105. Draw the circuit diagram of a noninverting Schmitt trigger circuit and derive the expression for the upper trip level (HTL) and lower trip level (LTL).
106. Mention some important applications of comparators.
107. What is over-temperature controlling circuit? Draw the schematic diagram of a over-temperature controlling circuit and explain its operation.
108. What is active filter? Give a comparison between passive filters and active filters.
109. Give the classification of filters depending on the pass-band/stop-band frequencies and draw their ideal and practical responses.
110. Give the classification of filters depending on the design techniques and draw their characteristic response graphs.
111. Give a comparison between analog and digital filters.
112. What do you mean by audio and radio filters?
113. What is the order of a filter? Draw the characteristic responses of first-order and second-order Butterworth filters.
114. Draw the circuit diagram of a first-order low-pass Butterworth active filter and its frequency response and explain.
115. Draw the circuit diagram of a first-order low-pass Butterworth active filter and derive the expression for the higher cut-off frequency.
116. Draw the circuit diagram of a 2nd-order low-pass Butterworth active filter and its frequency response and explain.
117. Draw the circuit diagram of a 2nd-order low-pass Butterworth active filter and derive the expression for the higher cut-off frequency.
118. Draw the circuit diagram of a first-order high-pass Butterworth active filter and its frequency response and explain.
119. Draw the circuit diagram of a first-order high-pass Butterworth active filter and derive the expression for the higher cut-off frequency.
120. Draw the circuit diagram of a 2nd-order high-pass Butterworth active filter and its frequency response and explain.
121. Draw the circuit diagram of a 2nd-order high-pass Butterworth active filter and derive the expression for the higher cut-off frequency.
122. Draw the circuit diagram of a third-order low-pass Butterworth filter and explain.
123. Draw the circuit diagram of a 4th-order low-pass Butterworth filter and explain.
124. What is band-pass filter? What are the types of band-pass filters - explain?
125. Draw the circuit diagram of a first-order wide band-pass Butterworth filter and explain its operation with its frequency response.
126. Draw the circuit diagram of a first-order narrow band-pass Butterworth filter and explain its operation with frequency response.
127. What is band-reject filter? Write the names of different types of band-reject filters and explain.
128. Draw the circuit diagram of a first-order wide band-reject Butterworth filter and explain its operation with its frequency response.

129. Draw the circuit diagram of a first-order narrow band-reject Butterworth filter and explain its operation with frequency response.
130. What is all-pass filter? What are the applications of this filter?
131. Draw the circuit diagram of a first-order all-pass filter and explain its operation with the help input and output waveforms.
132. Draw the circuit diagram of a first-order all-pass filter and derive the expression for the voltage gain and phase angle.
133. What is the switched capacitor filter? Why do we need this type of filter?
134. Explain how we can simulate resistance using switched capacitor system.
135. What is the switched capacitor system? Derive the expression for the equivalent resistance simulated by a switched capacitor.
136. With appropriate figures, explain how a first-order low-pass filter can be implemented using switched capacitor.
137. Draw the circuit diagram of a first-order low-pass filter implemented using switched capacitor and derive the expression for the higher cut-off frequency.

14.49 Exercise

1. For the differential amplifier of Fig.14-132 determine the DC operating point and the value of tail current.

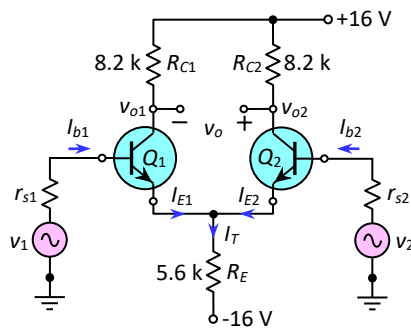


Fig.14-132: Differential amplifier for Ex. No. 1

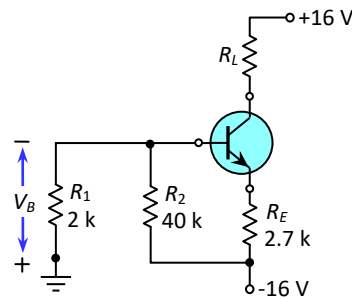


Fig.14-133: Constant current source for Ex. No. 3

2. For the differential amplifier of Ex. No. 1, calculate the input impedance, output impedance and gain in four modes of operation. Assume $h_{fe} = 120$, $r_o = 250 \text{ k}$.
3. The circuit of Fig.14-133 is a constant current source. Calculate the value of the constant current, and also determine the range of R_L for which the circuit will work as a constant current source.
4. For circuit of Fig.14-134, estimate the value of CMRR with and without the constant current source.

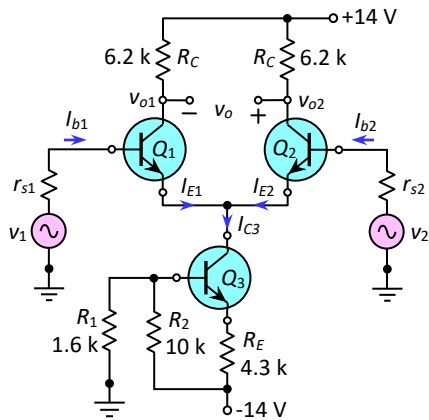


Fig.14-134: Differential amplifier with constant current source

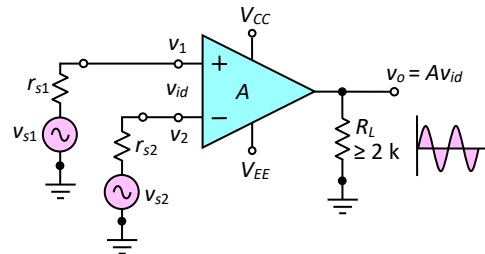


Fig.14-134: Open-loop differential amplifier

5. Calculate the DC operating condition of different stages of the circuit of Fig.14-28. Assume: $\beta = 150$, $I_s = 10^{-14}$ A for all NPN transistors and $\beta = 60$, $I_s = 10^{-14}$ A for all PNP transistors, but for Q_{14} and Q_{20} , $I_s = 4 \times 10^{-14}$ A. In Q_{13} , 30% collector current goes to output section, and 70% collector current flows downward. Assume that the supply voltage is ± 16 V.
6. OP97 is a precision op-amp. The typical values of large-signal differential voltage gain (A_{dm}) and the CMRR is 200 V/mV, and 150 dB, respectively. Calculate the value of common-mode voltage gain (A_{cm}) in absolute value and in dB.
7. For the circuit of Fig.14-47(a) determine the minimum value of input voltage for which the output will be saturated. Assume $A = 2.5 \times 10^6$ and the maximum output voltage swing is ± 14 V.
8. For the differential amplifier of Fig.14-135, assume that the open-loop gain $A = 2 \times 10^5$, $v_{s1} = 40 \mu\text{V}$ (RMS), $v_{s2} = 10 \mu\text{V}$ (RMS) and the maximum output voltage swing is ± 12 V. Neglecting the voltage dropped across the source resistance and assuming both the input signals have same frequency and same phase, determine the RMS output voltage and draw the waveform. Repeat the same process considering $v_{s1} = 100 \mu\text{V}$ (RMS).
9. Calculate and draw the output voltage for the inverting amplifier of Fig.14-135. If
 - c) $v_{s1} = 12 \mu\text{V} \sin(\omega t)$
 - d) $v_{s1} = 10 \text{ mV} \sin(\omega t)$
 Assume the open-loop gain of the op-amp is 2×10^5 , supply voltage is ± 16 V, and neglect the voltage drop across the source internal resistance, r_s .
10. For the non-inverting amplifier of Fig.14-136, determine closed-loop gain, closed-loop input impedance, closed-loop output impedance, and bandwidth. Assume $A = 2 \times 10^5$, $f_o = 5$ Hz, $Z_i = 1 \text{ M}\Omega$, and $Z_o = 75 \Omega$.

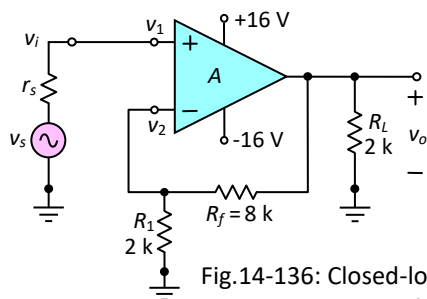


Fig.14-136: Closed-loop noninverting amplifier

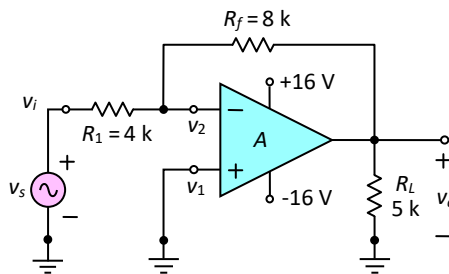


Fig.14-136: Closed-loop inverting amplifier

11. For the inverting amplifier of Fig.14-137, determine closed-loop gain, closed-loop input impedance, closed-loop output impedance, and bandwidth. assume $A = 2 \times 10^5$, $UGB = 1$ MHz, $Z_i = 1$ M Ω , and $Z_o = 75$ Ω
12. For the voltage follower of Fig.14-64, assume $A = 2 \times 10^5$, $f_o = 6$ Hz, $Z_i = 1.2$ M Ω , and $Z_o = 70$ Ω . Determine A_f , Z_{if} , Z_{of} and bandwidth.
13. For the differential amplifier of Fig.14-67, assume $R_1 = R_3 = 5$ k, $R_f = R_2 = 20$ k and the op-amps have $Z_i = 1.2$ M Ω , $Z_o = 100$ Ω , and $A = 2.5 \times 10^5$. Calculate the differential voltage gain, input impedance seen by the sources.
14. In the differential amplifier of Ex. No. 13, $v_x = (5$ mV) $\sin \omega t$ and $v_x = (2.5$ mV) $\sin \omega t$ are used. Calculate the output v_o . Draw the input and output waveforms.
15. An 8-input DAC is designed as the circuit of Fig.14-71 with $R_a = 1$ k, and $R_f = 2$ k. Calculate the value of the resistance of last (least significant) input R_h . Also calculate the step-size, and full-scale output voltage for this DAC. Assume, logic '1' = 5 V or logic '0' = 0 V.
16. Fig.14-138 shows a 4-bit binary to decimal converter. Assuming logic '1' = 1 V and logic '0' = 0 V, determine the output voltage for the binary input of 1011.

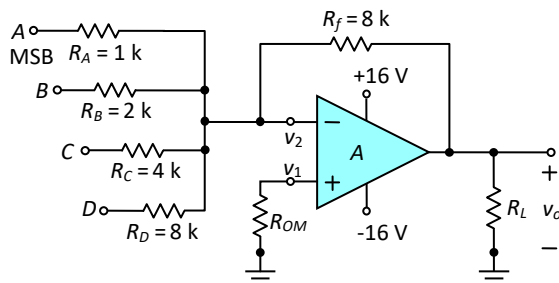


Fig.14-138: Binary to decimal converter

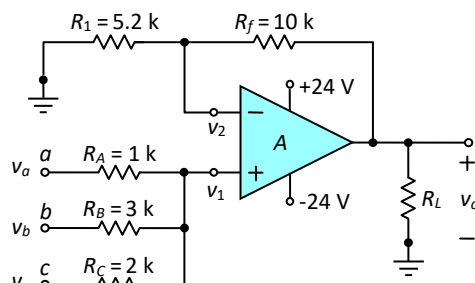


Fig.14-138: Noninverting summing amplifier

17. Calculate the voltage gain of the noninverting summing amplifier of Fig.14-139. Also calculate the output voltage for the input voltages of $v_a = 5$ V, $v_b = 3$ V and $v_c = 1$ V.
18. An instrumentation amplifier like that of Fig.14-73 is designed with $R_1 = R = 2$ k, $R_f = 120$ k, and a potentiometer in place of R_{gn} . If R_{gn} can be varied from 10 Ω to 150 k, calculate the lowest and highest gain of the instrumentation amplifier.
19. Suppose an integrator is so designed that its time constant is $R_1 C_f = 10$ s. Draw the input and output waveforms for

- i) A 14 V peak-to-peak square wave with the frequency of 50 Hz.
- ii) A 14 V peak-to-peak sinusoidal wave with the frequency of 50 Hz.
- iii) A 14 V peak-to-peak triangular wave with the frequency of 50 Hz.

Assume that the initial output of the integrator is +1 V.

20. Suppose a differentiator is designed with time constant is $R_1 C_f = 5$ s. Draw the input and output waveforms for
- iv) A 2 V peak-to-peak square wave with the frequency of 50 Hz.
 - v) A 2 V peak-to-peak sinusoidal wave with the frequency of 50 Hz.
 - vi) A 2 V peak-to-peak triangular wave with the frequency of 50 Hz.
- Assume that the initial output of the integrator is +1 V.
21. For the practical integrator circuit of Fig.14-140, draw the frequency response (graph). Assume that the open-loop gain of the op-amp is 100 dB.

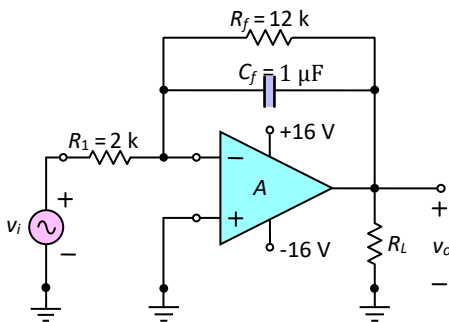


Fig.14-140: Practical integrator

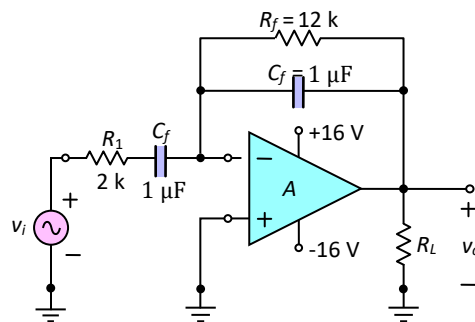


Fig.14-140: Practical differentiator

22. For the practical differentiator circuit of Fig.14-141, draw the frequency response (graph). Assume that the open-loop gain of the op-amp is 100 dB.
23. Design a practical integrator circuit for the lowest frequency of 10 kHz.
24. Design a practical differentiator circuit that will be used to differentiate signals of lowest frequency 5 kHz.
25. Determine the output voltage for the circuit of Fig.14-81, if $v_i = 1.2$ V, $R_1 = 5$ k, $I_s = 10$ nA. Also determine the output for inputs of 2.4 V and 4.8 V. Assume, $V_T \approx 26$ mV.
26. A closed-loop noninverting amplifier is designed with $A_f = 200$ using 741 operational amplifier that has slew-rate of 0.5 V/ μ s at this gain. The input signal is $v_i = (5 \text{ mV}) \sin \omega t$. Assuming that the amplifier does never saturate, determine the maximum frequency of the input signal and the output peak-to-peak value. Draw the output waveform if the signal frequency is 200 kHz considering slew-rate of ∞ and 0.5 V/ μ s.
27. For the inverting comparator of Fig.14-100(a), assume $V_{ref} = 2.5$ V, $R_1 = 5$ k, $R_f = 10$ k, and the maximum output voltage is $\pm V_{max} = \pm 10$ V.
- i) Calculate the value of UTL, LTL, and hysteresis. Draw the transfer characteristic curve and output waveform for a sinusoidal input of peak value 16 V.
 - ii) Repeat (i) considering $V_{ref} = 0$ V.
 - iii) Repeat (i) considering $V_{ref} = 0$ V, and $-V_{max} = 0$ V.
28. For the noninverting comparator of Fig.14-103(a), assume $V_{ref} = 3$ V, $R_1 = 5$ k, $R_f = 47$ k, and the maximum output voltage is $\pm V_{max} = \pm 15$ V.

- i) Calculate the value of UTL, LTL, and hysteresis. Draw the transfer characteristic curve and output waveform for a sinusoidal input of peak value 6 V.
 - ii) Repeat (i) considering $V_{ref} = 0$ V.
 - iii) Repeat (i) considering $V_{ref} = 0$ V, and $-V_{max} = 0$ V.
29. Fig.14-142 shows the transfer characteristics of three inverting comparator circuits. Design these comparator circuits.

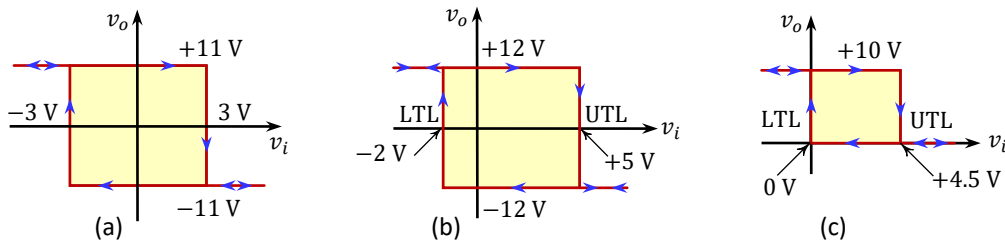


Fig.14-141: Transfer characteristics of Ex. No. 13-291

30. Design noninverting comparator circuits to achieve the transfer characteristics given in Fig.14-143.

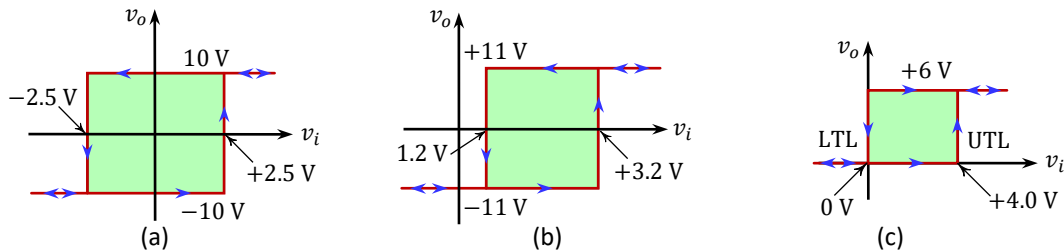


Fig.14-142: Transfer characteristics of Ex. No. 13-30

31. Design a first-order low-pass filter with the higher cut-off frequency of 15 kHz and the pass-band gain of 10.
32. Design a second-order Butterworth low-pass filter with the cut-off frequency of 12 kHz. Also, show the frequency response of the filter in a graph.
33. Design a second-order Butterworth high-pass filter with the cut-off frequency of 50 kHz. Also, show the frequency response of the filter in a Bode plot.
34. Design a band-pass filter with $f_L = 5$ kHz, $f_H = 50$ kHz, and a pass-band gain $A_{ft} = 8$. Draw the frequency response of the filter.
35. Design a narrow band-pass filter with $Q = 6$, $f_c = 8$ kHz, and $A_f = 12$.
36. Design a wide band-reject filter that can eliminate the range of frequencies 2 kHz to 10 kHz. The pass-band gain of the filter should be 10.
37. A band reject filter (as shown in Fig.14-126) is designed with $R = 10$ k Ω and $C = 1$ μ F. Determine its notch frequency f_N .
38. For the all-pass filters of Fig.14-139, calculate the voltage gain and phase shift. Draw the given input signal and the output signal waveforms.

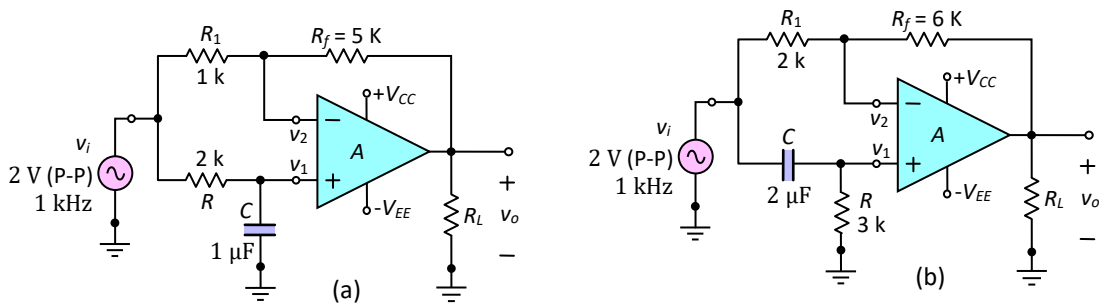


Fig.14-143: All-pass filter for Ex. No. 38

39. Calculate the equivalent resistance simulated by the switched capacitor system of Fig.14-145.
40. For the switched capacitor filter of Fig.14-146, calculate the range of voltage gain and higher cut-off frequency if the clock frequency (f_c) changes from 1 kHz to 100 kHz.

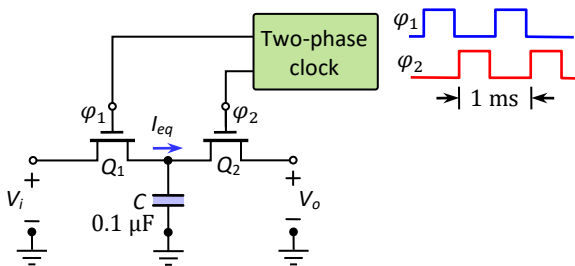


Fig.14-145: Switched capacitor system for Ex. No. 39

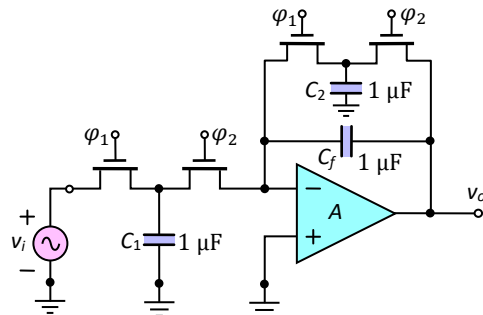


Fig.14-146: Switched capacitor system for Ex. No. 40

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